Torrent 12FFC Multi-Protocol PHY

TSMC 12nm FFC

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User Guide

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About this Document

This user guide describes the architectural features, hardware and software functionality, supported configurability and verification of the Torrent 12FFC Multi-Protocol PHY IP design. It is intended for those involved in the process of product design setup, configuration and verification.

This document is organized as follows:

Section 1 Overview on page 13

Section 2 PHY RTL Compile Options on page 21

Section 3 Configuration on page 23

Section 4 PHY Pin List on page 26

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Section 6 PHY Configuration on page 46

Section 7 Operation Procedures And Options on page 52

Section 8 PHY Operation on page 68

Section 9 PHY Register Descriptions on page 94

Section 10 PMA Configuration And Diagnostic Bus on page 116

Additional Publications

Torrent 12FFC Multi-Protocol PHY Quick Start Guide

Torrent 12FFC Multi-Protocol PHY Datasheet

Torrent 12FFC Multi-Protocol PHY Integration Guide

Table 1 lists possible acronyms and terms used in this document.

Term	Definition
AMBA™	Advanced Microcontroller Bus Architecture
APB	Advanced Peripheral Bus
ATB	Analog test bus
BIST	Built-in self test
CDB	Configuration and diagnostics bus in the PMA. (Same as the APB bus in the PHY)
CMN	Common module
CTLE	Continuous Time Linear Equalizer
DAC	Digital-to-analog converter
DFE	Decision Feedback Equalization
DSM	Delta Sigma Modulator
Gb/s	Gigabit per second
GDSII	Graphic Design System II
HVM	High Volume Manufacturing
ISI	Inter symbol interference
Lane	The digital and analog components of the PMA associated with one serial bit stream
LEC	Logic Equivalency Checking
LSB	Least significant bit
Macro	The entire PHY component being delivered
Mb/s	Megabit per second
Module	Module refers to any sub-component of the overall macro
MP-PHY	Multiprotocol PHY
PCS	Physical Coding Sub-layer
PD	Phase detector
PHY	Physical layer
PI	Phase Interpolator
PIPE	Physical Interface for the PCE Express
PLL	Phase Lock Loop
PMA	Physical Media Attachment
PPM	Parts Per Million
PRBS	Pseudo Random Binary Sequence
RX	Receiver
SDC	Synopsys™ Design Constraints
SoC	System on a Chip
SSC	Spread Spectrum Clocking
TAP	Test Access Port
TX	Transmitter
UI	Unit Interval - The amount of time taken for one serial bit of data to be transmitted
VCO	Voltage-Controlled Oscillator
VGA	Variable Gain Amplifier
XCVR	Transceiver

Table 1: Terms and Definitions

1 Overview

Cadence Torrent 12FFC Multi-Protocol PHY is designed to support multiple protocols using a single PMA hard macro with continuous frequency range support from 1Gbps to 10Gbps, and is compliant with PCIe (Gen 1/2/3), USB 3.1 (Gen1/2), XFI, XAUI, QSGMII/SGMII, USXGMII (rate compatible only, not fully electrically compliant),and SATA 3 (Gen 1/2/3) specifications. The PCS complies with the PIPE 4.4.1 standard and provides support for the dynamic equalization features of different protocols.

Figure 1 illustrates the Torrent 12FFC Multi-Protocol PHY IP components

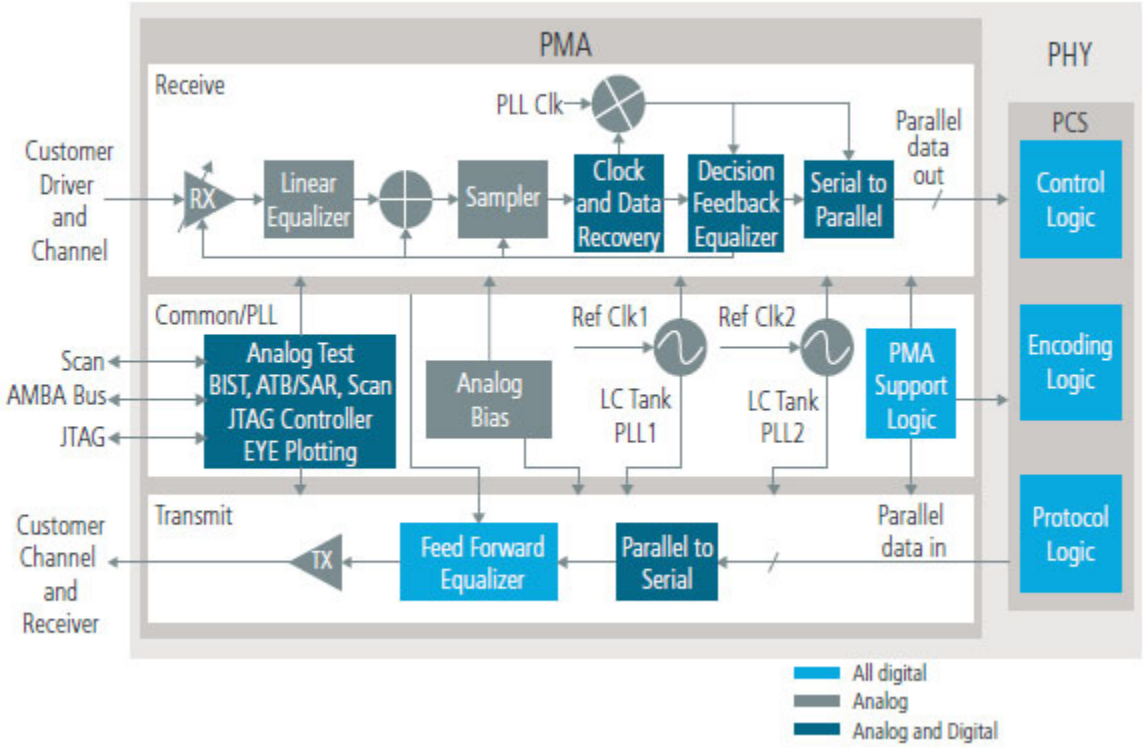


Figure 1: Torrent 12FFC Multi-Protocol PHY

1.1 Key Features

Multiple Interfaces

- PIPE 4.2 – Primary interface between PHY and controller for PCIe, USB3.2 and SATA.
  - Supports 16- or 32-bit PIPE data width.
- Raw SerDes - Provides direct connection to the PMA (bypassing the PIPE PCS) for Ethernet, DisplayPort Tx, SATA, JESD204B, and XFI.
  - 20-bit wide Tx and Rx data bus with separate Tx and Rx source synchronous clocks.
  - Data rate clock provided as separate PHY output for use as source of clock for Tx data.
- AMBA APB3 – Interface for access to PHY configuration, control and status registers.
  - 200 MHz (max) APB clock frequency.
  - 16-bit read/write data width.
  - 16-bit address width.
- JTAG – Standard TAP controller interface for silicon testing and alternate path to access APB registers.
  - Allows bypassing of APB interface (and clock) for APB register access during manufacturing test.

Diverse Functionality

- Supports the following standards
  - USB3.2 Gen 1 (5 Gbps) and Gen 2 (10 Gbps)
  - PCIe Gen 1 (2.5 Gbps), Gen 2 (5 Gbps) and Gen 3 (8 Gbps)
  - DisplayPort 1.4 RBR (1.62 Gbps), HBR (2.7 Gbps), HBR2 (5.4 Gbps) and HBR3 (8.1 Gbps)
  - Embedded DisplayPort 1.4b UI_Rate_1 through UI_Rate_8 (1.62 Gbps, 2.16 Gbps, 2.43 Gbps, 2.7 Gbps, 3.24 Gbps, 4.32 Gbps, 5.4 Gbps and 8.1 Gbps)
  - Ethernet: SGMII (1.25 Gbps), 2.5G Ethernet and 5G Ethernet
  - SATA Gen 1 (1.5 Gbps), Gen 2 (3 Gbps) and Gen 3 (6 Gbps)
  - JESD204B (various rates from 1.25 to 10.3125 Gbps)
  - XFI
- Supports lane configurations from 1 to 8 lanes.
- Multi-link support for simultaneous independent links of the same protocol:
  - Any supported combination of PCIe links with the only restriction being that lanes within a multi-lane link must be adjacent to each other. Thus, a 4-lane PHY can be configured to support 4x1 links, 1x4 link, 2x2 links, 2x1 + 1x2 links.
  - A USB3.2, SATA or Ethernet link per lane.
  - 2 DP / eDP Tx links where each link can be 1, 2 or 4 lanes per the total number of available lanes (with same restriction as PCIe that lanes in the same link must be adjacent).
  - 2 JESD204B links.
- Multi-protocol (max of 2), multi-link support for simultaneous independent links of the following combinations:
  - Single links of PCIe, DP/eDP and JESD204b (pick 2).
  - Single link of PCIe, eDP, JESD204b (pick 1) + multiple links of USB3.2, DP, SATA, Ethernet (pick 1).
  - Multiple PCIe + multiple USB3.2 links, if all PCIe and USB3.2 links use same reference clock and SSC configuration.

  - Multiple links of USB3.2, SATA and Ethernet (pick 2).
- Supports PCIExpress L1 substates.
- Robust clock/data recovery tracks static frequency offset of +/- 300 PPM, and spread spectrum up to 5000 PPM.
- 3-tap DFE and Tx FFT.
- Supports three input reference clock sources – one differential from bumps (external) and two single-ended from SoC (internal)
- External reference clock active detection.
- Supports SRIS and internal SSC generation.
  - 1 kB max payload size for PCIe.
- Single off-chip resistor required.
- Automatic calibration of on-chip termination resistors.

Testability and Debug Facilities

- IEEE 1149.1 DC JTAG boundary scan
- IEEE 1149.6 AC JTAG boundary scan
- BIST pattern generation and checking – PRBS7/15/23/31 plus user-defined pattern capability
- Programmable non-destructive receive data eye measurement
- Analog test bus with integrated 8-bit digitizer
- Digital test bus
- Jitter injection through internal loopback path
- Multiple internal loopbacks and programmable control overrides
- Separate PHY- and PMA-level signal isolation modes

1.2 PHY Architecture

The Torrent 12FFC Multi-Protocol PHY is designed with a lane-based architecture, providing greater control over floor-planning, placement, packaging, and I/O integration than other hard PHY solutions, while maintaining reliability and ease of use associated with GDSII macros.

1.3 Top Level Architecture

The Torrent 12FFC Multi-Protocol PHY consists of a Physical Media Attachment (PMA) layer and a Physical Coding Sub-layer (PCS). What follows is PHY level and PMA level block diagram with comprehensive description of their underlying building blocks.

1.4 PHY-Level Architecture

1.4.1 PHY Level Block Diagram

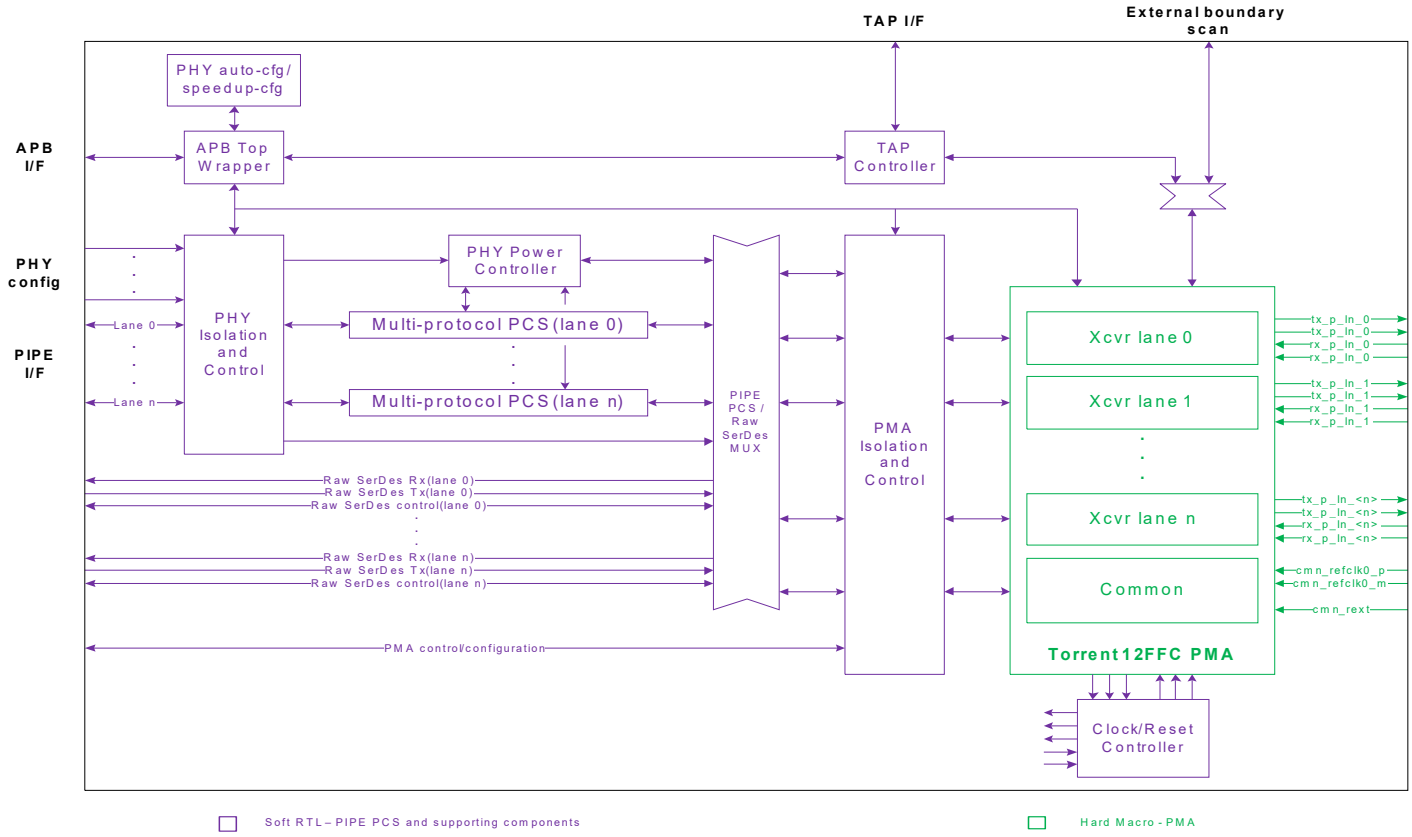


Figure 2: Torrent 12FFC Multi-Protocol PHY block diagram

1.5 PMA-Level Architecture

1.5.1 PMA Level Block Diagram

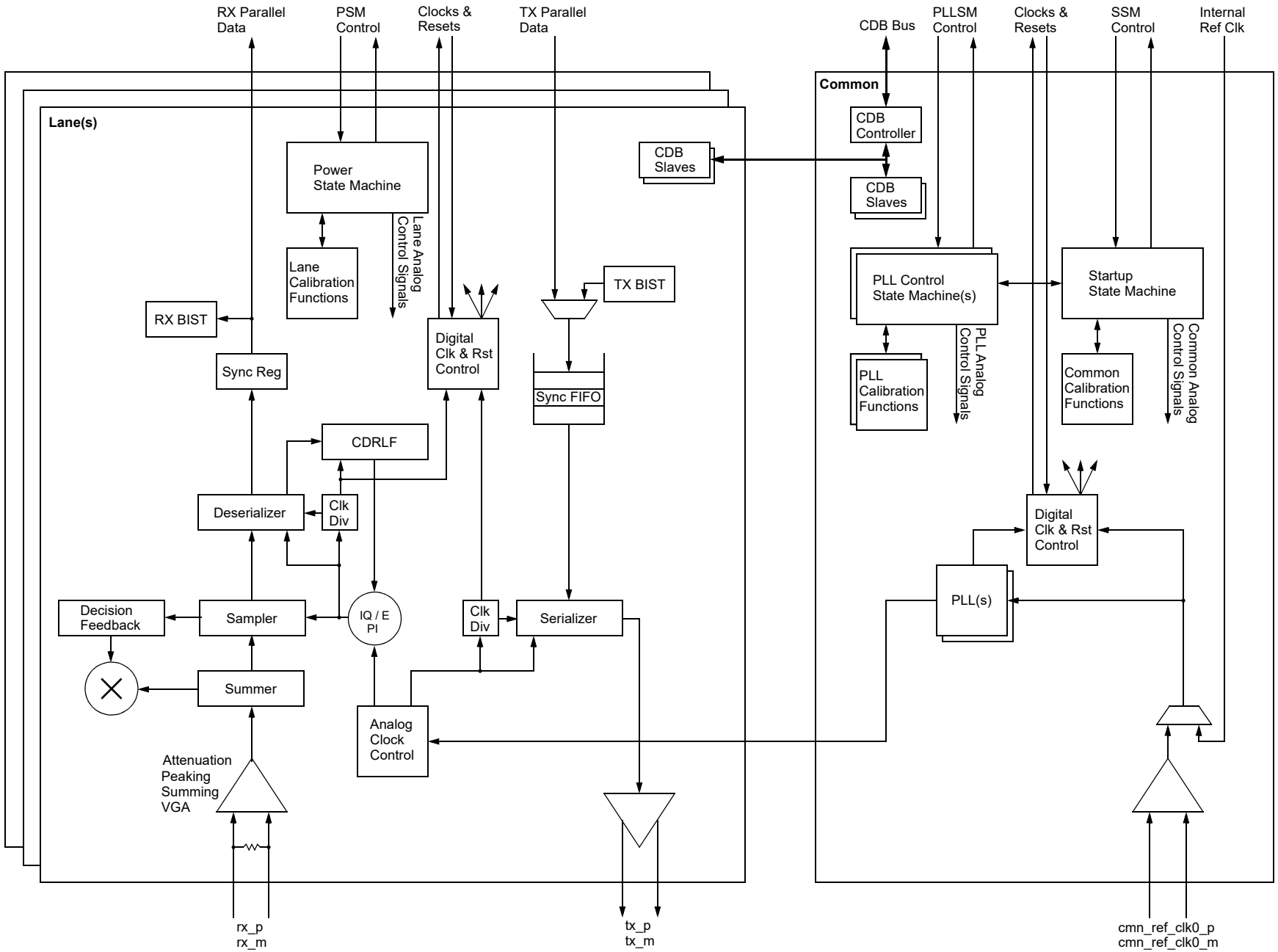


Figure 3: PMA macro block diagram

1.5.2 Common Area Target

- 324um x 720um hard macro common block

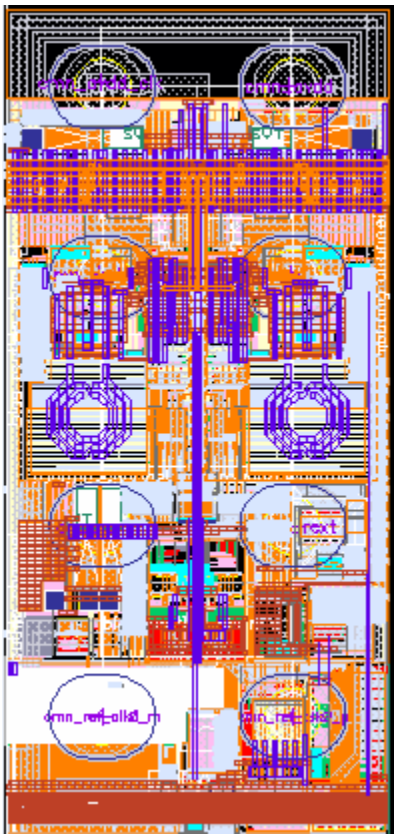


Figure 4: Common Area

1.5.3 Lane Area Target

324um x 720um hard macro transceiver block

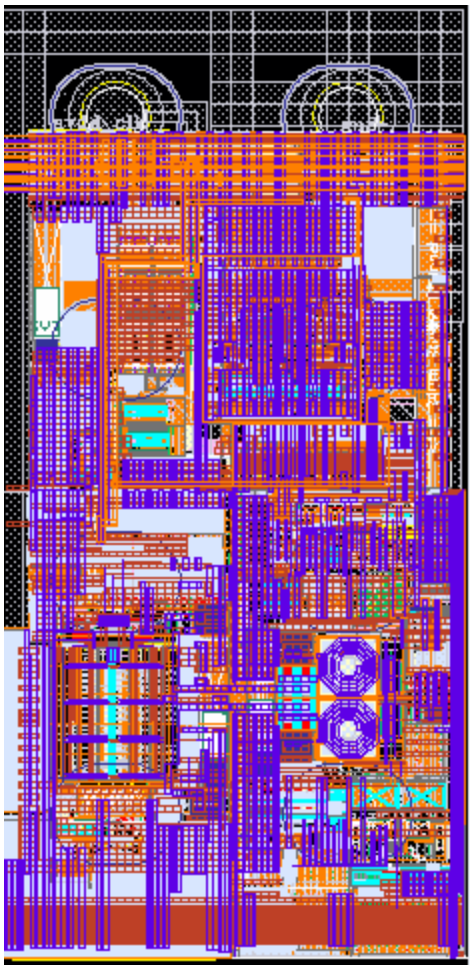


Figure 5: Lane Area

1.6 Reliability

1.6.1 Lifetime

- Mission profile MTBF reliability of 10 years lifetime at 110C and Vdd + 10%

1.6.2 ESD

Supply Domain	Min	Typ	Max	Units	Description
HBM	2.0			kV	Human Body Model Target
CDM	6.0			A	Charged-Device Model Target

Table 2: ESD Protection Targets

1.7 Process Flavor

- TSMC 12FFC technology

1.8 Components Used

- RVT, LVT, and SLVT devices

1.9 Metal Stack

- 2Xa1Xd_h_3Xe_vhv_2z_UT-APRDL
- 2Xa1Xd_h_4Xe_vhvh_2r_UT-APRDL
- 2Xa1Xd_h_3Xe_vhv_2y2r_UT-APRDL

1.10 Bump Pitch

- 162 um bump pitch

1.11 Poly Orientation

- North/South poly orientation

1.12 Third Party Licensing Requirement / Restriction

- tcbn12ffcllbwp12p90 Ver. 090a
- tcbn12ffcllbwp12p90lvt Ver. 090a
- tcbn12ffcllbwp12p90pm Ver. 090a
- tcbn12ffcllbwp12p90lvl Ver. 090a

TSMC12nm technology digital standard cell base and power management libraries; specific libraries and versions

2 PHY RTL Compile Options

2.1 PHY Types

The Torrent16/12 FFC PHY RTL code base can be configured to support multiple different PHY types based on which verilog defines are set (see Table 3). These PHY types include:

- Multi-protocol - PIPE interface + full Raw SerDes interface, supporting all PIPE standards (PCIe, USB3 and SATA) and all Raw SerDes interface standards (Ethernet, DP/eDP, SATA, etc.), with the following variations:
  - Reduced Raw SerDes interface, tailored to include only the interface pins required for Ethernet and SATA.
- Raw SerDes only, which eliminates the PIPE interface and its underlying PCS logic.
  - Supports the Reduced Raw SerDes interface option.
- PIPE only, which eliminates the Raw SerDes interface. PIPE only can be ‘tuned’ to support any combination of the PIPE standards.
- For all PHY configurations which support PIPE, the following options are provided:
  - The PIPE PCS can be ‘tuned’ for which protocols to support (i.e. PCIe only, SATA only, USB3 only, PCIe + USB3, PCIe + SATA, USB3 + SATA, or all 3).
  - Two clock configuration options are available: standard PIPE and source synchronous PIPE. For standard PIPE, all synchronous PIPE pins are aligned to pipe_l*_clk. For source synchronous PIPE, the synchronous PIPE output pins remain aligned to pipe_l*_clk, but the synchronous PIPE input pins are aligned to pipe_l*_clk_in.
- The PHY supports an option to include a per-lane state change monitor interrupt output pins(phy_interrupt_ln_<>). These interrupts are asserted high when a power state or PIPE data rate change request does not complete within the configured maximum timeout.

PHY Type	RTL Defines						
	CDN_MPPHY_ PIPE_REMOVE	CDN_MPPHY_ RAW_REMOVE	CDN_MPHY_ ETHERNET_ONLY	MP_PCIE	MP_USB	MP_SATA	MP_128B_PATH
Full Multi-protocol				X	X	X	X
Multi-protocol with reduced Raw			X	X	X	X	X
PIPE PCIe only		X		X			X
PIPE USB only		X			X		
PIPE SATA only		X			X		
PIPEonly		X		X	X	X	X
Raw SerDes-only	X						
Reduced Raw SerDes-only	X		X				

Table 3: PHY Configuration Define Settings for Various PHY Types

2.2 RTL Defines

2.2.1 Pre-delivery

The following defines set the IP configuration and are stripped from the customer deliverable RTL:

CDN_MPPHY_PIPE_REMOVE: When defined, the PIPE interface and underlying PIPE PCS logic is removed.

CDN_MPPHY_RAW_REMOVE: When defined, the Raw SerDes interface and underlying logic is removed.

CDN_MPPHY_ETHERNET_ONLY: When defined, the Raw SerDes interface pins are reduced to a sub-set required for Ethernet and SATA (i.e. the Rx equalization evaluation, Tx get coefficient, Tx de-emphasis, receiver detect, Tx margin, Tx swing and Rx termination interface pins are removed). If CDN_MPPHY_RAW_REMOVE is defined, this define has no effect.

MP_PCIE: When defined and CDN_MPPHY_PIPE_REMOVE is not defined, enables support for PCIe Gen1/2. When MP_128B_PATH is also defined, PCIE Gen3/4 is supported, as well.

MP_USB: When defined and CDN_MPPHY_PIPE_REMOVE is not defined, enables support for USB Gen1. When MP_128B_PATH is also defined, USB Gen2 is supported, as well.

MP_SATA: When defined and CDN_MPPHY_PIPE_REMOVE is not defined, enables support for SATA Gen1/2/3.

MP_128B_PATH: When defined, support for the 128b/13xb encoded PIPE data rates (PCIe Gen3/4 and USB Gen2) is enabled. Only applicable if CDN_MPPHY_PIPE_REMOVE is not defined and MP_PCIE and/or MP_USB are defined.

MP_SRC_SYNC_PCLK: When defined, source synchronous PIPE is enabled. The default configuration is standard PIPE (i.e. without the define set). Only applicable if CDN_MPPHY_PIPE_REMOVE is not defined.

MP_PHY_AUTO_SAFE_EN: When defined, the state change monitor interrupt outputs are included in the PHY pinout.

2.2.2 Post-delivery

The following defines are included in the customer deliverable RTL and are provided to allow customization of the IP by the customer:

CDN_MPPHY_LN{1:n}_SLAVE_ONLY: When defined for a lane, the lane will only operate as a slave lane. phy_link_cfg_ln_n shall not be driven high and must be tied low.

CDN_MPPHY_LN{1:n}_MASTER_ONLY: When defined for a lane, the lane will only operate as a slave lane. phy_link_cfg_ln_n shall not be driven low and must be tied high.

For any lane, only 1 (or neither) of these defines can be set.

CDN_MPPHY_PCIE_AUTO_CFG: When defined, PHY logic is included that automatically configures the PHY for PCIe single or multi-link operation per the PHY Programmer’s Guide at startup with 100 MHz reference clock, no SSC (i.e. no APB registers writes required from SoC).

3 Configuration

As described in the PHY Overview, the Torrent multi-protocol multi-link PHY supports multiple serial standards, many of which can operate simultaneously. This section describes the details of the PHY configuration. Note: The Ethernet links in these tables are 1G, 2.5G or 5G Ethernet.

3.1 Multi-link / multi-protocol support

The PHY supports simultaneous operation of independent links of the following PHY configurations:

PCIe	USB3	SATA	DP/ eDP Tx	Ethernet	JESD204B	XFI/ USXGMII	Description
1+							**PCIe only**: N x1 links[6]
	1+						**USB3 only**: N links
		1+					**SATA only**: N links
			1+				**DP/eDP Tx only**: Max of 2, where each link can have 1, 2 or 4 lanes.
				1+			**Ethernet only**: N links
					1+		**JESD204B only**: Max of 2 independent links
						1+	**XFI/USXGMII only**: N links
1	1+						**PCIe + USB3** (No SSC for PCIe): 1 PCIe link (1xM); P USB3 links.
1+	1+						**PCIe + USB3** (PCIe and USB3 same SSC): M lanes of x1 PCIe links[6] and P USB3 links.
1		1+					**PCIe + SATA**: 1 PCIe link (1xM) and P SATA links.
1			1				**PCIe + DP/eDP Tx**: 1 PCIe link (1xM) and 1 DP/eDP Tx link (1xP)
1+				1+			**PCIe (no SSC) + Ethernet**: M lanes of x1 PCIe (no SSC) links[6] and P Ethernet links
1				1+			**PCIe (SSC) + Ethernet**: 1 link of PCIe (with SSC) + P Ethernet links

Table 4: Supported PHY configurations

PCIe	USB3	SATA	DP/ eDP Tx	Ethernet	JESD204B	XFI/ USXGMII	Description
1					1		**PCIe + JESD204B**: 1 PCIe link (1xM) and 1 JESD204B link
1						1+	**PCIe + XFI/USXGMII**: 1 PCIe link (1xM) and P XFI/USXGMII links.
	1+	1+					**USB3 + SATA**: M USB3 links and P SATA links
	1+		1				**USB3 + DP/eDP Tx**: M USB3 links and 1 DP/eDP Tx link (1xP)
	1+			1+			**USB3 + Ethernet**: M USB3 links and P Ethernet links
	1+				1		**USB3 + JESD204B**: M USB3 links and 1 JESD204B link
	1+					1+	**USB3 + XFI/USXGMII**: M USB3 links and P XFI/USXGMII links
		1+	1				**SATA + DP/eDP Tx**: M SATA links and 1 DP/eDP Tx link (1xP)
		1+			1		**SATA + JESD204B**: M SATA links and 1 JESD204B link
		1+				1+	**SATA + XFI/USXGMII**: M SATA links and P XFI/USXGMII links
			1		1		**DP/eDP + JESD204B**: 1 DP/eDP Tx link (1xM) and 1 JESD204B link
			1			1+	**DP/eDP + XFI/USXGMII**: 1 DP/eDP Tx link (1xM) and P XFI/USXGMII links

Table 4: Supported PHY configurations (Continued)

Notes:
1. 1+ indicates the PHY can support 1 or more independent links for the protocol when in this configuration.
2. 1 indicates the PHY can support only 1 link for the protocol when in this configuration.
3. N = number of PHY lanes.
4. M = number of PHY lanes configured for the first selected protocol (left to right in the table above).
5. P = N-M (i.e. remaining lanes)
6. Or, fewer links, when 1 or more links are multi-lane links.

3.2 Multi-Lane Links

Multi-lane links (PCIe and DP/eDP) are configured using the phy_link_cfg_ln_<1:n> PHY inputs. For each multi-lane link, 1 lane must be designated as the link master. This must be the lowest lane number in the link. phy_link_cfg_ln_<> for this lane must be asserted high and phy_link_cfg_ln_<> for the other lanes must be de-asserted low. (Of note: phy_link_cfg_ln_<> must be asserted high for all single lane links.) There is no phy_link_cfg_ln_0 PHY input because it is unnecessary. Lane 0 must always be a link master.

For example, a 1x4 PCIe link can be created from any 4 adjacent PHY lanes, such as lanes 2, 3, 4 and 5. In this case, phy_link_cfg_ln_2 must be asserted high, while phy_link_cfg_ln_<3:5> are de-asserted low.

On the PIPE and Raw SerDes interfaces, certain PHY pins are designated as link signals. For these pins, only the master lane's signals are used and the pins for slave lanes are ignored.

3.3 PHY PLL Configuration

As described in the PMA specification, the PMA PLLs (PLL0 and PLL1) must be configured based on which standards will be enabled on each lane. Table 5 is a summary of the PLL mapping based on the PHY configuration.

PHY configuration	PLL0	PLL1
Single link PCIe only	All lanes	Not used/disabled
Multi-link PCIe only	PCIe Gen1/2 lanes	PCIe Gen 3 lanes
Multi-link USB3 only	All lanes	Not used/disabled
Multi-link SATA only	All lanes	Not used/disabled
Multi-link DP/eDP Tx only (2 links max)	DP/eDP link 0 lanes	DP/eDP link 1 lanes
Multi-link Ethernet only	All lanes	Not used/disabled
Multi-link JESD204B only (2 links max)	JESD204B link 0 lanes	JESD204B link 1 lanes
Multi-link XFI/USXGMII only	All lanes	Not used/disabled
Single link PCIe (non-SSC) + USB3	PCIe lanes	USB lanes
Multi-link PCIe + USB3 (all links use same SSC)	PCIe Gen1/2 USB3	PCIe Gen3
Single link PCIe + multi-link SATA	PCIe link	SATA links
Single link PCIe + single link DP/eDP Tx	PCIe link	DP/eDP link
Multi-link PCIe (non-SSC) + multiple Ethernet links	PCIe Gen1/2 Ethernet	PCIe Gen3
Single link PCIe (SSC) + multiple Ethernet links	PCIe link	Ethernet link
Single link PCIe + single link JESD204B	PCIe link	JESD204B link
Single link PCIe + multiple XFI/USXGMII links	PCIe link	XFI/USXGMII link
Multi-link USB3 + multi-link SATA	USB3 links	SATA links
Multi-link USB3 + single link DP/eDP Tx	USB3 links	DP/eDP link
Multi-link USB3 + multi-link Ethernet	USB3 links	Ethernet links
Multi-link USB3 + single link JESD204B	USB3 links	JESD204B link
Multi-link USB3 + multi-link XFI/USXGMII	USB3 links	XFI/USXGMII links
Multi-link SATA + single link DP/eDP Tx	SATA links	DP/eDP link
Multi-link SATA + single link JESD204B	SATA links	JESD204B link
Multi-link SATA + multi-link XFI/USXGMII	SATA links	XFI/USXGMII links
Single link DP/eDP Tx + single link JESD204B	DP/eDP link	JESD204B link
Single link DP/eDP Tx + multi-link XFI/USXGMII	DP/eDP link	XFI/USXGMII links

Table 5: PLL Mapping based on PHY Configuration

The PMA specification and Programmer's Guide describes how to configure the PLLs based on the PHY configuration.

3.4 PHY link configuration

The mode of operation of each link (PCIe, USB, SATA or Ethernet) is configured through the phy_l<00:nn>_mode pins and PMA registers. Non-SSC PCIe is the default register configuration for each lane. All other modes of operation require additional configuration and the PMA specification describes the register configurations required for each protocol.

At the PHY level, the PHY_PLL_CFG register must be configured based on the PHY configuration.

- PHY_PLL_CFG[0]: Set to 1 if PHY is configured with only a single PCIe link (such the all PCIe rates are driven by PLL0). Otherwise, clear to 0.
- PHY_PLL_CFG[1]: Set to 1 if PHY is using both PLL0 and PLL1. Otherwise, clear to 0, if only using PLL0.

The default is single-link PCIe with only PLL0 being used.

4 PHY Pin List

The following tables describe the pin list of the MP PHY. Note that several of these pins connect directly to the Torrent PMA. In these cases, this will be noted in the description that further description is provided in the PMA specification.

4.1 Power Pin List

Dir.	Signal Name	Type	External pin / bump or Internal core pin.	Description
input	gnd	power	bump	Macro ground.
input	cmn_avdd	power	bump	PMA common analog non high speed clock power
input	cmn_avdd_clk	power	bump	PMA common analog high speed clock power
input	cmn_avdd_h	power	bump	PMA common analog high voltage power
input	xcvr_avdd_ln_{n:0}	power	bump	PMA transmitter analog and non high speed clock power
input	xcvr_avdd_clk_ln_{n:0}	power	bump	PMA transceiver analog high speed clock power

Table 6: Multi-protocol PHY power pin list

4.2 External Pin / Bump List

Dir.	Signal Name	Type	Timing	Description
inout	cmn_refclk_p cmn_refclk_m	analog	n/a	PMA external reference clock differential input/output - configurable as reference clock input or as reference clock output. **Note: When configured as an output, a reference clock signal shall not be driven into these pins.**
output	tx_p_ln_{n:0} tx_m_ln_{n:0}	Diff	n/a	PMA transmitter serial data
input	rx_p_ln_{n:0} rx_m_ln_{n:0}	Diff	n/a	PMA receiver serial data
inout	cmn_rext	analog	n/a	PMA external calibration resistor

Table 7: Multi-protocol PHY external pin / bump list

4.3 On-Chip/SoC Functional Pin List

Notes:

* - Indicates a shared link signal. For multi-lane links (i.e. PCIe and DP/eDP Tx), only the master lane's pin is used for a shared link pin. For slave lanes, link output pins shall be ignored and link input pins can be tied high or low.

{nn:00} / {n:0} - Indicates a per-lane pin with nn / n indicating the highest lane number. For example for a 4 lane PHY, pipe_l{nn:00}_clk describes four pins: pipe_l00_clk, pipe_l01_clk, pipe_l02_clk and pipe_l03_clk.

Dir.	Signal Name	Type	Timing	Description
External APB interface				

Table 8: Multi-protocol PHY On-chip functional pin list

Dir.	Signal Name	Type	Timing	Description
input	apb_pclk	CMOS	n/a	APB bus clock.
input	apb_preset_n	CMOS	async	APB bus reset.
input	apb_penable	CMOS	apb_pclk	APB bus enable.
input	apb_psel	CMOS	apb_pclk	APB bus device select.
input	apb_pwrite	CMOS	apb_pclk	APB bus write.
input	apb_paddr[15:0]	CMOS	apb_pclk	APB bus address.
input	apb_pwdata[15:0]	CMOS	apb_pclk	APB bus write data.
output	apb_prdata[15:0]	CMOS	apb_pclk	APB bus read data.
output	apb_pready	CMOS	apb_pclk	APB bus ready.
PHY interface - These signals are shared by PIPE and Raw SerDes interfaces. (* - Indicates shared link signal. For multi-lane links, only the master lane is used for shared signals. Otherwise, the signal is per-lane.)				
input	phy_reset_n	CMOS	async	**PHY reset** : Asserting this signal low will reset all PHY logic for the entire PHY with the exception of the APB registers and TAP controller. Note: Upon de-assertion, all PHY inputs must be driven as described in the PIPE specification for a PIPE reset.
input	phy_link_cfg_ln_{n:1}	CMOS	static	**PHY link/lane configuration.** Used to map lanes to links. 0 - designates a lane as a link slave. 1 - designates a lane as a link master. Lanes are grouped into links from lowest lane number to highest. Adjacent slave lanes are part of multi-lane link down to the lowest numbered adjacent master lane. Lane 0 is always a master and thus does not require its own phy_link_cfg input. For example, for a 4 lane PHY can be configured as follows: {phy_link_cfg_ln_3, phy_link_cfg_ln_2, phy_link_cfg_ln_1} • 3'b111 = 4x1 configuration (i.e. 4 independent links) • 3'b000 = 1x4 configuration (i.e. single 4 lane link with lane 0 as the master lane) • 3'b010 = 2x2 configuration (i.e. 2 2-lane links with lane 0 and lane 2 as the master lanes) • 3'b011 = 2x1+1x2 configuration (i.e. 2 single lane links plus 1 2-lane link wiht lanes 0, 1 and 2 as the master lanes) • 3'b101 = 2x1+1x2 configuration (with lane 0, 1 and 3 as master lanes) • 3'b110 = 2x1+1x2 configuration (wiht lane 0, 2 and 3 as master lanes) • 3'b100 = 1x3+1x1 configuration (i.e. 1 3-lane link plus 1 single lane link with lane 0 and 3 as master lanes) • 3'b001 = 1x3+1x1 configuration (wiht lane 0 and 1 master lanes) Note: phy_reset_n and apb_preset_n must be asserted to modify this signal. The master lane of each link is that link's lane 0. It shall not be disabled unless all other lane's in the link are disabled. When each phy_link_cfg_ln_* = 0, the PHY is in a single-link configuration.
input	phy_l{nn:00}_reset_n *	CMOS	async	**PHY link reset** : Asserting this signal low will reset the PHY components (PCS and PMA) for the associated link, with the exception of the APB registers. For multi-lane links, only the master lane is valid. Disabled/ignored for slave lanes (per -link/shared signal). When in a single-link configuration, only phy_reset_n is required. phy_l00_reset_n can be tied high.

Table 8: Multi-protocol PHY On-chip functional pin list (Continued)

Dir.	Signal Name	Type	Timing	Description
input	phy_l{nn:00}_mode[1:0] *	CMOS	static	**PHY mode** : PHY mode of operation for the associated link. • 2'b00 = PCIe across PIPE I/F • 2'b01 = USB across PIPE I/F • 2'b10 = SATA across PIPE I/F • 2'b11 = Raw SerDes I/F mode These pins are only present when the PHY is configured to support both PIPE and RawSerDes I/Fs or PIPE only. For Raw SerDes only configuration, these pins are removed.
PIPE interface (* - Indicates shared link signal. For multi-lane links, only the master lane is used for shared signals. Otherwise, the signal is per-lane.) Note: If the PIPE interface is not required, a configuration option is available to remove the PIPE interface and underlying PHY PCS logic. When the PHY is configured for Raw SerDes operation, all PIPE I/F inputs are unused and can be driven low or high. The PIPE interface supports 2 configuration options: standard PIPE and source synchronous PIPE. For standard PIPE , all synchronous PIPE pins are aligned to pipe_l*_clk. For source synchronous PIPE, PIPE outputs remain aligned to pipe_l*_clk. But, all synchronous PIPE inputs are instead aligned to pipe_l*_clk_in, which is only present on the PIPE I/F in this configuration.				
input	phy_l{nn:00}_pcie_l1_ss_sel *	CMOS	static	**PIPE PCIe L1 substate enable**: When high, PCIe L1 SS is entered using protocol defined in PIPE 4.4 specification and described in the PHY operations chapter. PIPE-based L1 SS entry/exit uses the pipe_l*_powerdown and pipe_l*_phy_status pins instead of the phy_l*_ent_l1_x/ phy_l*_ack_l1_x pins. When low, the L1 substate side-band signaling protocol is used. The sideband protocol is described in the PHY operation chapter and uses the phy_l*_ent_l1_x/phy_l*_ack_l1_x pins. For both modes, the phy_l*_rx_elec_idle_det_en and phy_l*_tx_cmn_mode_en pins are used. However, the polarity of these pins is opposite depending on the mode. Note: These are 'strap' pins that are expected to be tied high or low depending on which L1 substate mode is supported by the PCIe controller(s) associated with the link. For multi-lane links, only the master lane is valid. Disabled/ ignored for slave lanes (per -link/shared signal).
output	pipe_l{nn:00}_clk *	CMOS	n/a	**PIPE clock**. For standard PIPE configuration, all synchronous PIPE signals are aligned to this clock. For source synchronous PIPE, all synchronous PIPE outputs are aligned to this clock. For multi-lane links, only the master lane is valid. Disabled/ignored for slave lanes (per -link/shared signal).
input	pipe_l{nn:00}_clk_in *	CMOS	n/a	**PIPE clock in**. This pin is not present for standard PIPE configuration. It is only present for source synchronous PIPE configuration. For source synchronous PIPE, all synchronous PIPE inputs are aligned to this clock. For multi-lane links, only the master lane is valid. Ignored for slave lanes (per -link/shared signal).
input	pipe_l{nn:00}_rate[1:0] *	CMOS	pipe_l*_clk / pipe_l*_clk_in	**PIPE link signaling rate**. Signals the PCIe link data rate. For multi-lane links , only the master lane is valid. Disabled/ ignored for slave lanes (per -link/shared signal). • 2'b00 : PCIe Gen 1 (2.5 Gbps) / USB3 Gen 1 (5 Gbps) / SATA Gen 1 (1.5 Gbps) • 2'b01 : PCIe Gen 2 (5 Gbps) / USB3 Gen 2 (10 Gbps) / SATA Gen 2 (3.0 Gbps) • 2'b10 : PCIe Gen 3 (8 Gbps) / USB3 Reserved / SATA Gen 3 (6 Gbps) • 3'b11 : Reserved

Table 8: Multi-protocol PHY On-chip functional pin list (Continued)

Dir.	Signal Name	Type	Timing	Description
input	pipe_l{nn:00}_powerdown[2:0] *	CMOS	pipe_l*_clk / pipe_l*_clk_in / async	**PIPE link power state**. Selects the PCIe link power state. For multi-lane links, only the master lane is valid. Disabled/ ignored for slave lanes (per -link/shared signal). • 3'b000 : PCIe P0 / USB3 P0 / SATA PhyRdy power state. • 3'b001 : PCIe P0s / USB3 P1 / SATA Partial power state. • 3'b010 : PCIe P1 / USB3 P2 / SATA Slumber power state. • 3'b011 : PCIe P2 / USB3 P3 / SATA DevSleep power state. • 3'b100 : PCIe P1.x (L1 sub-state) power state. Not valid for USB or SATA. Only valid when phy_pcie_l1_ss_sel is high. When phy_l*_pcie_l1_ss_sel is low, bit [2] should be tied low. • 3'b101 - 3'b111 : Reserved Timing is asynchronous in power states where pipe_l*_clk/ pipe_l*_clk_in is not running (i.e. PCIe P2/L1 SS, USB P3 or SATA DevSleep).
input	pipe_l{nn:00}_eb_mode *	CMOS	static	**PIPE elasticity buffer mode** : 0 = nominal half-full mode; 1 = nominal empty mode. Valid for USB Gen1 and SATA only. For PCIe and USB Gen 2 drive low. (PCIe and USB Gen 2 only supports nominal half-full operation.)
input	pipe_l{nn:00}_32bit_sel *	CMOS	static	**PIPE data width configuration** •0 - 16-bit pipe data width •1 - 32-bit pipe data width Note: Can only be changed when phy_reset_n is asserted low. For multi-lane links , only the master lane is valid. Disabled/ignored for slave lanes (per -link/shared signal).
output	pipe_l{nn:00}_phy_status	CMOS	pipe_l*_clk / async	**PIPE Phy Status** : Used to communicate the completion of several PHY functions (power state change, rate change and receiver detect). This communications completion protocol is described in detail in the PIPE specification. Timing is asynchronous in power states where the pipe_l*_clk is not running.
input	pipe_l{nn:00}_tx_data[31:0]	CMOS	pipe_l*_clk / pipe_l*_clk_in	**PIPE parallel transmit data**. In 16-bit interface mode only the 16 LSBs are used.
input	pipe_l{nn:00}_tx_data_k[3:0]	CMOS	pipe_l*_clk / pipe_l*_clk_in	**PIPE transmit data control** - Selects whether associated symbol is data (1'b0) or K-character (1'b1). In 16-bit interface mode only the 2 LSBs are used. Only used for PCIe Gen 1/2, USB3 Gen 1 and SATA. For PCIe Gen 3 and USB3 Gen 2 drive low.
input	pipe_l{nn:00}_tx_sync_header[3:0]	CMOS	pipe_l*_clk / pipe_l*_clk_in	**PIPE transmit data sync header** - 128 bit block header bits. Only used for PCIe Gen 3 and USB3 Gen 2 operation. Bits [3:2] are only used for USB Gen 2 operation, drive low for PCIe Gen 3. For PCIe Gen 1/2, USB3 Gen 1 and SATA drive low.
input	pipe_l{nn:00}_tx_start_block	CMOS	pipe_l*_clk / pipe_l*_clk_in	**PIPE transmit start block** - 1 = start word for a 128 bit block. Only used for PCIe Gen 3 and USB3 Gen 2 operation. For PCIe Gen 1/2, USB3 Gen 1 and SATA drive low.
input	pipe_l{nn:00}_tx_data_valid	CMOS	pipe_l*_clk / pipe_l*_clk_in	**PIPE transmit data valid :-** Indicates whether the data on pipe_l*_tx_data, pipe_l*_tx_data_k, pipe_l*_tx_start_block and pipe_l*_tx_sync_header is valid or not. When transmitting data, for PCIe Gen 3 and USB3 Gen 2, de-asserted for a single clock cycle every N data blocks (based on mode and data width). For PCIe Gen1/2, USB Gen 1 and SATA, never de-asserted while transmitting data.
input	pipe_l{nn:00}_tx_elec_idle	CMOS	pipe_l*_clk / pipe_l*_clk_in / async	**PIPE transmit electrical idle**: Used to place the TX lines into electrical idle state when in P0 power state. Timing is asynchronous in power states where pipe_l*_clk/ pipe_l*_clk_in is not running.

Table 8: Multi-protocol PHY On-chip functional pin list (Continued)

Dir.	Signal Name	Type	Timing	Description
input	pipe_l{nn:00}_tx_det_rx_lpbk *	CMOS	pipe_l*_clk / pipe_l*_clk_in	**PIPE TxDetectRx/Loopback**: Used to either start a receiver detection operation (in PCIe P1 and USB P2/P3 power states) or configures for loopback (in P0 power state). For non-USB multi-lane links, only the master lane is valid and is disabled/ ignored for slave lanes (per-link/shared signal). For USB multi-lane links, this is a per-lane signal.
input	pipe_l{nn:00}_tx_compliance	CMOS	pipe_l*_clk / pipe_l*_clk_in	**PIPE transmit compliance**: Sets the running disparity to negative for PCIe compliance patterns.
input	pipe_l{nn:00}_tx_ones_zeros	CMOS	pipe_l*_clk / pipe_l*_clk_in	**PIPE transmit Ones/Zeros**: 1 = transmit a repeating pattern of 80 1's followed by 80 0's. Used for USB3.2 Gen 1 compliance testing. For PCIe, USB3.2 Gen 2 and SATA drive low.
input	pipe_l{nn:00}_tx_pattern[1:0]	CMONS	pipe_l*_clk / pipe_l*_clk_in	**PIPE transmit pattern**: Selects the character to be transmitted for OOB. Used for SATA only. For PCIe and USB drive low. • 2'b00 = K28.5 (Align) • 2'b01 = D24.3 • 2'b10 = D10.2 • 2'b11 = Reserved
input	pipe_l{nn:00}_tx_128b_enc_byp	CMOS	pipe_l*_clk / pipe_l*_clk_in	**PIPE transmit 128-bit/13x-bit encoder bypass**: 1 = enables bypass fo 128b/132xb encoder. When asserted, the transmit sync header bits are not inserted into the transmit data stream and pipe_l*_tx_data_valid is no longer de-asserted every N blocks. 0 = normal 128b/13xb operation. Primary use is to support transmission of USB3.2 Gen 2 compliance patterns CP10, CP11 and CP12. For PCIe, USB3.2 Gen 1 and SATA operation, drive low.
input	pipe_l{nn:00}_tx_deemph[17:0]	CMOS	pipe_l*_clk / pipe_l*_clk_in	**PIPE transmitter de-emphasis level control**: For PCIe Gen 1/2 and USB3 Gen 1, only bits [1:0] are used: • 2'b00 = -6dB de-emphasis • 2'b01 = -3.5dB de-emphasis • 2'b10 = No de-emphasis • 2'b11 = Reserved For PCIe Gen 3: • [5:0] = C_{-1} • [11:6] = C_0 • [17:12] = C_{+1} For USB3 Gen 2, only bits [1:0] are used. Logic inside the PCS calculates the 18-bit de-emphasis value to the PMA (as defined above for PCIe Gen 3) based on the de-emphasis mode selected and the PMA's local FS value. For the default register configuration: • 2'b00 = CP13 (with pre-shoot, but no de-emphasis) • 2'b01 = normal mission mode and CP15 (with pre-shoot and de-emphasis) • 2'b10 = CP14 (with de-emphasis but no pre-shoot) • 2'b11 = CP16 (no pre-shoot or de-emphasis)
input	pipe_l{nn:00}_tx_get_local_preset_coef	CMOS	pipe_l*_clk / pipe_l*_clk_in	**PIPE transmitter get local preset coefficients**: For PCIe Gen 3 only when PHY is in P0 power state. For SATA, USB3 and PCIe Gen 1/2 (and when PHY is not in P0 power state) drive low. Asserted high for a single cycle to initiate a Get Local Preset Coefficient operation.
input	pipe_l{nn:00}_tx_local_preset_index[3:0]	CMOS	pipe_l*_clk / pipe_l*_clk_in	**PIPE transmitter local preset index**: For PCIe Gen 3 only. For SATA, PCIe Gen1/2 and USB3 drive low. Selects the preset index for the Get Coefficient operation. Must be valid upon assertion of pipe_l*_tx_get_local_preset_coef. Note: When in low swing mode (pipe_l*_swing == 1), only a subset of the index values are supported (as described in PMA specification).

Table 8: Multi-protocol PHY On-chip functional pin list (Continued)

Dir.	Signal Name	Type	Timing	Description
output	pipe_l{nn:00}_tx_local_tx_coef_valid	CMOS	pipe_l*_clk	**PIPE transmitter local coefficient valid**: Asserted high for a single clock cycle upon completion of the Get Local Preset Coefficient operation.
output	pipe_l{nn:00}_tx_local_tx_pre-set_coef[17:0]	CMOS	pipe_l*_clk	**PIPE transmitter local preset coefficient**: Tx deemphasis coefficient corresponding to the preset selected by pipe_l*_tx_local_preset_index. Valid upon assertion of pipe_l*_tx_local_tx_coef_valid.
output	pipe_l{nn:00}_tx_local_fs[5:0]	CMOS	pipe_l*_clk	**PIPE transmitter FS value**: Valid upon initial de-assertion of pipe_l*_phy_status.
output	pipe_l{nn:00}_tx_local_lf[5:0]	CMOS	pipe_l*_clk	**PIPE transmitter LF value**: Valid upon initial de-assertion of pipe_l*_phy_status.
input	pipe_l{nn:00}_tx_margin[2:0]	CMOS	static	**PIPE transmitter voltage margin control**: • 3'b000 - Maximum driver output swing • 3'b001 - 3'b110 - Incrementally decreasing driver output swing • 3'b111 - Reserved
input	pipe_l{nn:00}_tx_swing	CMSO	static	**PIPE transmitter voltage swing**: • 1'b0 - Full swing • 1'b1 - Low swing
output	pipe_l{nn:00}_rx_data[31:0]	CMOS	pipe_l*_clk	**PIPE parallel receive data** .In 16-bit interface mode only the 16 LSBs are valid.
output	pipe_l{nn:00}_rx_data_k[3:0]	CMOS	pipe_l*_clk	**PIPE receive data / K-character control**: Indicates whether the associated symbol is data (1'b0) or a K-character (1'b1). For 16-bit interface mode only the 2 LSBs are valid. Only valid for PCIe Gen 1/2, USB3 Gen 1 and SATA.
output	pipe_l{nn:00}_rx_sync_header[3:0]	CMOS	pipe_l*_clk	**PIPE receive data sync header**: 128 bit block header bits. Only valid for PCIe Gen 3 and USB Gen 2.
output	pipe_l{nn:00}_rx_start_block	CMOS	pipe_l*_clk	**PIPE receive start block**: 1 = start word for a 128 bit block. Only valid for PCIe Gen 3 and USB Gen 2.
output	pipe_l{nn:00}_rx_data_valid	CMOS	pipe_l*_clk	**PIPE recevie data valid**: 1 = data on pipe_l*_rx_data, pipe_l*_rx_data_k, pipe_l*_rx_sync_header and pipe_l*_rx-_start_block is valid.
output	pipe_l{nn:00}_rx_status[2:0]	CMOS	pipe_l*_clk	**PIPE receiver status**. • 3'b000 : Received data OK. • 3'b001 : 1 PCIe SKP symbol / USB SKPOS / SATA Align added. • 3'b010 : 1 PCIe SKP symbol / USB SKPOS / SATA Align removed. • 3'b011 : Receiver detected / SATA misalign detected. • 3'b100 : Both 8B10B and 128b/13xb decode error. • 3'b101 : Elastic buffer overflow. • 3'b110 : Elastic buffer underflow. • 3'b111 : Receiver disparity error.
output	pipe_l{nn:00}_rx_valid	CMOS	pipe_l*_clk	**PIPE Rx valid**: Used to indicate symbol lock to the Rx data stream.

Table 8: Multi-protocol PHY On-chip functional pin list (Continued)

Dir.	Signal Name	Type	Timing	Description
output	pipe_l{nn:00}_rx_elec_idle	CMOS	async	**PIPE Rx electrical idle** PCIe: Indicates receiver detection of electrical idle, when in the P0, P0s, P1 and upon entry into a L1 substate. Upon exit from a L1 substate, forced high for a configurable time while analog Rx signal detect logic is powered up. Time forced high depends on whether all links were powered down or not. USB: Indicates detection of LFPS signaling for all power states. Asserted low when receiving LFPS and high otherwise. SATA: Indicates receiver detection of electrical idle when in all power states. Used for detection of OOB signaling. Per the SATA specification, OOB signals are groupings of bursts of the ALIGNp primitive at the Gen 1 data rate combined with electrical idle gaps. Asserted high when receiving electrical idle and low otherwise (which includes when receiving ALIGNp primitive during OOB signaling).
output	pipe_l{nn:00}_align_detect	CMOS	pipe_l*_clk	**PIPE SATA align detect** : 1 = SATA align primitive received. Valid for SATA only.
input	pipe_l{nn:00}_rx_polarity	CMOS	pipe_l*_clk / pipe_l*_clk_in	**PIPE receive invert polarity** • 1'b0 : Polarity inversion disabled. • 1'b1 : Polarity inversion enabled. Receive data is inverted.
input	pipe_l{nn:00}_rx_termination	CMOS	async	**PIPE receiver termination**: 1= enable Rx termination for the lane; 0 = disable Rx termination for the lane. Used for USB only. For PCIe drive low. (For PCIe, Rx termination is enabled when the lane is enabled.)
input	pipe_l{nn:00}_rx_eq_training	CMOS	pipe_l*_clk / pipe_l*_clk_in	**PIPE receiver equalization training enable**: 1 = enable Rx equalization training; 0 = normal operation. Used for USB only. For PCIe and USB drive low.
input	pipe_l{nn:00}_rx_eq_eval	CMOS	pipe_l*_clk / pipe_l*_clk_in	**PIPE receiver equalization evaluation start**: 1 = perform far end transmitter equalization evaluation and report result on pipe_l*_rx_link_eval_fb_dir_change. Used for PCIe Gen 3 only. For USB, PCIe Gen1/2 and SATA drive low.
output	pipe_l{nn:00}_rx_link_eval_f-b_dir_change[5:0]	CMOS	pipe_l*_clk	**PIPE receiver link equalization feedback direction change**: Far-end transmitter equalization evaluation result. Valid upon assertion of pipe_l*_phy_status. Used for PCIe Gen 3 only.
input	phy_l{nn:00}_rx_elec_idle_det_en *	CMOS	async	**PHY Rx Electrical Idle detect enable / Rx Electrical Idle detect disable** : Controls the PHY signal detection circuitry during L1 substate entry. The polarity of this pin depends on the L1 SS mode selected: • phy_l*_pcie_l1_ss_sel == 0 : After assertion of phy_l*_ack_l1_x, de-assert low to complete entry into L1.1/L1.2. Assert high for all other power states. • phy_l*_pcie_l1_ss_sel == 1 : After de-assertion of pipe_l*_phy_status, drive high to complete entry into L1.1/L1.2. De-assert low for all other power states. For multi-lane links, only the master lane is valid. Disabled/ignored for slave lanes (per-link/shared signal). For USB3 or SATA lanes, tie high if phy_l*_pcie_l1_ss_sel is low and tie low if phy_l*_pcie_l1_ss_sel is high.

Table 8: Multi-protocol PHY On-chip functional pin list (Continued)

Dir.	Signal Name	Type	Timing	Description
input	phy_l{nn:00}_tx_cmn_mode_en *	CMOS	async	**PHY Tx common mode enable / Tx common mode disable**: Controls the PHY common mode circuitry during L1.2 entry. The polarity of this pin depends on the L1 SS mode selected: • phy_l*_pcie_l1_ss_sel == 0 : After assertion of phy_l*_ack_l1_x, de-assert low to complete entry into L1.2 (for L1.1 leave asserted high). Assert high for all other power states. • phy_l*_pcie_l1_ss_sel == 1 : After de-assertion of pipe_l*_phy_status, drive high to complete entry into L1.2 (for L1.1 leave low). De-assert low for all other power states. For multi-lane links, only the master lane is valid. Disabled/ignored for slave lanes (per-link/shared signal). For USB3 or SATA lanes, tie high if phy_l*_pcie_l1_ss_sel is low and tie low if phy_l*_pcie_l1_ss_sel is high.
input	phy_l{nn:00}_ent_l1_x *	CMOS	async	**PHY enter L1 substate (sideband)** : Asserted high to direct associated PHY link to enter a L1 substate. Once asserted or de-asserted, this signal shall not change state again until after the corresponding transaction on phy_ack_l1_x. Can only be asserted high when the link is in the P1 power state. For multi-lane links, only the master lane is valid. Disabled/ignored for slave lanes (per-link/shared signal). Only valid when phy_pcie_l1_ss_sel is low. When phy_l*_pcie_l1_ss_sel is high, tie low.
output	phy_l{nn:00}_ack_l1_x *	CMOS	async	**PHY acknowledge L1 substate (sideband)** : Acknowledgment from PHY that the requested L1_x entry or exit operation has been complete. Asserted high to indicate entry into L1 substate is complete. De-asserted low to indicate exit from L1 substate (back to P1 power state) is complete. For multi-lane links, only the master lane is valid. Disabled/ignored for slave lanes (per-link/shared signal). Only valid when phy_pcie_l1_ss_sel is low. When phy_l*_pcie_l1_ss_sel is high, leave unconnected
input	phy_en_refclk	CMOS	async	**PHY reference clock enable**: When cmn_refclk_<p/m> is configured as a reference clock output, 1 = glitch-less enable of the reference clock output. 0 = glitch-less turn off reference clock output. Used in L1.x entry/exit protocol. Note: When outputting a derived reference clock from the PLL, it is recommended to drive phy_en_refclk low until the PHY has completed start-up and de-asserted pipe_l*_reset_n for PIPE operation or asserted cmn_ready for Raw SerDes operation.. The clock will not be 'good' until the PHY has completed initial start-up.
output	phy_refclk_reqd	CMOS	async	**PHY reference clock input required**: 1 = input reference clock required. 0 = input reference clock may be turned off. Only valid when a valid reference clock is being provided to the PHY (i.e. upon L1.x exit, not valid until input reference clock re-started). Used in L1.x entry/exit protocol when reference clock being output on cmn_refclk_<p/m>.

Table 8: Multi-protocol PHY On-chip functional pin list (Continued)

Dir.	Signal Name	Type	Timing	Description
input	phy_pma_suspend_override	CMOS	async	**PHY PMA common suspend override enable**: 1 = disables suspending the PMA common when all links are in low power state (L1 SS, PCIe P2, USB P3, powerdown disabled, etc.). 0 = PMA common will be suspended when all links are in low power state. Driving this pin high has no effect if the PMA common is already suspended or is in the process of being suspended (i.e. it will not force the PMA common to resume). It only prevents the start of suspending the PMA common. However, if this pin is de-asserted when all links are in low power state, the PMA common will then be suspended.
Raw SerDes interface - This interface is used to support direct connection to the PMA, which enables support for Ethernet (i.e. SGMII, 2.5 Gbps and 5 Gbps), DP/eDP Tx and generic SerDes modes (i.e. JESD204B). This interface is used when the link is configured for Raw SerDes through phy_l*_mode pins or when the PHY is configured as Raw SerDes only. See the PMA specification for additional description of these pins • # indicates pins that are removed for an Reduced Raw SerDes configuration. • When the PHY is configured for PIPE operation, all Raw SerDes I/F inputs are unused and can be driven low or high and outputs shall be ignored. • RTL configurations are available to remove some or all of these pins.				
output	pma_cmn_ready	CMOS	async	**PMA common ready** : 1 = PMA common is ready for operation. Used as part of Raw SerDes startup sequence and power state changes.
input	pma_xcvr_pllclk_en_ln_{n:0} *	CMOS	async	**Link PLL clock enable** : Used to cleanly turn off the data rate clock in the PMA for all lanes in the associated link.
output	pma_xcvr_pllclk_en_ack_ln_{n:0} *	CMOS	async	**Link PLL clock enable acknowlegent** : Indicates whether the data rate clock in the PMA for the associated link is running or not.
input	pma_xcvr_power_state_req_ln_{n:0}[5:0] *	CMS	async	**Link power state request** : Used to change the link's power state. When the link has completed the transition to the requested power state, the requested state will be reflected on pma_xcvr_power_state_ack_ln_*. • 6'b000000 : Idle • 6'b000001 : A0 - Tx/Rx active • 6'b000010 : A1 - Powerdown 1 (equivalent to PCIe P0s / USB P1) • 6'b000100 : A2 - Powerdown 2 (equivalent to PCIe P1 / USB P2) • 6'b001000 : A3 - Powerdown 3 (equivalent to PCIe P2 / USB P3) • 6'b010000 : Reserved • 6'b100000 : Reserved This is a one hot encoded signal. A change in the requested state shall not occur until after the current request has been acknowledged and pma_xcvr_power_state_req_ln_* has been returned to 0. Upon reset release, either A0 or A2 must be requested.
output	pma_xcvr_power_state_ack_ln_{n:0}[5:0] *	CMS	async	**Link power state acknowledgement** : This signal provides indication that a power state change request has completed. • 6'b000000 : Idle • 6'b000001 : A0 • 6'b000010 : A1 • 6'b000100 : A2 • 6'b001000 : A3 • 6'b010000 : Reserved • 6'b000000 : Idle This is a one hot encoded signal. Once a power state is acknowledged, the value will remain unchanged until a new power state is requested and the link has completed the transition to the new power state.

Table 8: Multi-protocol PHY On-chip functional pin list (Continued)

Dir.	Signal Name	Type	Timing	Description
input	pma_xcvr_standard_-mode_ln_{n:0}[1:0] *	CMOS	static	**Link mode of operation** : Selects the mode of operation 9i.e. data rate) for each lane in the link. This value can only be changed per the Raw SerDes mode of operation change or data rate change procedures. See PMA specification and Programmer's Guide for details.
input	pma_xcvr_data_width_ln_{n:0}[2:0] *	CMOS	static	**Link parallel data width** : Selects the width of the parallel Tx and Rx data buses for this link. The data width required depends on the mode of operation. This value can only be changed per the Raw SerDes mode of operation change or data rate change procedures. • 3'b101 : 20-bit • 3'b001 : 16-bit • All other encodings : Reserved
input	pma_tx_td_ln_{n:0}[19:0]	CMOS	pma_tx_td_-clk_ln_{n:0}	**PMA Tx data** : Transmit data for the associated lane. The valid bits are based on pma_xcvr_data_width_ln_* (either bits [19:0] or [15:0]). Unused bits are driven low.
input	pma_tx_td_clk_ln_{n:0}	CMOS	n/a	**PMA Tx data clock** : Transmit data clock for the associated lane. Transmit data (pma_tx_td_ln_*) is synchronous to this clock. Shall be driven from the lane's pma_pllclk_datart_ln_* PHY clock output.
input	pma_tx_elec_idle_ln_{n:0}	CMOS	pma_tx_td_-clk_ln_{n:0}	**PMA Tx electrical idle** : 1 = Tx differential output placed into electrical idle for the associated lane. 0 = transmit data.
output	pma_rx_rd_ln_{n:0}[19:0]	CMOS	pma_rx_rd_-clk_ln_{n:0}	**PMA Rx data** : Receive data for the associated lane. The valid bits are based on pma_xcvr_data_width_ln_* (either bits [19:0] or [15:0]).
output	pma_rx_rd_clk_ln_{n:0}	CMOS	n/a	**PMA Rx data clock** ; Receive data clock for the associated lane. Receive data (pma_rx_rd_ln_*) is synchronous to this clock.
output	pma_rx_rd_clk2x_ln_{n:0}	CMOS	n/a	**PMA 2x Rd data clock** : 2x clock frequency of pma_rx_rd_-clk_ln_* when lane is configured for 20-bit data width. Clock is frequency aligned, but not edge aligned. Clock is gated by default. Must be enabled through APB register write. (Max clock frequency = 625 MHz.)
output	pma_rx_signal_detect_ln_{n:0}	CMOS	async	**PMA receiver signal detect** : Asserted high upon detection of high speed signal on Rx differential inputs.
input	pma_tx_rcv_detect_en_ln_{n:0} (#)	CMOS	async	**PMA transmitter receiver detect enable** - Drives PMA input tx_rcv_detect_en_ln_* for the associated lane. Drive low upon de-assertion of phy_reset_n.
output	pma_tx_rcv_detect_done_ln_{n:0} (#)	CMOS	async	**PMA transmitter receiver detect done** - Driven from PMA output tx_rcv_detect_done_ln_* for the associated lane.
output	pma_tx_rcv_detected_ln_{n:0} (#)	CMOS	async	**PMA transmitter receiver detected** - Driven from PMA output tx_rcv_detect_done_ln_* for the assoicated lane.
input	pma_tx_deemphasis_ln_{n:0}[17:0] (#)	CMOS	async	**PMA transmitter deemphasis level control** - Drives PMA input tx_deemphasis_ln_* for the associated lane. Drives with desired initial deemphasis setting upon de-assertion of phy_reset_n.
input	pma_tx_get_local_preset_co-ef_ln_{n:0} (#)	CMOS	async	**PMA transmitter get local preset coefficient** - Drives PMA input tx_get_local_preset_coef_ln_* for the associated lane. Drive low upon de-asertion of phy_reset_n.
input	pma_tx_local_preset_index-_ln_{n:0}[3:0] (#)	CMOS	async	**PMA transmitter local preset index** - Drives PMA input tx_local_preset_index_ln_* for the associated lane. Drive low upon de-assertion of phy_reset_n.
output	pma_tx_local_tx_coef_valid_ln_{n:0} (#)	CMOS	async	**PMA transmitter local coefficient valid** - Driven from PMA output tx_local_tx_coef_valid_ln_* for the associated lane.
output	pma_tx_local_tx_preset_co-ef_ln_{n:0}[17:0] (#)	CMOS	async	**PMA transmitter local preset/init coefficients** - Driven from PMA output tx_local_tx_preset_coef_ln_* for the associated lane.
input	pma_tx_vmargin_ln_{n:0}[2:0] (#)	CMOS	async	**PMA transmitter voltage margin control** - Drives PMA input tx_vmargin_ln_* for the associated lane. Drive with desired initial transit margin upon de-assertion of phy_reset_n.

Table 8: Multi-protocol PHY On-chip functional pin list (Continued)

Dir.	Signal Name	Type	Timing	Description
input	pma_tx_low_power_swing_en_ln_{n:0} (#)	CMOS	async	**PMA transmitter low power voltage swing enable** - Drives PMA input tx_low_power_swign_en_ln_* for the associated lane. Drive with desired initial transmit swing upon de-assertion of phy_reset_n
input	pma_rx_termination_ln_{n:0} (#)	CMOS	async	**PMA receiver termination enable** - Drives PMA input rx_termination_ln_* for the associated lane. Drive with desired initial Rx termination setting upon de-assertion of phy_reset_n.
input	pma_rx_eq_training_data_valid_ln_{n:0} (#)	CMOS	async	**PMA receiver equalization training data valid** - Drvies PMA input rx_eq_training_data_valid_ln_* for the associated lane. Drive high upon de-assertion of phy_reset_n.
input	pma_rx_eq_training_ln_{n:0} (#)	CMOS	async	**PMA receiver equalization training enable** - Drives PMA input rx_eq_training_ln_* for the associated lane. Drive low upon de-assertion of phy_reset_n.
input	pma_rx_eq_eval_ln_{n:0} (#)	CMOS	async	**PMA receiver equalization enable** - Drives PMA input rx_eq_eval_ln_* for the associated lane. Drive low upon de-assertion of phy_reset_n.
output	pma_rx_eq_eval_status_ln_{n:0} (#)	CMOS	async	**PMA receiver equalization status** - Driven from PMA output rx_eq_eval_status_ln_* for the associated lane.
output	pma_rx_eq_eval_complete_ln_{n:0} (#)	CMOS	async	**PMA receiver equalization complete** - Driven from PMA output rx_eq_eval_complete_ln_* for the associated lane.
output	pma_rx_link_eval_fb_dir_change_ln_{n:0}[5:0] (#)	CMOS	async	**PMA receiver equalization feedback direction change** - Driven from rx_link_fb_dir_change_ln_* for the associated lane. These pins provide increment / decrement indication for how to adjust the far end transmitter's de-emphasis setting (pre-emphasis, main and post-emphasis). Increment / decrement is wrt to the magnitude of the value. As pre- and post-emphasis are negative values, an increment would make the value more negative and a decrement would make the value less negative.
input	mac_src_sel_ln_{n:0}[1:0]	CMOS	static	**MAC clock source select** : Selects which PMA clock to use as clock couse for pcs_mac_clk*_ln_*. • 2'b00 - PMA output xcvr_pll_clk_datart_ln_* for the associated lane. • 2'b01 - PMA output xcvr_pll_clk_fullrt_ln_* for the assocaited lane. • 2'b10 - PMA output cmn_ref_clk_rcv. • 2'b11 - Reserved
input	mac_div_sel0_ln_{n:0}[6:0]	CMOS	static	**MAC clock divider ratio select** : Selects the divider ratio for pcs_mac_clk_divx0_ln_*. • 7'd0 : Reserved • 7'd1 : Divide by 1 • 7'd2 : Divide by 2 • 7'd3 : Divide by 3 ... • 7'dn : Divide by n n = 127 (maximum)
input	mac_div_sel1_ln_{n:0}[8:0]	CMOS	static	**MAC clock divider ratio select** : Selects the divider ratio for pcs_mac_clk_divx1_ln_*. • 9'd0 : Reserved • 9'd1 : Divide by 1 • 9'd2 : Divide by 2 • 9'd3 : Divide by 3 ... • 9'dn : Divide by n n = 511 (maximum)
output	pcs_mac_clk_ln_{n:0}	CMOS	n/a	**PCS MAC clock** : Selected MAC clock (mac_src_sel_ln_*)
output	pcs_mac_clk_divx0_ln_{n:0}	CMOS	n/a	**Divide by X PCS MAC clock** : Selected MAC clock divided by configured divide ratio (mac_div_sel0_ln_*).
output	pcs_mac_clk_divx1_ln_{n:0}	CMOS	n/a	**Divide by X PCS MAC clock** : Selected MAC clock divided by configured divide ratio (mac_div_sel1_ln_*).

Table 8: Multi-protocol PHY On-chip functional pin list (Continued)

Dir.	Signal Name	Type	Timing	Description
External Reference Clock Active Detected				
input	pma_cmn_ext_refclk_detected_cfg	CMOS	static	**External reference clock active detection configuration:** • 1 = Perform an external reference clock active detection operation upon de-assertion of apb_preset_n • 0 = Do not perform detection operation upon de-assertion of apb_preset_n
output	pma_cmn_ext_refclk_detected	CMOS	async	**External reference clock active detection result**: • 1 = External reference clock detected • 0 = Not detected
output	pma_cmn_ext_refclk_detected_valid	CMOS	async	**External reference clock active detection result valid:** • 1 = pma_cmn_ext_refclk_detected value is valid (i.e. detection operation has completed) • 0 = Not valid.
PHY Interrupt - A PHY configuration option is available to include or exclude the PHY Interrupt pins. These pins are equired for PHYs used in applications requiring automotive safety.				
output	phy_interrupt_ln_{n:0}	CMOS	async	**PHY interrupt output (per lane)**: Asserted high when a power state or data rate change does not complete within a timeout. Cleared upon reading the associated lane's interrupt status register. Only valid for the master lane of the link. Slave lanes should be ignored.
PHY/PMA interface and PMA clocks - See PMA specification for further description of these pins.				
input	pma_cmn_refclk_int	CMOS	n/a	**PMA common internally generated reference clock input -** Single-ended internal (i.e. SoC-side) reference clock input.
input	pma_cmn_refclk_dig_sel	CMOS	constant	**PMA common reference clock select -** Selects the reference clock source for the digital logic between cmn_refclk_<p/m> and pma_cmn_refclk_int. Must be set before the de-assertion of apb_preset_n/phy_reset_n. • 0 - cmn_refclk_<p/m> • 1 - pma_cmn_refclk_int
input	pma_cmn_refclk_mode[1:0]	CMOS	constant	**PMA common differential reference clock mode -** Sets the mode of operation for differential reference clock input. Must be set before the de-assertion of apb_preset_n/phy_reset_n. • 2'b00: Differential reference clock. • 2'b01: Single ended DC coupled test clock reference clock. • 2'b10: Reserved. • 2'b11: Reserved
input	pma_cmn_refclk_int_mode[1:0]	CMOS	constant	**PMA common internal reference clock mode** - Sets the mode of operation for internal reference clock input. Must be set before the de-assertion of apb_preset_n/phy_reset_n. • 2'b00 - Reserved • 2'b01 - 100 MHz and greater reference clock • 2'b10 - Reserved • 2'b11 - Less than 100 MHz reference clock
input	pma_cmn_refclk_dig_div[1:0]	CMOS	constant	**PMA common reference clock digital divide ratio select -** Must be set before the de-assertion of apb_preset_n/phy_reset_n. • 2'b00 - Divide by 1 (set for reference clock in the 19.2 to 27 MHz range) • 2'b01 - Divide by 2 (Reserved) • 2'b10 - Divide by 4 (set for 100 MHz reference clock) • 2'b11 - Divide by 8 (Reserved)
input	pma_cmn_refclk_term_en	CMOS	constant	**PMA common differential reference clock termination enable** - enables/disables termination for difference reference clock input (cmn_refclk_<p/m>). Must be set before the de-assertion of apb_preset_n/phy_reset_n. • 0 = termination disabled • 1 = termination enabled

Table 8: Multi-protocol PHY On-chip functional pin list (Continued)

Dir.	Signal Name	Type	Timing	Description
input	pma_cmn_pll0_refclk_sel pma_cmn_pll1_refclk_sel	CMOS	constant	**PMA PLL reference clock select** - selects which reference clock input to be used by the associated PLL. 0 = cmn_refclk_<p/m>, 1 = pma_cmn_refclk_int.
output	pma_refclk	CMOS	n/a	**PMA common reference clock output** - Driven from cmn_ref_clk PMA clock output. Provided for diagnostic purposes.
output	pma_refclk_gated	CMOS	n/a	**PMA common reference clock gated output** - Driven from cmn_ref_clk_gated PMA clock output. Provided for diagnostic purposes.
output	pma_refclk_rcv_out	CMOS	n/a	**PMA common reference clock receiver output** - Clock output from the analog reference clock receiver circuits. Frequency is always the same as the selected reference clock input. Once enabled, this clock is free-running and is active in all power states (provided the input reference clock is running). Note - This clock is enabled by setting the PMA cmn_refclk_rcv_out_en bit ([6]) in the on page 102.
output	pma_refclk_der / pma_refclk_der_n	CMOS	n/a	**PMA derived differential reference clock output** - Only available with a reference clock input frequencyin the 19.2 to 27.0 MHz range. When enabled, a 100 MHz spread-spectrum clock is output on this pin. Default is disabled. These outputs can be used to drive a common reference clock to another IC(s).
output	pma_pllclk_datart_ln_{n:0}	CMOS	n/a	PMA PLL data rate clock output - Divided down version of the high speed PLL clock going to receiver and transmitter lanes. Frequency is based on the PMA configuration and data rate.
output	pma_pllclk_fullrt_ln_{n:0}	CMOS	n/a	**PMA PLL full rate clock output** - Divided down version of the high speed PLL clock going to the receiver and transmitter lanes. Frequency is based on the PMA configuration and data rate. Additional clock division through pma_fullrt_div.
input	pma_fullrt_div_ln_{n:0}[1:0]	CMOS	static	**PMA PLL full rate clock divider select** - divide ratio for pma_pllclk_fullrt_ln_*. • 00 = Divide by 1 • 01 = Divide by 2 • 10 = Divide by 4 • 11 = Divide by 8
input	pma_cmn_clock_stop_req	CMOS	n/a	**Clock Stop Request**: Asserting high causes all clocks in the PMA macro to stop, such that the value of all of the digital F/Fs in the PMA can be scanned out.
output	pma_cmn_clock_stop_ack	CMOS	n/a	**Clock Stop Acknowledgement** - Asserted high once all clocks in the PMA macro have stopped in response to assertion of pma_cmn_clock_stop_req. Note: This signal is only valid when pma_cmn_clock_stop_req is asserted high. At times such as power on reset, this signal may be high. Additionally, if the PMA is suspended (i.e. all links in L1.x), the internal PMA reference clock is disabled. Thus, upon assertion of pma_cmn_clock_stop_req, pma_cmn_clock_stop_ack will not be asserted high, until the PMA re-enables its internal reference clock upon exit from suspend.

Table 8: Multi-protocol PHY On-chip functional pin list (Continued)

4.4 Manufacturing Test And Diagnostics Related Pins

Dir.	Signal Name	Type	Timing	Description
input	iddq_en	CMOS	async	IDDQ mode enable : When asserted high, the PMA is placed into IDDQ mode with all analog logic powered down.
TAP interface				
input	tap_tck	CMOS	n/a	
input	tap_tms	CMOS	tap_tck	
input	tap_trst_n	CMOS	tap_tck	
input	tap_tdi	CMOS	tap_tck	
output	tap_tdo	CMOS	tap_tck	
output	tap_tdoen	CMOS	tap_tck	
Rx Data Debug (optional interface)				
input	dbg_rx_rd_lane_sel[2:0]	CMOS	static	Receiver data debug lane select
output	dbg_rx_rd_clk	CMOS	n/a	Receiver parallel data clock
output	dbg_rx_rd[19:0]	CMOS	dbg_rx_rd_clk	Receiver parallel data out
output	dbg_rx_bist_sync	CMOS	dbg_rx_rd_clk	Rx BIST sync - asserted high upon BIST logic has synchronize with ithe ncoming Rx data (loopback or from serial pins) and is checking for errors. (See PMA specification for further description of this signal.)
output	dbg_rx_bist_status	CMOS	dbg_rx_rd_clk	Rx BIST status - asserted high upon BIST logic detecting an error in the Rx data. (See PMA specification for further description of this signal.)
output	dbg_rx_bist_err_toggle	CMOS	dbg_rx_rd_clk	RX BIST error toggle - toggled upon BIST logic detecting an error in the Rx data. (See PMA specification for further description of this signal.)
Digital Test Bus (optional interface - see section 4.6 “Digital Test Bus” on page 41 for more detailed description)				
input	dtb_mux_sel[4:0]	CMOS	async	Digital test bus MUX select.
output	dtb_data[31:0]	CMOS	async	Digital test bus data out
SoC Analog Test Bus				
inout	pma_atb_core_0 pma_atb_core_1	CMOS	n/a	Core-side ATB signals - When enabled, the same signals driven on the top-level ATB bumps are driven on this signal.
Digital scan interface				
input	scanmode	CMOS	n/a	Scan mode signal for PHY.
input	scanen	CMOS	n/a	Scan enable signal for PHY.
input	scanclk_refclk	CMOS	n/a	Scan test clock for reference clock rate.
input	scanclk_psmclk	CMOS	n/a	Scan test clock for PSM clock rate.
input	scanclk_pipeclk_ln_{n:0}	CMOS	n/a	Scan test clock for pipe_clk for each lane.
input	cmn_scanclk_serclk	CMOS	n/a	Scan test clock for serializer and deserializer rate in common.
input	cmn_scanclk_pllclk	CMOS	n/a	Scan test clock for PLL rate in common.
input	xcvr_scanclk_serclk_ln_{n:0}	CMOS	n/a	Scan test clock for serializer/deserializer rate for each lane.
input	xcvr_scanclk_pllclk_ln_{n:0}	CMOS	n/a	Scan test clock for PLL rate for each lane.
input	scanclk_macclk_ln_{n:0}	CMOS	n/a	Scan test clock for pcs_mac_clk_ln_,, pcs_mac_clk_divx-0_ln_<> and pcs_mac_clk_divx1_ln_<>. Only present when PHY configuration supports Raw SerDes I/F.
input	scanen_cg	CMOS	n/a	Clock gate test bypass enable signal - overrides all clock gating logic in PHY, including the PMA.
input	phy_scanin[15:0]	CMOS	Scan clocks	Scan test input pins for all the PHY (non-PMA) components. The number of chains is dependant on the defined setting and is customer dependant. (16 chains is default)
output	phy_scanout[15:0]	CMOS	Scan clocks	Scan test output pins for all the PHY (non-PMA) components. The number of chains is dependant on the defined setting and is customer dependant. (16 chains is default)

Table 9: PHY manufacturing test and diagnostics related pin list

Dir.	Signal Name	Type	Timing	Description
input	pma_cmn_scanin[39:0]	CMOS	Scan clocks	Scan test input pins for all the Torrent PMA common.
output	pma_cmn_scanout[39:0]	CMOS	Scan clocks	Scan test output pins for all the Torrent PMA common.
input	pma_xcvr_scanin_ln_{n:0}[63:0]	CMOS	Scan clocks	Scan test input pins for the Torrent PMA lane.
output	pma_xcvr_scanout_ln_{n:0}[63:0]	CMOS	Scan clocks	Scan test output pins for all the Torrent PMA lane.
External boundary scan control interface (see PMA spec for further description)				
input	bscan_ext_select	CMOS	const	PMA external boundary scan select (1 = external boundary scan interface is selected, 0 = internal TAP controller selected)
input	tx_bscan_ext_shiftdr	CMOS	tx_bscan_ext_clockdr	PMA TX boundary scan enable
input	tx_bscan_ext_clockdr	CMOS	async	PMA TX boundary scan data sample clock
input	tx_bscan_ext_updatedr	CMOS	async	PMA Tx boundary scan data update clock
input	tx_bscan_ext_mode_c	CMOS	async	PMA Tx boundary scan mode control
input	tx_bscan_ext_ac	CMOS	async	PMA Tx boundary scan AC test mode select
input	tx_bscan_ext_acpulse	CMOS	async	PMA Tx boundary scan AC test pulse
output	tx_bscan_ext_tdo	CMOS	async	PMA Tx boundary scan test out - connected to tx_b-scan_tdo output for highest lane number in PMA macro. For each lane, the rx_bscan_tdo output is tied to tx_bscan_tdi input and between adjacent lanes the tx_bscan_tdo output is tied to the next higher lanes rx_bscan_tdi input.
input	rx_bscan_ext_shiftdr	CMOS	rx_bscan_ext_clockdr	PMA Rx boundary scan enable
input	rx_bscan_ext_clockdr	CMOS	async	PMA Rx boundary scan data sample clock
input	rx_bscan_ext_pc	CMOS	async	PMA Rx boundary scan preload clock
input	rx_bscan_ext_ac	CMOS	async	PMA Rx boundary scan AC mode
input	rx_bscan_ext_tdi	CMOS	async	PMA Rx boundary scan test in - connected to rx_bscan_tdi input for lane 0
Boundary scan control interface (Internal and External)				
input	xcvr_bscan_mode_en	CMOS	async	PMA Transceiver boundary scan mode enable : This signal enables boundary scan mode.

Table 9: PHY manufacturing test and diagnostics related pin list (Continued)

4.5 Un-supported PHY/PIPE Interface Signals

The following PIPE interface signals are not supported by the PHY:

- BlockAlignControl
- EncodeDecodeBypass (for 8b/10b data, supported for 128b/13xb data)
- RxStandBy / RxStandByStatus - Turning off Rx in P0 or PCIe P0s not supported.
- RxPresetHint[2:0] - Not recommended for use by PIPE specification.
- FS[5:0] / LF[5:0] - Not needed by PHY for far-end transmitter equalization.
- LinkEvaluationFeedbackFigureMerit[7:0] - LinkEvaluationFeedbackDirection implemented instead.
- RxEqInProgress - Not required by PHY.
- SRISEnable - By default, PHY is enabled for SRIS operation.
- PCLK Rate - the PCLK rate is determined based on the selected data rate and PIPE data bus width (16- or 32-b).
- PowerPresent
- PclkChangeOk / PclkChangeAck - Not needed as PCLK rate change not supported by PHY.
- ElasticBufferLocation[N:0]
- AsyncPowerChangeAck - Not needed as PHY does not support a power state change where PCLK is not active in either power state.

- CLKREQ# This pin is defined in the PCIe spec. It is expected to be implemented and controlled outside of the PHY.

4.6 Digital Test Bus

The digital test bus is an optional interface that allow monitoring of selected PMA interface signals and clocks. The following table describes the PMA signals provided based on the MUX selection.

• nnn - selects the lane number.
• All dtb_mux_sel values not in the above table will select the same dtb_data output as dtb_mux_sel == 5'b0000.

dtb_data [bit]	dtb_mux_sel		
	5'b00000	5'b00001	5'b00010
0	cmn_ssm_state[0]	cmn_pllsm0_state[0]	tx_sfifo_align_bump_in_ln_0
1	cmn_ssm_state[1]	cmn_pllsm0_state[1]	tx_sfifo_align_bump_out_ln_0
2	cmn_ssm_state[2]	cmn_pllsm0_state[2]	tx_sfifo_enq_reset_n_in_ln_0
3	cmn_ssm_state[3]	cmn_pllsm0_state[3]	tx_sfifo_enq_reset_n_out_ln_0
4	cmn_ssm_state[4]	cmn_pllsm0_state[4]	tx_sfifo_align_bump_in_ln_1
5	cmn_ssm_state[5]	cmn_pllsm0_state[5]	tx_sfifo_align_bump_out_ln_1
6	cmn_ssm_state[6]	cmn_pllsm0_state[6]	tx_sfifo_enq_reset_n_in_ln_1
7	cmn_ssm_state[7]	cmn_pllsm0_state[7]	tx_sfifo_enq_reset_n_out_ln_2
8	cmn_ssm_state[8]	cmn_pllsm0_state[8]	tx_sfifo_align_bump_in_ln_2
9	cmn_reset_n	cmn_pllsm0_state[9]	tx_sfifo_align_bump_out_ln_2
10	cmn_ready	cmn_pllsm0_state[10]	tx_sfifo_enq_reset_n_in_ln_2
11	cmn_refclk_active	cmn_pllsm0_state[11]	tx_sfifo_enq_reset_n_out_ln_2
12	cmn_refclk_disable	cmn_pllsm1_state[0]	tx_sfifo_align_bump_in_ln_3
13	cmn_macro_suspend_req	cmn_pllsm1_state[1]	tx_sfifo_align_bump_out_ln_3
14	cmn_macro_suspend_ack	cmn_pllsm1_state[2]	tx_sfifo_enq_reset_n_in_ln_3
15	cmn_ref_clk0_clk_gate_en	cmn_pllsm1_state[3]	tx_sfifo_enq_reset_n_out_ln_3
16	cmn_pll0_en	cmn_pllsm1_state[4]	tx_sfifo_align_bump_in_ln_4
17	cmn_pll0_locked	cmn_pllsm1_state[5]	tx_sfifo_align_bump_out_ln_4
18	cmn_pll0_disabled	cmn_pllsm1_state[6]	tx_sfifo_enq_reset_n_in_ln_4
19	cmn_pll0_ready	cmn_pllsm1_state[7]	tx_sfifo_enq_reset_n_out_ln_4
20	cmn_pll0_mode_sel	cmn_pllsm1_state[8]	tx_sfifo_align_bump_in_ln_5
21	cmn_pll1_en	cmn_pllsm1_state[9]	tx_sfifo_align_bump_out_ln_5
22	cmn_pll1_locked	cmn_pllsm1_state[10]	tx_sfifo_enq_reset_n_in_ln_5
23	cmn_pll1_disabled	cmn_pllsm1_state[11]	tx_sfifo_enq_reset_n_out_ln_5
24	cmn_pll1_ready	cmn_reset_n	tx_sfifo_align_bump_in_ln_6
25	cmn_pll1_mode_sel	cmn_ready	tx_sfifo_align_bump_out_ln_6
26	cmn_ref_clk_dig_sel	cmn_macro_suspend_req	tx_sfifo_enq_reset_n_in_ln_6
27	cmn_ref_clk_int_mode[0]	cmn_macro_suspend_ack	tx_sfifo_enq_reset_n_out_ln_6
28	cmn_ref_clk_int_mode[1]	cmn_pll0_en	tx_sfifo_align_bump_in_ln_7
29	cmn_ref_clk0_mode[0]	cmn_pll1_en	tx_sfifo_align_bump_out_ln_7
30	cmn_ref_clk0_mode[1]	cmn_pll0_ready	tx_sfifo_enq_reset_n_in_ln_7
31	0	cmn_pll1_ready	tx_sfifo_enq_reset_n_out_ln_7

Table 10: Digital Test Bus A

dtb_data [bit]	dtb_mux_sel		
	5'b01nnn	5'b10nnn	5'b11nnn
0	xcvr_psm_state_ln_{nnn}[0]	xcvr_power_state_req_ln_{nnn}[0]	tx_elec_idle_ln_{nnn}
1	xcvr_psm_state_ln_{nnn}[1]	xcvr_power_state_req_ln_{nnn}[1]	tx_lfps_en_ln_{nnn}
2	xcvr_psm_state_ln_{nnn}[2]	xcvr_power_state_req_ln_{nnn}[2]	tx_get_local_preset_coef_ln_{nnn}
3	xcvr_psm_state_ln_{nnn}[3]	xcvr_power_state_req_ln_{nnn}[3]	tx_local_preset_index_ln_{nnn}[0]
4	xcvr_psm_state_ln_{nnn}[4]	xcvr_power_state_req_ln_{nnn}[4]	tx_local_preset_index_ln_{nnn}[1]
5	xcvr_psm_state_ln_{nnn}[5]	xcvr_power_state_ack_ln_{nnn}[0]	tx_local_preset_index_ln_{nnn}[2]
6	xcvr_psm_state_ln_{nnn}[6]	xcvr_power_state_ack_ln_{nnn}[1]	tx_local_preset_index_ln_{nnn}[3]
7	xcvr_psm_state_ln_{nnn}[7]	xcvr_power_state_ack_ln_{nnn}[2]	tx_local_tx_coef_valid_ln_{nnn}

Table 11: Digital Test Bus B

dtb_data [bit]	dtb_mux_sel		
	5’b01nnn	5’b10nnn	5’b11nnn
8	xcvr_psm_state_ln_{nnn}[8]	xcvr_power_state_ack_ln_{nnn}[3]	tx_vmargin_ln_{nnn}[0]
9	xcvr_psm_state_ln_{nnn}[9]	xcvr_power_state_ack_ln_{nnn}[4]	tx_vmargin_ln_{nnn}[1]
10	xcvr_psm_state_ln_{nnn}[10]	xcvr_standard_mode_ln_{nnn}[0]	tx_vmargin_ln_{nnn}[2]
11	xcvr_psm_state_ln_{nnn}[11]	xcvr_standard_mode_ln_{nnn}[1]	tx_low_power_swing_en_ln_{nnn}
12	xcvr_psm_state_ln_{nnn}[12]	xcvr_data_width_ln_{nnn}[0]	tx_cmn_mode_en_ext_ln_{nnn}
13	xcvr_psm_state_ln_{nnn}[13]	xcvr_data_width_ln_{nnn}[1]	tx_rcv_detect_en_ln_{nnn}
14	xcvr_psm_state_ln_{nnn}14[]	xcvr_data_width_ln_{nnn}[2]	tx_rcv_detect_done_{nnn}
15	xcvr_psm_state_ln_{nnn}[15]	tx_sfifo_align_bump_in_ln_{nnn}	tx_rcv_detected_ln_{nnn}
16	xcvr_psm_state_ln_{nnn}[16]	tx_sfifo_align_bump_out_ln_{nnn}	rx_signal_detect_ln_{nnn}
17	xcvr_psm_state_ln_{nnn}[17]	tx_sfifo_enq_reset_n_in_ln_{nnn}	rx_lfps_detect_ln_{nnn}
18	xcvr_psm_state_ln_{nnn}[18]	tx_sfifo_enq_reset_n_out_ln_{nnn}	rx_termination_ln_{nnn}
19	xcvr_psm_state_ln_{nnn}[19]	cmn_ready	rx_sig_det_en_ext_ln_{nnn}
20	xcvr_psm_state_ln_{nnn}[20]	cmn_macro_suspend_req	rx_eq_training_data_valid_ln_{nnn}
21	xcvr_psm_state_ln_{nnn}[21]	cmn_macro_suspend_ack	rx_eq_training_ln_{nnn}
22	xcvr_psm_state_ln_{nnn}[22]	cmn_pll0_en	rx_eq_eval_ln_{nnn}
23	xcvr_pm_state_ln_{nnn}[23]	cmn_pll0_ready	rx_eq_eval_status_ln_{nnn}
24	xcvr_psm_state_ln_{nnn}[24]	cmn_pll1_en	rx_eq_eval_complete_ln_{nnn}
25	xcvr_psm_state_ln_{nnn}[25]	cmn_pll0_ready	rx_link_eval_fb_dir_change_ln_{nnn}[0]
26	cmn_reset_n	cmn_refclk_active	rx_link_eval_fb_dir_change_ln_{nnn}[1]
27	xcvr_link_reset_n_ln_{nnn}	cmn_refclk_disable	rx_link_eval_fb_dir_change_ln_{nnn}[2]
28	xcvr_lane_suspend_ln_{nnn}	xcvr_link_reset_n_ln_{nnn}	rx_link_eval_fb_dir_change_ln_{nnn}[3]
29	xcvr_pll_clk_en_ln_{nnn}	xcvr_lane_suspend_ln_{nnn}	rx_link_eval_fb_dir_change_ln_{nnn}[4]
30	xcvr_pll_clk_en_ack_ln_{nnn}	xcvr_pll_clk_en_ln_{nnn}	rx_link_eval_fb_dir_change_ln_{nnn}[5]
31	cmn_ready	xcvr_pll_clk_en_ack_ln_{nnn}	rx_cdrlf_fphl_locked_ln_{nnn}

Table 11: Digital Test Bus B (Continued)

5 PHY Internal CDB Access

The PHY macro contains an internal APB bus, which is known as the CDB bus. This is used to access configuration and diagnostic registers throughout the macro. There are two methods for accessing this CDB bus. The primary method of accessing the CDB bus is using the top level APB pins. The alternate method is using the TAP by reading and writing the PHY macro diagnostic control register on page 46. Arbitration between these two methods for access to the CDB is controlled by the Asynchronous APB Interface Slave module. Either the APB or the TAP can access the CDB, with the TAP having highest priority. Note that, activating either the apb_preset_n top level pin, or the diagnostic mode APB reset bit in the PHY macro diagnostic control register on page 46, will result in the CDB being reset.

This section assumes the reader is familiar with both the APB and TAP protocols. What is provided here is the necessary information to address the specific regions of the CDB address space. Additionally, the full register map for the macro top level CDB registers is provided. The full register address map for the PMA specific CDB registers is not described in this specification. See the PMA specification for details on these registers.

5.1 CDB Address Encoding

In this implementation, the APB address bus is 16 bits wide. Table 12 on page 43 illustrates the address mapping at the PHY level. Note that for PMA address space, one should see the PMA specification.

apb_paddr bits	Internal CDB address region
15:14	Internal CDB address space. • 2’b00 : Torrent PMA common module registers. • 2’b01 : Torrent PMA transmitter module registers. • 2’b10 : Torrent PMA receiver module registers. • 2’b11 : PCIe PHY macro top level registers.
13	When the PHY macro top level register space is selected, this bit selects PHY level PCS or PMA registers. • 1’b0 : PHY level PCS registers. • 1’b1 : PHY level PMA registers.
12	When the PHY macro top level register space is selected, this bit selects PHY level common or lane registers. • 1’b0 : PHY level common registers. • 1’b1 : PHY level lane registers.
11	Reserved - must be set to 1’b0.
10:8	When the PHY macro top level register space is selected, and lane registers are selected, these bits specify the lane being addressed. • 3’b000 : Lane 0 • 3’b001 : Lane 1 • 3’b010 - 3’b111 : Lane 2 - 7 When PHY level common registers are selected, these bits must be set to 4’b0000.
7:6	When the PHY macro top level register space is selected, these bits are reserved, and must be set to 2’b00.
5:0	When the PHY macro top level register space is selected, these bits specify up to 32 addressable registers for a given lane or common.

Table 12: CDB high level address space mapping

5.2 Top Level Register Address Space

The following table describes the address map for each of the registers within the macro top level register address region.

Address	Register name and location of detailed description
PHY/PCIe PCS Common Registers	
16'b11_0_0_0000_0000_0000	PIPE common control1 register on page 94
16'b11_0_0_0000_0000_0001	PIPE common control2 register on page 95
16'b11_0_0_0000_0000_0010	PIPE comma lock configuration1 register on page 95
16'b11_0_0_0000_0000_0011	PIPE comma lock configuration2 register on page 95
16'b11_0_0_0000_0000_0100	PIPE EIEOS lock configuration register on page 96
16'b11_0_0_0000_0000_0101	PIPE lane disable register on page 96
16'b11_0_0_0000_0000_0110	PIPE receiver detect inhibit register on page 96
16'b11_0_0_0000_0000_0111	PIPE RX electrical idle L1.x exit delay register on page 97
16'b11_0_0_0000_0000_1000	PHY common control isolation register on page 105
16'b11_0_0_0000_0000_1010	PHY state change monitor timeout on page 97
16'b11_0_0_0000_0000_1110	PHY PLL configuration register on page 97
16'b11_0_0_0000_0000_1111	PHY speedup control register on page 98
16'b11_0_0_0000_0001_0000	PHY External Reference Clock Detect Low Threshold register on page 98
16'b11_0_0_0000_0001_0001	PHY External Reference Clock Detect High Threshold register on page 98
16'b11_0_0_0000_0001_0010	PHY External Reference Clock Detect Measurement Interval register on page 98
16'b11_0_0_0000_0001_0011	PHY External Reference Clock Detect Delay register on page 99
16'b11_0_0_0000_0001_0100	PHY External Reference Clock Detect Isolation Control register on page 99
16'b11_0_0_0000_0010_0000	PHY USB3 Gen 2 pre-shoot configuration 0 register on page 101
16'b11_0_0_0000_0010_0001	PHY USB3 Gen 2 pre-shoot configuration 1 register on page 101
16'b11_0_0_0000_0010_0010	PHY USB3 Gen 2 de-emphasis configuration 0 register on page 101
16'b11_0_0_0000_0010_0011	PHY USB3 Gen 2 de-emphasis configuration 1 register on page 102
PHY/PCIe PCS Lane Registers **Note that there is a set of these register for each PMA lane (where nnn = 000 for lane 0).**	
16'b11_0_1_0nnn_0000_0000	PIPE TX control signal isolation register on page 106
16'b11_0_1_0nnn_0000_0001	PIPE Tx local preset coefficients low isolation register on page 106
16'b11_0_1_0nnn_0000_0010	PIPE Tx local preset coefficients high isolation register on page 106
16'b11_0_1_0nnn_0000_0011	PIPE TX Deemphasis low isolation register on page 107
16'b11_0_1_0nnn_0000_0100	PIPE Tx Deemphasis high isolation register on page 107
16'b11_0_1_0nnn_0000_0101	PIPE Tx local FS/LF isolation register on page 107
16'b11_0_1_0nnn_0000_0110	PCS PIPE TX Data low isolation register on page 107
16'b11_0_1_0nnn_0000_0111	PCS PIPE TX Data high isolation register on page 107
16'b11_0_1_0nnn_0000_1000	PCS PIPE RX control signal isolation register on page 108
16'b11_0_1_0nnn_0000_1001	PIPE Rx equalization evaluation isolation register on page 108
16'b11_0_1_0nnn_0000_1010	PHY link configuration isolation register on page 109
16'b11_0_1_0nnn_0000_1011	PCS PIPE Link control isolation register on page 109
16'b11_0_1_0nnn_0000_1100	PIPE USB loopback slave BER count register on page 109
16'b11_0_1_0nnn_0000_1110	PCS PIPE RX data low isolation register on page 110
16'b11_0_1_0nnn_0000_1111	PCS PIPE RX data high isolation register on page 110
16'b11_0_1_0nnn_0001_0000	Ethernet MAC clock configuration isolation register on page 110
16'b11_0_1_0nnn_0001_0001	Ethernet MAC clock divider isolation register on page 110
16'b11_0_1_0nnn_0001_0010	PHY interrupt status register on page 100
Torrent PMA Common Registers	
16'b11_1_0_0000_0000_0000	PMA common control1 register on page 102
16'b11_1_0_0000_0000_0001	PMA common control2 register on page 102
16'b11_1_0_0000_0000_0010	PMA SSM current state register on page 103
16'b11_1_0_0000_0000_0011	PMA PLL Raw SerDes control register on page 103
16'b11_1_0_0000_0000_0100	PMA common control signal isolation register on page 111
16'b11_1_0_0000_0000_0101	PMA PLL control0 isolation register on page 111
16'b11_1_0_0000_0000_0110	PMA PLL control1 isolation register on page 111
16'b11_1_0_0000_0000_1011	PMA PLL0 State Machine current state register on page 103
16'b11_1_0_0000_0000_1100	PMA PLL1 State Machine current state reg on page 103
16'b11_1_0_0000_0000_1111	PMA control isolation register on page 112
Torrent PMA Lane Registers **Note that there is a set of these register for each PMA lane (where nnn = 000 for lane 0).**	
16'b11_1_1_0nnn_0000_0000	PMA transceiver control register on page 104
16'b11_1_1_0nnn_0000_0001	PMA loopback control register on page 104
16'b11_1_1_0nnn_0000_0010	PMA PI position register on page 104

Table 13: Macro top level CDB register address map

Address	Register name and location of detailed description
16'b11_1_1_0nnn_0000_0011	PMA transceiver control isolation register on page 112
16'b11_1_1_0nnn_0000_0100	PMA Tx local preset coefficient low isolation register on page 113
16'b11_1_1_0nnn_0000_0101	PMA Tx local preset coefficient high isolation register on page 113
16'b11_1_1_0nnn_0000_0110	PMA Tx de-emphasis low isolation register on page 113
16'b11_1_1_0nnn_0000_0111	PMA Tx de-emphasis high isolation register on page 113
16'b11_1_1_0nnn_0000_1000	PMA Tx FS/LF isolation register on page 114
16'b11_1_1_0nnn_0000_1001	PMA Tx margin isolation register on page 114
16'b11_1_1_0nnn_0000_1010	PMA Link Mode isolation register on page 114
16'b11_1_1_0nnn_0000_1011	PMA Link Power State Control isolation register on page 114
16'b11_1_1_0nnn_0000_1101	PMA Rx equalization control isolation register on page 115
16'b11_1_1_0nnn_0000_1110	PMA low data isolation register on page 115
16'b11_1_1_0nnn_0000_1111	PMA high data isolation register on page 115
16'b11_1_1_0nnn_0001_0000	PMA Transceiver PSM current state low register on page 104
16'b11_1_1_0nnn_0001_0001	PMA Transceiver PSM current state high register on page 105

Table 13: Macro top level CDB register address map

5.3 PMA Setup

Refer to section 5 of the PMA specification for any PMA register configuration required (as necessary), which is different from the default.

6 PHY Configuration

6.1 PHY Configuration Overview

This PHY is designed to support multiple SerDes standards, as well as different reference clock rates. A given standard or reference clock rate may require registers and / or pins to be set to values other than their reset values. The required values for the registers and / or pins are specified using the Torrent programmers guide spreadsheet.

Not only can the PHY support multiple standards, but also support multiple standards simultaneously. Due to this flexibility, there are a large number of programming options that exist. It is recommended that the user read the following sections of this document in order to understand the general process for setting the register and pin values in the PHY. Once this information is understood, the Torrent programmers guide spreadsheet can be used to determine the correct programming of this macro to configure it to support the desired standards.

• Section 6.2 “PHY Initial Setup” on page 46
• Section 6.3 “PIPE Rate Change” on page 46
• Section 6.4 “Raw SerDes Rate Change” on page 46

6.2 PHY Initial Setup

As described in the PHY specification, the PHY must be configured/set (registers and configuration input pins) prior to release of phy_reset_n to the PHY. Registers and pins shall be configured for the selected PHY configuration, as specified in the tables in the programmers guide spreadsheet, between de-assertion of apb_preset_n and phy_reset_n.

6.3 PIPE Rate Change

For links using the PIPE interface, the link rate is changed using the pipe_l<nn:00>_rate PHY input pins for the master lane of the link using the PIPE protocol.

6.4 Raw SerDes Rate Change

For links using the Raw SerDes interface, the link rate is changed following the protocol described in the PHY specification:

- Place the link in the A3 power state (using pma_xcvr_power_state_req_ln_{n:0}/pma_xcvr_power_state_ack_ln_{n:0}) and disable PLL datarate clock (using pma_xcvr_pllclk_en_ln_{n:0}/pma_xcvr_pllclk_en_ack_ln_{n:0}).
- Set pma_xcvr_data_width_ln_{n:0} and pma_xcvr_standard_mode_ln_{n:0} for the new rate.
- Re-configure the PMA for the new rate. If this requires reconfiguring the PLL driving the link, follow the procedure in the PHY specification to:
    - Disable the PLL using PHY_PMA_PLL_RAW_CTRL[1:0].
    - Wait for the associated PLL to be disabled.
    - Reconfigure the required PMA registers.
    - Re-enable the PLL and wait for ready for operation.
- Re-enable the link.
-

6.5 Torrent Programmers Guide Spreadsheet Usage

This section describes how to use the Torrent programmers guide spreadsheet to generate the correct programming of this macro to configure it to support the desired standards, when not using the tables provided in this section. Note that the user should not deviate from the instructions for using this spreadsheet, by modifying any fields, other than those described in this procedure. Doing so can result in incorrect macro behavior.

1. Open the spreadsheet (Torrent_prg_gd.xlsm) with Microsoft Excel. Note that there are VBA scripts associated with this spreadsheet, so executable code must be allowed to run.
2. Select the “Config” sheet.
3. Click on the “Select Physical Implementation” button, which will trigger the “Select physical implementation” menu box to appear.
    1. In the “Select technology implementation” drop down box, select the technology implementation of the device.
    2. In the “Select revision number” drop down box, select the revision number of the device.
    3. In the “Select configuration option” drop down box, select the physical configuration of the device.
    4. Click on the OK button.
4. Click on the “Select Physical Lane Configuration” button, which will trigger the “Physical lane configuration” menu box to appear. Note: At this time, this menu box does not have an affected on the generated results of this spreadsheet, but may in the future.
    1. Use the buttons in this menu box to select the lanes that are physically implemented in the device.
    2. Click on the OK button.
5. Click on the “Select Standards” button, which will trigger the “Standard selection” menu box to appear.
    1. In the “Select standard 0” drop down box, select the desired standard to configure.
    2. In the “Select standard 1” drop down box, select the desired standard to configure (when configuring the device to support multiple standards simultaneously). Note that only specific standards can be selected to be paired with the standard selected using the “Select standard 0” drop down box. Therefore, the list in the “Select standard 1” drop down box may be a smaller list.
6. If Display Port, SRIO, or JESD are selected for either standard 0 or standard 1, click on the respective “Select Standard 0/1 Data Rate” button, which will trigger the respective “data rate”, menu box to appear.
    1. Select or enter the required data rate for the respective standard.
    2. Click on the OK button.
7. Click on the “Assign Standards To Lanes” button, which will trigger the “Assign selected standards to physical lanes” menu box to appear. Note: At this time, this menu box does not have an affected on the generated results of this spreadsheet, but may in the future.
    1. Use the buttons in this menu box to specify the physical lanes that the selected standard will be implemented on, or if the lane is disabled.
    2. Click on the OK button.
8. Click on the “Reference Clock Configuration” button, which will trigger the “Reference clock configuration” menu box to appear.
    1. In the “Select reference clock frequency” drop down box, select the desired reference clock frequency.
    2. In the “Select internal or external reference clock” box, select internal or external.
    3. In the “Select spread spectrum requirement” box, select the required spread spectrum setting.
9. Click on the “Generate Register Sheet” button, to generate the register and pin settings for the configuration settings that have been selected.
    - At this point, the required register write and pin information is available in the “Gen_Regs” sheet. The following points should be noted in regards to this sheet.
        - The register mnemonics column specifies the register to be written, or pin to be set.
        - For lane (not common) registers, the register in each lane where a given standard is to be used must be written.
        - The applicable technology tags and applicable revision tags columns are diagnostic columns that provide a quick check that the correct registers for a given technology and revision are generated.
        - The boolean column provides a logical equation as to when the register write should be applied. If the entry in this column is “True”, it applies in all cases.
        - The PHY / Common / Lane column identifies if the register or pin is associated with the PHY, common, or lane. In the case where the lanes are specified, the respective lanes relative to a selected standard will be identified.
        - The final columns contain the register write or pin values. Note that if “RST_VAL” is noted in a specific cell, no write is required.

10. Clicking on the “Generate Register CSV” button, will generate a comma separated value (CSV) file of the information in the “Gen_Regs” sheet. The following should be noted with regards to this generated file.
    - When reading this file into a tool like MS Excel, some rows will be more than one row of characters tall. It may be necessary to adjust the height of these rows in the tool to see the entire contents.
    - When reading this file into a tool like MS Excel, it may be necessary to adjust the width of the columns, in the tool to see the entire contents.
    - Notes exist in the file that indicate sections of registers that are associated with common functions and functions that are specific to a selected standard.

6.5.1 Generated Register CSV File Naming

The following tables describe the format of the file name for the generated register CSV file, and the information associated with each field that makes up the files name.

Programmers guide spreadsheet version 0.22 and earlier.

File Name Format:	Gen_Regs_<tech>_<rel>_<std_0>_<std_1>_<ref_clk_freq>_<ref_clk_src>_<ref_clk_ss>_<pll_0_ss>_<pll_1_ss>.csv
Field Name	**Field Description**
tech	**Technology tag**: Tag name used for the selected technology. Examples of this are: “”
rel	**Revision tag**: Tag name used for the selected revision number. Examples of this are: “R100”, “R110”, and “R120”.
std_0	**Standard 0 tag**: Tag name used for the standard selected as standard 0. The following are possible tag names. • PCIe_SL: PCIe Gen 1-3 single link • PCIe_ML: PCIe Gen 1-3 Multiple Links • USB_G1: USB 3.0 / 3.1 (Gen 1 Only) • USB: USB 3.0 / 3.1 • USB_TCAM_G1: USB 3.0 / 3.1 (Gen 1 Only) Type C Alt Mode • USB_TCAM: USB 3.0 / 3.1 Type C Alt Mode • DP<_data rate>: Display Port • XAUI_LP: XAUI (1X Only) • XAUI: XAUI • SGMII: SGMII • ENET: Ethernet • SATA_LP: SATA (1.5 and 3.0 Only) • SATA: SATA • SRIO<_data rate>: SRIO • JESD<_data rate>: JESD • XFI: XFI • USXGMII: USXGMII • 10GKR: 10G-KR • GEN_10_5: Generic 10 Gbps 5 Gbps
std_1	**Standard 1 tag**: Tag name used for the standard selected as standard 1. See the description for field std_0 for the list of possible tag names.
ref_clk_freq	**Reference clock frequency tag**: Tag name for the reference clock frequency. The following are examples of this tag and the respective frequency. • 100p0: 100 MHz • 19p2: 19.2 MHz • 20p0: 20.0 MHz • 24p0: 24.0 MHz • 25p0: 25.0 MHz • 26p0: 26.0 MHz • 27p0: 27.0 MHz • 37p5: 37.5 MHz • 75p0: 75.0 MHz • 125p0: 125.0 MHz • 150p0: 150.0 MHz • 156p25: 156.25 MHz
ref_clk_src	**Reference clock selected source**: Tag name for the source that is selected for the reference clock. The following are possible tag names. • Ext: External reference clock bumps • Int: Core side reference clock
ref_clk_ss	**Reference clock spread spectrum present**: Tag name that indicates if spread spectrum is present on the reference clock source. The following are possible tag names. • rcsst: Spread spectrum is present • rcssf: Spread spectrum is not present
pll_0_ss	**PLL 0 spread spectrum generation enable**: Tag name that indicates if spread spectrum generation is enabled on PLL 0. The following are possible tag names. • p0sst: Spread spectrum generation is enabled • p0ssf: Spread spectrum generation is disabled
pll_1_ss	**PLL 1 spread spectrum generation enable**: Tag name that indicates if spread spectrum generation is enabled on PLL 1. The following are possible tag names. • p1sst: Spread spectrum generation is enabled • p1ssf: Spread spectrum generation is disabled

Table 14: Generated register CSV file naming (version 0.22 and earlier)

Programmers guide spreadsheet version 0.23 and later.

File Name Format:	Gen_Regs_<tech>_<rel>_<std_0>_<std_1>_<ext_0_ref_clk>_<int_ref_clk>_ <dig_ref_clk>_<pll_0_ref_clk>_<pll_1_ref_clk>_<rcv_0_ref_clk>.csv
Field Name	**Field Description**
tech	**Technology tag**: Tag name used for the selected technology. Examples of this are: “”
rel	**Revision tag**: Tag name used for the selected revision number. Examples of this are: “R100”, “R110”, and “R120”.
std_0	**Standard 0 tag**: Tag name used for the standard selected as standard 0. The following are possible tag names. • PCIe_SL: PCIe Gen 1-3 single link • PCIe_ML: PCIe Gen 1-3 Multiple Links • USB_G1: USB 3.0 / 3.1 (Gen 1 Only) • USB: USB 3.0 / 3.1 • USB_TCAM_G1: USB 3.0 / 3.1 (Gen 1 Only) Type C Alt Mode • USB_TCAM: USB 3.0 / 3.1 Type C Alt Mode • DP: Display Port • XAUI_LP: XAUI (1X Only) • XAUI: XAUI • SGMII: SGMII • ENET: Ethernet • SATA_LP: SATA (1.5 and 3.0 Only) • SATA: SATA • SRIO: SRIO • JESD: JESD • XFI: XFI • USXGMII: USXGMII • 10GKR: 10G-KR • GEN_10_5: Generic 10 Gbps 5 Gbps
std_1	**Standard 1 tag**: Tag name used for the standard selected as standard 1. See the description for field std_0 for the list of possible tag names.

ext_0_ref_clk	**External reference clock 0 information tag**: Tag name for the information associated with external reference clock 0, when it is used. If this clock is not used, this tag name will not be present in the file name. The following is the format of this tag name, and the respective definitions of each sub field.
	• Ext0_<ss><freq>
	  • ss: If spread spectrum is present on the clock, “ss_” will be in the name.
	  • freq: Tag name for the frequency of this reference clock. The following are examples of this tag and the respective frequency.
	  • 100p0: 100 MHz
	  • 19p2: 19.2 MHz
	  • 20p0: 20.0 MHz
	  • 24p0: 24.0 MHz
	  • 25p0: 25.0 MHz
	  • 26p0: 26.0 MHz
	  • 27p0: 27.0 MHz
	  • 37p5: 37.5 MHz
	  • 75p0: 75.0 MHz
	  • 125p0: 125.0 MHz
	  • 150p0: 150.0 MHz
	  • 156p25: 156.25 MHz

int_ref_clk	**Internal reference clock information tag**: Tag name for the information associated with internal reference clock, when it is used. If this clock is not used, this tag name will not be present in the file name. The following is the format of this tag name, and the respective definitions of each sub field.
	• Int_<ss><freq>
	  • ss: If spread spectrum is present on the clock, “ss_” will be in the name.
	  • freq: Tag name for the frequency of this reference clock. See the description of the ext_0_ref_clk for examples of this tag.

dig_ref_clk	**Selected digital reference clock tag**: Tag name used for the reference clock selected to be used for the digital functions. The following are examples of this tag and the respective selected reference clock. • drc_e0: External reference clock 0 • drc_i: Internal reference clock • no tag: No reference clock selected
pll_0_ref_clk	**Selected PLL 0 reference clock tag**: Tag name used for the reference clock selected to be used for PLL 0. The following are examples of this tag and the respective selected reference clock. • p0rc_e0: External reference clock 0 • p0rc_i: Internal reference clock • no tag: No reference clock selected • If spread spectrum generation is enabled, _ss will be added to this tag.

Table 15: Generated register CSV file naming (version 0.23 and later)

File Name Format:	Gen_Regs_<tech>_<rel>_<std_0>_<std_1>_<ext_0_ref_clk>_<int_ref_clk>_ <dig_ref_clk>_<pll_0_ref_clk>_<pll_1_ref_clk>_<rcv_0_ref_clk>.csv
Field Name	**Field Description**
pll_1_ref_clk	**Selected PLL 1 reference clock tag**: Tag name used for the reference clock selected to be used for PLL 1. The following are examples of this tag and the respective selected reference clock. • p1rc_e0: External reference clock 0 • p1rc_i: Internal reference clock • no tag: No reference clock selected • If spread spectrum generation is enabled, _ss will be added to this tag.
rcv_0_ref_clk	**Selected reference clock receiver 0 reference clock tag**: Tag name used for the reference clock selected to be used for reference clock receiver 0. The following are examples of this tag and the respective selected reference clock. • r0rc_e0: External reference clock 0 • r0rc_i: Internal reference clock • no tag: No reference clock selected

Table 15: Generated register CSV file naming (version 0.23 and later)

7 Operation Procedures And Options

This section describes various operational procedures for the Torrent 12FFC Multi-Protocol PHY macro. Some are required and some are optional, which will be identified in the respective descriptions.

7.1 Procedure: External control of TX common mode and RX signal detect in A3

7.1.1 Versions impacted by issue

• All versions of Torrent 12FFC Multi-Protocol PHY .

7.1.2 Description of procedure

In some customer implementations, it is desirable to use the A3 power state (PIPE P2 power state) for both L2 and L1.x. With such implementations, it is necessary to control the TX common mode and RX signal detect when the PMA is in the A3 power state (PIPE P2 power state). This function is implemented by changing the state of a couple PMA CDB registers. After this, the phy_lnn_tx_cmn_mode_en and phy_lnn_rx_elec_idle_det_en inputs control the TX common mode and RX signal detect respectfully, in the same way that these pins control these functions in the L1 substates.

7.1.3 Details of the procedure

When writing PMA registers, write the following register bits.
• PMA: TX_PSC_A3[11] = 1’b1
• PMA: TX_PSC_A3[5] = 1’b0
• PMA: TX_PSC_A3[3] = 1’b0
• PMA: TX_PSC_A3[2] = 1’b0
• PMA: TX_PSC_A3[1] = 1’b0
• PMA: RX_PSC_A3[11] = 1’b1
• PMA: RX_PSC_A3[12] = 1’b1

7.2 Procedure: Reference clock driver setup

7.2.1 Versions impacted by issue

• All versions of Torrent 12FFC Multi-Protocol PHY .

7.2.2 Description of procedure

This procedure describes how to set up and enable the optional reference clock driver for the reference clock 0 bumps. By default, the driver is disabled. Note, the only time the registers associated with setting up this driver should be changed is while the PHY is in reset.

7.2.3 Details of the procedure

When writing PMA registers, write the following register bits.

- Select which reference clock will be driven, by writing the clock select bit in the PMA:CMN_CDIAG_REFCLK_DRV0_CTRL register. Note that when the received reference clock is selected, this is the internal reference clock driven on the pma_cmn_refclk_int pin.
- Set any of the remaining tuning register bits for the reference clock driver by writing the respective bits in the PMA:CMN_CDIAG_REFCLK_DRV0_CTRL register.
- If the derived reference clock is selected, enable the derived reference by setting the derived reference clock enable bit in the PMA:CMN_CDIAG_REFCLK_OVRD register.
- Enable the driver, by writing the driver enable override enable bit in the PMA:CMN_CDIAG_REFCLK_DRV0_CTRL register to 1’b0, and the RefClk enable input ignore bit in PHY_PIPE_CMN_CTRL1 register to 1'b1. At this point the driver is configured to drive the reference clock. The following points should be noted.
    - **The driver must never be enabled when a reference clock signal is being driven to the reference clock bumps by an external source.**
    - The PHY level phy_en_refclk pin can be used to gate the driven reference clock on and off, as described in the pin description.

7.3 Procedure: Power island control state machine input override

7.3.1 Versions impacted by issue

• All versions of Torrent 12FFC Multi-Protocol PHY .

7.3.2 Description of procedure

Each of the power island control state machines in the PMA have an option to override the inputs of these state machines in order to force the respective power island to the switched on or switched off state. This procedure describes how these input overrides are used. The following points should be noted in regards to this procedure.

- The diagnostic power island is not directly controlled by hardware, and can only be enabled by the procedure described here.
- The register writes described in this procedure must only be performed while the PMA is held in reset by the cmn_reset_n pin. The only exception to this are register writes to enable the diagnostic power island. These register writes can also be performed when a given lane is in the A0 power state.
- Each power island control state machine has a power island control override and power island control status register associated with it. This description will be a function of accesses to these registers.
- The reference clock is required to be active for this procedure.
- If any of the switched power islands other than the CDB registers power island are forced to be switched on, the CDB registers power island must first be forced to be switched on, this includes the diagnostic power island.
- Logic exists in the design that prevents the CDB register power island, as well as other power islands from being released from reset until after an opportunity exists to write registers in the common control module control and diagnostic registers space. The reason for this is, there are registers that can be used to tune the analog reference clock receiver circuits, and if these need to be written to, this must happen before any other function uses the reference clock. To complete this procedure, any register outside of the common control module control and diagnostic registers space can be accessed. This will trigger the release of reset of all the power island control circuits.
- The override values for the PSM calibration power island and transceiver data path power island must be the same for all the lanes that make up a link.

7.3.3 Details of the procedure

After the PHY CDB has been released from reset and before the PHY has been released from reset using the PIPE reset, perform the following steps to use the power island control state machine input override function.

1. If any of the analog reference clock receiver circuit tuning registers are required to be written, write these registers.
2. Access any register outside of the common control module control and diagnostic registers space. A simple register read is sufficient for this.
3. If either the transceiver calibration or transceiver data path power islands are being controlled by this procedure, force the PSM clock gate to be enabled by writing the force PSM gated clock on bit in the XCVR_PSM_USER_DEF_CTRL on page 209 to 1’b1, for the lane where these power island are being forced on.
4. Set the power island controller input override enable bit of the power island control override register to 1’b1.
5. Set either the power suspend request override or the power recover request override bit in the power island control override register to 1’b1.
6. For the power islands except for CDB registers and diagnostic power islands, release the PHY from reset using the PIPE reset. After this reset is released, these power islands will sequence through the requested process.
7. When the power suspend acknowledge and power recover acknowledge bits in the power island control override register are set to 1’b1, this will indicate that the power island control state machine has completed the requested process. Note that both of these bits will be set, because this is required for internal state machines that normally control the power island control state machine to function correctly.

If for some reason, it is desired to switch the forced state of the power island to the opposite state (forced on to off or forced off to on). The following steps should be followed before repeating the above procedure to force the new state.

1. Set the power suspend request override and the power recover request override bit in the power island control override register to 1’b0.
2. Set the power island controller input override enable bit of the power island control override register to 1’b0.

3. Wait for the suspend acknowledge bit in the power island control override register that is the opposite of the originally forced request to transition to 1’b0.

7.4 Procedure: Power island control state machine output override

7.4.1 Versions impacted by issue

• All versions of Torrent 12FFC Multi-Protocol PHY .

7.4.2 Description of procedure

Each of the power island control state machines in the PMA have an option to directly override the outputs from these state machines, that control the power islands switching functions. This override option provides the ability to force the respective power island to the switched on or switched off state, completely under register control. This procedure describes how these output overrides are used. The following points should be noted in regards to this procedure.

- These output overrides are intended to be for debug of catastrophic conditions, where the reference clock may not be running.
- Due to the nature of this override function being intended for catastrophic debug, there may be related functions that do not work correctly. Therefore, there are likely to be limitations as to what can be used when a power island is controlled using this override option. For example, if there are clocks that are not functioning correctly in the design, reset synchronizers that the power island control state machine drives may not allow the respective reset to be released.
- One should be familiar with the sequencing of the output control signals when using this override function. These signal sequence in a specific order, and this order needs to be adhered to when using this override function.
- Each power island control state machine has a power island control override and power island control status register associated with it. This description will be a function of accesses to these registers.
- The override values for the PSM calibration power island and transceiver data path power island must be the same for all the lanes that make up a link.

7.4.3 Details of the procedure

Perform the following steps to use the power island control state machine output override function.

1. Read the power island control status register to determine the current state of the power island control state machine output signals.
2. Reference the description of the power island controller output override field in the power island control override register. In this description there is a prescribed sequence of values that can be written to this field. Use the value that was read in the previous step to determine where in the sequence the power island control state machine currently is.
3. Write the value that was read in the first step to the power island controller output override field in the power island control override register.
4. Set the power island controller output override enable bit of the power island control override register to 1’b1.
5. Referencing the description of the power island controller output override field in the power island control override register, identify the sequence of values that are required to be written to transition from the current state of the power island control state machine outputs to the desired state.
6. Write the identified sequence in the previous step to the power island controller output override field of the power island control override register, one value at a time.

7.5 Procedure: Change PMA TX Emphasis (PCIe Gen 1 and 2, and USB Gen 1)

7.5.1 Versions impacted by issue

• All versions of Torrent 12FFC Multi-Protocol PHY .

7.5.2 Description of procedure

In some cases, it may be desirable to change the default TX emphasis settings of -6, -3.5, and 0 dB, for PCIe Gen 1 and 2 and USB3 Gen 1, that are under the control of the tx_deemphasis[1:0] pin. This can be done by changing the default values of the following PMA registers to the values specified in table 135 “Additional optional calculated emphasis values” on page 499 of the Torrent 12FFC Multi-Protocol PHY PMA specification.

• PMA: TX_TXCC_CPRE_MULT_00
• PMA: TX_TXCC_CPRE_MULT_01
• PMA: TX_TXCC_CPRE_MULT_10
• PMA: TX_TXCC_CPRE_MULT_11
• PMA: TX_TXCC_CPOST_MULT_00
• PMA: TX_TXCC_CPOST_MULT_01
• PMA: TX_TXCC_CPOST_MULT_10
• PMA: TX_TXCC_CPOST_MULT_11

7.5.3 Details of the procedure

When writing PMA registers, write any of the registers noted above to the required value for the desired level of emphasis. It is also possible to write these registers at other times. It is not advisable to write these registers while data is actually being transmitted as the emphasis levels will change immediately.

7.6 Procedure: Display Port TX Swing And Emphasis Level Control

7.6.1 Versions impacted by issue

• All versions of Torrent 12FFC Multi-Protocol PHY .

7.6.2 Description of procedure

Display port swing level and emphasis level control is unique compared to other standards supported by this macro. When changing the swing and emphasis level, multiple register writes are required. The issue of concern here is that the analog driver can get in a state that is not valid while in the middle of writing the sequence of registers that are required to change the swing and emphasis levels. For this reason, latches exist in the macro to hold the state of the driver, while these register writes take place, then when the writes are complete, the latches are switched to the transparent state.

7.6.3 Details of the procedure

1. Using the programmers guide spreadsheet, generate the CSV file for the required Display Port data rate.
2. Using the Display Port standard, identify the desired swing level (0, 1, 2, or 3). This is identified using the SL_DP variable in the generated CSV file.
3. Using the Display Port standard, identify the desired de-emphasis level (0, 1, 2, or 3). This is identified using the DE_DP variable in the generated CSV file.
4. Using the Display Port standard, identify the desired differential voltage level (low or high). This is identified using the VDIFF_DP variable in the generated CSV file.
5. Write TX_DIAG_ACYA[0] to 1’b1 to freeze the current state of the analog TX driver.
6. Write TX_TXCC_CTRL to the value associated with the required SL_DP, DE_DP, and VDIFF_DP variable settings.
7. Write DRV_DIAG_TX_DRV to the value associated with the required SL_DP, DE_DP, and VDIFF_DP variable settings.
8. Write TX_TXCC_MGNFS_MULT_000 to the value associated with the required SL_DP, DE_DP, and VDIFF_DP variable settings.
9. Write TX_TXCC_CPOST_MULT_00 to the value associated with the required SL_DP, DE_DP, and VDIFF_DP variable settings.
10. Write TX_DIAG_ACYA[0] to 1’b0 to allow the state of the analog TX driver to reflect the new programmed values.

7.7 Procedure: Change PMA Margin Settings

7.7.1 Versions Impacted by Issue

• All versions of Torrent 12FFC Multi-Protocol PHY .

7.7.2 Description of Procedure

The full swing and half swing margin settings for the PMA are defined as required for PCIe. For this mode of operation, full swing is defined as 1.0V Vppd. In some modes of operation, it may be desirable to change the margin settings for different voltage swings. This can be done by changing the default values of the coefficient calculator margin multiplier registers, defined in section 10.2.19 “TX Coefficient Calculator Registers” on page 210 of the PMA specification. The table here describes an example of margin register settings that can be used for different I/O driver swing range.

tx_ vmargin	Full Swing Margin	Full Swing Nominal Voltage Vppd (mV)	Half Swing Margin
3’b000	RC * 0/128	1200	RC * 0/128
3’b001	RC * 2/128	1143	RC * 5/128
3’b010	RC * 5/128	1086	RC * 11/128
3’b011	RC * 8/128	1029	RC * 17/128
3’b100	RC * 11/128	971	RC * 23/128
3’b101	RC * 14/128	914	RC * 29/128
3’b110	RC * 17/128	857	RC * 35/128
3’b111	RC * 20/128	800	RC * 41/128

Table 16: Example alternate margin calculation values

7.7.3 Details of the procedure

When writing PMA registers, write any of the registers noted above to the required value for the desired level of margin. It is also possible to write these registers at other times. It is not advisable to write these registers while data is actually being transmitted as the emphasis levels will change immediately.

7.8 Procedure: Running Eye Surf

7.8.1 Versions Impacted by Issue

• All versions of Torrent 12FFC Multi-Protocol PHY .

7.8.2 Description of Procedure

The eye surf function is used to measure the RX eye height and width at the analog data sampler. This is a non destructive process, which means it can be performed in mission mode and not create any bit errors in the received data. The eye surf function can measure any point in a the eye space consisting of 32 steps in the east or west direction from center (the eye width) and 32 steps in the north or south direction from center (the eye height)

By running the eye surf function multiple times over different points in the eye space, different types of diagnostic tests can be performed. The following are examples of such tests.

• Bathtub curve: Setting the north / south coordinate to 0, and then running multiple tests to sweep the east / west coordinate.
• Full eye plot: Running multiple tests to sweep both the north / south and east / west coordinates.

7.8.3 Details of the procedure

After the PHY CDB has been released from reset and before the PHY has been released from reset using the PIPE reset, perform the following.

1. Enable the analog receiver E path in the A0 and A1 power states by writing the RX E path enable bit in the PMA: RX_PSC_A0 and PMA: RX_PSC_A1 registers to 1'b1.
2. Enable the diagnostic switched power island by first writing the power island controller input override enable bit in the PMA: CMN_CDIAG_DIAG_PWRI_OVRD register to 1'b1, then writing the power recovery request override bit in the PMA: CMN_CDIAG_DIAG_PWRI_OVRD register to 1'b1. Note that even though these bits are in the same register, they must be set with two separate writes.

Proceed as normal to receive data in the A0 power state. At the point that it is desired to run the eye surf function, perform the following. It should be noted that any TX equalization process should not be running when running the eye surf function.

1. Disable all receiver equalization functions by writing the enable bit of the PMA: RX_REE_GCSM1_CTRL, PMA: RX_REE_GCSM2_CTRL, and PMA: RX_REE_PERGCSM_CTRL REE state machine control registers to 1'b0.
2. Write the PMA: RX_EYESURF_TMR_DELLOW register to 16'h000F.
3. Determine the desired number of RX clock cycles the eye surf function should be run for each test point. The number of UI tested for each RX clock cycle is equivalent to the parallel data width for the standard mode being tested. Up to 2^{32} RX clock cycles can be tested. A good starting value for this is 0x2000. Once the desired number of RX clock cycles is determined, write this value into the PMA: RX_EYESURF_TMR_TESTHIGH and PMA: RX_EYESURF_TMR_TESTLOW registers.
4. For each coordinate point being tested, perform the following.
    1. Set the north / south coordinate for the point being tested by writing the PMA: RX_EYESURF_NS_COORD register. Note that each step in the offset field of this register corresponds to approximately 8 mV.
    2. Set the east / west coordinate for the point being tested by writing the PMA: RX_EYESURF_EW_COORD register. Note that each step in the offset field of this register corresponds to 1/32 UI.
    3. Initiate the eye surf process for the test point by writing the eye surf run bit of the PMA: RX_EYESURF_CTRL register to 1'b1.
    4. Wait for confirmation that the eye surf process is complete by reading the PMA: RX_EYESURF_CTRL register. When the eye surf done bit in this register is set to 1'b1, the process is complete.
    5. Read the PMA: RX_EYESURF_ERRCNT register to report the number of bit errors that occurred at the test point during the test.
    6. Write the eye surf run bit of the PMA: RX_EYESURF_CTRL register to 1'b0.
    7. Read the PMA: RX_EYESURF_CTRL register until the eye surf done bit is set to 1'b0.
5. Once the data for all the desired test points have been collected, the receiver equalization functions can be enabled again, by writing the enable bit of the PMA: RX_REE_GCSM1_CTRL, PMA: RX_REE_GCSM2_CTRL, and PMA: RX_REE_PERGCSM_CTRL REE state machine control registers to 1'b1.
6. If no further eye surf functions are to be run, the analog receiver E path can be disabled in the power state controller by writing the RX E path enable bit in the PMA: RX_PSC_A0 and PMA: RX_PSC_A1 registers to 1'b0.
7. If no further eye surf functions are to be run, the diagnostic switched power island can be switched off by writing the power recovery request override bit in the PMA: CMN_CDIAG_DIAG_PWRI_OVRD register to 1'b0.

7.9 Procedure: Enabling PMA Loopback Functions

7.9.1 Versions impacted by issue

• All versions

7.9.2 Description of procedure

The PMA contains multiple loopback functions. The following procedure describes the process for enabling the loopback functions. The following additional points should be noted in regards to this.

• Only one loopback function for a lane should be enabled at any given point in time.
• A lane must be in the reset state when enabling any loopback function.
• One should read the testability section of the PMA specification for further details about each loopback function.

7.9.3 Details of the procedure

1. Confirm that the lanes that make up a link are held in reset by driving the PIPE reset active, and the CDB reset is currently driven inactive.
2. Enable the desired loopback function by setting the corresponding bit in the PHY_PMA_XCVR_LPBK register to 1’b1. Note that only one loopback function for a given lane should be enabled at any given point in time.
3. Release the lanes that make up a link from reset by driving the PIPE reset inactive.

7.10 Procedure: Using reference clock output for bifurcation

7.10.1 Versions impacted by issue

• All versions

7.10.2 Description of procedure

For cases when multiple instances of this macro are instantiated in a bifurcated PHY, it is common to use the cmn_ref_-clk_rcv from a single PMA to drive the reference clock input of other PMAs. The cmn_ref_clk_rcv output is disabled by default. Therefore, in these implementations, it is critical to enable this output prior to accessing registers in the PMAs that use this clock. The reason for this is, the reference clock is required to enable the CDB registers power island in the PMA.

7.10.3 Details of the procedure

Prior to accessing registers in the PMA, drive the cmn_ref_clk_rcv_en signal active by writing the respective bit in the PHY_PMA_CMN_CTRL1 register.

7.11 Procedure: Running BIST

7.11.1 Versions impacted by issue

• All versions

7.11.2 Description of procedure

This procedure describes how to use the built in BIST functions found in each lane of the PMA. This procedure assumes that the PMA is set up with one of the TX to RX loopback functions, and data is generated by the TX BIST and checked by the RX BIST.

7.11.3 Details of the procedure

After the PHY CDB has been released from reset and before the PHY has been released from reset using the PIPE reset, perform the following.

1. Enable the diagnostic switched power island by first writing the power island controller input override enable bit in the PMA: CMN_CDIAG_DIAG_PWRI_OVRD register to 1’b1, then writing the power recovery request override bit in the PMA: CMN_CDIAG_DIAG_PWRI_OVRD register to 1’b1. Note that even though these bits are in the same register, they must be set with two separate writes.

Proceed as normal to the A0 power state. At the point that it is desired to run the BIST function, perform the following.

1. Set the BIST receiver sync count to 250 by writing the receiver BIST sync count field of the PMA: RX_BIST_SYNCCNT register to 16’h00FA.
2. Select the desired BIST data pattern by writing the transmitter BIST mode field of the PMA: TX_BIST_CTRL register to the respective value.
3. If the selected data pattern is user defined data FIFO, perform the following.
    1. Clear the contents of the user defined data FIFO, by writing the transmitter BIST user defined data FIFO clear bit in the PMA: TX_BIST_CTRL register to 1’b1.
    2. Write the desired user defined data pattern to the PMA: TX_BIST_UDDWR register.
4. Select the expected BIST data pattern by writing the receiver BIST mode field of the PMA: RX_BIST_CTRL register to the respective value. Note that this value must match the value written to the transmitter BIST mode field of the PMA: TX_BIST_CTRL register.
5. If the selected data pattern is user defined data FIFO, perform the following.
    1. Clear the contents of the user defined data FIFO, by writing the receiver BIST user defined data FIFO clear bit in the PMA: RX_BIST_CTRL register to 1’b1.
    2. Write the desired user defined data pattern to the PMA: RX_BIST_UDDWR register.
6. Enable the transmitter BIST by writing the transmitter BIST enable bit of the PMA: TX_BIST_CTRL register to 1’b1.
7. Enable the receiver BIST by writing the receiver BIST enable bit of the PMA: RX_BIST_CTRL register to 1’b1.
8. Determine if the RX BIST has synced to the received data pattern, by observing the PMA: rx_bist_sync pin. This pin can be observed by the PHY: PHY_PMA_XCVR_CTRL register.
9. The status of the RX BIST can be monitored using the PMA: rx_bist_status and PMA: rx_bist_err_toggle pins. These pins can be observed by the PHY: PHY_PMA_XCVR_CTRL register.
10. The receiver error count can be determined by reading the PMA: RX_BIST_ERRCNT register.
11. When BIST testing is complete, disable the receiver BIST by writing the receiver BIST enable bit of the PMA: RX_BIST_CTRL register to 1’b0.
12. When BIST testing is complete, disable the transmitter BIST by writing the transmitter BIST enable bit of the PMA: TX_BIST_CTRL register to 1’b0.
13. The error tracking information in the receiver can be cleared by writing the receiver BIST error reset bit in the PMA: RX_BIST_CTRL register to 1’b1 then 1’b0.

7.12 Procedure: Procedure for not using the external calibration resistor

7.12.1 Versions impacted by issue

• All versions of Torrent 12FFC Multi-Protocol PHY .

7.12.2 Description of procedure

When multiple instances of the Torrent PHY are used in the same SOC, the option exists to only populate the external calibration resistor in one instance of the Torrent PHY, and share the calibration information across each of the other instances of the Torrent PHY. For the purposes of this description, the instance with an external resistor attached will be named Torrent_ExtR, and the instances without an external resistor attached will be named Torrent_NoExtR.

When using this procedure, the Torrent_ExtR instance must be enabled at the start of day, while keeping all Torrent_NoExtR instances disabled. This will allow the calibrations to be run on Torrent_ExtR, such that the calibration codes can be read from the Torrent_ExtR instance, and written to all the instances of the Torrent PHY. After the writes are complete, the all the instances of the Torrent PHY can go through a normal enable process.

7.12.3 Details of the procedure

1. For each Torrent_NoExtR:
   - Hold each CDB reset active.
   - Hold each PIPE reset active.
2. For the Torrent_ExtR:
   1. Release the CDB reset.
   2. Program the PHY and PMA registers for the required operation (reference clock frequency), as specified by the programmers guide spreadsheet.
   3. Force the CDB power island to remain enabled, by first writing the power island controller input override enable bit of the PMA: CMN_CDIAG_CDB_PWRI_OVRD register to 1’b1, and then writing the power recover request override bit of the PMA: CMN_CDIAG_CDB_PWRI_OVRD register to 1’b1.
   4. Release the PIPE reset.
   5. Wait for the PMA cmn_ready signal to be driven active.
   6. Save the external bias current calibration code, by reading the calibration code field of the PMA: CMN_IBCAL_CTRL register.
   7. Save the transmitter pull up calibration code, by reading the calibration code field of the PMA: CMN_TXPUCAL_CTRL register.
   8. Save the transmitter pull down calibration code, by reading the calibration code field of the PMA: CMN_TXPDCAL_CTRL register.
   9. Save the receiver calibration code, by reading the calibration code field of the PMA: CMN_RXCAL_CTRL register.
   10. Activate the PIPE reset.
   11. Activate the CDB reset.
3. For each Torrent to be enabled for use.
   1. Release the CDB reset.
   2. Program the PHY and PMA registers for the required operation, as specified by the programmers guide spreadsheet.
   3. Use the saved value for the external bias current calibration code as the functional calibration code for the external bias current calibration, by writing the value to the calibration code override value field, and writing the calibration code override enable bit to 1’b1 in the PMA: CMN_IBCAL_OVRD register.
   4. Use the saved value for the transmitter pull up calibration code as the functional calibration code for the transmitter pull up calibration, by writing the value to the calibration code override value field, and writing the calibration code override enable bit to 1’b1 in the PMA: CMN_TXPUCAL_OVRD register.
   5. Use the saved value for the transmitter pull down calibration code as the functional calibration code for the transmitter pull down calibration, by writing the value to the calibration code override value field, and writing the calibration code override enable bit to 1’b1 in the PMA: CMN_TXPDCAL_OVRD register.

6. Use the saved value for the receiver calibration code as the functional calibration code for the receiver calibration, by writing the value to the calibration code override value field, and writing the calibration code override enable bit to 1’b1 in the PMA: CMN_RXCAL_OVRD register.
7. Release the PIPE reset, and use the PHYs as normal.

8 PHY Operation

This section provides description and timing diagrams for PHY operation. Unless otherwise described in this specification, the macro's PIPE interface is compliant with the timing diagrams described in the PIPE v4.0 specification.

# indicates the master lane of a link.

8.1 Start-up Sequence Timing Diagram

The below diagrams assume 2 single lane links (i.e. PHY configured as 2x1, 2 independent single-lane links), in which one link is enabled at start-up and the other link is enabled after the first link has been enabled. The first diagram shows 2 PIPE links and the second 2 Raw SerDes links. A link (whether single or multi-lane) is enabled at start-up when phy_l*_reset_n is high upon de-assertion of phy_reset_n (high). A link is disabled at start-up by asserting phy_l*_reset_n low upon de-assertion of phy_reset (high).

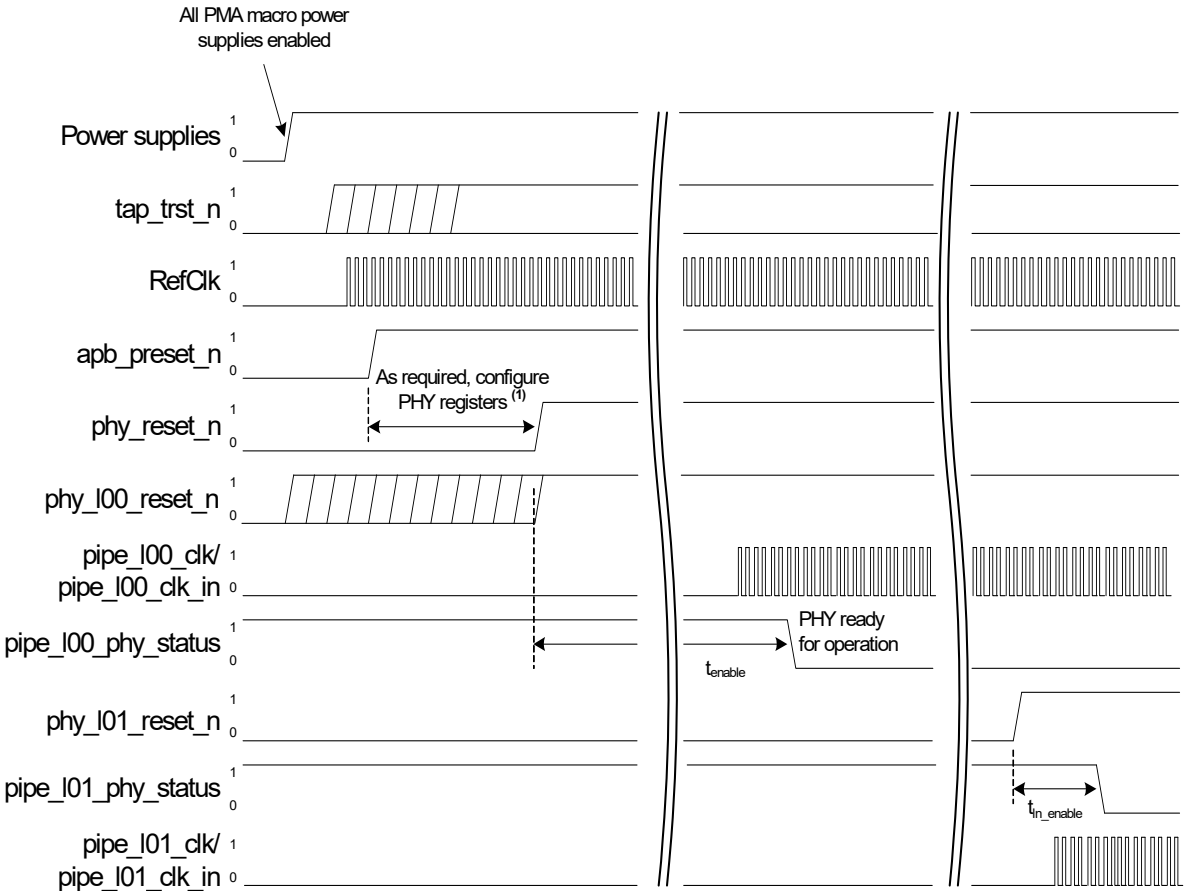


Figure 6: PIPE Start-up sequence

(1) Configure PHY and PMA APB registers as required for the selected PHY modes of operation to be supported between de-assertion of apb_preset_n and de-assertion of phy_reset_n.

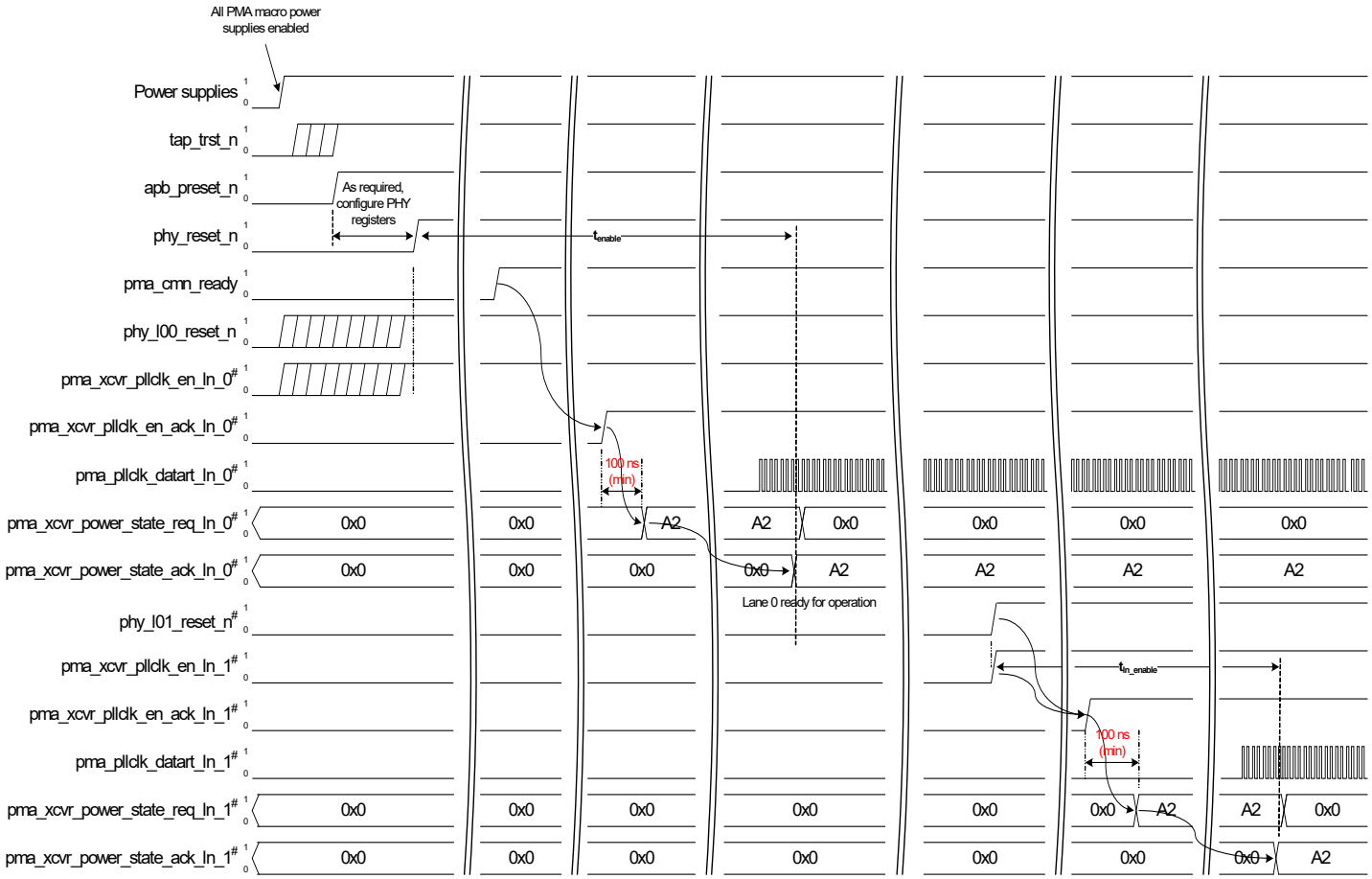


Figure 7: Raw SerDes Start-up sequence

Upon release of reset for a link (i.e. phy_reset_n when the link is enabled at start-up or phy_l*_reset_n otherwise), pma_xcvr_power_state_req_ln_* shall be set to 0x0 for a Raw SerDes link and pma_xcvr_pllclk_en_ln_* shall be asserted high. Upon assertion high of pma_xcvr_pllclk_en_ack_ln_* (following assertion of pma_cmn_ready), pma_xcvr_power_state_req_ln_* can be set to A2. Upon entry into the A2 power state, the link power state can be changed to the A0 power state following Raw SerDes I/F power state change procedure described in 8.4 “Power state change” on page 70.

(1) Configure PHY and PMA APB registers as required for the selected PHY modes of operation to be supported after de-assertion of apb_preset_n and before de-assertion of phy_reset_n.

pma_xcvr_data_width_ln_* and pma_xcvr_standard_mode PHY inputs for the link must be set for the selected Raw SerDes mode prior to enabling the link.

8.2 PHY disable description

To disable the PHY and place it in its lowest power state, assert apb_preset_n and phy_reset_n low and disable cmn_refclk_<p/m> and apb_pclk. To re-enable the PHY follow the start-up sequence described above.

To disable a link, assert phy_l*_reset_n low for the link’s master lane. This will disable the link and place all PMA lanes of the link in their lowest power state (i.e. the powerdown disabled state).

When a link is disabled, its inputs must be driven with their initial/start-up/reset values as described in the PIPE specification, in Figure 6 and in Figure 7.

To re-enable the link, de-assert phy_l*_reset_n for the link’s master lane and follow the start-up sequence described above for enabling the second link.

For a single-link configuration, there is no difference between the PHY and the link. As such, phy_l00_reset_n operates the same as phy_reset_n.. In this case, the link cannot be disabled separate from the PHY. The above PHY disable procedure is used for either the link or the PHY when configured as a single-link.

8.3 Disabling reference clock to PHY

The reference clock to the PHY can be disabled (gated off low) when all links are either powerdown disabled (with phy_l*_reset_n driven low for the master lane of the link) or for PCIe links in the L1 substate (L1.x). When the last link is PCIe and is entering L1.x, the reference clock can be disabled to the PHY upon assertion high of phy_l*_ent_l1_x and phy_l*_ack_l1_x for the master lane of the link. When the last link is being powerdown disabled, the reference clock can be gated off 2 usec after the assertion of phy_l*_reset_n for the master lane of the link.

The reference clock must be enabled upon de-assertion low of phy_l*_ent_l1_x for the master lane of any PCIe link that is exiting L1.x or de-assertion high of phy_l*_reset_n for the master lane of any link that is exiting powerdown disabled.

When enabling/disabling the reference clock and the PHY is not in reset (phy_reset_n == 0), the clock must be cleanly turned on and off, such that no runt pulses are received by the PHY. Upon signaling exit from L1.x or upon enabling a link, the reference clock does not have to be active. However, the L1.x exit latency / link enable latency does not start until the reference clock does.

8.4 Power state change

8.4.1 PIPE

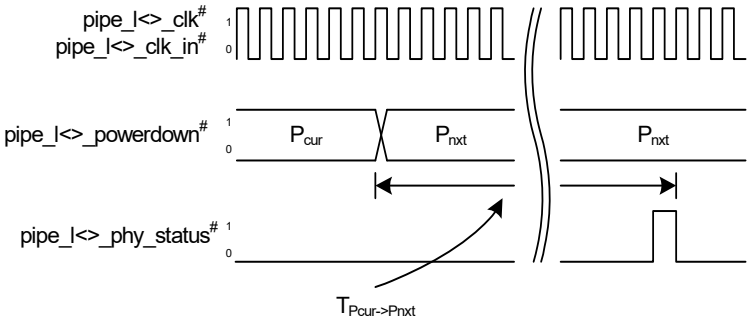


Figure 8: PIPE mode power state change (without PIPE clock stop)

For PCIe, this corresponds to P0/P0s/P1 -> P0/P0s/P1 power state change. For USB, this corresponds to P0/P1/P2 -> P0/P1/P2 power state change.

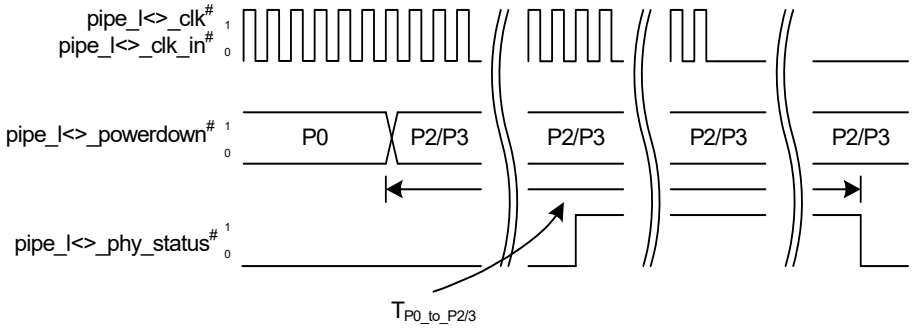


Figure 9: PIPE power state change (with PIPE clock stop)

For PCIe, this corresponds to P0/P1 -> P2 power state change. For USB, this corresponds to P0 to P3 power state change.

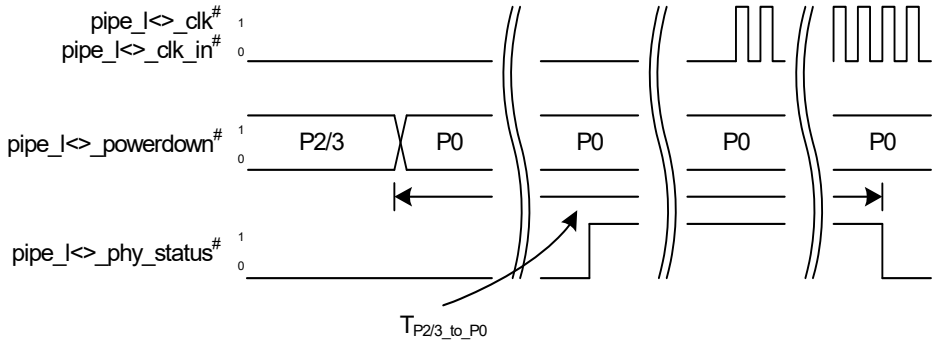


Figure 10: PIPE power state change (with PIPE clock enable)

For PCIe, this corresponds to P2 -> P0/P1 power state change. For USB, this corresponds to P3 -> P0/P2 power state change.

8.4.1.1 PCIe L1 Sub-state Entry/Exit Timing Diagrams

The PCIe L1 sub-states are entered either through either the side-band interface (phy_l<>_ent_l1_x / phy_l<>_ack_l1_x) when phy_l<>_pcie_l1_ss_sel is low or using PIPE powerdown and PhyStatus signals when phy_l<>_pcie_l1_ss_sel is high. 2 sets of signals are used for both modes: phy_l<>_rx_elec_idle_det_en and phy_l<>_tx_cmn_mode_en. There are 2 use cases (same for each mode) depending on whether the PHY is outputting a reference clock or not. When not outputting a reference clock, phy_en_refclk is tied low and not used as part of the L1.x entry and exit protocols.

A link can only use one of the L1.x entry/exit interfaces as selected with phy_l<>_pcie_l1_ss_sel. phy_l<>_pcie_l1_ss_sel is a static PHY input and can only be changed when the PHY and/or link is disabled.

8.4.1.1.1 L1.x entry (sideband signaling)

The PHY must be in the P1 power state prior to initiating L1.x entry when using sideband signaling. L1.x entry is initiated by asserting phy_l<>_ent_l1_x high.

The L1.x entry procedure is dependent upon the PHY's mode of operation based upon whether the PHY is outputting a reference clock on cmn_refclk_<p/m> or not. When not outputting a reference clock, phy_en_refclk is ignored and the input reference clock can be turned off upon assertion of phy_l<>_ack_l1_x (when all other links are also powered down). When outputting a reference clock, phy_en_refclk cannot be de-asserted until after phy_l<>_ack_l1_x is asserted high (when all other links are also powered down). Upon assertion of phy_l<>_ack_l1_x, the output reference clock can be turned off by de-assertion low of phy_en_refclk. After de_assertion of phy_en_refclk, the PHY reference clock input (i.e. pma_cmn_refclk_int) can be turned off only upon the de-assertion low of phy_refclk_reqd.

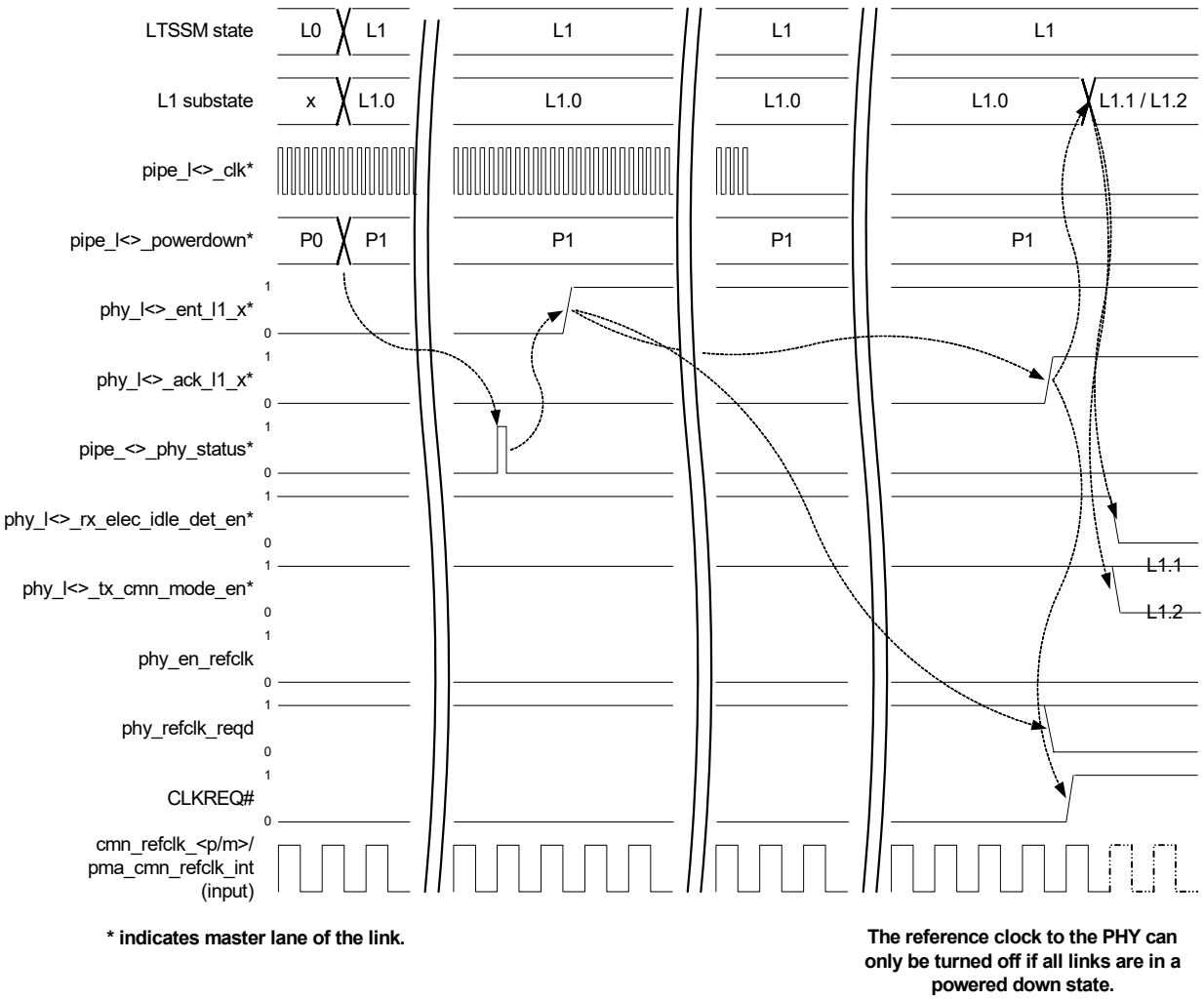


Figure 11: PCIe L1.1 entry sideband timing (without reference clock output)

For L1.1, phy_l<>_tx_cmn_mode_en is left asserted high. For L1.2, phy_l<>_tx_cmn_mode_en is de-asserted upon entering L1.2 Idle sub-state.

The input reference clock can only be turned off if all links are powered down (i.e. disabled, in PCIe P2, USB P3, PCIe L1.x, etc.). This can be determined based on each link's power state or based on the state of phy_refclk_reqd. When phy_refclk_reqd is high, the input reference clock must be left running. Upon de-assertion, the input reference clock may be turned off.

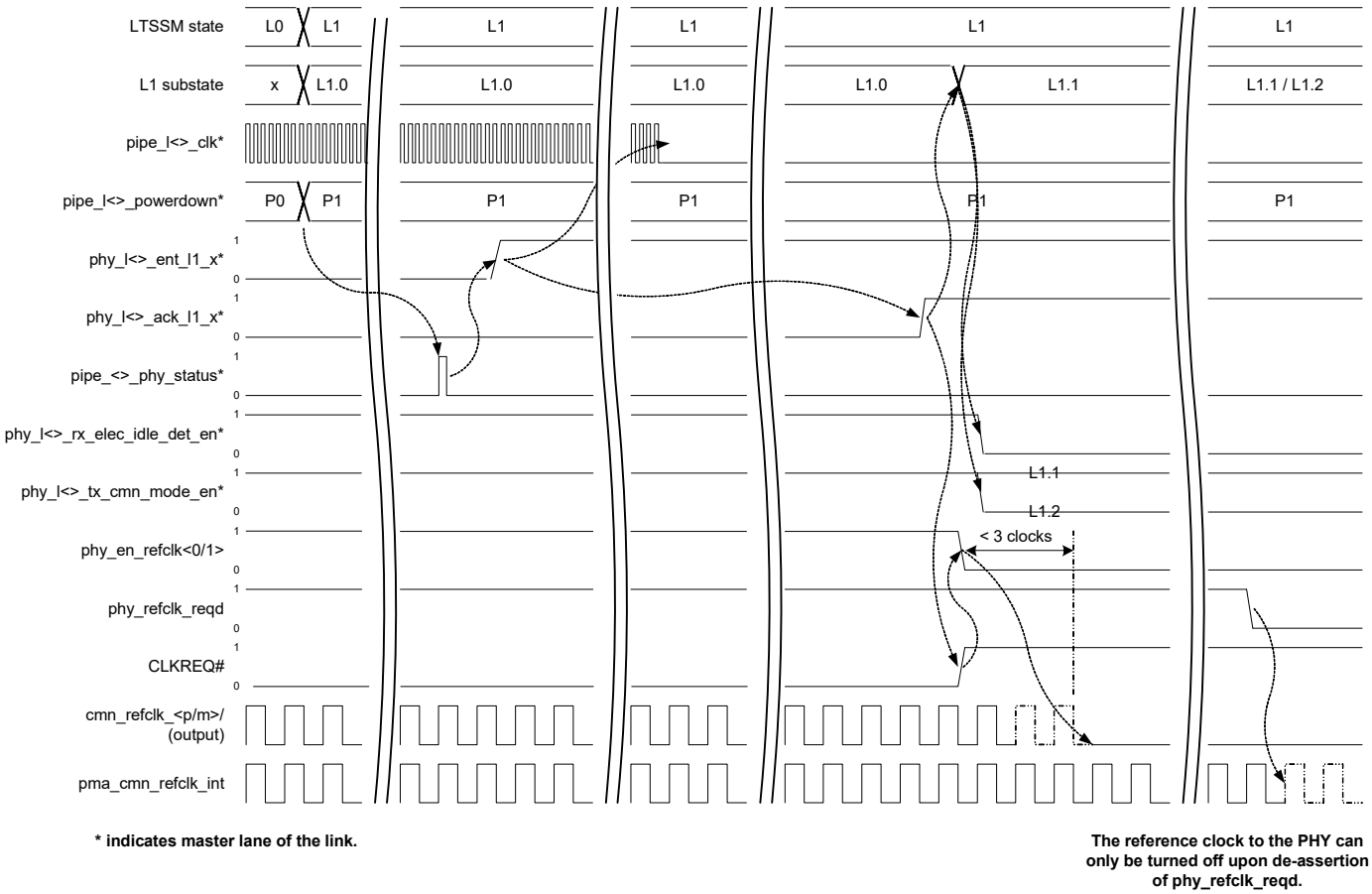


Figure 12: L1.x entry sideband timing (with reference clock output)

For this case, phy_en_refclk is used in conjunction with phy_l<>_ent_l1_x / phy_l<>_ack_l1_x. phy_en_refclk has 2 functions: (1) it signals when to gate the output reference clock and (2) to complete the powerdown of the PHY. phy_en_refclk shall not be de-asserted unless all links are in a powered down state. The input reference clock shall not be turned off until phy_refclk_reqd de-asserts low.

8.4.1.1.2 L1.x entry (PIPE-based signaling)

The PHY can be in either the P0 or P1 power states prior to initiating L1.x entry when using PIPE-based signaling. L1.x entry is initiated by setting pipe_l<>_powerdown to P1.x.

The PIPE-based L1.x entry procedure has the same dependency upon whether outputting a reference clock as sideband signaling mode. When not outputting a reference clock, phy_en_refclk is ignored and the input reference clock can be turned off upon assertion and de-assertion of pipe_l<>_phy_status (when all other links are also powered down). When outputting a reference clock, phy_en_refclk cannot be de-asserted until after assertion and de-assertion of pipe_l<>_phy_status (when all other links are also powered down). After de_assertion of phy_en_refclk, the PHY reference clock input (i.e. pma_cmn_refclk_int) can be turned off only upon the de-assertion low of phy_refclk_reqd.

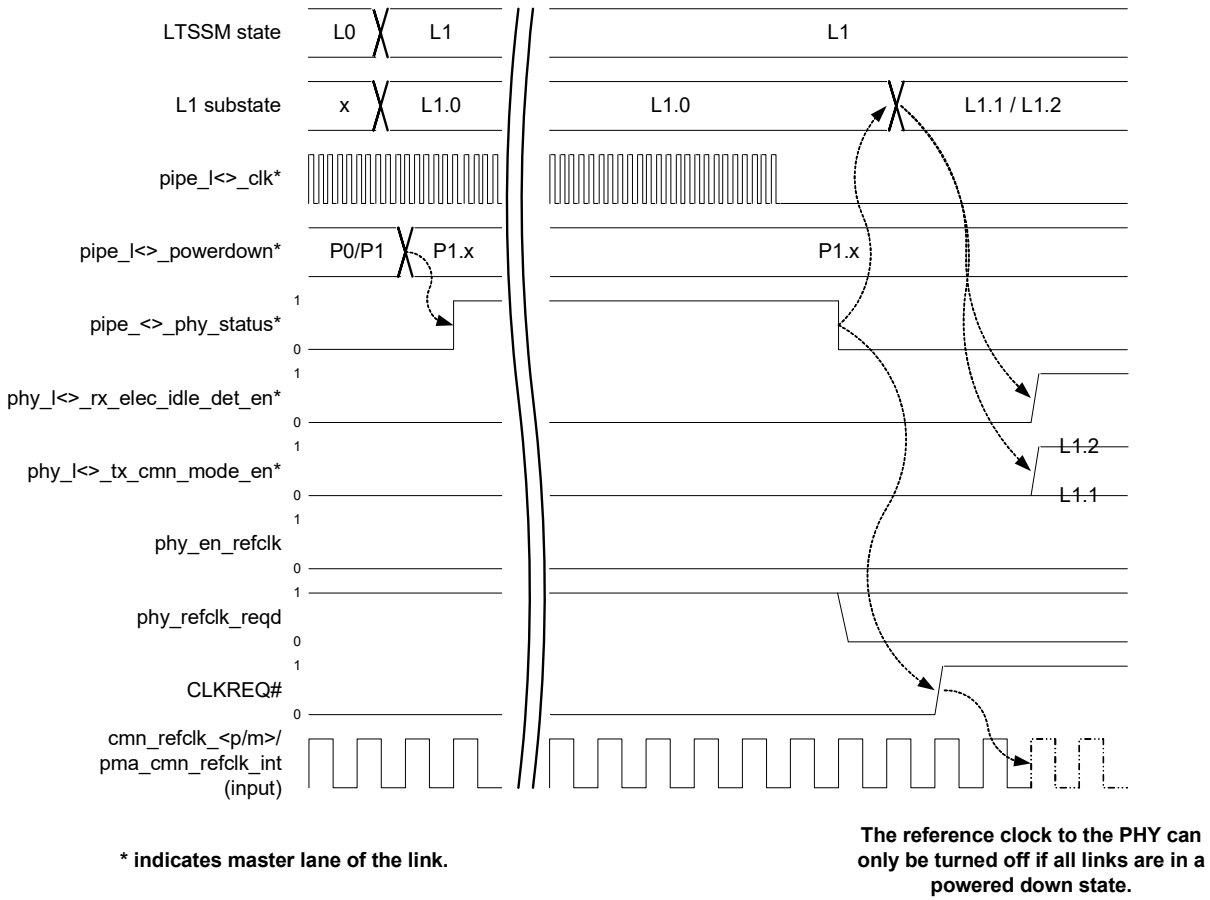


Figure 13: PCIe L1.1 entry PIPE timing (without reference clock output)

For L1.1, phy_l<>_tx_cmn_mode_en is left de-asserted low. For L1.2, phy_l<>_tx_cmn_mode_en is asserted high upon entering L1.2 Idle sub-state.

The input reference clock can only be turned off if all links are powered down (i.e. disabled, in PCIe P2, USB P3, PCIe L1.x, etc.). This can be determined based on each link's power state or based on the state of phy_refclk_reqd. When phy_refclk_reqd is high, the input reference clock must be left running. Upon de-assertion, the input reference clock may be turned off.

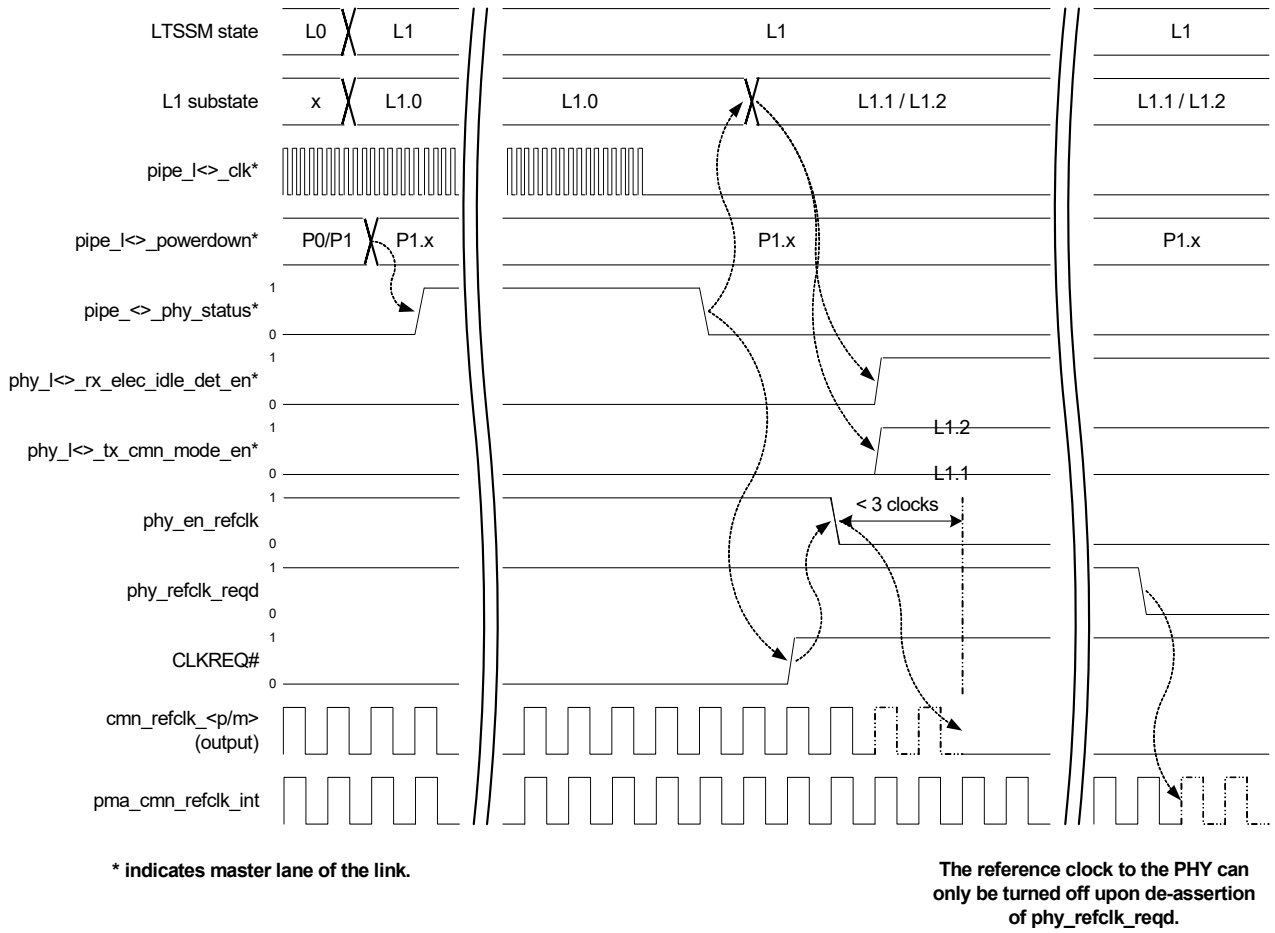
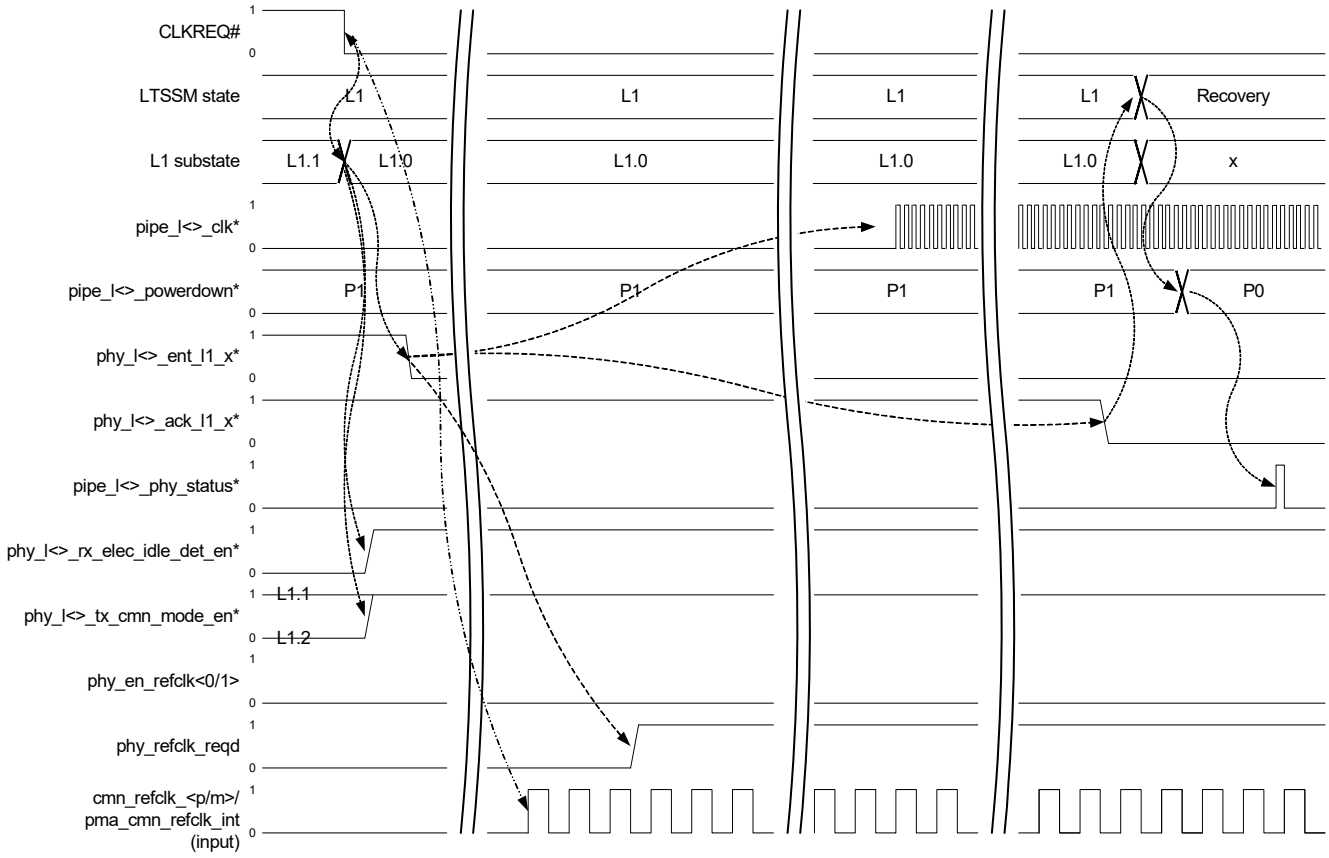


Figure 14: L1.x entry PIPE timing (with reference clock output)

For this case, phy_en_refclk is used in conjunction with pipe_l<>_powerdown and pipe_l<>_phy_status. phy_en_refclk has 2 functions: (1) it signals when to gate the output reference clock and (2) to complete the powerdown of the PHY. phy_en_refclk shall not be de-asserted unless all links are in a powered down state. The input reference clock shall not be turned off until phy_refclk_reqd de-asserts low.

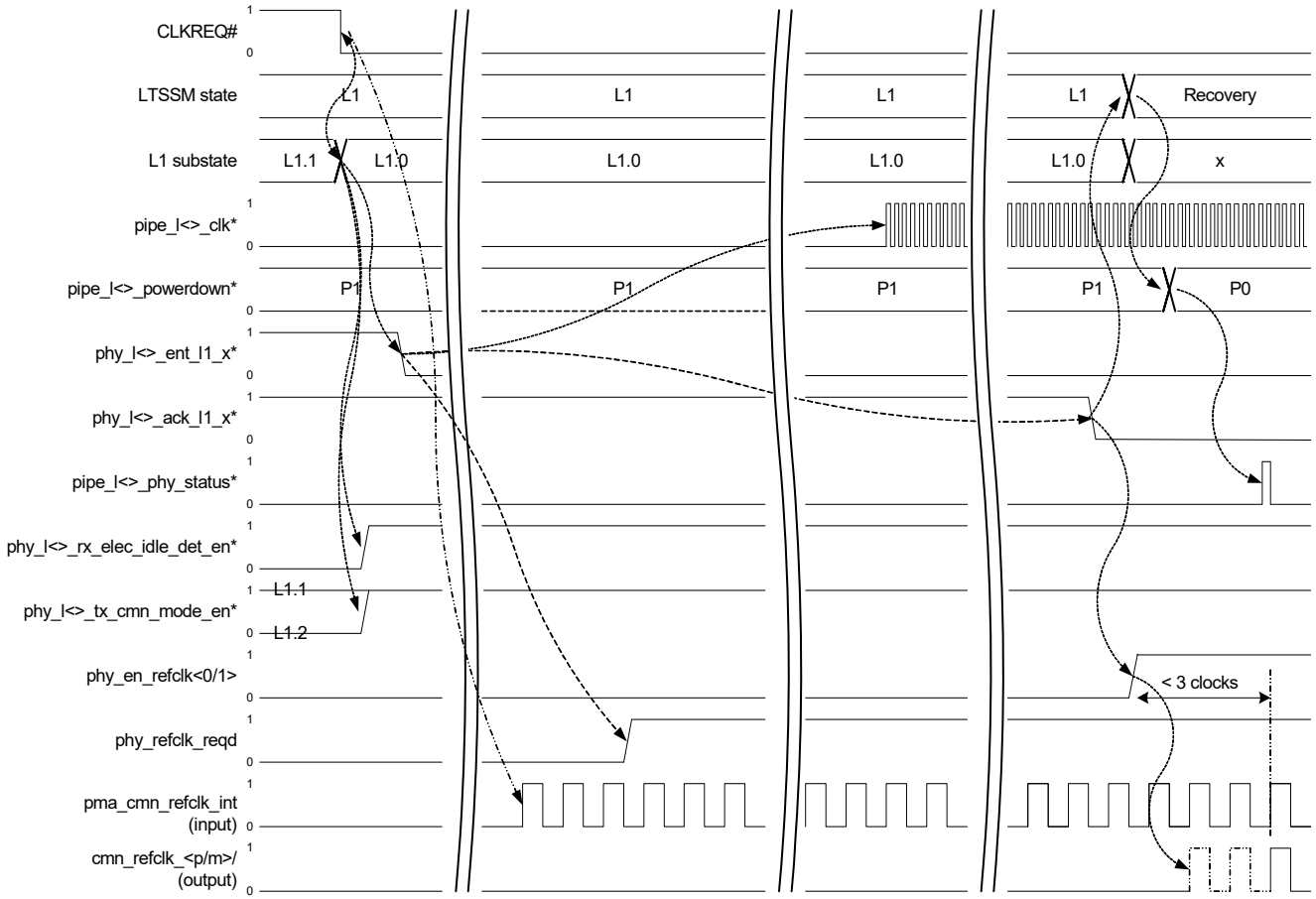
8.4.1.1.3 L1.x exit (sideband signaling)

Exit from L1.x when using sideband signaling is initiated by the de-assertion low of phy_l<>_ent_l1_x. The only difference between L1.1 and L1.2 exit is that phy_l<>_tx_cmn_mode_en must be re-asserted high for L1.2 and not for L1.1.



*** indicates master lane of the link.**

Figure 15: PCIe L1.x exit sideband timing (without reference clock output)



*** indicates master lane of the link.**

Figure 16: PCIe L1.x exit sideband timing (with reference clock output)

In this case, phy_en_refclk shall not be asserted high until phy_l<>_ack_l1_x is de-asserted low.

8.4.1.1.4 L1.x exit (PIPE-based signaling)

Exit from L1.x when using PIPE-based signaling is initiated by the changing pipe_l<>_powedown from P1.x to P1. The only supports exit from P1.x to the P1 power state. Exit directly to P0 is not supported. The only difference between L1.1 and L1.2 exit is that phy_l<>_tx_cmn_mode_en must be re-asserted high for L1.2 and not for L1.1.

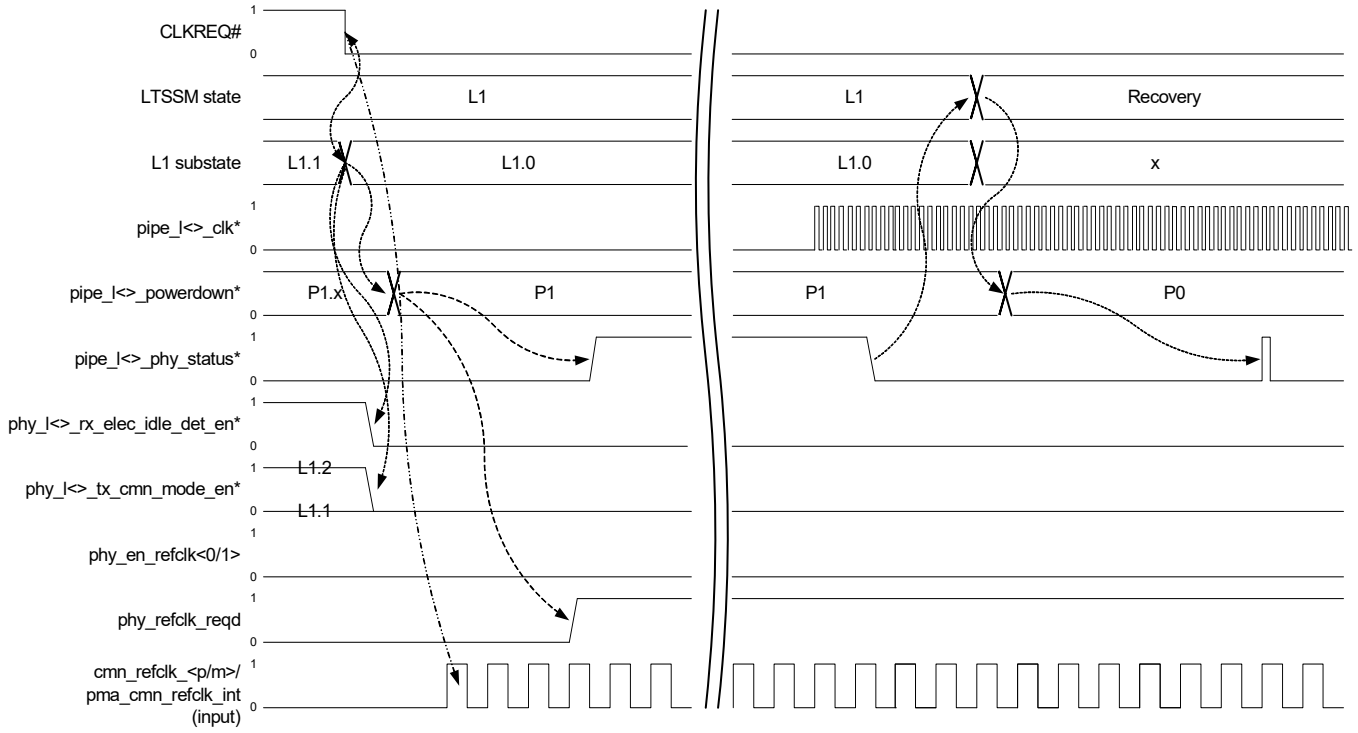


Figure 17: PCIe L1.x exit PIPE timing (without reference clock output)

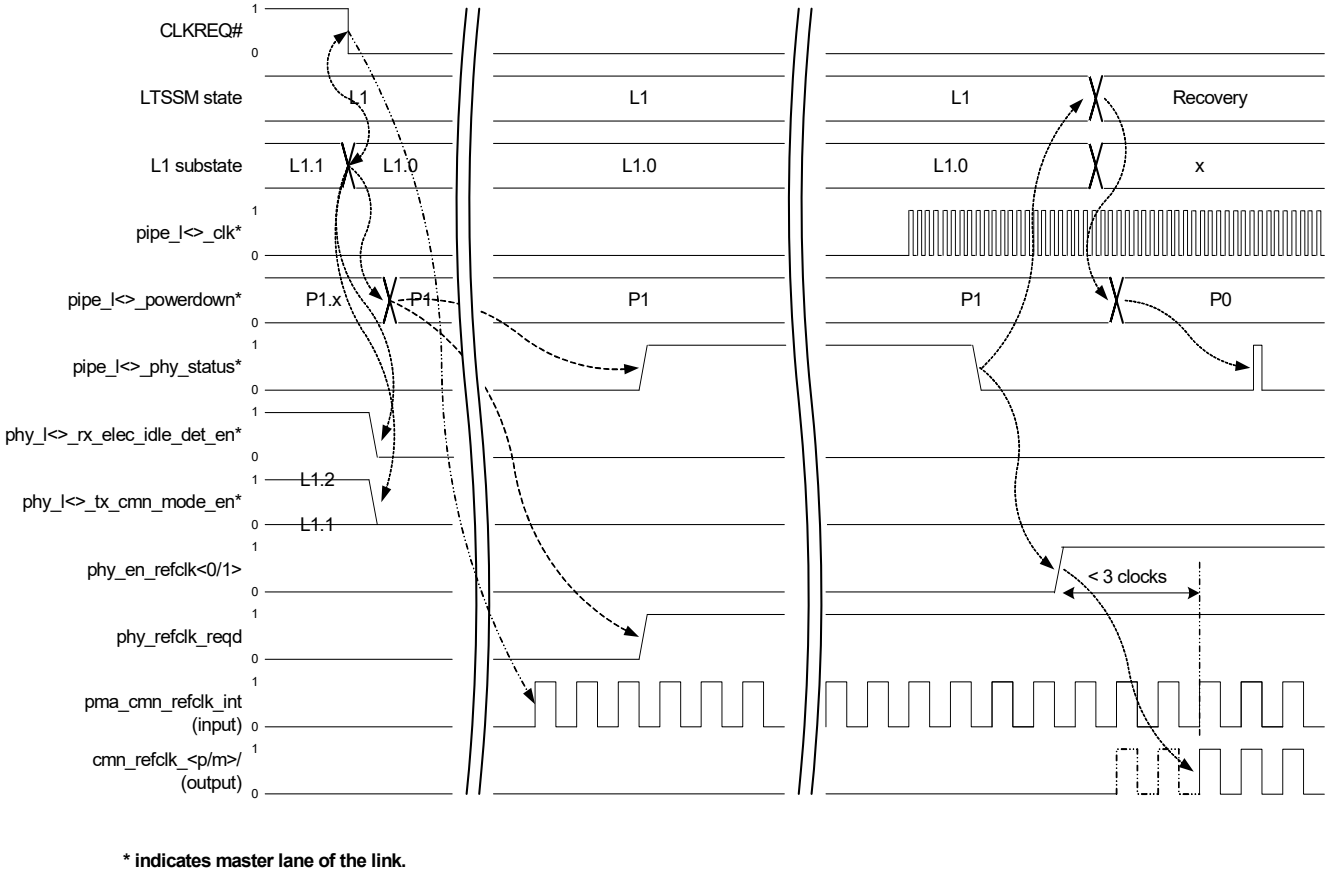


Figure 18: PCIe L1.x exit PIPE timing (with reference clock output)

In this case, phy_en_refclk shall not be asserted high until phy_l<>_ack_l1_x is de-asserted low.

8.4.2 Raw SerDes

A Raw SerDes link's power state is selected through the pma_xcvr_power_state_req_ln_* and pma_xcvr_power_state_ack_ln_* pins. (PIPE links use the pipe_l*_powerdown pins.) The following timing diagrams describe the procedure for changing the power state of an Ethernet link.

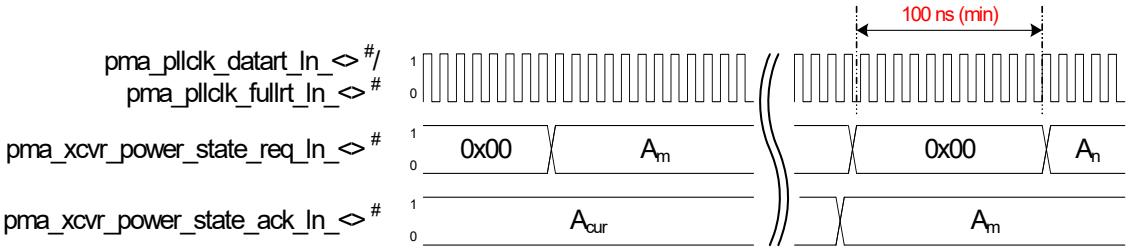


Figure 19: Raw SerDes interface link A0/A1/A2 power state change timing

When pma_xcvr_power_state_ack_ln_* == pma_xcvr_power_state_req_ln_*, the link is in the new power state. Return pma_xcvr_power_state_req_ln_* to 0x00 upon the link being in the new power state. A new power state change request cannot be signaled for at least 100 nsec after pma_xcvr_power_state_req_ln_* == 0x00.

For multi-lane links, pma_xcvr_power_state_ack_ln_<> may go to zero prior to reflecting the 'next power state' value. This is due to cases where not all of the lanes in the multi-lane link complete the power state change on the same clock cycle.

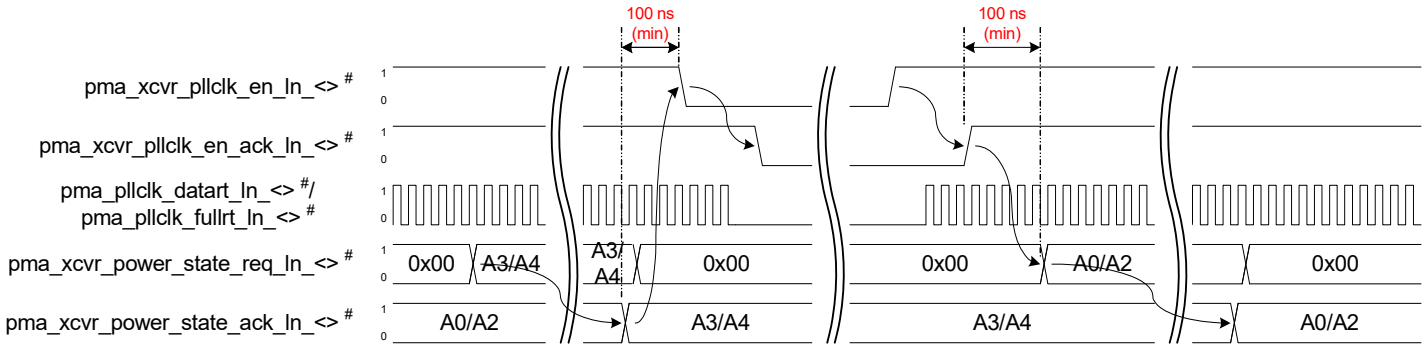


Figure 20: Raw SerDes interface link A3/A4 power state change timing

When changing the power state between A0, A1 and A2, the datarate and fullrate clocks are not gated. However, they are gated after entering A3/A4 and enabled prior to exiting A3/A4. The A3 power state can only be entered from the A0 or A2 power states and the A4 power state can only be entered from the A2 power state.

There is a minimum 100 nsec time on A3/A4 entry between pma_power_state_ack_ln_* == A3/A4 and de-assertion of pma_xcvr_pllclk_en_ln_*. And for A3/A4 exit, there is a minimum 100 nsec time between pma_xcvr_pllclk_en_ack_ln_* assertion and pma_xcvr_power_state_req_ln_* == A0/A2.

8.4.3 PHY/PMA common logic powerdown

When all links are in a power down state (i.e. Raw SerDes A3 or PIPE PCIe P2/L1.x or PIPE USB P3 or powerdown disabled), the PHY/PMA common logic (including PLLs) will be automatically suspended and placed in its lowest powered state. Upon any link exiting from one of these power down states, the PMA common is resumed.

8.5 Data Rate change

8.5.1 PIPE interface

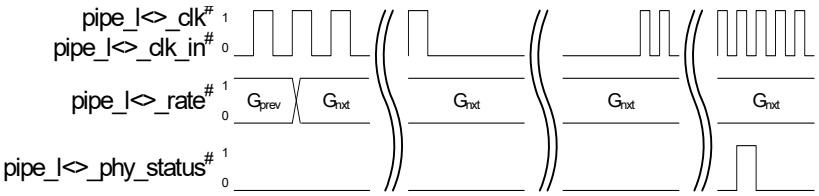


Figure 21: PIPE data rate change timing

8.5.2 Raw SerDes interface

The following timing diagram shows the steps required to change a link's data rate using the Raw SerDes interface:

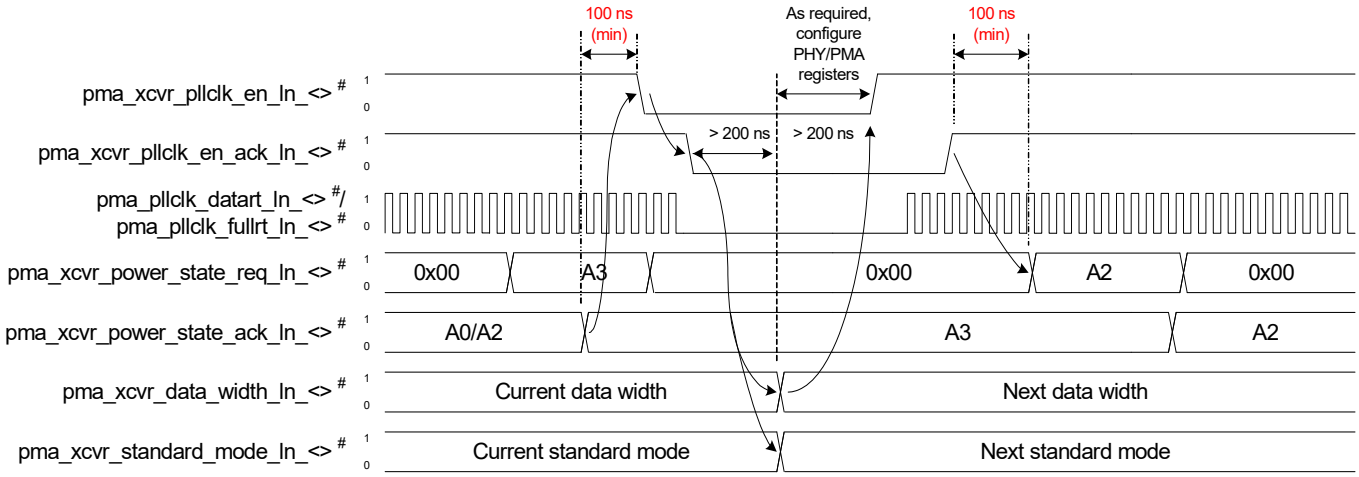


Figure 22: Raw SerDes interface data rate change

Note: The link must be placed into the A3 power state and the data rate clock disabled prior to changing the pma_xcvr_-data_width_ln_*/pma_xcvr_standard_mode_ln_* signals for the new data rate. Upon changing pma_xcvr_standard_-mode_ln_<>/pma_xcvr_data_width_ln_<>, wait at least 200 nsec before asserting pma_xcvr_pllclk_en_ln_* and exiting A3 power state.

If PLL reconfiguration is required for the new data rate, perform the following steps:

- Disable the associated PLL by clearing PHY_PMA_PLL_RAW_CTRL[0] for PLL0 and bit [1] for PLL1.
- Wait for the associated PLL to be disabled. For PLL0, wait for PHY_PMA_CMN_CTRL2[2] == 1. For PLL1, wait for PHY_PMA_CMN_CTRL2[3] == 1.
- Reconfigure PMA registers for associated PLL for the selected data rate.
- Re-enable the associated PLL by setting PHY_PMA_PLL_RAW_CTRL[0] or [1].
- Wait for the associated PLL to be ready for operation. For PLL0, wait for PHY_PMA_CMN_CTRL2[0] == 1. For PLL1, wait for PHY_PMA_CMN_CTRL2[1] == 1.
- Re-enable the link as described by above timing diagram (i.e. assert pma_xcvr_pllclk_en_ln_<> high, wait for pma_x-cvr_pllclk_en_ack_ln_<> == 1 and change link power state to A2). The next power state after a data rate change is A2. The power state can be subsequently change to A0 as required.

8.6 Changing the mode of operation when using the Raw Ser-Des interface

To change a Raw SerDes mode of operation, the link must be placed in reset:

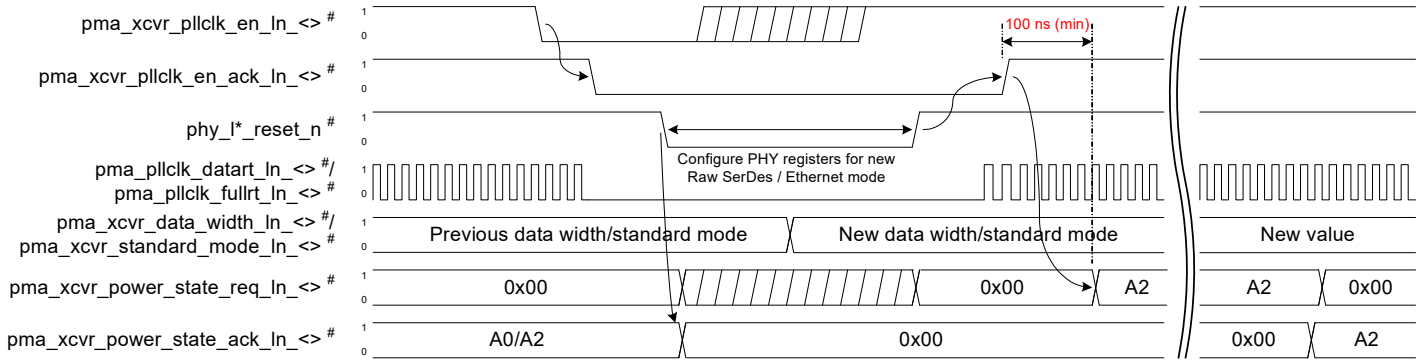


Figure 23: Raw SerDes interface mode of operation change

Of note:

- The link's datarate and fullrate clocks shall be gated prior to asserting link reset. Upon de-assertion of pma_xcvr_pll-clk_en_ack_ln_*, phy_l*_reset_n may be asserted low.
- Assert pma_xcvr_pllclk_en_ln_* high prior to de-asserting phy_l*_reset_n.
- While the link is in reset, write the PHY/PMA registers as required to select the new Raw SerDes / Ethernet mode. pma_xcvr_data_width_ln_* and pma_xcvr_standard_mode_ln_* must also be set for the selected mode prior to the release of reset.
- After reset release, upon pma_xcvr_power_state_ack_ln_* == pma_xcvr_power_state_req_ln_*, the link will be operating in the new Raw SerDes / Ethernet mode.

8.7 Transmit Data Path Timing Diagrams

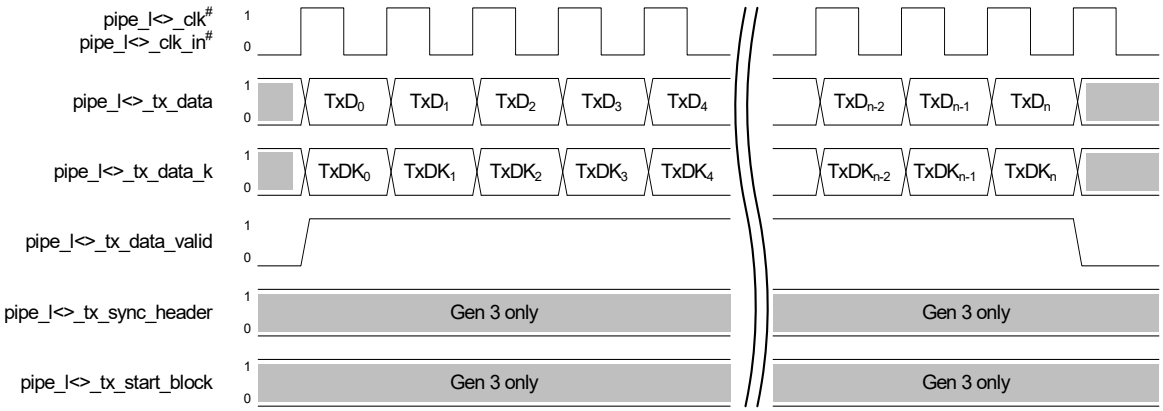


Figure 24: PIPE PCIe Gen1/2 and USB3.2 Gen 1 transmit data path

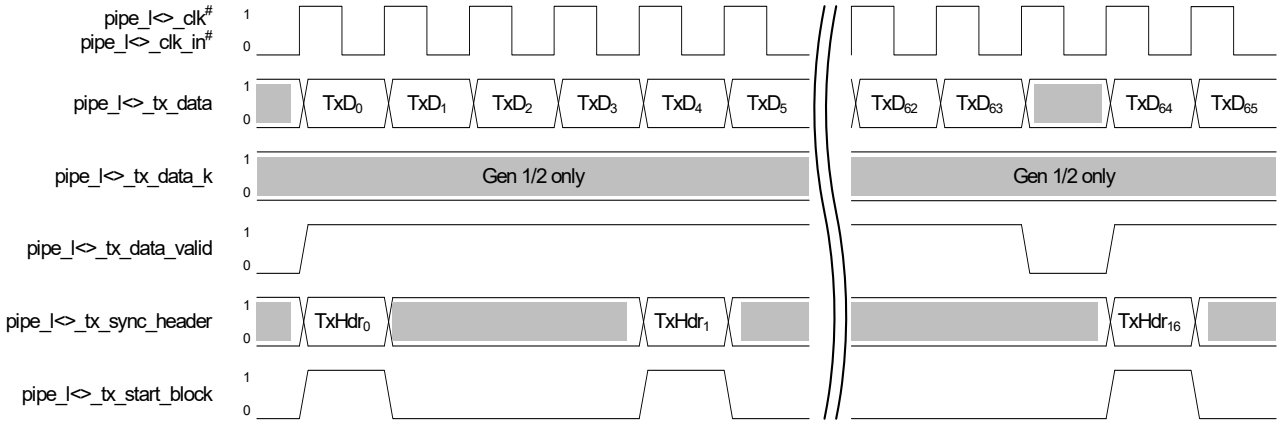


Figure 25: PIPE PCIe Gen3 and USB3.2 Gen 2 transmit data path

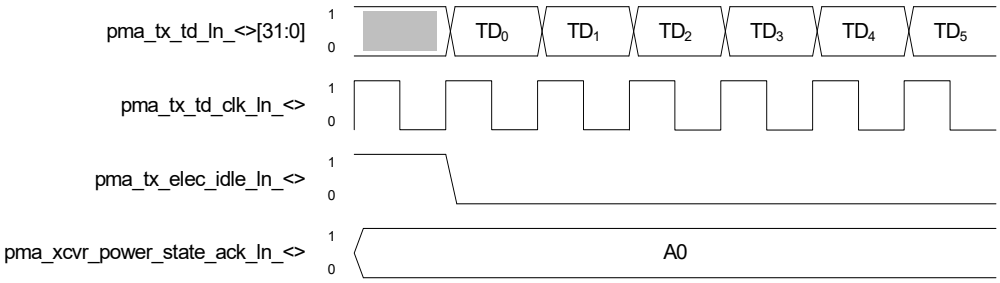


Figure 26: Raw SerDes transmit data path

8.8 Receiver detection

The receiver detection interface is used to perform a receiver detection sequence, which provides indication of whether a receiver is connected to the transmitter for the lane (for those standards that support, such as PCIe and USB). Note: The selected reference clock to the PMA must be running to perform a receiver detection operation regardless of the mode.

8.8.1 PIPE (PCIe or USB)

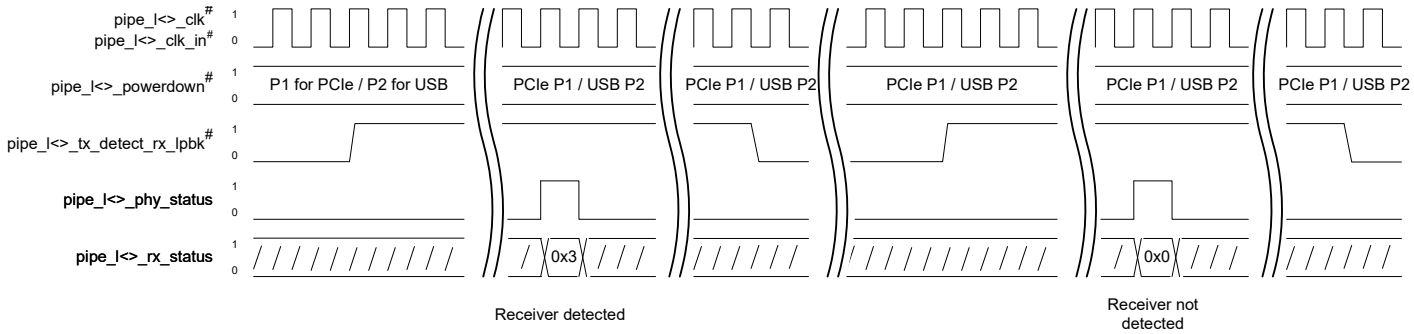


Figure 27: PIPE PCIe and USB (P2) recevier detection timing

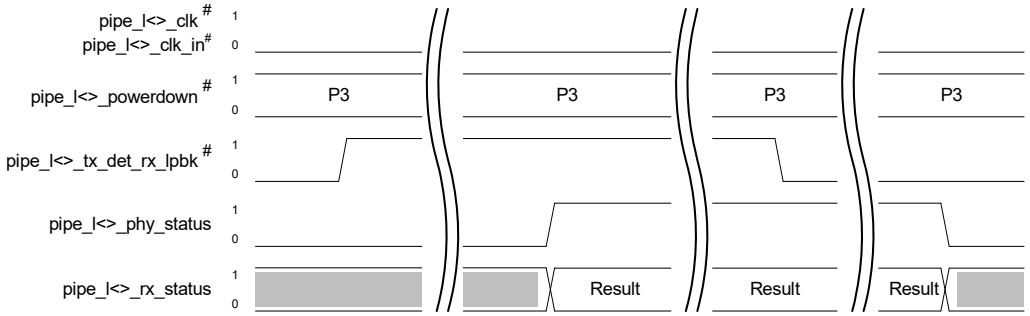


Figure 28: PIPE USB (P3) recevier detection timing

8.8.2 Raw SerDes

For the Raw SerDes receiver detection inteface, a full asynchronous handshake protocol between the pma_tx_rcv_detect_en_ln_<> PHY input and pma_tx_rcv_detect_done_ln_<> PHY output is used to initiate and signal completion of the receiver detect operation. The pma_tx_rcv_detected_ln_<> PHY output is valid upon assertion of pma_tx_rcv_detect_done_ln_<> and indicates whether a receiver was detected or not (1 = detected, 0 = no receiver detected).

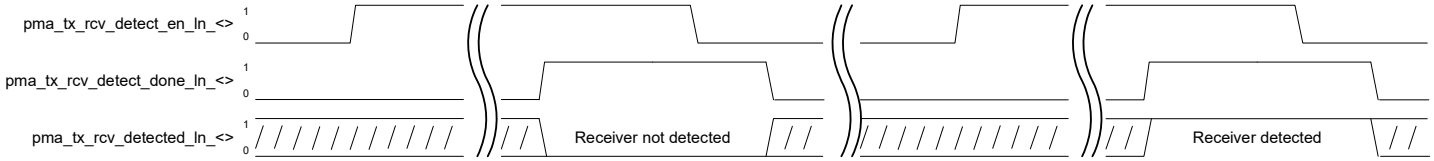


Figure 29: Raw SerDes recevier detection timing

- pma_tx_rcv_detect_en_ln_<> shall not be asserted high while pma_tx_rcv_detect_done_ln_<> is high.
- After asserting pma_tx_rcv_detect_en_ln_<> high, it shall not be de-asserted low unitl pma_tx_rcv_detect_done_ln_<> has been asserted high.

8.9 Transmit deemphasis

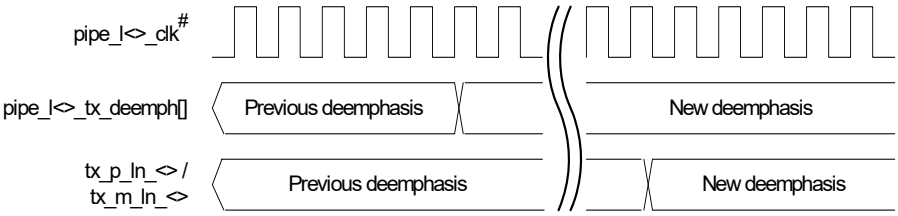


Figure 30: PIPE mode Tx deemphasis change timing

For all PIPE rates except USB3 Gen 2, pipe_l<>_tx_deemph[] drives the associated lane's PMA tx_deemphasis_ln_<> pins directly.

For USB3 Gen2, pipe_l<>_tx_deemph[1:0] is used to map to 1 of the 4 required de-emphasis settings: 0b01 = with pre-shoot and de-emphasis (mission mode and compliance pattern CP15), 0b00 = with pre-shoot, but no de-emphasis (CP13), 0b10 = with de-emphasis, but no pre-shoot (CP14) and 0b11 = no pre-shoot or de-emphasis (CP16), with the default register configuration. Logic in the PCS, uses the local Tx FS value from the PMA to calculate the C_{-1}, C_0 and C_1 values to apply to the PMA based on the de-emphasis mode selected.

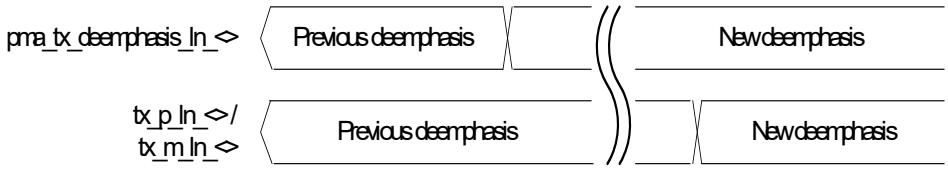


Figure 31: Raw SerDes mode Tx deemphasis change timing

In each case, the new de-emphasis setting will take effect within 128 nsec.

8.10 Receive Data Path Timing Diagrams

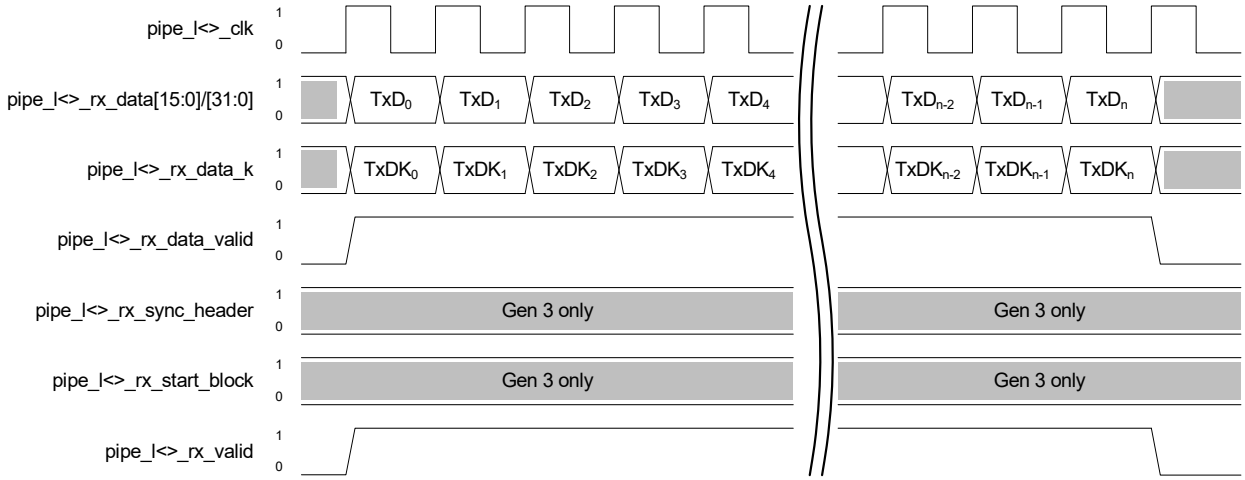


Figure 32: PIPE PCIe Gen1/2 and USB3.2 Gen 1 receive data path

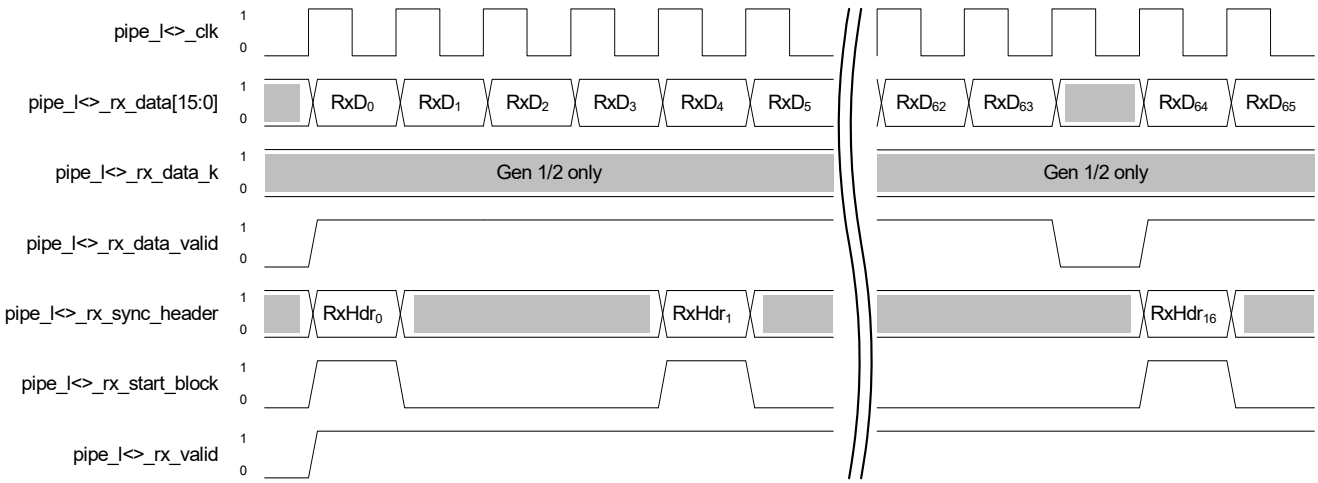


Figure 33: PIPE PCIe Gen3 and USB3.2 Gen 2 receive data path

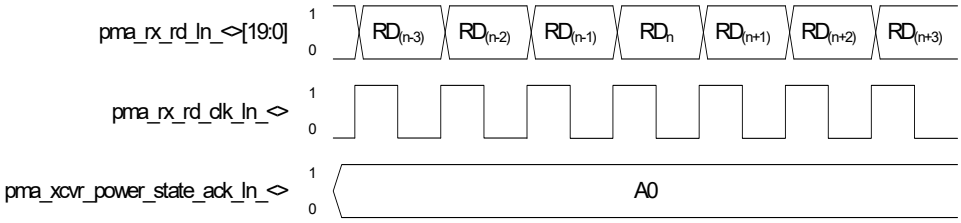


Figure 34: Raw SerDes receive data path

8.11 Transmitter Equalization Timing Diagrams

8.11.1 PIPE PCIe

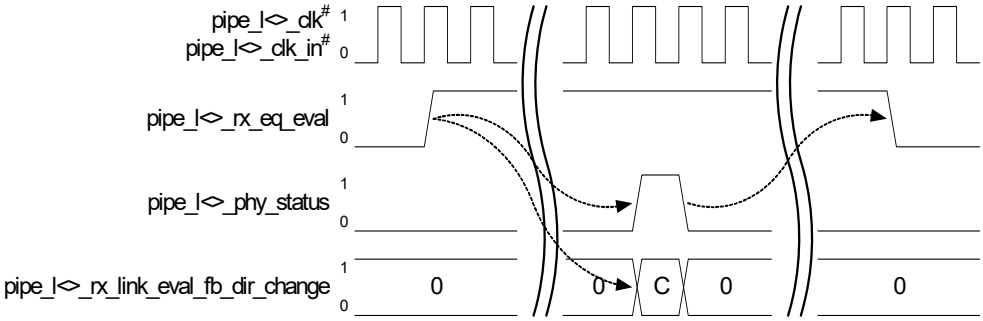


Figure 35: Receiver equalization evaluation (typical)

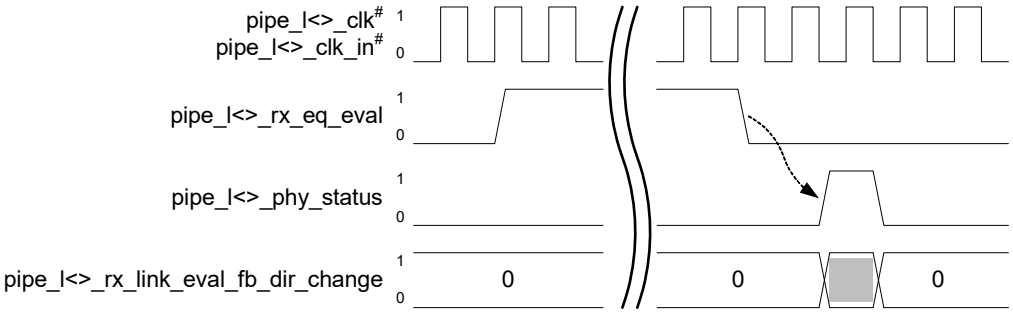


Figure 36: Receiver equalization evaluation (aborted)

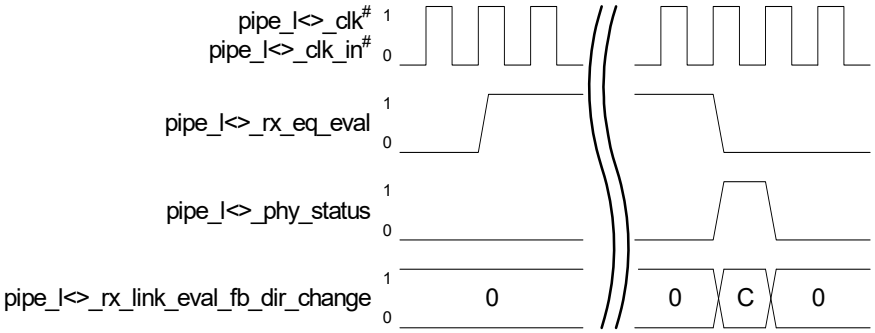


Figure 37: Receiver equalization evaluation (aborted with race condition)

Note: In each of the above cases, a new receiver equalization evaluation shall not be initiated until pipe_l<> phy_status has been asserted for the previous evaluation (whether aborted or not).

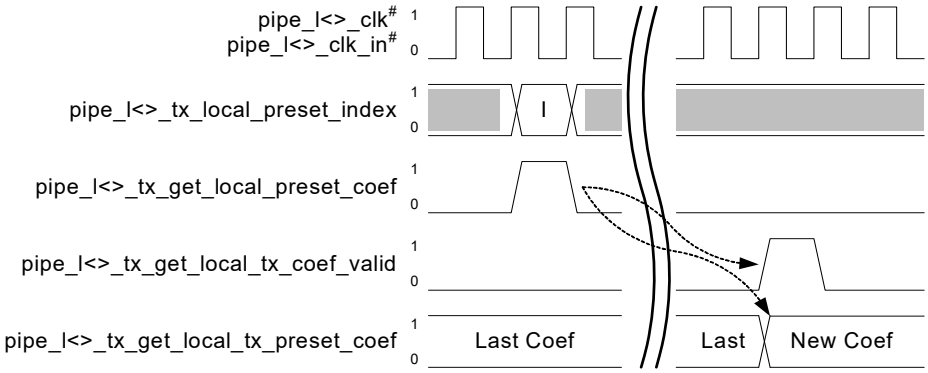


Figure 38: Get local transmit preset coefficients

8.11.2 Raw SerDes

The Rx equalization evaluation interface is used to ‘evaluate’ the far-end transmitter’s de-emphasis settings as part of transmitter equalization. For the Raw SerDes interface a full asynchronous handshake protocol between pma_rx-_eq_eval_ln_<> PHY input pin and pma_rx_eq_eval_status_ln_<> PHY output pin is used to initiate and signal completion of the operation. pma_rx_eq_eval_complete_ln_<> and pma_rx_link_eval_fb_dir_change_ln_<>[] PHY outputs are valid upon the assertion (high) of pma_rx_eq_eval_status_ln_<>. pma_rx_eq_eval_complete_ln_<> indicates whether the evaluation has completed (when high) or should be continued (when low). pma_rx_link_eval_fb_dir_change_ln_<> provides the evaluation result.

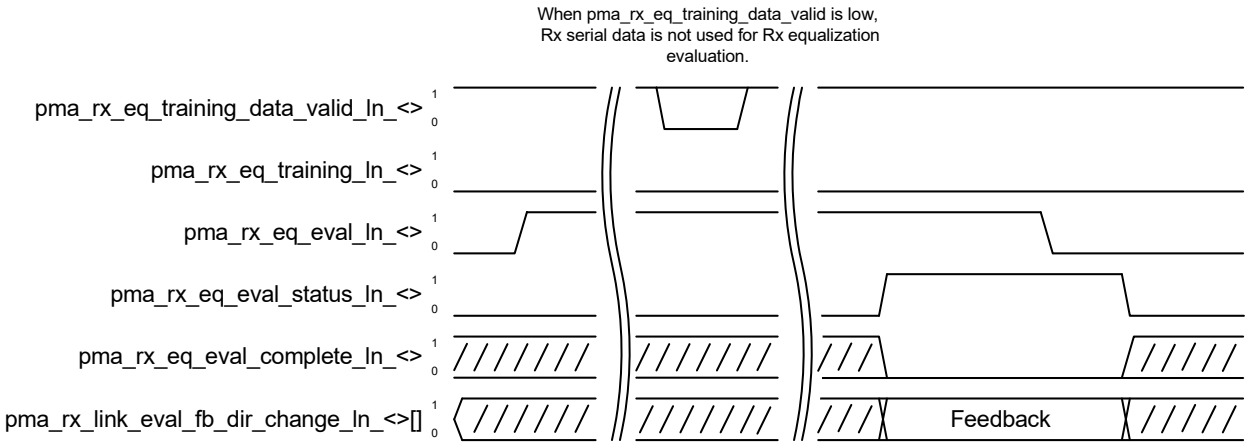


Figure 39: Raw SerDes Rx equalization evaluation (not complete)

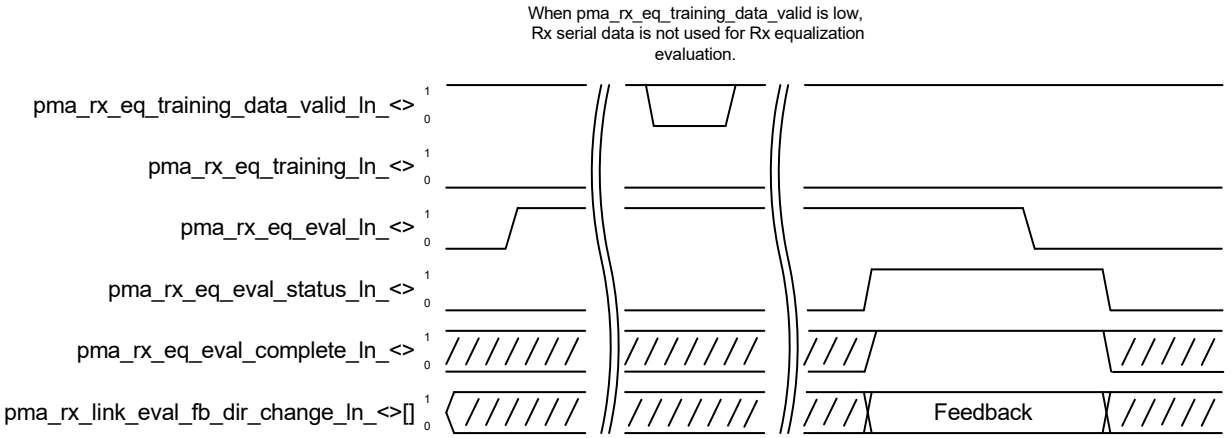


Figure 40: Raw SerDes Rx equalization evaluation (complete)

As shown in Figure 39 and Figure 40, pma_rx_eq_training_data_valid_ln_<> can be de-asserted low during an evaluation. This prevents the incoming serial data from being used for the evaluation.

- pma_rx_eq_eval_ln_<> shall not be asserted high while rx_eq_eval_status_ln_<> is asserted high.
- After asserting pma_rx_eq_eval_ln_<> high, it shall not be de-asserted low until pma_rx_eq_eval_status_ln_<> has been asserted high.

The Get Tx coefficient interface provides access to the PHY's preset transmitter coefficients. A full asynchronous handshake protocol is used between the pma_tx_get_local_preset_coef_ln_<> PHY input pins and the pma_tx_local_tx_coef_valid_ln_<> PHY output pin to initiate and signal completion of the get transmitter coefficient operation. When performing a Get Preset Coefficient, pma_tx_local_preset_index_ln_<>[] provides the coefficient preset value. pma_tx_local_tx_preset_coef_ln_<>[] is valid with the preset transmitter coefficient upon assertion (high) of pma_tx_local_tx_coef_valid_ln_<>.

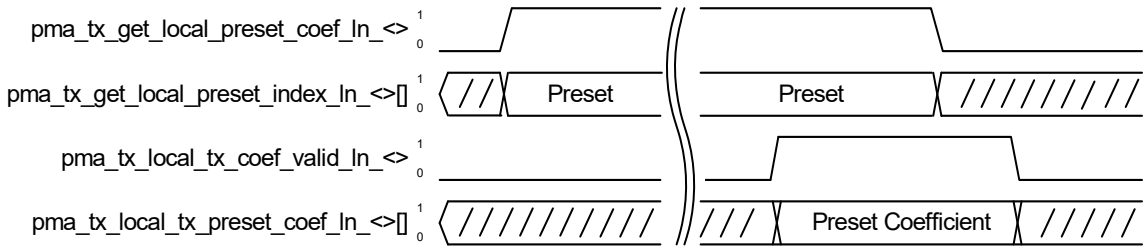


Figure 41: Raw SerDes Get Preset transmitter coefficient

- pma_tx_get_local_preset_coef_ln_<> shall not be asserted high while pma_tx_local_tx_coef_valid_ln_<> is asserted high.
- After asserting pma_tx_get_local_preset_coef_ln_<> high, it shall not be de-asserted low until pma_tx_local_tx_coef_valid_ln_<> has been asserted high.
- When performing a Get Preset Coefficient operation, pma_tx_local_preset_index_ln_<> shall remain valid while pma_tx_get_local_preset_coef_ln_<> is asserted high.

8.12 Disabling Unused Lane(s) in a multi-lane PCIe link

As defined in chapter 9 of the PIPE specification, a lane(s) is disabled by asserting pipe_l{nn:00}_tx_elec_idle and pipe_l{nn:00}_tx_compliance for the selected lane(s). This method can be used in any power state where PIPE clock is active. Recovery of a disabled lane(s) requires reseting the link (i.e. by asserting phy_l*_reset_n low for the link) and performing a normal link startup. For further description, please refer to the PIPE specification.

8.13 External reference clock active detection

The PHY supports detection of whether an external reference clock applied to the cmn_refclk_<p/m> pins is active or not. The logic supports 2 modes of operation: (1) automatic at start of day and (2) manual through APB register access. A requirement for operation (including start of day) is that the apb_pclk (between 10 and 200 MHz) is provided for the duration of the measurement interval.

For automatic mode, drive pma_cmn_ext_refclk_detected_cfg PHY pin high. Upon de-assertion of apb_preset_n, a detection operation will automatically be performed. Depending on the frequency of apb_pclk, the detection operation takes between 125 usec (10 MHz apb_pclk) and 6 usec (200 Mhz apb_pclk). The default for this mode is to detect a 100 MHz external reference clock with an APB clock range from 10 to 200 MHz. Because of the wide range of supported apb_pclk frequencies, this makes false detection of a non-100 MHz clock signal on cmn_refclk_<p/m> possible. For a 10 MHz APB clock, the detection range is from 4.5 MHz to 100.5 MHz. For a 200 MHz APB clock, the detection range is from 90 MHz to 2 GHz. If this is a concern, manual mode can be used to narrow the detection range based on the APB clock frequency.

Figure 42 illustrates detected automatic operation while Figure 43 illustrates non-detected automatic operation.

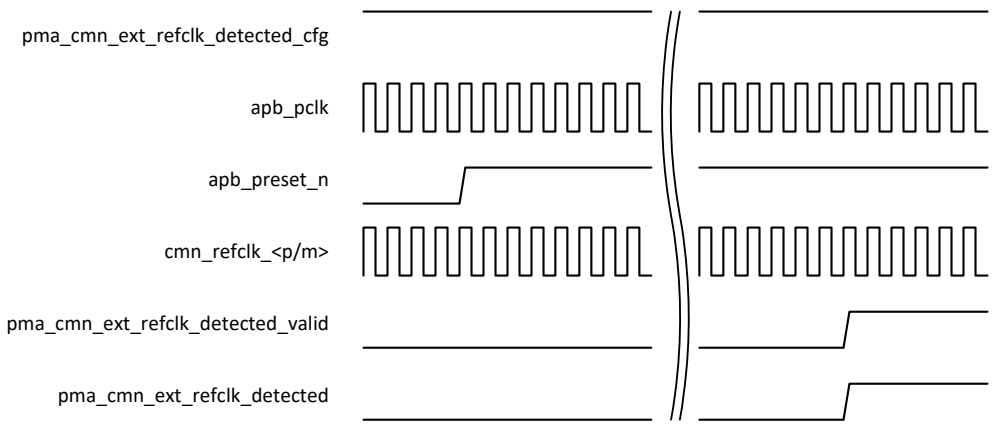


Figure 42: Detected Automatic Operation Timing Diagram

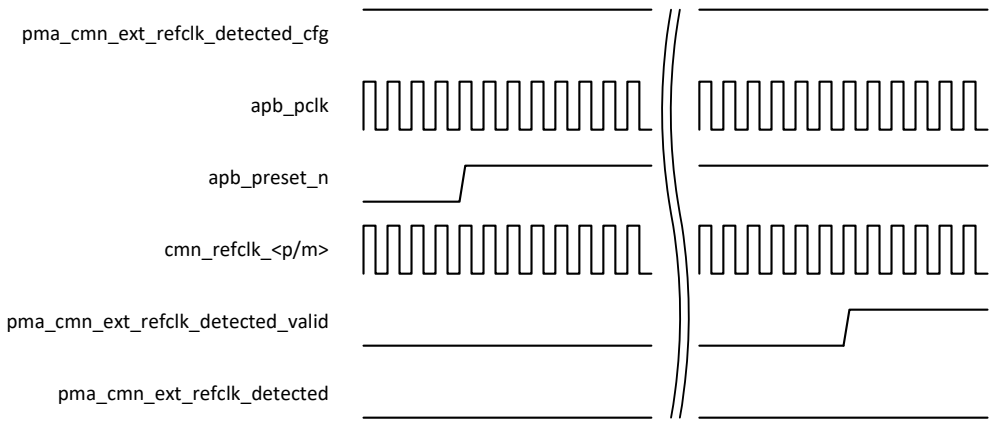


Figure 43: Non-Detected Automatic Operation Timing Diagram

In manual mode, a detection operation is initiated through an APB register write. A manual detection operation is initiated by writing 1 to PHY_REFCLK_DET_ISO_CTRL[0]. A manual detection operation can only be initiated when any previous operation (automatic or manual) has completed. The reference clock frequency detection range can be narrowed by adjusting the detection threshold registers (PHY_REFCLK_DET_THRES_LOW and PHY_REFCLK_DET_THRES_HIGH) based on the actual APB clock frequency. For example, for a 200 MHz APB clock frequency, PHY_REFCLK_DET_-THRES_HIGH can be configured with a value of 550. This would narrow the detection range to 90 to 110 MHz. This range can be narrowed further by narrowing the gap between the low and high thresholds.

Figure 44 illustrates detected manual operation while Figure 45 illustrates non-detected manual operation.

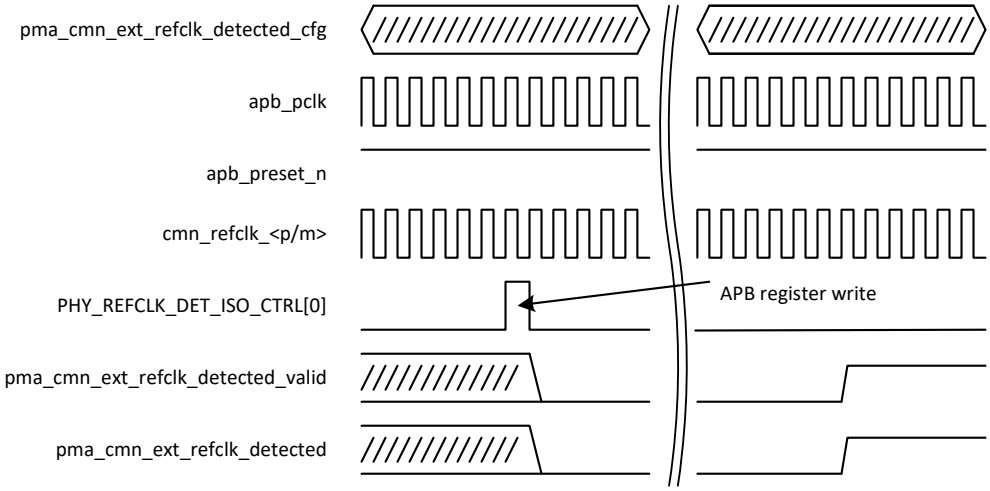


Figure 44: Detected Manual Operation Timing Diagram

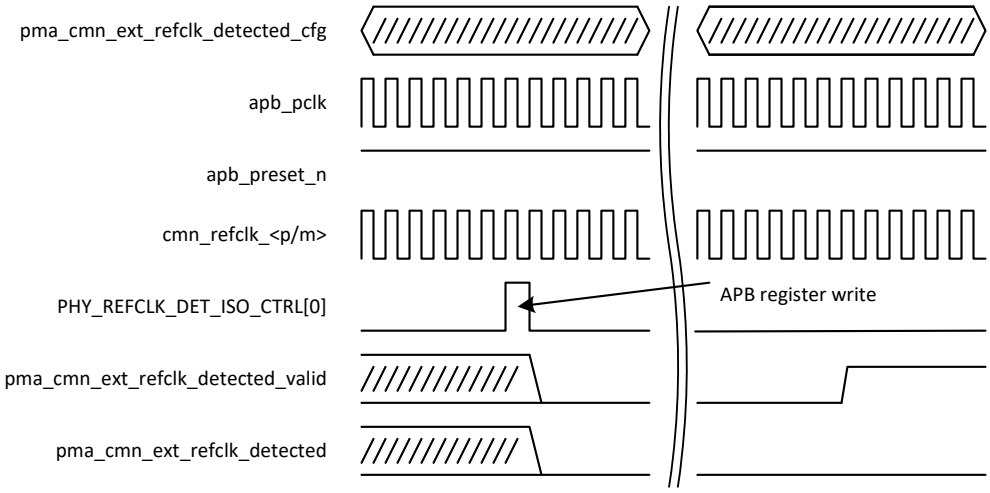


Figure 45: Non-Detected Manual Operation Timing Diagram

The detection thresholds are based on the expected number of external reference clock cycles in the measurement interval. The measurement interval is a configured number of APB clock cycles set in PHY_REFCLK_DET_INTERVAL register.

The measurement interval is set such that the corresponding reference clock count will be between the low and high thresholds upon completion of the measurement interval, when the external reference clock is active. The maximum measurement interval is based on ensuring the corresponding reference clock count will not exceed 2^16 - 1. The minimum measurement interval is set based on the desired measurement accuracy and to minimize the chance of a false positive indication due to noise on the reference clock input.

8.14 Rx Data Debug Timing Diagrams

The Rx Data Debug interface is an optional debug interface of the PHY. It provides a means to capture the receive BIST data, while monitoring for BIST errors. For a detailed description of BIST, see the PMA specification.

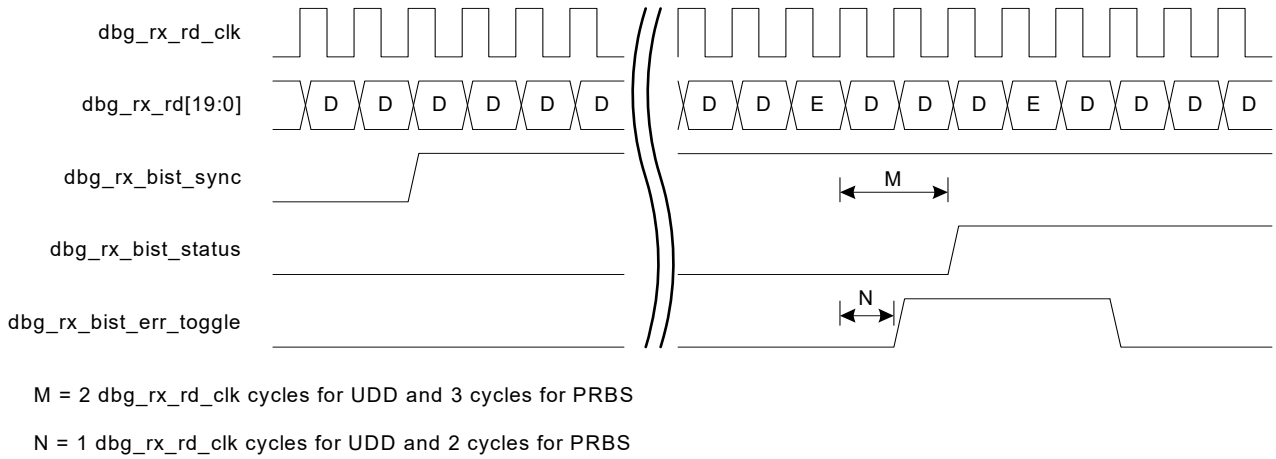


Figure 46: Rx Data Debug interface

8.14.1 Operation

Place the lane/link in the A0 power state (P0 PIPE power state), configure the desired Tx-to-Rx loopback mode and enable PMA Tx and Rx BIST. Upon assertion (high) of dbg_rx_bist_sync, the Rx BIST logic has aligned itself to the received BIST data stream (BIST error detection is disabled until dbg_rx_bist_sync is asserted). For each Rx data word received with at least 1 corrupted bit, dbg_rx_bist_err_toggle will ‘toggle’ (high-to-low or low-to-high). For the first Rx data word received with at least 1 corrupted bit, dbg_rx_bist_status will be asserted (high) and remain asserted until reset. There is a slight clock cycle offset between the corrupted Rx data word (dbg_rx_rd) and the assertion/toggle of dbg_rx_bist_status/dbg_rx-_bist_err_toggle due to pipelining in the PMA’s Rx BIST logic.

All PRBS BIST modes and UDD has a unique and repeated pattern. To determine which Rx data word bit(s) has been corrupted, one or more uncorrupted data words must be captured immediately prior to the corrupted word. These ‘good’ data words are used to synchronize to the data stream. For PRBS BIST, only 1 ‘good’ data word is required to determine the index into the PRBS pattern. For UDD, the number of ‘good’ data words depends on whether all data words in the pattern are unique or not. If all possible 16/20-bit data patterns are unique, only a single ‘good’ word is required. Otherwise, the number of data words captured must enable unique determination of position in the UDD data pattern.

8.15 PHY state change monitor operation

The PHY contains logic that monitors whether power state (signaled on the PIPE or Raw SerDes I/Fs) and data rate (signeled on the PIPE I/F only) changes complete within a specified maximum timeout.

State change failures are signaled by setting the interrupt status bit in the PHY_INTERRUPT_STS register for the master lane of the link and (when included in the PHY configuration) the assertion of phy_interrupt_ln_<> pin () for the master lane of the link. Reading PHY_INTERRUPT_STS clears the interrupt status bit and de-asserts phy_interrupt_ln_<>.

The timeout value is provided in the PHY_STATE_CHG_TIMEOUT register (bits [14:0]). The register value is the number of APB clock cycles divided by 32. For the maximum APB clock frequency (200 MHz), this provides a maximum state change timeout of 5.2 msec. The default timeout is set at 1 msec for a 200 MHz APB clock.

The state monitor for each link is enabled/disabled with PHY_INTERRUPT_STS[15]. For each link, the master lane’s enable bit must be set high to enable the internal logic to monitor the PHY state changes on that link.

The PHY's APB clock must be active and free-running when the state change monitor logic for any link is enabled.

Procedure:

- Based on the frequency of the APB clock provided to the PHY configure PHY_STATE_CHG_TIMEOUT for a 1 msec timeout value.
- Enable the state change monitor logic for the associated link by setting PHY_INTERRUPT_STS[15] to 1. It is recommended to wait until the link has completed startup and is in the active state before enabling.
- Monitor phy_interrupt_ln_<> for the master lanes of the enabled links.

9 PHY Register Descriptions

Depending on the PHY configuration, some registers and register fields described below are removed and replaced with R/O registers/fields (with a return value of 0). The following convention indicates whether a register or field is present or not based on the configuration:

@ - indicates registers/fields that are present only for PHY configurations which include the PIPE interface.

& - indicates registers/fields that are present only for PHY configurations which include the Raw SerDes interface.

9.1 PHY and PCS Configuration and Control

This section describes the top level PHY and PCIe PCS configuration and control registers.

Register Name:			PIPE common control1 register
Register Address:			16'b11_0_0_0000_0000_0000
Register Mnemonic:			PHY_PIPE_CMN_CTRL1
Bits	**Type**	**Reset**	**Description**
15:13	R/O	3'd0	Reserved
12	RC	1'b0	**PHY APB access timeout**: When set, an APB read/write request to PHY registers failed (i.e. timed out). When set, this bit is cleared upon read.
11	R/O	1'b0	Reserved
10	R/W	1'b1	**PCIe PCS Comma realign**: This field controls the comma alignment state machine to re-align to new bit position without going to loss of sync state. The requirement of the new bit position should meet the number of COMMAs as per Symbol unlock count register definition. When new bit position is identified the comma alignment state machine remains in sync state with the alignment now locked to the new bit position. This field needs to be programmed during the PHY initialization routine before training sequences are received. The effect here is that pipe_rx_valid is not de-asserted upon re-alignment. When this bit is 0, pipe_rx_valid will be de-asserted upon loss of COMMA lock and subsequent re-alignment. Applies for PCIe Gen 1/2 and USB3.2 Gen 1 only. (@)
9	R/W	1'b0	**Block alignment clear on EIOS** : When set, upon receiving a PCIe EIOS, 128b/130b block alignment is reset regardless of Rx signal detect from the PMA (applies for PCIe Gen 3 only). (@)
8	R/W	1'b0	**Comma alignment clear on EIOS** : When set, upon receiving a PCIe EIOS, Comma Alignment is reset regardless of Rx signal detect from the PMA (applies for PCIe Gen 1/2 only). (@)
7	R/W	1'b0	**Block alignment ignore Rx SigDetect** : When set, 128b/13xb block alignment will not be reset due to loss of signal detection from the PMA (applies for PCIe Gen 3 and USB3.2 Gen 2 only). (@)
6	R/W	1'b0	**Comma alignment ignore Rx SigDetect** : When set, Comma alignment will not be reset due to loss of signal detection from the PMA (applies for PCIe Gen 1/2 and USB3.2 Gen 1 only). (@)
5:4	R/W	2'd0	**Rx signal detect delay** : Selects the number of clock cycles of delay to add to the PMA signal detect when the bit alignment blocks should be reset after losing signal. (@) • 2'b00 = 1 Rx data rate clock cycle • 2'b01 = 2 Rx data rate clock cycles • 2'b10 = 3 Rx data rate clock cycles • 2'b11 = 4 Rx data rate clock cycles
3:2	R/O	2'd0	Reserved
1	R/W	1'b0	**RefClk disable override**: 1 = overrides turning off reference clock receiver by forcing cmn_refclk_disable PMA input low. 0 = normal control of cmn_refclk_disable PMA input by PHY logic. (@)
0	R/W	1'b0	**PHY RefClk enable input ingnore** : 0 = ignore phy_en_refclk PHY input (forces low internally). 1 = phy_en_refclk_used as specified for controling enable/disable of cmn_refclk_<p/m>. (@)

Table 17: PIPE common control1 register

Register Name:			PIPE common control2 register (@)
Register Address:			16'b11_0_0_0000_0000_0001
Register Mnemonic:			PHY_PIPE_CMN_CTRL2
Bits	**Type**	**Reset**	**Description**
15:12	R/W	4'd11	**USB SuperSpeed Tx LFPS Stretch** : Minimum number of data rate clock cycles in which PMA tx_lfps_en signal is asserted for USB SuperSpeed rate. Number of data rate clock cycles must be > 1 PMA RefClk cycle.
11:8	R/W	4'd13	**USB SuperSpeedPlus Tx LFPS Stretch** : Minimum number of data rate clock cycles in which PMA tx_lfps_en signal is asserted for USB SuperSpeedPlus rate. Number of data rate clock cycles must be > 1 PMA RefClk cycle.
7	R/O	1'b0	Reserved
6	R/W	1'b1	**PCIe PCS TX electrical idle pre release**:When this bit is set, the TX electrical idle release to the PMA is advanced 1 cycle to allow the adjustment of the datapath timing
5	R/W	1'b0	**RX equaliser complete mask**: When this bit is cleared, the PHY will return direction change of 0 when PMA indicates evaluation complete. Subsequent evaluation requests would clear the PMA iteration counters. When set high, the PMA equalization complete signal is ignored.
4	R/W	1'b1	**PCIe PCS EIOS cycle error mask**: When this bit is enabled and the pipe rx interface is outputting an EIOS symbol, decode errors will be masked out
3	R/W	1'b0	**USB Gen 2 Bit Error Correction Disable**: When this bit is high, bit error correction on SKP and SDS symbols is disabled.
2	R/W	1'b0	**USB PIPE3 Compatibility Mode enable** : When this bit is set to 1, USB PIPE3 compatibility mode is enabled. In this mode, when operating in nominal empty Elasticity Buffer mode, when the EB buffer goes empty, instead of de-asserting PIPE RxDataValid, a USB SKIP OS is inserted into the data stream. This is the behavior as defined in PIPE version 3. When this bit is low, PIPE RxDataValid is de-asserted when the EB buffer goes empty, as recommended by PIPE version 4.
1	R/W	1'b0	**USB Loopback Slave Error Count disable**: When this bit is set to 1, disables the error count for US loopback slave, such that the error count is not inserted into the BCNT OS.
0	R/W	1'b1	**USB Elasticity Buffer Re-align enable**: When this bit is set to 1, when Rx for a USB link is initially started, the elasticity buffer is re-aligned to its idle point upon seeing 3 consecutive COMMAs (i.e. from TS1/TS2s) in the same relative bit position. The purpose of this is to re-align the elasticity buffer (i.e. CTC) after receiving the TSEQ data, which contains no SKIP OSs.

Table 18: PIPE common control2 register

Register Name:			PIPE comma lock configuration1 register (@)
Register Address:			16'b11_0_0_0000_0000_0010
Register Mnemonic:			PHY_PIPE_COM_LOCK_CFG1
Bits	**Type**	**Reset**	**Description**
15:12	R/W	4'h4	**PCIe PCS Comma unlock count:** The number of COMMA symbols that need to be seen in the wrong bit position before the comma alignment state machine will transition to RESYNC or LOS state
11:0	R/W	12'h400	**PCIe PCS Comma full lock count:** The number of COMMA symbols that need to be seen in the same bit position for the comma alignment state machine to lock. The field is used for initial reset lock.

Table 19: PIPE comma lock configuration1 register

Register Name:			PIPE comma lock configuration2 register (@)
Register Address:			16'b11_0_0_0000_0000_0011
Register Mnemonic:			PHY_PIPE_COM_LOCK_CFG2
Bits	**Type**	**Reset**	**Description**
15:8	R/W	8'h8	**PCIe PCS Comma lock count fast:** The number of COMMA symbols that needs to be seen in the same bit position for the comma state machine to lock. This field is used while the PCS is in P0 state after an EIOS has been seen, ie detecting FTS
7:0	R/W	8'h20	**PCIe PCS Comma lock count:** The number of COMMA symbols that needs to be seen in the same bit position for the comma state machine to lock. This field is used while the PCS is transitioning back to the P0 power state.

Table 20: PIPE comma lock configuration2 register

Register Name:			PIPE EIEOS lock configuration register (@)
Register Address:			16'b11_0_0_0000_0000_0100
Register Mnemonic:			PHY_PIPE_EIE_LOCK_CFG
Bits	**Type**	**Reset**	**Description**
15:12	R/W	4'h1	**EIE lock count fast:** The number of EIEOS blocks that need to be seen in the same bit postion for the alignment state machine to lock for Gen3/4. The field is used while the PCS is in P0 state after an EIEOS has been seen
11:8	R/W	4'h3	**EIE lock count :** The number of EIEOS blocks that need to be seen in the same bit postion for the alignment state machine to lock for Gen3/4. The field is used while the PCS is transitioning out of a power state change and not performing a rate change
7:0	R/W	8'h7F	**EIE full lock count:** The number of EIEOS blocks that need to be seen in the same bit postion for the alignment state machine to lock for Gen3/4. The field is used for initial after reset lock or lock after a rate change

Table 21: PIPE EIEOS lock configuration register

Register Name:			PIPE lane disable register (@)
Register Address:			16'b11_0_0_0000_0000_0101
Register Mnemonic:			PHY_PIPE_LANE_DSBL
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'd0	Reserved
7:0	R/W	8'd0	**PIPE lane disable:** Each bit corresponds to a lane (i.e. bit [0] -> lane 0, bit [1] -> lane 1, etc). When set the corresponding PIPE lane is disabled. Lanes that are disabled will transmit electrical idle and will not return any data on PIPE Rx interface. Used to disable unused lanes in a multi-lane PCIe link. (The PMA transceiver corresponding to lane will be placed in suspend.)

Table 22: PIPE lane disable register

Register Name:			PIPE receiver detect inhibit register (@)
Register Address:			16'b11_0_0_0000_0000_0110
Register Mnemonic:			PHY_PIPE_RCV_DET_INH
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h3d09	**PCS Receiver Detect Inhibit Counter Load Value:** Counter load value to delay receiver detection request to PMA until PMA common mode is within the required range. The timer (running on divided reference clock from PMA) starts once the PMA common has completed startup. If receiver detect request is received while timer has not expired, the PCS will wait until the timer expires before signaling the request to the PMA. Load value is specified in multiples of 4x the divider reference clock period (typically 40 nsec for 100 MHz reference clock input frequency) for a 2.5 msec inhibit time.

Table 23: PIPE receiver detect inhibit register

Register Name:			PIPE Rx electrical idle L1.x exit delay register
Register Address:			16'b11_0_0_0000_0000_0111
Register Mnemonic:			PHY_PIPE_RX_ELEC_IDLE_DLY
Bits	**Type**	**Reset**	**Description**
15:10	R/W	6'd15	**PCIe PCS L1.x exit Rx electrical idle force fast count :** Counter load value to hold PIPE Rx Electrical Idle high upon exit from L1.x. Counter is loaded and starts counting down after phy_l*_rx_elec_idle_det_en is asserted high. Default is 500 nsec. This counter accounts for time to power on just the analog Rx signal detect block. Based on reference clock cycles (25 MHz default). Minimum value = 6'd0.
9:0	R/W	10'd150	**PCIe PCS L1.x exit Rx electrical idle force full count :** Counter load value to hold PIPE Rx Electrical Idle high upon exit from L1.x when the PMA common was powered down. Counter is loaded and starts counting down upon de-assertion of PMA cmn_ref_clk_disable (upon de-assertion of phy_l*_ent_l1_x for first link). Default is 6 usec. This counter accounts for time to power on the bias and decap and only takes effect if the PMA was suspended. Based on reference clock cycles (25 MHz default). Minimum value = 10'd1.

Table 24: PIPE RX electrical idle L1.x exit delay register

Register Name:			PHY PLL configuration register
Register Address:			16'b11_0_0_0000_0000_1110
Register Mnemonic:			PHY_PLL_CFG
Bits	**Type**	**Reset**	**Description**
15:2	R/O	14'd0	Reserved
1	R/W	1'b0	**PLL configuration:** 0 = PHY configured to only use PLL0 / PLL1 disabled. 1 = PHY configured to use both PLL0 and PLL1.
0	R/W	1'b1	**Single link PCIe configuration :** 1 = PHY configured such that there is only a single PCIe link (1xN) and all PCIe rates will be driven by PLL0. Any other links are not PCIe and will not use PLL0. 0 = ALl other configurations. This is a static configuration, which can only be changed while phy_reset_n is asserted low.

Table 25: PHY PLL configuration register

Register Name:			PHY state change monitor timeout
Register Address:			16'b11_0_0_0000_0000_1010
Register Mnemonic:			PHY_STATE_CHG_TIMEOUT
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h30D4	**State change timeout:** Bits [19:4] of the state change timeout (bits [3:0] are zero). The timeout is the maximum number of APB clock cycles (abp_pclk) that are allowed for completion of a PHY power state change for the link. If the timeout expires, phy_interrupt_ln_<> for the associated lane is asserted and interrupt status register bit is set. For maximum allowed APB clock frequency, this provides 5.2 msec maximum timeout. Default is set for 1 msec for the maximum APB clock frequency.

Table 26: PHY state change monitor timeout

Register Name:			PHY speedup control register
Register Address:			16'b11_0_0_0000_0000_1111
Register Mnemonic:			PHY_AUTO_CFG_SPDUP
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12'd0	Reserved
3	R/O	1'b0	**Speedup configuration complete:** 1 = PHY speedup configuration is complete, 0 = not complete.
2	R/W	1'b0	**Speedup configuration stall:** 1 = upon completion of PHY speedup configuration, do not release reset to PMA until this bit is cleared, 0 = release reset to PMA upon completion of PHY speedup configuration. Do not set high when [1] = 0.
1	R/W	1'b0	**Speedup configuration enable:** If set to 1 upon de-assertion (high) of phy_reset_n, the PHY will be configured for simulation speedup. Only for use in RTL or gate-level simulations. Not for use in silicon.
0	R/O	1'b0	Reserved

Table 27: PHY speedup control register

Register Name:			PHY external reference clock detect low threshold register
Register Address:			16'b11_0_0_0000_0001_0000
Register Mnemonic:			PHY_REFCLK_DET_THRES_LOW
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'd450	**External Reference Clock Active Detect Low Threshold:** This is the minimum number of external reference clock cycles which must be counted during the measurement interval to indicate a valid clock detected. The default value is based on 100MHz reference clock external frequency and 200 MHz (max) apb_pclk frequency.

Table 28: PHY External Reference Clock Detect Low Threshold register

Register Name:			PHY external reference clock detect high threshold register
Register Address:			16'b11_0_0_0000_0001_0001
Register Mnemonic:			PHY_REFCLK_DET_THRES_HIGH
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'd10050	**External Reference Clock Active Detect High Threshold:** This is the maximum number of external reference clock cycles which must be counted during the measurement interval to indicate a valid clock detected. The default value is based on 100MHz reference clock frequency and 10 MHZ apb_pclk frequency.

Table 29: PHY External Reference Clock Detect High Threshold register

Register Name:			PHY external reference clock detect measurement interval register
Register Address:			16'b11_0_0_0000_0001_0010
Register Mnemonic:			PHY_REFCLK_DET_INTERVAL
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'd1000	**External Reference Clock Active Detect Measurement Interval:** This is the number of apb_pclk cycles in which to count external reference clock cycles. The default corresponds to 5 us for 200 MHz apb_clk and 100 us for 10 MHz apb_clock.

Table 30: PHY External Reference Clock Detect Measurement Interval register

Register Name:			PHY external reference clock detect delay register
Register Address:			16'b11_0_0_0000_0001_0011
Register Mnemonic:			PHY_REFCLK_DET_OP_DELAY
Bits	**Type**	**Reset**	**Description**
15:8	R/W	8'd16	**External Reference Clock Active Detect End Delay:** This is the number of apb_pclk cycles to wait upon completion of measurement interval before capturing the result (accounts for synchronization delays).
7:0	R/W	8'd100	**External Reference Clock Active Detect Start Delay:** This is the number of apb_pclk cycles to wait prior to start of measurement interval (accounts for enable delay of reference clock in PMA).

Table 31: PHY External Reference Clock Detect Delay register

Register Name:			PHY external reference clock detect isolation control register
Register Address:			16'b11_0_0_0000_0001_0100
Register Mnemonic:			PHY_REFCLK_DET_ISO_CTRL
Bits	**Type**	**Reset**	**Description**
15:13	R/O	3'd0	Reserved
12	R/O	1'b0	Captures the current value of the pma_cmn_ext_refclk_detected_cfg PHY input.
11:10	R/O	2'd0	Reserved
9	R/O	1'b0	Current value of pma_cmn_ext_refclk_detected PHY output.
8	R/O	1'b0	Current value of pma_cmn_ext_refclk_detected_valid PHY output.
7:1	R/O	7'd0	Reserved
0	W/O	1'b0	**External Reference Clock Active Detect Start:** Write with 1 to initiate an external reference clock active detect operation. Any previous operation must have completed prior to writing with 1.

Table 32: PHY External Reference Clock Detect Isolation Control register

Register Name:			PHY interrupt status register
Register Address:			16'b11_0_1_0nnn_0001_0010
Register Mnemonic:			PHY_INTERRUPT_STS
Bits	**Type**	**Reset**	**Description**
15	R/W	1'b0	**State change monitor enable** - 1 = state change monitor enabled, 0 = state change monitor disabled. Note: Only the master lane's state change monitor for a link is enabled. The state change monitor for a slave lane is disabled regardless of the state of this bit.
14:11	R/O	4'd0	Reserved
10:8	R/O	3'd0	**Next power state/data rate** - Only valid when one of the interrupt status bits is set. Indicates the requested power state or data rate for the state change failure. For PIPE requests, loaded based on the following: • For a data rate change - {1'b0, requested pipe_rate[1:0]}. • For a power state change using PIPE powerdown - requested pipe_powerdown[2:0] • For sideband L1.x entry - 3'b100. • For sideband L1.x exit - 3'b010. For Raw SerDes power state change requests, loaded based on following: • 3'b000 = power state change with A0 as requested power state • 3'b001 = power state change with A1 as requested power state • 3'b010 = power state change with A2 as requested power state • 3'b011 = power state change with A3 as requested power state • 3'b100 = power state change with A4 as requested power state • 3'b101 = power state change with A5 as requested power state • 3'b110 = PLL clock enable asserted • 3'b111 = PLL clock enable de-asserted
7	R/O	1'b0	Reserved
6:4	R/O	3'd0	**Current power state/data rate** - Only valid when one of the interrupt status bits is set. Indicates the starting power state or data rate for the state change failure. For PIPE requests, loaded based on the following: • For a data rate change - {1'b0, starting pipe_rate[1:0]}. • For a power state change using PIPE powerdown - starting pipe_powerdown[2:0] • For sideband L1.x entry - 3'b010. • For sideband L1.x exit - 3'b100. For Raw SerDes power state change requests, loaded based on following: • 3'b000 = power state change with A0 as starting power state • 3'b001 = power state change with A1 as starting power state • 3'b010 = power state change with A2 as starting power state • 3'b011 = power state change with A3 as starting power state • 3'b100 = power state change with A4 as starting power state • 3'b101 = power state change with A5 as starting power state • 3'b110 = PLL clock enable asserted • 3'b111 = PLL clock enable de-asserted
3:2	R/O	2'd0	Reserved
1	WRC	1'b0	**Data rate state change interrupt status** - Set to 1 upon data rate change timeout. Cleared upon read. Bit is writeable to allow the interrupt to be set manually for test purposes.Only set high by data rate change timeout if bit[0] == 0.
0	WRC	1'b0	**Power state change interrupt status** - Set to 1 upon power state change timeout. Cleared upon read. Bit is writeable to allow the interrupt to set manually for test purposes. Only set high by power state change timeout if bit[1] == 0.

Table 33: PHY interrupt status register

Register Name:			PHY USB3 Gen 2 pre-shoot configuration 0 register (@)
Register Address:			16'b11_0_0_0000_0010_0000
Register Mnemonic:			PHY_PIPE_USB3_GEN2_PRE_CFG0
Bits	**Type**	**Reset**	**Description**
15:8	R/W	8'h0a	**USB3 Gen 2 transmit pre-shoot multiplier configuration 1:** For a lane configured for USB3 mode at Gen 3 data rate, sets the multiplier configuration for the Tx de-emphasis pre-shoot (C_{-1}) when pipe_l<>_tx-_deemphasis[1:0] == 0b01 for the associated lane. Default: 5/64.
7:0	R/W	8'h0a	**USB3 Gen 2 transmit pre-shoot multiplier configuration 0:** For a lane configured for USB3 mode at Gen 3 data rate, sets the multiplier configuration for the Tx de-emphasis pre-shoot (C_{-1}) when pipe_l<>_tx-_deemphasis[1:0] == 0b00 for the associated lane. The multiplier is set based on: • Bit 7 = Enable multiply by 1.0 • Bit 6 = Enable multiply by 0.5 (1/2) • Bit 5 = Enable multiply by 0.25 (1/4) • Bit 4 = Enable multiply by 0.125 (1/8) • Bit 3 = Enable multiply by 0.0625 (1/16) • Bit 2 = Enable multiply by 0.03125 (1/32) • Bit 1 = Enable multiply by 0.015625 (1/64) • Bit 0 = Enable multiply by 0...78125 (1/128) Default: 5/64.

Table 34: PHY USB3 Gen 2 pre-shoot configuration 0 register

Register Name:			PHY USB3 Gen 2 pre-shoot configuration 1 register (@)
Register Address:			16'b11_0_0_0000_0010_0001
Register Mnemonic:			PHY_PIPE_USB3_GEN2_PRE_CFG1
Bits	**Type**	**Reset**	**Description**
15:8	R/W	8'h00	**USB3 Gen 2 transmit pre-shoot multiplier configuration 3:** For a lane configured for USB3 mode at Gen 3 data rate, sets the multiplier configuration for the Tx de-emphasis pre-shoot (C_{-1}) when pipe_l<>_tx-_deemphasis[1:0] == 0b11 for the associated lane. Default: 0.
7:0	R/W	8'h00	**USB3 Gen 2 transmit pre-shoot multiplier configuration 2:** For a lane configured for USB3 mode at Gen 3 data rate, sets the multiplier configuration for the Tx de-emphasis pre-shoot (C_1) when pipe_l<>_tx-_deemphasis[1:0] == 0b10 for the associated lane. Default: 0.

Table 35: PHY USB3 Gen 2 pre-shoot configuration 1 register

Register Name:			PHY USB3 Gen 2 de-emphasis configuration 0 register (@)
Register Address:			16'b11_0_0_0000_0010_0010
Register Mnemonic:			PHY_PIPE_USB3_GEN2_POST_CFG0
Bits	**Type**	**Reset**	**Description**
15:8	R/W	8'h10	**USB3 Gen 2 transmit pre-shoot multiplier configuration 1:** For a lane configured for USB3 mode at Gen 3 data rate, sets the multiplier configuration for the Tx de-emphasis (C_1) when pipe_l<>_tx_deempha-sis[1:0] == 0b01 for the associated lane. Default: 1/8.
7:0	R/W	8'h00	**USB3 Gen 2 transmit pre-shoot multiplier configuration 0:** For a lane configured for USB3 mode at Gen 3 data rate, sets the multiplier configuration for the Tx de-emphasis (C_1) when pipe_l<>_tx_deempha-sis[1:0] == 0b00 for the associated lane. Default: 0.

Table 36: PHY USB3 Gen 2 de-emphasis configuration 0 register

Register Name:			PHY USB3 Gen 2 de-emphasis configuration 1 register (@)
Register Address:			16'b11_0_0_0000_0010_0011
Register Mnemonic:			PHY_PIPE_USB3_GEN2_POST_CFG1
Bits	**Type**	**Reset**	**Description**
15:8	R/W	8'h00	**USB3 Gen 2 transmit pre-shoot multiplier configuration 3:** For a lane configured for USB3 mode at Gen 3 data rate, sets the multiplier configuration for the Tx de-emphasis (C_1) when pipe_l<>_tx_deemphasis[1:0] == 0b11 for the associated lane. Default: 0.
7:0	R/W	8'h10	**USB3 Gen 2 transmit pre-shoot multiplier configuration 2:** For a lane configured for USB3 mode at Gen 3 data rate, sets the multiplier configuration for the Tx de-emphasis (C_1) when pipe_l<>_tx_deemphasis[1:0] == 0b10 for the associated lane. Default: 1/8.

Table 37: PHY USB3 Gen 2 de-emphasis configuration 1 register

9.2 PMA Configuration and Control

This section describes PMA configuration and control registers.

Register Name:			PMA common control1 register
Register Address:			16'b11_1_0_0000_0000_0000
Register Mnemonic:			PHY_PMA_CMN_CTRL1
Bits	**Type**	**Reset**	**Description**
15:7	R/O	9'b0	Reserved
6	R/W	1'b0	Drives cmn_refclk_rcv_out_en PMA input
5	R/O	1'b0	Current value of cmn_macro_suspend_ack PMA output
4	R/O	1'b0	Current value of cmn_refclk_active PMA output
3:1	R/O	3'd0	Reserved
0	R/O	1'b0	Current value of cmn_ready pin PMA output

Table 38: PMA common control1 register

Register Name:			PMA common control2 register
Register Address:			16'b11_1_0_0000_0000_0001
Register Mnemonic:			PHY_PMA_CMN_CTRL2
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'd0	Reserved
7	R/O	1'b0	Current value of cmn_pll1_locked PMA output
6	R/O	1'b0	Current value of cmn_pll0_locked PMA output
5	R/O	1'b0	Current value of cmn_pll1_clk_en_ack PMA output
4	R/O	1'b0	Current value of cmn_pll0_clk_en_ack PMA output
3	R/O	1'b1	Current value of cmn_pll1_disabled PMA output
2	R/O	1'b1	Current value of cmn_pll0_disabled PMA output
1	R/O	1'b0	Current value of cmn_pll1_ready PMA output
0	R/O	1'b0	Current value of cmn_pll0_ready PMA output

Table 39: PMA common control2 register

Register Name:			PMA SSM current state register
Register Address:			16'b11_1_0_0000_0000_0010
Register Mnemonic:			PHY_PMA_SSM_STATE
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'd0	Reserved
8:0	R/O	9'h000	**PMA SSM** : Current state of the PMA startup state machine. PMA output (Not re-synchronized to apb_p-clk)

Table 40: PMA SSM current state register

Register Name:			PMA PLL Raw SerDes control register
Register Address:			16'b11_1_0_0000_0000_0011
Register Mnemonic:			PHY_PMA_PLL_RAW_CTRL
Bits	**Type**	**Reset**	**Description**
15:2	R/O	14'd0	Reserved
1	R/W	1'b1	**Raw SerDes PLL1 control** : When set to 1, cmn_pll1_en PMA input is controlled by PHY logic. When set to 0, cmn_pll1_en PMA input is forced low. When PLL1 is driving lanes configured for Raw SerDes operation, this bit allows 'manual' enable/disable of PLL1 to allow reconfiguration of the PLL for a different data rate. The PHY logic drives cmn_pll1_en high unless the PHY is configured to operation on PLL0 only (PHY_PLL_CFG[1] == 0).
0	R/W	1'b1	**Raw SerDes PLL0 control** : When set to 1, cmn_pll0_en PMA input is controlled by PHY logic. When set to 0, cmn_pll0_en PMA input is forced low. When PLL0 is driving lanes configured for Raw SerDes operation, this bit allows 'manual' enable/disable of PLL0 to allow reconfiguration of the PLL for a different data rate. The PHY logic drives cmn_pll0_en high except when the PHY is configured for single link PCIe mode and is performing a rate change between PCIe Gen 1/2 and PCIe Gen 3 (PHY_PLL_CFG[0] == 1). However, PCIe is only supported for PIPE mode and not applicable when PLL0 is driving lanes configured for Raw SerDes operation.

Table 41: PMA PLL Raw SerDes control register

Register Name:			PMA PLL0 State Machine current state register
Register Address:			16'b11_1_0_0000_0000_1011
Register Mnemonic:			PHY_PMA_PLL0_SM_STATE
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'd0	Reserved
11:0	R/O	12'h300	Current value of cmn_pllsm0_state[11:0]. PMA output (Debug only: Not re-synchronized)

Table 42: PMA PLL0 State Machine current state register

Register Name:			PMA PLL1 State Machine current state register
Register Address:			16'b11_1_0_0000_0000_1100
Register Mnemonic:			PHY_PMA_PLL1_SM_STATE
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'd0	Reserved
11:0	R/O	12'h300	Current value of cmn_pllsm1_state[11:0]. PMA output (Debug only: Not re-synchronized)

Table 43: PMA PLL1 State Machine current state reg

Register Name:			PMA transceiver control register
Register Address:			16'b11_1_1_0nnn_0000_0000
Register Mnemonic:			PHY_PMA_XCVR_CTRL
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'd0	Reserved
8	R/W	1'b0	Drives the tx_differential_invert PMA input for the associated lane.
7:5	R/O	3'h0	Reserved
4	R/O	1'b0	Current value of rx_cdrlf_fphl_locked PMA output for the associated lane.
3	R/O	1'b0	Current value of rx_bist_status PMA output for the associated lane.
2	R/O	1'b0	Current value of rx_bist_err_toggle PMA output for the associated lane.
1	R/O	1'b0	Current value of rx_bist_sync PMA output for the associated lane.
0	R/W	1'b0	Drives the rx_differential_invert PMA input for the associated lane.

Table 44: PMA transceiver control register

Register Name:			PMA loopback control register
Register Address:			16'b11_1_1_0nnn_0000_0001
Register Mnemonic:			PHY_PMA_XCVR_LPBK
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'd0	Reserved
8	R/W	1'b0	Drives the tx_bist_hold PMA input for all lanes in the associated link (i.e. the bit associated with the master lane of the link drives all lanes in the link). Synchronized to transmit data rate clock.
7:6	R/O	2'd0	Reserved
5	R/W	1'b0	Drives the xcvr_lpbk_fe_parallel_en PMA input for the associated lane.
4	R/W	1'b0	Drives the xcvr_lpbk_ne_parallel_en PMA input for the associated lane.
3	R/W	1'b0	Drives the xcvr_lpbk_recovered_clk_en PMA input for the associated lane.
2	R/W	1'b0	Drives the xcvr_lpbk_line_en PMA input for the associated lane.
1	R/W	1'b0	Drives the xcvr_lpbk_isi_gen_en PMA input for the associated lane.
0	R/W	1'b0	Drives the xcvr_lpbk_serial_en PMA input for the associated lane.

Table 45: PMA loopback control register

Register Name:			PMA PI position register
Register Address:			16'b11_1_1_0nnn_0000_0010
Register Mnemonic:			PHY_PMA_PI_POS
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h00	Current value of rx_pi_val PMA output for the associated lane.
7:0	R/O	8'h00	Current value of rx_eye_plot_pi_val PMA output for the associated lane.

Table 46: PMA PI position register

Register Name:			PMA PSM current state lower register
Register Address:			16'b11_1_1_0nnn_0001_0000
Register Mnemonic:			PHY_PMA_PSM_STATE_LO
Bits	**Type**	**Reset**	**Description**
15:0	R/O	16'h0000	Current value of xcvr_psm_state[15:0] for the associated lane - PMA power state machine state. (Not re-synchronized to apb_pclk)

Table 47: PMA Transceiver PSM current state low register

Register Name:			PMA PSM current state higher register
Register Address:			16'b11_1_1_0nnn_0001_0001
Register Mnemonic:			PHY_PMA_PSM_STATE_HI
Bits	**Type**	**Reset**	**Description**
15:13	R/O	3'd0	Reserved
12	R/O	1'b0	Current value of xcvr_psm_ready for the associated lane.
11:10	R/O	2'd0	Reserved
9:0	R/O	10'd0	Current value of xcvr_psm_state[25:16] for the associated lane - PMA power state machine state. (Not re-synchronized to apb_pclk)

Table 48: PMA Transceiver PSM current state high register

9.3 PHY and PMA Isolation

Isolation mode is a mode that allows all of the macro control input pins and some internal signals to be driven directly by internal CDB register bits, in the event for instance, that components outside of the PHY macro are not functioning correctly. In addition to controlling the input signals, this mode also allows for selected output pins to be monitored using CDB registers as well. This allows for a significant amount of the macro to be independently verified.

There are two isolation modes supported in this macro. These are :

- Full PHY macro isolation mode, where PHY top level macro control inputs and outputs can be controlled and observed.
- PMA only isolation mode, where PMA specific inputs and outputs can be controlled and observed, such that the PMA can be controlled independently from the PCS.

Isolation mode is controlled by the isolation_en in the on page 112. When isolation_en is set, the isolation_mode_sel bit in the on page 112 selects the type of isolation. When isolation_mode_sel is low, PHY Macro isolation mode is selected. When isolation_mode_sel is high, PMA Isolation Mode is selected. The registers defined in this section control the specified signals when in PHY and PMA isolation modes. In the description of each register, one will find a note indicating in which isolation mode(s) the register can be used.

9.3.1 PHY isolation

Register Name:			PHY common control signal isolation register (@)
Register Address:			16'b11_0_0_0000_0000_1000
Register Mnemonic:			PHY_ISO_CMN_CTRL
Bits	**Type**	**Reset**	**Description**
15:13	R/O	3'd0	Reserved
12	R/O	1'b0	Current value of phy_refclk_reqd PHY output.
11:9	R/O	3'd0	Reserved
8	R/W	1'b0	Drives phy_en_refclk PHY input when in PHY macro and PMA isolation mode.
7:6	R/O	2'd0	Reserved
5	R/W	1'b0	Drives phy_pma_suspend_override PHY input when in PHY macro and PMA isolation mode.
4	R/W	1'b0	Drives refclk_rcvr_pwrdn internal PHY signal when in PHY macro and PMA isolation mode (1 = powers down the reference clock receiver). During normal operation, refclk_rcvr_pwrdn is driven from ~abp_preset_n, which powers down the reference clock receiver when the PHY is disabled.
3:1	R/O	3'd0	Reserved
0	R/W	1'b1	Drives phy_reset_n PHY input when in PHY macro and PMA isolation mode.

Table 49: PHY common control isolation register

Register Name:			PIPE TX control signal isolation register (@)
Register Address:			16'b11_0_1_0nnn_0000_0000
Register Mnemonic:			PHY_PIPE_ISO_TX_CTRL
Bits	**Type**	**Reset**	**Description**
15:12	R/W	4'h0	Drives pipe_tx_data_k PHY input for the associated lane when in PHY macro and PMA isolation modes.
11:9	R/O	3'd0	Reserved
8	R/W	1'b0	Drives pipe_tx_ones_zeros input for the associated lane when in PHY macro and PMA isolation modes.
7:5	R/O	3'd0	Reserved
4	R/W	1'b0	Drives pipe_tx_elec_idle PHY input for the associated lane when in PHY macro and PMA isolation modes
3	R/W	1'b0	Drives pipe_tx_128b_enc_byp PHY input for the associated lane when in PHY macro and PMA isolation modes.
2	R/W	1'b0	Drives pipe_tx_compliance PHY input for the associated lane when in PHY macro and PMA isolation modes.
1:0	R/W	2'd0	Drives pipe_tx_pattern PHY input for the associated lane when in PHY macro and PMA isolation modes.

Table 50: PIPE TX control signal isolation register

Register Name:			PIPE TX local preset coefficients low isolation register (@)
Register Address:			16'b11_0_1_0nnn_0000_0001
Register Mnemonic:			PHY_PIPE_ISO_TX_LPC_LO
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'b00	Reserved
13:8	R/O	6'h00	Current value of pipe_tx_local_tx_preset_coefficients[11:6] for the associated lane when PHY_PCS_ISO_TX_LPC_HI[15] == 1. Otherwise, 0.
7:6	R/O	2'b00	Reserved
5:0	R/O	6'h00	Current value of pipe_tx_local_tx_preset_coefficients[5:0] for the associated lane when PHY_PCS_ISO_TX_LPC_HI[15] == 1. Otherwise, 0.

Table 51: PIPE Tx local preset coefficients low isolation register

Register Name:			PIPE TX local preset coefficients high isolation register (@)
Register Address:			16'b11_0_1_0nnn_0000_0010
Register Mnemonic:			PHY_PIPE_ISO_TX_LPC_HI
Bits	**Type**	**Reset**	**Description**
15	R/O	1'b0	Set upon assertion of pipe_tx_local_tx_coeff_vld PHY output for the associated lane. Cleared upon writing PHY_PCS_ISO_TX_LPC_HI[12] with a 0.
14:13	R/O	2'b00	Reserved
12	R/W	1'b0	Drives pipe_tx_get_local_preset_coef PHY output for the associated lane when in PHY macro and PMA isolation modes
11:8	R/W	4'b0000	Drives pipe_tx_local_preset_index PHY output for the associated lane when in PHY macro and PMA isolation modes.
7:6	R/O	2'b00	Reserved
5:0	R/O	6'h00	Current value of pipe_tx_local_tx_preset_coefficients[17:12] for the associated lane when PHY_PCS_ISO_TX_LPC_HI[15] == 1. Otherwise, 0.

Table 52: PIPE Tx local preset coefficients high isolation register

Register Name:			PCS TX PIPE deemphasis low isolation register (@)
Register Address:			16'b11_0_1_0nnn_0000_0011
Register Mnemonic:			PHY_PCS_ISO_TX_DMPH_LO
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'd0	Reserved
13:8	R/W	6'h00	Drives pipe_tx_deemph[11:6] PHY input for the associated lane when in PHY macro and PMA isolation mode.
7:6	R/O	2'h0	Reserved
5:0	R/W	6'h00	Drives pipe_tx_deemph[5:0] PHY input for the associated lane when in PHY macro and PMA isolation mode.

Table 53: PIPE TX Deemphasis low isolation register

Register Name:			PIPE TX deemphasis high isolation register (@)
Register Address:			16'b11_0_1_0nnn_0000_0100
Register Mnemonic:			PHY_PIPE_ISO_TX_DMPH_HI
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h000	Reserved
5:0	R/W	6'h0	Drives pipe_tx_deemph[17:12] PHY input for the associated lane when in PHY macro and PMA isolation modes.

Table 54: PIPE Tx Deemphasis high isolation register

Register Name:			PIPE TX local FS and LF isolation register (@)
Register Address:			16'b11_0_1_0nnn_0000_0101
Register Mnemonic:			PHY_PIPE_ISO_TX_FSLF
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'b00	Reserved
13:8	R/O	6'h00	Current value of pipe_tx_local_fs PHY output for the associated lane. (Not re-synchronized to apb_pclk)
7:6	R/O	2'b00	Reserved
5:0	R/O	6'h00	Current value of pipe_tx_local_lf PHY output for the associated lane. (Not re-synchronized to apb_pclk)

Table 55: PIPE Tx local FS/LF isolation register

Register Name:			PCS TX PIPE data low isolation register (@)
Register Address:			16'b11_0_1_0nnn_0000_0110
Register Mnemonic:			PHY_PCS_ISO_TX_DATA_LO
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h0000	Drives pipe_tx_data[15:0] PHY input for the associated lane when in PHY macro and PMA isolation mode.

Table 56: PCS PIPE TX Data low isolation register

Register Name:			PCS TX PIPE data high isolation register (@)
Register Address:			16'b11_0_1_0nnn_0000_0111
Register Mnemonic:			PHY_PCS_ISO_TX_DATA_HI
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h0000	Drives pipe_tx_data[31:16] PHY input for the associated lane when in PHY macro and PMA isolation mode.

Table 57: PCS PIPE TX Data high isolation register

Register Name:			PCS RX control signal isolation register (@)
Register Address:			16'b11_0_1_0nnn_0000_1000
Register Mnemonic:			PHY_PCS_ISO_RX_CTRL
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Current value of pipe_rx_data_k PHY output for the associated lane, when PHY_PCS_ISO_RX_CTRL[5] == 1. Otherwise, 0. (Not re-synchronized to apb_pclk)
11:10	R/O	d'h0	Reserved
9	R/W	1'b0	Drives pipe_rx_eq_training PHY input for the associated lane when in PHY macro and PMA isolation modes.
8	R/W	1'b0	Drives pipe_rx_termination PHY input for the associated lane when in PHY macro and PMA isolation modes.
7	R/W	1'b0	Drives pipe_rx_polarity PHY input for the associated lane when in PHY macro and PMA isolation modes.
6	R/O	1'b0	Reserved
5	R/O	1'b0	Current value of pipe_rx_valid PHY output for the associated lane.
4	R/O	1'b1	Current value of pipe_rx_elec_idle PHY output for the associated lane.
3	R/O	1'b0	Current value of pipe_align_detect PHY output for the associated lane.
2:0	R/O	3'b000	Current value of pipe_rx_status PHY output for the associated lane. Holds the highest priority pipe_rx-_status value seen, since the last register read. Cleared upon read.

Table 58: PCS PIPE RX control signal isolation register

Register Name:			PIPE RX equalization evaluation isolation register (@)
Register Address:			16'b11_0_1_0nnn_0000_1001
Register Mnemonic:			PHY_PIPE_ISO_RX_EQ_EVAL
Bits	**Type**	**Reset**	**Description**
15:13	R/O	3'h0	Reserved
12	R/W	1'b0	Drives pipe_invalid_request for the associated lane when in PHY macro and PMA isolation modes.
11	R/W	1'b0	pipe_link_eval_dir_change[5:4] bit reversal enable. When low, no bit reversal. When high, the bit positions for pipe_link_eval_dir_change[5:4] PHY output for the associated lane are reversed.
10	R/W	1'b0	pipe_link_eval_dir_change[3:2] bit reversal enable. When low, no bit reversal. When high, the bit positions for pipe_link_eval_dir_change[3:2] PHY output for the associated lane are reversed.
9	R/W	1'b0	pipe_link_eval_dir_change[1:0] bit reversal enable. When low, no bit reversal. When high, the bit positions for pipe_link_eval_dir_change[1:0] PHY output for the associated lane are reversed.]
8	R/W	1'b0	Drives pipe_rx_eval PHY input for the associated lane when in PHY macro and PMA isolation modes.
7	R/O	1'b0	Reserved
6	R/O	1'b1	Captures pipe_phy_status for Rx equalization evaluation PHY output for the associated lane (does not include power state change signaling). Set when pipe_phy_status is high and pipe_rx_eq_eval to the PCS is high (i.e. after the isolation MUX). Cleared upon read.
5:0	R/O	6'h00	pipe_link_eval_dir_change PHY output for the associated lane (prior to bit reversal logic) upon completion of Rx equalization evaluation. Captured upon assertion of Rx equalization eval pipe_phy_status when pipe_rx_eq_eval to the PCS (i.e. after the isolation MUX) is high. Cleared upon read. (Not re-synchronized to apb_pclk)

Table 59: PIPE Rx equalization evaluation isolation register

Register Name:			PHY link configuration isolation register
Register Address:			16'b11_0_1_0nnn_0000_1010
Register Mnemonic:			PHY_ISO_LINK_CFG
Bits	**Type**	**Reset**	**Description**
15	R/W	1'b0	Drives phy_link_cfg_ln_{nnnn} PHY input when in PHY macro and PMA isolation modes.
14:13	R/O	2'h0	Reserved
12	R/W	1'b0	Drives pipe_l{nnnn}_32bit_sel PHY input when in PHY macro and PMA isolation modes.
11:10	R/O	2'd0	Reserved
9:8	R/W	2'd0	Drives pma_fullrt_div_ln_{nnnn} PHY input when in PHY macro and PMA isolation modes.
7:6	R/O	2'd0	Reserved
5	R/W	1'b0	Drives pipe_l{nnnn}_pcie_l1_ss_sel PHY input when in PHY macro and PMA isolation mode.
4	R/W	1'd0	Drives pipe_l{nnnn}_eb_mode PHY input when in PHY macro and PMA isolation modes.
3:2	R/O	2'd0	Reserved
1:0	R/W	2'h0	Drives phy_l{nnnn}_mode PHY input when in PHY macro and PMA isolation modes.

Table 60: PHY link configuration isolation register

Register Name:			PCS Link control isolation register
Register Address:			16'b11_0_1_0nnn_0000_1011
Register Mnemonic:			PHY_PCS_ISO_LINK_CTRL
Bits	**Type**	**Reset**	**Description**
15:14	R/O	4'h0	Reserved
13	R/O	1'b0	Current value of phy_l*_ack_l1_x PHY output for the associated lane. (@)
12	R/W	1'b0	Drives the phy_l*_ent_l1_x PHY input for the associated lane when in PHY macro and PMA isolation modes. (Link signal - for multi-lane links, master lane used for all lanes in link.)
11	R/W	1'b0	Drives the phy_l*_rx_elec_idle_det_en PHY input for the associated lane when in PHY macro and PMA isolation modes. (Link signal - for multi-lane links, master lane used for all lanes in link.)
10	R/W	1'b0	Drives the phy_l*_tx_cmn_mode_en PHY input for the associated lane when in PHY macro and PMA isolation modes. (Link signal - for multi-lane links, master lane used for all lanes in link.)
9:8	R/W	2'b00	Drives the pipe_l*_rate PHY input for the associated lane when in PHY macro and PMA isolation modes. (Link signal - for multi-lane links, master lane used for all lanes in link.)
7	R/O	1'b0	Reserved
6:4	R/W	3'b010	Drives the pipe_l*_powerdown PHY input for the associated lane when in PHY macro and PMA isolation modes. (Link signal - for multi-lane links, master lane used for all lanes in link.)
3	R/O	1'b0	Reserved
2	R/W	1'b0	Drives the pipe_l*_tx_det_rx_lpbk PHY input for the associated lane when in PHY macro and PMA isolation modes. (Link/lane signal - for non-USB multi-lane links, master lane used for all lanes in link. For USB multi-lane links, used only for the associated lane).
1	R/O	1'b1	Captures pipe_l*_phy_status (for power state and rate change) PHY output for the associated lane. Set when pipe_phy_status is high and cleared upon read if pipe_phy_status is low. (@)
0	R/W	1'b1	Drives the phy_l*_reset_n PHY input for the associated lane when in PHY macro and PMA isolation modes. (Link signal - for multi-lane links, master lane used for all lanes in link.)

Table 61: PCS PIPE Link control isolation register

Register Name:			PIPE USB Gen 1 loopback slave BER count register (@)
Register Address:			16'b11_0_1_0nnn_0000_1100
Register Mnemonic:			PHY_PIPE_ISO_USB_BER_CNT
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'd0	Reerved
7:0	R/O	8'd0	Current value of USB 3.1 Gen 1 loopback slave Bit Error Count from the PCS.

Table 62: PIPE USB loopback slave BER count register

Register Name:			PCS RX data low isolation register (@)
Register Address:			16'b11_0_1_0nnn_0000_1110
Register Mnemonic:			PHY_PCS_ISO_RX_DATA_LO
Bits	**Type**	**Reset**	**Description**
15:0	R/O	16'h0000	Current value of pipe_rx_data[15:0] PHY output. (Not re-synchronized to apb_pclk)

Table 63: PCS PIPE RX data low isolation register

Register Name:			PCS RX data high isolation register (@)
Register Address:			16'b11_0_1_0nnn_0000_1111
Register Mnemonic:			PHY_PCS_ISO_RX_DATA_HI
Bits	**Type**	**Reset**	**Description**
15:0	R/O	16'h0000	Current value of pipe_rx_data[31:16] PHY output. (Not re-synchronized to apb_pclk)

Table 64: PCS PIPE RX data high isolation register

Register Name:			Ethernet MAC clock configuration isolation register (&)
Register Address:			16'b11_0_1_0nnn_0001_0000
Register Mnemonic:			PHY_ETH_ISO_MAC_CLK_CFG
Bits	**Type**	**Reset**	**Description**
15:2	R/O	14'h0000	Reserved
1:0	R/W	2'h0	Drives mac_src_sel PHY input for the associated lane when in PHY macro and PMA isolation mode.

Table 65: Ethernet MAC clock configuration isolation register

Register Name:			Ethernet MAC clock divider isolation register (&)
Register Address:			16'b11_0_1_0nnn_0001_0001
Register Mnemonic:			PHY_ETH_ISO_MAC_CLK_DIV
Bits	**Type**	**Reset**	**Description**
15:7	R/W	9'd1	Drives mac_div_sel1 PHY input for the associated lane when in PHY macro and PMA isolation mode.
6:0	R/W	7'd1	Drives mac_div_sel0 PHY input for the associated lane when in PHY macro and PMA isolation mode.

Table 66: Ethernet MAC clock divider isolation register

9.3.2 PMA isolation

Register Name:			PMA common control signal isolation register
Register Address:			16'b11_1_0_0000_0000_0100
Register Mnemonic:			PHY_PMA_ISO_CMN_CTRL
Bits	**Type**	**Reset**	**Description**
15	R/W	1'b0	Drives cmn_ref_clk_term_en PMA input when in PHY macro or PMA isolation modes
14	R/W	1'b0	Drives cmn_ref_clk_dig_sel PMA input when in PHY macro or PMA isolation modes
13:12	R/W	2'h0	Drives cmn_ref_clk_dig_div PMA input when in PHY macro or PMA isolation modes
11:10	R/W	2'd0	Drives cmn_ref_clk_int_mode PMA input when in PHY macro and PMA isolation modes.
9:8	R/W	2'd0	Drives cmn_ref_clk0_mode PMA input when in PHY macro and PMA isolation modes.
7	R/O	1'b0	Current value of cmn_clock_stop_ack PMA output.
6	R/W	1'b0	Drives cmn_clock_stop_req PMA input when in PMA isolation mode
5	R/O	1'b0	Reserved
4	R/W	1'b0	Drives cmn_ref_clk0_clk_gate_en PMA input when in PMA isolation mode.
3	R/W	1'b0	Drives cmn_refclk_disable PMA input when in PMA isolation mode.
2	R/W	1'b0	Drives cmn_macro_suspend_req PMA input when in PMA isolation mode.
1	R/O	1'b0	Reserved
0	R/W	1'b1	Drives cmn_reset_n PMA input when in PMA isolation mode.

Table 67: PMA common control signal isolation register

Register Name:			PMA PLL control0 isolation register
Register Address:			16'b11_1_0_0000_0000_0101
Register Mnemonic:			PHY_PMA_ISO_PLL_CTRL0
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'd0	Reserved
7	R/W	1'b0	Drives cmn_pll1_ref_clk_sel PMA input when in PHY macro or PMA isolation mode.
6	R/W	1'b0	Drives cmn_pll0_ref_clk_sel PMA input when in PHY macro or PMA isolation mode.
5	R/W	1'b0	Drives cmn_pll1_mode_sel PMA input when in PMA isolation mode.
4	R/W	1'b0	Drives cmn_pll0_mode_sel PMA input when in PMA isolation mode.
3	R/W	1'b0	Drives cmn_pll1_clk_en PMA input when in PMA isolation mode
2	R/W	1'd0	Drives cmn_pll0_clk_en PMA input when in PMA isolation mode
1	R/W	1'd1	Drives cmn_pll1_en PMA input when in PMA isolation mode
0	R/W	1'd1	Drives cmn_pll0_en PMA input when in PMA isolation mode

Table 68: PMA PLL control0 isolation register

Register Name:			PMA PLL control1 isolation register
Register Address:			16'b11_1_0_0000_0000_0110
Register Mnemonic:			PHY_PMA_ISO_PLL_CTRL1
Bits	**Type**	**Reset**	**Description**
15:12	R/W	4'h1	Drives cmn_pll1_clk_datart1_div PMA input when in PMA isolation mode
11:8	R/W	4'h1	Drives cmn_pll1_clk_datart0_div PMA input when in PMA isolation mode
7:4	R/W	4'h2	Drives cmn_pll0_clk_datart1_div PMA input when in PMA isolation mode
3:0	R/W	4'h4	Drives cmn_pll0_clk_datart0_div PMA input when in PMA isolation mode

Table 69: PMA PLL control1 isolation register

Register Name:			PMA Isolation control register
Register Address:			16'b11_1_0_0000_0000_1111
Register Mnemonic:			PHY_PMA_ISOLATION_CTRL
Bits	**Type**	**Reset**	**Description**
15	R/W	1'b0	PHY/PMA isolation enable (isolation_en) - When set, enables isolation (PHY or PMA).
14	R/W	1'b0	PHY/PMA common isolation enable (cmn_isolation_en) - When in PHY Macro Isolation Mode, the PHY common isolation register(s) are selected. When in PMA Isolation Mode, the PMA common isolation register(s) are selected.
13	R/O	1'd0	Reserved
12	R/W	1'b0	PHY/PMA isolation mode select (isolation_mode_sel) - When isolation_en is set, this bit selects between PHY isolation mode and PMA isolation mode. 0 = PHYMacro isolation mode; 1 = PMA isolation mode.
11:8	R/O	4'd0	Reserved
7:0	R/W	8'd0	PHY/PMA lane isolation enable (ln_isolation_en) - When in PHY Macro Isolation Mode, the selected PHY lane(s) isolation registers are selected. When in PMA Isolation Mode, the selected PMA lane(s) isolation registers are selected.

Table 70: PMA control isolation register

Register Name:			PMA Transceiver control isolation register
Register Address:			16'b11_1_1_0nnn_0000_0011
Register Mnemonic:			PHY_PMA_ISO_XCVR_CTRL
Bits	**Type**	**Reset**	**Description**
15	R/O	1'b0	Current value of xcvr_pll_clk_en_ack PMA output for the associated lane.
14	R/O	1'b0	Reserved
13	R/W	1'b0	Drives tx_lfps_en PMA input for the associated lane when in PMA isolation mode.
12	R/W	1'b0	Drives tx_elec_idle PMA input for the associated lane when in PMA isolation mode or PHY isolation mode and lane is configured for Raw SerDes mode.
11	R/O	1'b0	Current value of tx_rcv_detected PMA output for the associated lane. (Not re-synchronized to apb_pclk)
10	R/O	1'b0	Current value of tx_rcv_detect_done PMA ouptut for the associated lane.
9	R/W	1'b0	Drives tx_rcv_detect_en PMA input for the associated lane when in PMA isolation mode or PHY isolation mode and lane is configured for Raw SerDes mode.
8	R/W	1'b0	Drives xcvr_link_reset_n PMA input for the associated lane when in PMA isolation mode.
7	R/W	1'b0	Drives xcvr_pll_clk_en PMA input for the associated lane when in PMA isolation mode or PHY isolation mode and lane is configured for Raw SerDes mode.
6	R/O	1'b0	Reserved
5	R/W	1'b0	Drives xcvr_lane_suspend PMA input for the associated lane when in PMA isolation mode.
4	R/O	1'b0	Current value of rx_lfps_detect PMA output for the associated lane.
3	R/O	1'b0	Current value of rx_signal_detect PMA output for the associated lane.
2	R/O	1'b0	Reserved
1	R/W	1'b0	Drives rx_termination PMA input for the associated lane when in PMA isolation mode or PHY isolation mode and lane is configured for Raw SerDes mode.
0	R/O	1'b0	Reserved

Table 71: PMA transceiver control isolation register

Register Name:			PMA transmitter local preset coefficient low isolation register
Register Address:			16'b11_1_1_0nnn_0000_0100
Register Mnemonic:			PHY_PMA_ISO_TX_LPC_LO
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'b00	Reserved
13:8	R/O	6'h00	Value of tx_local_tx_preset_coef[11:6] PMA output for the associated lane captured upon assertion of tx_local_preset_coef_valid for the associated lane. Cleared upon read.
7:6	R/O	2'b00	Reserved
5:0	R/O	6'h00	Value of tx_local_tx_preset_coef[5:0] PMA output for the associated lane captured upon assertion of tx_local_preset_coef_valid for the associated lane. Cleared upon read.

Table 72: PMA Tx local preset coefficient low isolation register

Register Name:			PMA transmitter local preset coefficient high isolation register
Register Address:			16'b11_1_1_0nnn_0000_0101
Register Mnemonic:			PHY_PMA_ISO_TX_LPC_HI
Bits	**Type**	**Reset**	**Description**
15	R/O	1'b0	Current value of tx_local_preset_coef_valid PMA output for the associated lane.
14:13	R/O	2'd0	Reserved
12	R/W	1'b0	Drives tx_get_local_preset_coef PMA input for the associated lane when in PMA isolation mode or PHY isolation mode and lane is configured for Raw SerDes mode.
11:8	R/W	4'h0	Drives tx_local_preset_index PMA input for the associated lane when in PMA isolation mode or PHY isolation mode and lane is configured for Raw SerDes mode.
7:6	R/O	2'b00	Reserved
5:0	R/O	6'h00	Value of tx_local_tx_preset_coef[17:12] PMA output for the associated lane captured upon assertion of tx_local_preset_coef_valid for the associated lane. Cleared upon read.

Table 73: PMA Tx local preset coefficient high isolation register

Register Name:			PMA Tx de-emphasis low isolation register
Register Address:			16'b11_1_1_0nnn_0000_0110
Register Mnemonic:			PHY_PMA_ISO_TX_DMPH_LO
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'd0	Reserved
13:8	R/W	6'd0	Drives tx_deemphasis[11:6] PMA input for the associated lane when in PMA isolation mode or PHY isolation mode and lane is configured for Raw SerDes mode.
7:6	R/O	2'd0	Reserved
5:0	R/W	6'd0	Drives tx_deemphasis[5:0] PMA input for the associated lane when in PMA isolation mode or PHY isolation mode and lane is configured for Raw SerDes mode.

Table 74: PMA Tx de-emphasis low isolation register

Register Name:			PMA Tx de-emphasis high isolation register
Register Address:			16'b11_1_1_0nnn_0000_0111
Register Mnemonic:			PHY_PMA_ISO_TX_DMPH_HI
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'd0	Reserved
5:0	R/W	6'd0	Drives tx_deemphasis[17:12] PMA input for the associated lane when in PMA isolation mode or PHY isolation mode and lane is configured for Raw SerDes mode.

Table 75: PMA Tx de-emphasis high isolation register

Register Name:			PMA Tx FS/LF isolation register
Register Address:			16'b11_1_1_0nnn_0000_1000
Register Mnemonic:			PHY_PMA_ISO_TX_FSLF
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'd0	Reserved
13:8	R/O	6'd0	Current value of tx_local_fs PMA ouptut for the associated lane. (Not re-synchronized to apb_pclk) Note: In a RTL simulation without UPF, the reset value of this field will be 0x2D.
7:6	R/O	2'd0	Reserved
5:0	R/O	6'd0	Current value of tx_local_lf PMA ouptut for the associated lane. (Not re-synchronized to apb_pclk) Note: In a RTL simulation without UPF, the reset value of this field will be 0x0F.

Table 76: PMA Tx FS/LF isolation register

Register Name:			PMA TX margin register
Register Address:			16'b11_1_1_0nnn_0000_1001
Register Mnemonic:			PHY_PMA_ISO_TX_MGN
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h00	Reserved
8	R/W	1'b0	Drives tx_low_power_swing_en PMA input for the associated lane when in PMA isolation mode or PHY isolation mode and lane is configured for Raw SerDes mode.
7:3	R/O	5'h00	Reserved
2:0	R/W	3'b000	Drives tx_vmargin PMA input for the associated lane when in PMA isolation mode or PHY isolation mode and lane is configured for Raw SerDes mode.

Table 77: PMA Tx margin isolation register

Register Name:			PMA Isolation mode control register
Register Address:			16'b11_1_1_0nnn_0000_1010
Register Mnemonic:			PHY_PMA_ISO_LINK_MODE
Bits	**Type**	**Reset**	**Description**
15	R/W	1'b1	tx_reset_n_ln_{nnnn} PMA input when in PMA isolation mode.
14	R/W	1'b1	rx_reset_n_ln_{nnnn} PMA input when in PMA isolation mode.
13:6	R/O	8'd0	Reserved
5:4	R/W	2'd0	Drives xcvr_standard_mode_ln_{nnnn} PMA input when in PMA isolation mode or PHY isolation mode and lane is configured for Raw SerDes mode.
3	R/O	1'b0	Reserved
2:0	R/W	3'd0	Drives xcvr_data_width_ln_{nnnn} PMA input when in PMA isolation mode or PHY isolation mode and lane is configured for Raw SerDes mode.

Table 78: PMA Link Mode isolation register

Register Name:			PMA Isolation power state control register
Register Address:			16'b11_1_1_0nnn_0000_1011
Register Mnemonic:			PHY_PMA_ISO_PWRST_CTRL
Bits	**Type**	**Reset**	**Description**
15	R/W	1'b0	rx_sig_det_en_ext_ln_{nnnn} PMA input when in PMA isolation mode. (Used for PCIe)
14	R/W	1'b0	tx_cmn_mode_en_ext_ln_{nnnn} PMA input when in PMA isolation mode. (Used for PCIe)
13:8	R/O	6'h00	Current value of xcvr_power_state_ack_ln_{nnnn} PMA output.
7:6	R/O	2'b00	Reserved
5:0	R/W	6'h00	Drives xcvr_power_state_req_ln_{nnnn} PMA input when in PMA isolation mode or PHY isolation mode and lane is configured for Raw SerDes mode..

Table 79: PMA Link Power State Control isolation register

Register Name:			PMA RX equalization control isolation register
Register Address:			16'b11_1_1_0nnn_0000_1101
Register Mnemonic:			PHY_PMA_ISO_RX_EQ_CTRL
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'd0	Reserved
13	R/W	1'b0	Drives rx_eq_training_data_valid PMA input for the associated lane when in PMA isolation mode or PHY isolation mode and lane is configured for Raw SerDes mode.
12	R/W	1'b0	Drives rx_eq_training PMA input for the associated lane when in PMA isolation mode or PHY isolation mode and lane is configured for Raw SerDes mode.
11:10	R/O	2'd0	Reserved
9:4	R/O	6'h00	The value of rx_link_eval_fb_dir_change PMA output for the associated lane upon assertion of rx_eq_eval_status to PMA. Cleared upon read.
3	R/O	1'b0	The value of rx_eq_eval_complete PMA output for the associated lane upon assertion of rx_eq_eval_status. Cleared upon read.
2	R/W	1'b0	Drives rx_invalid_request PMA input for the associated lane when in PMA isolation mode.
1	R/O	1'b0	Current value of rx_eq_eval_status PMA output for the associated lane.
0	R/W	1'b0	Drives rx_eq_eval PMA input for the associated lane when in PMA isolation mode or PHY isolation mode and lane is configured for Raw SerDes mode.

Table 80: PMA Rx equalization control isolation register

Register Name:			PMA low data isolation register
Register Address:			16'b11_1_1_0nnn_0000_1110
Register Mnemonic:			PHY_PMA_ISO_DATA_LO
Bits	**Type**	**Reset**	**Description**
15:0	R/O	16'h0000	Current value of rx_rd[15:0] PMA output for the current lane. (Not re-synchronized to apb_pclk). This register can be written and the value will drive tx_td[15:0] for the associated lane when in PMA isolation mode or PHY isolation mode and lane is configured for Raw SerDes mode.

Table 81: PMA low data isolation register

Register Name:			PMA high data isolation register
Register Address:			16'b11_1_1_0nnn_0000_1111
Register Mnemonic:			PHY_PMA_ISO_DATA_HI
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12'h000	Reserved
3:0	R/O	4'h0	Current value of rx_rd[19:16] PMA output for the current lane. (Not re-synchronized to apb_pclk). This register can be written and the value will drive tx_td[19:16] for the associated lane when in PMA isolation mode or PHY isolation mode and lane is configured for Raw SerDes mode.

Table 82: PMA high data isolation register

10 PMA Configuration And Diagnostic Bus

10.1 Overview

The configuration and diagnostics bus (CDB) is a simple parallel bus interface, used to read and write registers inside the macro for configuration and diagnostics purposes. This bus protocol is compliant with the AMBA 3 APB bus protocol specification (v1.0). The top level pins that make up this bus are described in section 2.3 “On-Chip Functional Pin List” on page 39, and all have the prefix “cdb_”. Figures 110 on page 658 and 111 on page 658 illustrate timing diagrams for accessing the internal registers using the CDB. Note that the registers on this bus can be written and read after cdb_preset_n is released, and prior to cmn_reset_n is released. Most registers can also be written prior to the switched power islands being switched on. The exception to this is the CDB registers associated with TX and RX BIST functions.

The bits that make up the address bus are divided up into logical groups that serve different functions in the address decoding process. These logical groups, and their respective functions are described in table 83 on page 116. The address decoding for the entire CDB address space is described in detail in section 10.2 “CDB Address Map” on page 117.

cdb_paddr bits	Decode Function
15:14	These bits specify the module type that is being addressed. • 2’b00 : Common module. • 2’b01 : Transceiver and transmitter lane module. • 2’b10 : Receiver lane module. • 2’b11 : Reserved.
13	This bit can be used to write to the same register across all the lanes in a given link (link multi-write). When using this bit to perform a link multi-write, bits 9 through 12 must specify the link ID specified on the xcvr_link_id for each lane in the link being addressed. For implementations where this PMA is instantiated within a bifurcated PHY, this multi-write function is not supported, and this bit must be tied to 1’b0 for all accesses. • 1’b0 : Link multi-write access disabled (required for bifurcated PHY solutions). • 1’b1 : Link multi-write access enabled. Note: The link multi-write function is a function of the xcvr_link_id pin. Therefore, this pin must be stable for a given link prior to attempting a link multi-write. This bit should not be used when performing an access to the common module. This bit should not be used when performing register reads.
12:9	When a transmitter or receiver lane is selected, using bits 15 and 14, these bits specify the particular lane that is being addressed, when not performing a link multi-write. • 4’b0000 : Lane 0 • 4’b0001 : Lane 1 • 4’b0010 : Lane 2 • 4’b0011 : Lane 3 • 4’b0100 : Lane 4 • 4’b0101 : Lane 5 • 4’b0110 : Lane 6 • 4’b0111 : Lane 7 • 4’b1000 - 4’b1111 : Reserved Note: When the common module is addressed, these bits must be set to 4’b0000. When a transmitter or receiver lane is selected, using bits 15 and 14, and a link multi-write is being performed, these bits specify the particular link that is being addressed.
8:5	These bits specify a sub-module within the selected lane or common module.
4:0	These bits specify up to 32 addressable registers within a selected sub-module.

Table 83: CDB High Level Address Decoding

10.2 CDB Address Map

The following table specifies the top level address space for the macro.

Address bits 15:9	Addressed Region
00_0_0000	Common Components Address Space on page 118
00_0_0001 - 00_1_1111	Reserved
01_m_nnnn	General lane addressing : m = multi-write, nnnn = lane number. Transceiver And Transmitter Components Address Space on page 118
01_0_0000	Transmitter lane 0 : Transceiver And Transmitter Components Address Space on page 118
01_0_0001	Transmitter lane 1 : Transceiver And Transmitter Components Address Space on page 118
01_0_0010	Transmitter lane 2 : Transceiver And Transmitter Components Address Space on page 118
01_0_0011	Transmitter lane 3 : Transceiver And Transmitter Components Address Space on page 118
01_0_0100	Transmitter lane 4 : Transceiver And Transmitter Components Address Space on page 118
01_0_0101	Transmitter lane 5 : Transceiver And Transmitter Components Address Space on page 118
01_0_0110	Transmitter lane 6 : Transceiver And Transmitter Components Address Space on page 118
01_0_0111	Transmitter lane 7 : Transceiver And Transmitter Components Address Space on page 118
01_0_1000 - 01_0_1111	Reserved
01_1_0000	Transceiver and transmitter write only to the same register in all the lanes of a link where lane 0 is the master lane : Transceiver And Transmitter Components Address Space on page 118
01_1_0001	Transceiver and transmitter write only to the same register in all the lanes of a link where lane 1 is the master lane : Transceiver And Transmitter Components Address Space on page 118
01_1_0010	Transceiver and transmitter write only to the same register in all the lanes of a link where lane 2 is the master lane : Transceiver And Transmitter Components Address Space on page 118
01_1_0011	Transceiver and transmitter write only to the same register in all the lanes of a link where lane 3 is the master lane : Transceiver And Transmitter Components Address Space on page 118
01_1_0100	Transceiver and transmitter write only to the same register in all the lanes of a link where lane 4 is the master lane : Transceiver And Transmitter Components Address Space on page 118
01_1_0101	Transceiver and transmitter write only to the same register in all the lanes of a link where lane 5 is the master lane : Transceiver And Transmitter Components Address Space on page 118
01_1_0110	Transceiver and transmitter write only to the same register in all the lanes of a link where lane 6 is the master lane : Transceiver And Transmitter Components Address Space on page 118
01_1_0111	Transceiver and transmitter write only to the same register in all the lanes of a link where lane 7 is the master lane : Transceiver And Transmitter Components Address Space on page 118
10_m_nnnn	General lane addressing : m = multi-write, nnnn = lane number. Receiver Components Address Space on page 119
10_0_0000	Receiver lane 0 : Receiver Components Address Space on page 119
10_0_0001	Receiver lane 1 : Receiver Components Address Space on page 119
10_0_0010	Receiver lane 2 : Receiver Components Address Space on page 119
10_0_0011	Receiver lane 3 : Receiver Components Address Space on page 119
10_0_0100	Receiver lane 4 : Receiver Components Address Space on page 119
10_0_0101	Receiver lane 5 : Receiver Components Address Space on page 119
10_0_0110	Receiver lane 6 : Receiver Components Address Space on page 119
10_0_0111	Receiver lane 7 : Receiver Components Address Space on page 119
10_0_1000 - 10_0_1111	Reserved
10_1_0000	Receiver write only to the same register in all the lanes of a link where lane 0 is the master lane : Receiver Components Address Space on page 119
10_1_0001	Receiver write only to the same register in all the lanes of a link where lane 1 is the master lane : Receiver Components Address Space on page 119
10_1_0010	Receiver write only to the same register in all the lanes of a link where lane 2 is the master lane : Receiver Components Address Space on page 119
10_1_0011	Receiver write only to the same register in all the lanes of a link where lane 3 is the master lane : Receiver Components Address Space on page 119
10_1_0100	Receiver write only to the same register in all the lanes of a link where lane 4 is the master lane : Receiver Components Address Space on page 119
10_1_0101	Receiver write only to the same register in all the lanes of a link where lane 5 is the master lane : Receiver Components Address Space on page 119
10_1_0110	Receiver write only to the same register in all the lanes of a link where lane 6 is the master lane : Receiver Components Address Space on page 119
10_1_0111	Receiver write only to the same register in all the lanes of a link where lane 7 is the master lane : Receiver Components Address Space on page 119
10_1_1000 - 11_1_1111	Reserved

Table 84: Macro Top Level Address Space

The following table specifies the address map for the common components. Also noted in this table are the respective power island a given region of registers are on.

Base address bits 15:9 :	00_0_0000	
Address bits 8:5	**Addressed Region**	**Power Island**
0000	Macro ID Registers Address Space on page 120	CDB register
0001	Startup State Machine Registers Address Space on page 125	Always on
0001	PLL 0 Control State Machine Registers Address Space on page 126	CDB register
0001	PLL 1 control state machine registers address space on page 128	CDB register
0010	Common control module control and diagnostic registers address space on page 130	Always on
0011	Bandgap calibration controller registers address space on page 150	CDB register
0011	External bias current calibration controller registers address space on page 153	CDB register
0100	PLL 0 controller registers address space (slave 0) on page 156	CDB register
0101	PLL 0 controller registers address space (slave 1) on page 156	CDB register
0110	PLL 1 controller registers address space (slave 0) on page 164	CDB register
0111	PLL 1 controller registers address space (slave 1) on page 164	CDB register
1000	Transmitter pull up resistor calibration controller registers address space on page 170	CDB register
1000	Transmitter pull down resistor calibration controller registers address space on page 170	CDB register
1000	Receiver resistor calibration controller registers address space on page 176	CDB register
1001	Signal detect clock calibration registers address space on page 180	CDB register
1010	Reserved	N/A
1011	Reserved	N/A
1100	Common clock measurement module registers address space on page 183	Diagnostic
1101	Common PLL 0 control and diagnostic registers address space on page 184	CDB register
1110	Common PLL 1 control and diagnostic registers address space on page 192	CDB register
1111	Common functions control and diagnostic registers address space on page 196	Always on

Table 85: Common Components Address Space

The following table specifies the address map for the transmitter components. Also noted in this table are the respective power island a given region of registers are on.

Base address bits 15:9 :	01_m_nnnn	
Address bits 8:5	**Addressed Region**	**Power Island**
0000	Powers state machine registers address space on page 202	CDB register
0001	Reserved	N/A
0010	TX coefficient calculator registers address space (slave 0) on page 210	CDB register
0011	TX coefficient calculator registers address space (slave 1) on page 211	CDB register
0100	Reserved	N/A
0101	Reserved	N/A
0110	Driver controller control and diagnostic registers address space on page 244	CDB register
0111	Transceiver control and diagnostic registers address space on page 233	CDB register
1000	Transmitter power state controller registers address space on page 247	CDB register
1001	Receiver detect registers address space on page 251	Always on
1010	Transmitter BIST controller registers address space on page 252	Diagnostic
1011	Reserved	N/A
1100	Reserved	N/A
1101	Reserved	N/A
1110	Reserved	N/A
1111	Transmitter control and diagnostics registers address space on page 254	Always on

Table 86: Transceiver And Transmitter Components Address Space

The following table specifies the address map for the receiver components. Also noted in this table are the respective power island a given region of registers are on.

Base address bits 15:9 :	10_m_nnnn	
Address bits 8:5	**Addressed Region**	**Power Island**
0000	Receiver power state controller registers address space on page 256 (Not present in TX only lanes)	CDB register
0001	Reserved	N/A
0001	Reserved	N/A
0010	Signal detect calibration 0 registers address space on page 260 (Not present in TX only lanes)	CDB register
0010	Signal detect calibration 1 registers address space on page 263 (Not present in TX only lanes)	CDB register
0010	Sampler error DAC control registers address space on page 265 (Not present in TX only lanes)	CDB register
0011	Receiver sampler latch calibration registers address space on page 266 (Not present in TX only lanes)	CDB register
0100	CDRLF configuration registers address space on page 321 (Not present in TX only lanes)	CDB register
0100	Receiver signal detect filter registers address space on page 327 (Not present in TX only lanes)	Always on
0100	Receiver LFPS detect filter registers address space on page 329 (Not present in TX only lanes)	Always on
0101	Eye surf registers address space on page 324 (Not present in TX only lanes)	Diagnostic
0101	Receiver BIST controller registers address space on page 326	Diagnostic
0110	Reserved	N/A
0111	Reserved	N/A
1000	Receiver equalizer engine registers address space (slave 0) on page 273 (Not present in TX only lanes)	CDB register
1001	Receiver equalizer engine registers address space (slave 1) on page 275 (Not present in TX only lanes)	CDB register
1010	Receiver equalizer engine registers address space (slave 2) on page 276 (Not present in TX only lanes)	CDB register
1011	Receiver equalizer engine registers address space (slave 3) on page 277 (Not present in TX only lanes)	CDB register
1100	Reserved	N/A
1101	Reserved	N/A
1110	Transceiver clock measurement module registers address space on page 330	Diagnostic
1111	Receiver control and diagnostics registers address space on page 332	Always on

Table 87: Receiver Components Address Space

10.2.1 Special Register Conventions

Depending on the PMA configuration, some registers and register fields described below are removed and replaced with R/O registers/fields (with a return value of 0). The following convention indicates whether a register or field is present or not based on the configuration:

- !: Indicates registers / fields that are not present in a PMA transmitter only lane. A read of any of these registers in a PMA transmitter only lane, will return all 0s.

The following conventions are used for the reset values of specific register fields.

- n’hX: Where n is an integer value. In these cases, the reset values are not necessarily known during reset. An example of this would be cases where the register field is a signal driven from the analog.
- n’hM: Where n is an integer value. This notation is used for some of the macro ID register fields, where the information varies between releases. An example of this is the metal stack register fields. In these cases, a given release my have multiple metal stack options available. In these scenarios, this notation will be used, and the possible values will be provided in the register field description.

10.2.2 Macro ID Registers

Base address bits 15:5 :	00_0_0000 : 0000
Address bits 4:0	**Register name and location of detailed description**
00000	Product type ID register on page 120
00001	Reserved
00010	Reserved
00011	Product number ID register on page 120
00100	Product revision ID register on page 120
00101 - 00111	Reserved
01000	Product technology manufacturer ID register on page 121
01001	Product technology process node ID register on page 121
01010	Product technology process flavor ID register 0 on page 121
01011	Product technology process flavor ID register 1 on page 121
01100	Product I/O voltage ID register on page 121
01101	Product SerDes lanes ID register on page 122
01110 - 01111	Reserved
10000	Product metal layers ID register 0 on page 122
10001	Product metal layers ID register 1 on page 122
10010	Product metal layers ID register 2 on page 123
10011	Product metal layers ID register 3 on page 123
10100	Product metal layer direction ID register on page 124
10100 - 11111	Reserved

Table 88: Macro ID Registers Address Space

Register Name:			Product type ID register
Register Mnemonic:			CMN_PID_TYPE
Register Address:			00_0_0000 : 0000 : 00000
Bits	**Type**	**Reset**	**Description**
15:0	R/O	16’h7364	**Product type** : This field contains the ASCII codes that represent the product type “sd” for SerDes.

Register Name:			Product number ID register
Register Mnemonic:			CMN_PID_NUM
Register Address:			00_0_0000 : 0000 : 00011
Bits	**Type**	**Reset**	**Description**
15:0	R/O	16’hM	**Product number** : This field contains the binary coded decimal numbers that represent the product number. The following are the possible reset values for this field. • 16’h0801: SD0801 • 16’h0802: SD0802 • 16’h0803: SD0803 • 16’h0804: SD0804 • 16’h0805: SD0805

Register Name:			Product revision ID register
Register Mnemonic:			CMN_PID_REV
Register Address:			00_0_0000 : 0000 : 00100
Bits	**Type**	**Reset**	**Description**
15:0	R/O	16’h0130	**Product revision** : This field contains the binary coded decimal numbers that represent the product revision.

Register Name:			Product technology manufacturer ID register
Register Mnemonic:			CMN_PID_MFG
Register Address:			00_0_0000 : 0000 : 01000
Bits	**Type**	**Reset**	**Description**
15:0	R/O	16'h0074	**Product technology manufacturer** : This field contains the ASCII codes that represent the product technology manufacturer. • TSMC: "t" •

Register Name:			Product technology process node ID register
Register Mnemonic:			CMN_PID_NODE
Register Address:			00_0_0000 : 0000 : 01001
Bits	**Type**	**Reset**	**Description**
15:0	R/O	16'h0012	**Product technology process node** : This field contains the binary coded decimal numbers that represent the product technology node

Register Name:			Product technology process flavor ID register 0
Register Mnemonic:			CMN_PID_FLV0
Register Address:			00_0_0000 : 0000 : 01010
Bits	**Type**	**Reset**	**Description**
15:0	R/O	16'h6666	**Product technology flavor** : This field contains the ASCII codes that represent the first two characters of the product technology flavor. • "ff"

Register Name:			Product technology process flavor ID register 1
Register Mnemonic:			CMN_PID_FLV1
Register Address:			00_0_0000 : 0000 : 01011
Bits	**Type**	**Reset**	**Description**
15:0	R/O	16'h6300	**Product technology flavor** : This field contains the ASCII codes that represent the second two characters of the product technology flavor. • "c"

Register Name:			Product I/O voltage ID register
Register Mnemonic:			CMN_PID_IOV
Register Address:			00_0_0000 : 0000 : 01100
Bits	**Type**	**Reset**	**Description**
15:0	R/O	16'h0120	**Product I/O voltage** : This field contains the binary coded decimal numbers that represent the product I/O voltage. The most significant byte represents the value to the left of the decimal point, and the least significant byte represents the value to the right of the decimal point. For example, 1.5V is represented as 0x0150.

Register Name:			Product SerDes lanes ID register
Register Mnemonic:			CMN_PID_LANES
Register Address:			00_0_0000 : 0000 : 01101
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'hM	**Product SerDes lanes left of common** : This field contains the binary coded decimal numbers that represent the number of lanes implemented in this SerDes product on the left side of the common module. The following are the possible reset values for this field. • 8'h04: 4 lanes on the left side of common. • 8'h03: 3 lanes on the left side of common. • 8'h02: 2 lanes on the left side of common. • 8'h01: 1 lanes on the left side of common. • 8'h00: 0 lanes on the left side of common.
7:0	R/O	8'hM	**Product SerDes lanes right of common** : This field contains the binary coded decimal numbers that represent the number of lanes implemented in this SerDes product on the right side of the common module. The following are the possible reset values for this field. • 8'h04: 4 lanes on the right side of common. • 8'h03: 3 lanes on the right side of common. • 8'h02: 2 lanes on the right side of common. • 8'h01: 1 lanes on the right side of common. • 8'h00: 0 lanes on the right side of common.

Register Name:			Product metal layers ID register 0
Register Mnemonic:			CMN_PID_METAL0
Register Address:			00_0_0000 : 0000 : 10000
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'hM	**Product xd metal layers** : This field contains the binary coded decimal number that represent the number of xd metal layers used for this product. The following are the possible reset values for this field. • 4'h1: For all metal stacks
11:8	R/O	4'hM	**Product xc metal layers** : This field contains the binary coded decimal number that represent the number of xc metal layers used for this product. The following are the possible reset values for this field. • 4'h0: For all metal stacks
7:4	R/O	4'hM	**Product xa metal layers** : This field contains the binary coded decimal number that represent the number of xa metal layers used for this product. The following are the possible reset values for this field. • 4'h2: For all metal stacks
3:0	R/O	4'hM	**Product x metal layers** : This field contains the binary coded decimal number that represent the number of x metal layers used for this product. The following are the possible reset values for this field. • 4'h0: For all metal stacks

Register Name:			Product metal layers ID register 1
Register Mnemonic:			CMN_PID_METAL1
Register Address:			00_0_0000 : 0000 : 10001
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:8	R/O	4'h0	Reserved
7:4	R/O	4'hM	**Product xy metal layers** : This field contains the binary coded decimal number that represent the number of xy metal layers used for this product. The following are the possible reset values for this field. • 4'h0: For all metal stacks

3:0	R/O	4'hM	**Product xe metal layers** : This field contains the binary coded decimal number that represent the number of xe metal layers used for this product. The following are the possible reset values for this field.
			• 4'h3: For the following metal stacks:
			  • 2xa1xdh3xevhv2z
			  • 2xa1xdh3xevhv2y2r
			  • 2xa1xdh3xevhv2y2yy2r
			• 4'h4: For all other metal stacks

Register Name:			Product metal layers ID register 2
Register Mnemonic:			CMN_PID_METAL2
Register Address:			00_0_0000 : 0000 : 10010
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'hM	**Product yb metal layers** : This field contains the binary coded decimal number that represent the number of yb metal layers used for this product. The following are the possible reset values for this field. • 4'h0: For all metal stacks
11:8	R/O	4'hM	**Product yy metal layers** : This field contains the binary coded decimal number that represent the number of yy metal layers used for this product. The following are the possible reset values for this field. • 4'h2: For the following metal stacks: • 2xa1xdh3xevhv2y2yy2r • 4'h0: For all other metal stacks
7:4	R/O	4'hM	**Product ya metal layers** : This field contains the binary coded decimal number that represent the number of ya metal layers used for this product. The following are the possible reset values for this field. • 4'h0: For all metal stacks

3:0	R/O	4'hM	**Product y metal layers** : This field contains the binary coded decimal number that represent the number of y metal layers used for this product. The following are the possible reset values for this field.
			• 4'h2: For the following metal stacks:
			  • 2xa1xdh3xevhv2y2r
			  • 2xa1xdh3xevhv2y2yy2r
			• 4'h0: For all other metal stacks

Register Name:			Product metal layers ID register 3
Register Mnemonic:			CMN_PID_METAL3
Register Address:			00_0_0000 : 0000 : 10011
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'hM	**Product yx metal layers** : This field contains the binary coded decimal number that represent the number of yx metal layers used for this product. The following are the possible reset values for this field. • 4'h0: For all metal stacks
11:8	R/O	4'hM	**Product u metal layers** : This field contains the binary coded decimal number that represent the number of u metal layers used for this product. The following are the possible reset values for this field. • 4'h0: For all metal stacks
7:4	R/O	4'hM	**Product r metal layers** : This field contains the binary coded decimal number that represent the number of r metal layers used for this product. The following are the possible reset values for this field. • 4'h0: For the following metal stacks: • 2xa1xdh3xevhv2z • 4'h2: For all other metal stacks
3:0	R/O	4'hM	**Product z metal layers** : This field contains the binary coded decimal number that represent the number of z metal layers used for this product. The following are the possible reset values for this field. • 4'h2: For the following metal stacks: • 2xa1xdh3xevhv2z • 4'h0: For all other metal stacks

Register Name:			Product metal layer direction ID register
Register Mnemonic:			CMN_PID_METALD
Register Address:			00_0_0000 : 0000 : 10100
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14	R/O	1'h0	Reserved
13	R/O	1'h0	Reserved
12	R/O	1'hM	Metal 12 direction (when used) : 1'b0 = Vertical, 1'b1 = Horizontal The following are the possible reset values for this field. • 1'h1: For the following metal stacks: • 2xa1xdh3xevhv2y2yy2r • 1'h0: For all other metal stacks:
11	R/O	1'h0	Metal 11 direction (when used) : 1'b0 = Vertical, 1'b1 = Horizontal
10	R/O	1'hM	Metal 10 direction (when used) : 1'b0 = Vertical, 1'b1 = Horizontal. The following are the possible reset values for this field. • 1'h0: For the following metal stacks: • 2xa1xdh3xevhv2z • 2xa1xdh4xevhvh2r (Note that this bit is incorrect for this metal layer. It should be a 1'b1, indicating a horizontal layer) • 1'h1: For the following metal stacks: • 2xa1xdh3xevhv2y2r • 2xa1xdh3xevhv2y2yy2r
9	R/O	1'h0	Metal 9 direction (when used) : 1'b0 = Vertical, 1'b1 = Horizontal
8	R/O	1'h1	Metal 8 direction : 1'b0 = Vertical, 1'b1 = Horizontal
7	R/O	1'h0	Metal 7 direction : 1'b0 = Vertical, 1'b1 = Horizontal
6	R/O	1'h1	Metal 6 direction : 1'b0 = Vertical, 1'b1 = Horizontal
5	R/O	1'h0	Metal 5 direction : 1'b0 = Vertical, 1'b1 = Horizontal
4	R/O	1'h1	Metal 4 direction : 1'b0 = Vertical, 1'b1 = Horizontal
3	R/O	1'h0	Metal 3 direction : 1'b0 = Vertical, 1'b1 = Horizontal
2	R/O	1'h1	Metal 2 direction : 1'b0 = Vertical, 1'b1 = Horizontal
1	R/O	1'h0	Metal 1 direction : 1'b0 = Vertical, 1'b1 = Horizontal
0	R/O	1'h1	Metal 0 direction : This layer does not have a direction associated with it.

10.2.3 Startup State Machine Registers

Base address bits 15:5 :	00_0_0000 : 0001
Address bits 4:0	**Register name and location of detailed description**
00000	Startup state machine control register on page 125
00001	Bandgap enable timer register on page 125
00010	Bias enable timer register on page 125
00011 - 00110	Reserved
00111	Startup state machine user defined control register on page 126

Table 89: Startup State Machine Registers Address Space

Register Name:			Startup state machine control register
Register Mnemonic:			CMN_SSM_SM_CTRL
Register Address:			00_0_0000 : 0001 : 00000
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7	R/W	1'h0	**Bandgap enable override enable** : When active (1'b1), the bandgap enable override bit in this register will drive the ssmda_bandgap_en pin from the SSM directly.
6	R/W	1'h0	**Bandgap enable override** : When enabled by the bandgap enable override enable bit in this register, this bit will drive the ssmda_bandgap_en pin from the SSM directly. **Note: The bandgap enable hold enable bit in the Startup state machine user defined control register on page 126 must also be set to 1'b0 in order for this register to function correctly.**
5	R/W	1'h0	**Bias enable override enable** : When active (1'b1), the bias enable override bit in this register will drive the ssmda_bias_en pin from the SSM directly.
4	R/W	1'h0	**Bias enable override** : When enabled by the bias enable override enable bit in this register, this bit will drive the ssmda_bias_en pin from the SSM directly.
3:2	R/O	2'h0	Reserved
1	R/W	1'h1	**Skip post bandgap enable re-calibration** : When this bit is active (1'b1), the post bandgap enable calibration state will be skipped if it was previously run, unless the macro is disabled or reset.
0	R/W	1'h1	**Skip auto re-calibration** : When this bit is active (1'b1), the auto calibration state will be skipped if it was previously run, unless the macro is disabled or reset.

Register Name:			Bandgap enable timer register
Register Mnemonic:			CMN_SSM_BANDGAP_TMR
Register Address:			00_0_0000 : 0001 : 00001
Bits	**Type**	**Reset**	**Description**
15:5	R/O	11'h0	Reserved
4:0	R/W	5'h01	**Bandgap enable state timer value** : Value used for the timer when the startup state machine is in the bandgap enable state. This timer delay is specified as the number of reference clocks to count. This value creates a delay of 40 nSec. Note, when exiting common suspend mode, a total of 1000 nSec is required for the bandgap to be enabled. This total delay is what is ultimately used to determine what the delay needs to be in the bandgap enable state. Of this total time, 960 nSec of this time is accounted for in the suspend recovery state. The remainder is accounted for here.

Register Name:			Bias enable timer register
Register Mnemonic:			CMN_SSM_BIAS_TMR
Register Address:			00_0_0000 : 0001 : 00010
Bits	**Type**	**Reset**	**Description**
15:7	R/O	9'h0	Reserved
6:0	R/W	7'h19	**Bias enable state timer value** : Value used for the timer when the startup state machine is in the bias enable state. This timer delay is specified as the number of reference clocks to count. This value creates a delay of 1 uSec.

Register Name:			Startup state machine user defined control register
Register Mnemonic:			CMN_SSM_USER_DEF_CTRL
Register Address:			00_0_0000 : 0001 : 00111
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:2	R/W	6'h00	Reserved - spare
1	R/W	1'h0	**Force SSM gated clock on**: Setting this bit to 1'b1 will force the SSM gated clock on, independent of the internal SSM state machine clock gate controls.
0	R/W	1'h1	**Bandgap enable hold enable**: This bit enables the bandgap enable hold function, which holds the bandgap enable active in the common suspend state until the signal detect function is switched off.

10.2.4 PLL 0 Control State Machine Registers

Base address bits 15:5 :	00_0_0000 : 0001
Address bits 4:0	**Register name and location of detailed description**
01000	PLL 0 control state machine control register on page 126
01001	PLL 0 enable timer register on page 126
01010	PLL 0 pre-charge timer register on page 127
01011	PLL 0 VREF delay timer register on page 127
01100	PLL 0 lock delay timer register on page 127
01101	PLL 0 clock disable delay timer register on page 127
01110	Reserved
01111	PLL 0 control state machine user defined control register on page 127

Table 90: PLL 0 Control State Machine Registers Address Space

Register Name:			PLL 0 control state machine control register
Register Mnemonic:			CMN_PLLSM0_SM_CTRL
Register Address:			00_0_0000 : 0001 : 01000
Bits	**Type**	**Reset**	**Description**
15:10	R/O	6'h0	Reserved
9	R/W	1'h0	**PLL enable override enable** : When active (1'b1), the PLL enable override bit in this register will drive the pllsmda_pll_en pin from the PLLSM directly.
8	R/W	1'h0	**PLL enable override** : When enabled by the PLL enable override enable bit in this register, this bit will drive the pllsmda_pll_en pin from the PLLSM directly.
7	R/W	1'h0	**PLL reset override enable** : When active (1'b1), the PLL reset override bit in this register will drive the pllsmda_pll_rst_n pin from the PLLSM directly.
6	R/W	1'h0	**PLL reset override** : When enabled by the PLL reset override enable bit in this register, this bit will drive the pllsmda_pll_rst_n pin from the PLLSM directly.
5	R/W	1'h0	**PLL pre charge override enable** : When active (1'b1), the PLL pre charge override bit in this register will drive the pllsmda_pll_pre_charge pin from the PLLSM directly.
4	R/W	1'h0	**PLL pre charge override** : When enabled by the PLL pre charge override enable bit in this register, this bit will drive the pllsmda_pll_pre_charge pin from the PLLSM directly.
3:1	R/O	3'h0	Reserved
0	R/W	1'h1	**Skip PLL re-calibration** : When this bit is active (1'b1), the PLL calibration state will be skipped if it was previously run, unless the PLL is disabled or resetting the state machine.

Register Name:			PLL 0 enable timer register
Register Mnemonic:			CMN_PLLSM0_PLLEN_TMR
Register Address:			00_0_0000 : 0001 : 01001
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12'h0	Reserved
3:0	R/W	4'h4	**PLL enable state timer value** : Value used for the timer when the startup state machine is in the PLL enable state. This timer delay is specified as the number of reference clocks to count.

Register Name:			PLL 0 pre-charge timer register
Register Mnemonic:			CMN_PLLSM0_PLLPRE_TMR
Register Address:			00_0_0000 : 0001 : 01010
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h32	**PLL pre-charge state timer value** : Value used for the timer when the startup state machine is in the PLL pre-charge state. This timer delay is specified as the number of reference clocks to count. This value creates a delay of 2 uSec.

Register Name:			PLL 0 VREF delay timer register
Register Mnemonic:			CMN_PLLSM0_PLLVREF_TMR
Register Address:			00_0_0000 : 0001 : 01011
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12'h0	Reserved
3:0	R/W	4'h1	**PLL VREF delay state timer value** : Value used for the timer when the startup state machine is in the PLL VREF delay state. This timer delay is specified as the number of reference clocks to count.

Register Name:			PLL 0 lock delay timer register
Register Mnemonic:			CMN_PLLSM0_PLLLOCK_TMR
Register Address:			00_0_0000 : 0001 : 01100
Bits	**Type**	**Reset**	**Description**
15:10	R/O	6'h0	Reserved
9:0	R/W	10'h0D1	**PLL lock delay state timer value** : Value used for the timer when the startup state machine is in the PLL lock delay state. This timer delay is specified as the number of reference clocks to count.

Register Name:			PLL 0 clock disable delay timer register
Register Mnemonic:			CMN_PLLSM0_PLLCLKDIS_TMR
Register Address:			00_0_0000 : 0001 : 01101
Bits	**Type**	**Reset**	**Description**
15:2	R/O	14'h0	Reserved
1:0	R/W	2'h1	**PLL clock disable delay state timer value** : Value used for the timer when the startup state machine is in the PLL clock disable delay state. This timer delay is specified as the number of reference clocks to count.

Register Name:			PLL 0 control state machine user defined control register
Register Mnemonic:			CMN_PLLSM0_USER_DEF_CTRL
Register Address:			00_0_0000 : 0001 : 01111
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:1	R/W	7'h0	Reserved - spare
0	R/W	1'h0	**PLL lock override**: When active (1'b1), this bit will force the PLL lock indication active.

10.2.5 PLL 1 Control State Machine Registers

Base address bits 15:5 :	00_0_0000 : 0001
Address bits 4:0	**Register name and location of detailed description**
10000	PLL 1 control state machine control register on page 128
10001	PLL 1 enable timer register on page 128
10010	PLL 1 pre-charge timer register on page 128
10011	PLL 1 VREF delay timer register on page 129
10100	PLL 1 lock delay timer register on page 129
10101	PLL 1 clock disable delay timer register on page 129
10110	Reserved
10111	PLL 1 control state machine user defined control register on page 129

Table 91: PLL 1 control state machine registers address space

Register Name:			PLL 1 control state machine control register
Register Mnemonic:			CMN_PLLSM1_SM_CTRL
Register Address:			00_0_0000 : 0001 : 10000
Bits	**Type**	**Reset**	**Description**
15:10	R/O	6'h0	Reserved
9	R/W	1'h0	**PLL enable override enable** : When active (1'b1), the PLL enable override bit in this register will drive the pllsmda_pll_en pin from the PLLSM directly.
8	R/W	1'h0	**PLL enable override** : When enabled by the PLL enable override enable bit in this register, this bit will drive the pllsmda_pll_en pin from the PLLSM directly.
7	R/W	1'h0	**PLL reset override enable** : When active (1'b1), the PLL reset override bit in this register will drive the pllsmda_pll_rst_n pin from the PLLSM directly.
6	R/W	1'h0	**PLL reset override** : When enabled by the PLL reset override enable bit in this register, this bit will drive the pllsmda_pll_rst_n pin from the PLLSM directly.
5	R/W	1'h0	**PLL pre charge override enable** : When active (1'b1), the PLL pre charge override bit in this register will drive the pllsmda_pll_pre_charge pin from the PLLSM directly.
4	R/W	1'h0	**PLL pre charge override** : When enabled by the PLL pre charge override enable bit in this register, this bit will drive the pllsmda_pll_pre_charge pin from the PLLSM directly.
3:1	R/O	3'h0	Reserved
0	R/W	1'h1	**Skip PLL re-calibration** : When this bit is active (1'b1), the PLL calibration state will be skipped if it was previously run, unless the PLL is disabled or resetting the state machine.

Register Name:			PLL 1 enable timer register
Register Mnemonic:			CMN_PLLSM1_PLLEN_TMR
Register Address:			00_0_0000 : 0001 : 10001
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12'h0	Reserved
3:0	R/W	4'h4	**PLL enable state timer value** : Value used for the timer when the startup state machine is in the PLL enable state. This timer delay is specified as the number of reference clocks to count.

Register Name:			PLL 1 pre-charge timer register
Register Mnemonic:			CMN_PLLSM1_PLLPRE_TMR
Register Address:			00_0_0000 : 0001 : 10010
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h32	**PLL pre-charge state timer value** : Value used for the timer when the startup state machine is in the PLL pre-charge state. This timer delay is specified as the number of reference clocks to count.

Register Name:			PLL 1 VREF delay timer register
Register Mnemonic:			CMN_PLLSM1_PLLVREF_TMR
Register Address:			00_0_0000 : 0001 : 10011
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12'h0	Reserved
3:0	R/W	4'h1	**PLL VREF delay state timer value** : Value used for the timer when the startup state machine is in the PLL VREF delay state. This timer delay is specified as the number of reference clocks to count.

Register Name:			PLL 1 lock delay timer register
Register Mnemonic:			CMN_PLLSM1_PLLLOCK_TMR
Register Address:			00_0_0000 : 0001 : 10100
Bits	**Type**	**Reset**	**Description**
15:10	R/O	6'h0	Reserved
9:0	R/W	10'h0D1	**PLL lock delay state timer value** : Value used for the timer when the startup state machine is in the PLL lock delay state. This timer delay is specified as the number of reference clocks to count.

Register Name:			PLL 1 clock disable delay timer register
Register Mnemonic:			CMN_PLLSM1_PLLCLKDIS_TMR
Register Address:			00_0_0000 : 0001 : 10101
Bits	**Type**	**Reset**	**Description**
15:2	R/O	14'h0	Reserved
1:0	R/W	2'h1	**PLL clock disable delay state timer value** : Value used for the timer when the startup state machine is in the PLL clock disable delay state. This timer delay is specified as the number of reference clocks to count.

Register Name:			PLL 1 control state machine user defined control register
Register Mnemonic:			CMN_PLLSM1_USER_DEF_CTRL
Register Address:			00_0_0000 : 0001 : 10111
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:1	R/W	7'h0	Reserved - spare
0	R/W	1'h0	**PLL lock override**: When active (1'b1), this bit will force the PLL lock indication active.

10.2.6 Common Control Module Control And Diagnostics Registers

Base address bits 15:5 :	00_0_0000 : 0010
Address bits 4:0	**Register name and location of detailed description**
00000	Common power island control timer register on page 130
00001	Common CDB power island control override register on page 131
00010	Common CDB power island control status register on page 133
00011	Common PLL controller power island control override register on page 135
00100	Common PLL controller power island control status register on page 136
00101	Common common calibration power island control override register on page 138
00110	Common common calibration power island control status register on page 139
00111	Common transceiver controller power island control override register on page 141
01000	Common transceiver controller power island control status register on page 142
01001	Common diagnostic power island control override register on page 144
01010	Common diagnostic power island control status register on page 145
01011	PLL full rate and data rate clock control register on page 146
01100	Reference clock receiver override register on page 146
01101	Reference clock receiver test register on page 147
01110	Power state machine clock receiver control register on page 147
01111	Signal detect clock oscillator control register on page 147
10000	Reference clock bump driver 0 control register on page 148
10001 - 11011	Reserved
11100	Common control CDB diagnostic register on page 148
11101	Common control reset diagnostic register on page 148
11110	Reserved
11111	Common control cover your alternatives register on page 149

Table 92: Common control module control and diagnostic registers address space

Register Name:			Common power island control timer register
Register Mnemonic:			CMN_CDIAG_PWRI_TMR
Register Address:			00_0_0000 : 0010 : 00000
Bits	**Type**	**Reset**	**Description**
15:11	R/O	5’h0	Reserved
10:8	R/W	3’h1	**Power enable phase 2 timer value**: This specifies the number of reference clock cycles the power island control state machines in common will wait in the power phase 2 enable states, in order to allow enough time for the second phase of the switched domain to power up, before deactivating the isolation functions. The default value of this register corresponds to a delay of 40 nSec. Note: A value of 0 in this field is not valid, and will result in a delay of 1 clock cycle.
7:3	R/O	5’h0	Reserved
2:0	R/W	3’h1	**Power enable phase 1 timer value**: This specifies the number of reference clock cycles the power island control state machines in common will wait in the power phase 1 enable states, in order to allow enough time for the first phase of the switched domain to power up, before enabling the second phase of the power up. The default value of this register corresponds to a delay of 40 nSec. Note: A value of 0 in this field is not valid, and will result in a delay of 1 clock cycle.

Register Name:			Common CDB power island control override register **Note: This register must not be part of any register read write test where patterns other than those described in this register are written to this register.**
Register Mnemonic:			CMN_CDIAG_CDB_PWRI_OVRD
Register Address:			00_0_0000 : 0010 : 00001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Power island controller input override enable**: When enabled, the power island control state machine input override bits in this register will drive the power island control state machine directly, and override any control from other state machines in the macro. **Note: The CDB power island must be triggered to be enabled, by accessing a register outside of the common control module control and diagnostic registers space, prior to enabling this function.** **Note: This bit must never change state in the same register write that any of the override bits it controls in this register are changed.** • 1'b0: Override disabled • 1'b1: Override enabled
14	R/W	1'h0	**Power island controller output override enable**: When enabled, the power island control state machine output override bits in this register will drive the power island control state machine outputs, and override any control from power island control state machine. **Note: This bit must never change state in the same register write that any of the override bits it controls in this register are changed.** • 1'b0: Override disabled • 1'b1: Override enabled
13:12	R/O	2'h0	Reserved
11	R/W	1'h0	**Power suspend request override**: When enabled, this bit will override the power_suspend_req input of the power island control state machine. **Note: Whenever changing the state of this bit, no other CDB register access should be initiated for 1 uSec, to keep the CDB clock off while the power island state change transition is in progress.**
10	R/O	1'h0	**Power suspend acknowledge**: This is the current state of the power_suspend_ack output from the power island control state machine. **Note: This bit will always be set to 1'b1 after the power island is enabled, as a function of the power island controller input override enable bit in this register.**
9	R/W	1'h0	**Power recover request override**: When enabled, this bit will override the power_recover_req input of the power island control state machine. **Note: Whenever changing the state of this bit, no other CDB register access should be initiated for 1 uSec, to keep the CDB clock off while the power island state change transition is in progress.**
8	R/O	1'h1	**Power recover acknowledge**: This is the current state of the power_recover_ack output from the power island control state machine. **Note: This bit will always be set to 1'b1 after the power island is enabled, as a function of the power island controller input override enable bit in this register.**

Register Name:			Common CDB power island control override register **Note: This register must not be part of any register read write test where patterns other than those described in this register are written to this register.**
Register Mnemonic:			CMN_CDIAG_CDB_PWRI_OVRD
Register Address:			00_0_0000 : 0010 : 00001
Bits	**Type**	**Reset**	**Description**
7:0	R/W	8’h24	**Power island controller output override:** When enabled, the bits in this field will override the output signals from the power island control state machine. The following are the power island control state machine output signals, and the bits in this field that they are associated with. • Bit 7: state_sandh • Bit 6: state_ret_save • Bit 5: state_ret_restore (active low) • Bit 4: power_en_ph_1 • Bit 3: power_en_ph_2 • Bit 2: power_isolation • Bit 1: power_por_reset_n • Bit 0: power_rstr_reset_n The default value of the bits in this register are such that the power island is not forced to be switched on when the power island controller output override enable bit in this register is enabled. When using this register to switch this power island on and off, the bits in this field are required to change in a specific order with individual register writes. The writes must happen in the order specified below. • 8’b00100100 : Power island is not being forced on • 8’b10100100 • 8’b10110100 • 8’b10111100 • 8’b10111111 • 8’b10111011 • 8’b00111011 : Power island is being forced on • 8’b10111011 • 8’b10111111 • 8’b11111111 • 8’b10111111 • 8’b10111110 • 8’b10100110 : Power island is being forced off, and in state retention • 8’b10110110 • 8’b10111110 • 8’b10111111 • 8’b10011111 • 8’b10111111 • 8’b10111011 • 8’b00111011 : Power island is being forced on, with state restored **Note: When using this output override option in conditions when the reference clock is off, the CDB registers on the power island will only be readable, and latency will be much longer. The reason for this is the reference clock is normally used to synchronize control functions associated with this power island. Therefore these control functions will not proceed as they would in normal operation. This results in writes not being capable, and reads completing under the control of the watchdog timer.**

Register Name:			Common CDB power island control status register
Register Mnemonic:			CMN_CDIAG_CDB_PWRI_STAT
Register Address:			00_0_0000 : 0010 : 00010
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/O	8'h3B	**Power island controller output status:** This field indicates the current state of the output signals from the power island control state machine. The following are the power island control state machine output signals, and the bits in this field that they are associated with. • Bit 7: state_sandh • Bit 6: state_ret_save • Bit 5: state_ret_restore (active low) • Bit 4: power_en_ph_1 • Bit 3: power_en_ph_2 • Bit 2: power_isolation • Bit 1: power_por_reset_n • Bit 0: power_rstr_reset_n

Register Name:			Common PLL controller power island control override register **Note: This register must not be part of any register read write test where patterns other than those described in this register are written to this register.**
Register Mnemonic:			CMN_CDIAG_PLLC_PWRI_OVRD
Register Address:			00_0_0000 : 0010 : 00011
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Power island controller input override enable**: When enabled, the power island control state machine input override bits in this register will drive the power island control state machine directly, and override any control from other state machines in the macro. **Note: The CDB power island must be triggered to be enabled, by accessing a register outside of the common control module control and diagnostic registers space, prior to enabling this function.** **Note: This bit must never change state in the same register write that any of the override bits it controls in this register are changed.** • 1'b0: Override disabled • 1'b1: Override enabled
14	R/W	1'h0	**Power island controller output override enable**: When enabled, the power island control state machine output override bits in this register will drive the power island control state machine outputs, and override any control from power island control state machine. **Note: This bit must never change state in the same register write that any of the override bits it controls in this register are changed.** • 1'b0: Override disabled • 1'b1: Override enabled
13:12	R/O	2'h0	Reserved
11	R/W	1'h0	**Power suspend request override**: When enabled, this bit will override the power_suspend_req input of the power island control state machine.
10	R/O	1'h1	**Power suspend acknowledge**: This is the current state of the power_suspend_ack output from the power island control state machine. **Note: This bit will always be set to 1'b1 after the power island is enabled, as a function of the power island controller input override enable bit in this register.**
9	R/W	1'h0	**Power recover request override**: When enabled, this bit will override the power_recover_req input of the power island control state machine.
8	R/O	1'h0	**Power recover acknowledge**: This is the current state of the power_recover_ack output from the power island control state machine. **Note: This bit will always be set to 1'b1 after the power island is enabled, as a function of the power island controller input override enable bit in this register.**

Register Name:			Common PLL controller power island control override register **Note: This register must not be part of any register read write test where patterns other than those described in this register are written to this register.**
Register Mnemonic:			CMN_CDIAG_PLLC_PWRI_OVRD
Register Address:			00_0_0000 : 0010 : 00011
Bits	**Type**	**Reset**	**Description**
7:0	R/W	8'h24	**Power island controller output override:** When enabled, the bits in this field will override the output signals from the power island control state machine. The following are the power island control state machine output signals, and the bits in this field that they are associated with. • Bit 7: state_sandh • Bit 6: state_ret_save • Bit 5: state_ret_restore (active low) • Bit 4: power_en_ph_1 • Bit 3: power_en_ph_2 • Bit 2: power_isolation • Bit 1: power_por_reset_n • Bit 0: power_rstr_reset_n The default value of the bits in this register are such that the power island is not forced to be switched on when the power island controller output override enable bit in this register is enabled. When using this register to switch this power island on and off, the bits in this field are required to change in a specific order with individual register writes. The writes must happen in the order specified below. • 8'b00100100 : Power island is not being forced on • 8'b10100100 • 8'b10110100 • 8'b10111100 • 8'b10111111 • 8'b10111011 • 8'b00111011 : Power island is being forced on • 8'b10111011 • 8'b10111111 • 8'b11111111 • 8'b10111111 • 8'b10111110 • 8'b10100110 : Power island is being forced off, and in state retention • 8'b10110110 • 8'b10111110 • 8'b10111111 • 8'b10011111 • 8'b10111111 • 8'b10111011 • 8'b00111011 : Power island is being forced on, with state restored •

Register Name:			Common PLL controller power island control status register
Register Mnemonic:			CMN_CDIAG_PLLC_PWRI_STAT
Register Address:			00_0_0000 : 0010 : 00100
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/O	8'h24	**Power island controller output status:** This field indicates the current state of the output signals from the power island control state machine. The following are the power island control state machine output signals, and the bits in this field that they are associated with. • Bit 7: state_sandh • Bit 6: state_ret_save • Bit 5: state_ret_restore (active low) • Bit 4: power_en_ph_1 • Bit 3: power_en_ph_2 • Bit 2: power_isolation • Bit 1: power_por_reset_n • Bit 0: power_rstr_reset_n

Register Name:			Common common calibration power island control override register **Note: This register must not be part of any register read write test where patterns other than those described in this register are written to this register.**
Register Mnemonic:			CMN_CDIAG_CCAL_PWRI_OVRD
Register Address:			00_0_0000 : 0010 : 00101
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Power island controller input override enable**: When enabled, the power island control state machine input override bits in this register will drive the power island control state machine directly, and override any control from other state machines in the macro. **Note: The CDB power island must be triggered to be enabled, by accessing a register outside of the common control module control and diagnostic registers space, prior to enabling this function.** **Note: This bit must never change state in the same register write that any of the override bits it controls in this register are changed.** • 1'b0: Override disabled • 1'b1: Override enabled
14	R/W	1'h0	**Power island controller output override enable**: When enabled, the power island control state machine output override bits in this register will drive the power island control state machine outputs, and override any control from power island control state machine. **Note: This bit must never change state in the same register write that any of the override bits it controls in this register are changed.** • 1'b0: Override disabled • 1'b1: Override enabled
13:12	R/O	2'h0	Reserved
11	R/W	1'h0	**Power suspend request override**: When enabled, this bit will override the power_suspend_req input of the power island control state machine.
10	R/O	1'h1	**Power suspend acknowledge**: This is the current state of the power_suspend_ack output from the power island control state machine. **Note: This bit will always be set to 1'b1 after the power island is enabled, as a function of the power island controller input override enable bit in this register.**
9	R/W	1'h0	**Power recover request override**: When enabled, this bit will override the power_recover_req input of the power island control state machine.
8	R/O	1'h0	**Power recover acknowledge**: This is the current state of the power_recover_ack output from the power island control state machine. **Note: This bit will always be set to 1'b1 after the power island is enabled, as a function of the power island controller input override enable bit in this register.**

Register Name:			Common common calibration power island control override register **Note: This register must not be part of any register read write test where patterns other than those described in this register are written to this register.**
Register Mnemonic:			CMN_CDIAG_CCAL_PWRI_OVRD
Register Address:			00_0_0000 : 0010 : 00101
Bits	**Type**	**Reset**	**Description**
7:0	R/W	8'h24	**Power island controller output override:** When enabled, the bits in this field will override the output signals from the power island control state machine. The following are the power island control state machine output signals, and the bits in this field that they are associated with. • Bit 7: state_sandh • Bit 6: state_ret_save • Bit 5: state_ret_restore (active low) • Bit 4: power_en_ph_1 • Bit 3: power_en_ph_2 • Bit 2: power_isolation • Bit 1: power_por_reset_n • Bit 0: power_rstr_reset_n The default value of the bits in this register are such that the power island is not forced to be switched on when the power island controller output override enable bit in this register is enabled. When using this register to switch this power island on and off, the bits in this field are required to change in a specific order with individual register writes. The writes must happen in the order specified below. • 8'b00100100 : Power island is not being forced on • 8'b10100100 • 8'b10110100 • 8'b10111100 • 8'b10111111 • 8'b10111011 • 8'b00111011 : Power island is being forced on • 8'b10111011 • 8'b10111111 • 8'b11111111 • 8'b10111111 • 8'b10111110 • 8'b10100110 : Power island is being forced off, and in state retention • 8'b10110110 • 8'b10111110 • 8'b10111111 • 8'b10011111 • 8'b10111111 • 8'b10111011 • 8'b00111011 : Power island is being forced on, with state restored •

Register Name:			Common common calibration power island control status register
Register Mnemonic:			CMN_CDIAG_CCAL_PWRI_STAT
Register Address:			00_0_0000 : 0010 : 00110
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/O	8'h24	**Power island controller output status:** This field indicates the current state of the output signals from the power island control state machine. The following are the power island control state machine output signals, and the bits in this field that they are associated with. • Bit 7: state_sandh • Bit 6: state_ret_save • Bit 5: state_ret_restore (active low) • Bit 4: power_en_ph_1 • Bit 3: power_en_ph_2 • Bit 2: power_isolation • Bit 1: power_por_reset_n • Bit 0: power_rstr_reset_n

Register Name:			Common transceiver controller power island control override register **Note: This register must not be part of any register read write test where patterns other than those described in this register are written to this register.**
Register Mnemonic:			CMN_CDIAG_XCVRC_PWRI_OVRD
Register Address:			00_0_0000 : 0010 : 00111
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Power island controller input override enable**: When enabled, the power island control state machine input override bits in this register will drive the power island control state machine directly, and override any control from other state machines in the macro. **Note: The CDB power island must be triggered to be enabled, by accessing a register outside of the common control module control and diagnostic registers space, prior to enabling this function.** **Note: This bit must never change state in the same register write that any of the override bits it controls in this register are changed.** • 1'b0: Override disabled • 1'b1: Override enabled
14	R/W	1'h0	**Power island controller output override enable**: When enabled, the power island control state machine output override bits in this register will drive the power island control state machine outputs, and override any control from power island control state machine. **Note: This bit must never change state in the same register write that any of the override bits it controls in this register are changed.** • 1'b0: Override disabled • 1'b1: Override enabled
13:12	R/O	2'h0	Reserved
11	R/W	1'h0	**Power suspend request override**: When enabled, this bit will override the power_suspend_req input of the power island control state machine.
10	R/O	1'h1	**Power suspend acknowledge**: This is the current state of the power_suspend_ack output from the power island control state machine. **Note: This bit will always be set to 1'b1 after the power island is enabled, as a function of the power island controller input override enable bit in this register.**
9	R/W	1'h0	**Power recover request override**: When enabled, this bit will override the power_recover_req input of the power island control state machine.
8	R/O	1'h0	**Power recover acknowledge**: This is the current state of the power_recover_ack output from the power island control state machine. **Note: This bit will always be set to 1'b1 after the power island is enabled, as a function of the power island controller input override enable bit in this register.**

Register Name:			Common transceiver controller power island control override register **Note: This register must not be part of any register read write test where patterns other than those described in this register are written to this register.**
Register Mnemonic:			CMN_CDIAG_XCVRC_PWRI_OVRD
Register Address:			00_0_0000 : 0010 : 00111
Bits	**Type**	**Reset**	**Description**
7:0	R/W	8'h24	**Power island controller output override:** When enabled, the bits in this field will override the output signals from the power island control state machine. The following are the power island control state machine output signals, and the bits in this field that they are associated with. • Bit 7: state_sandh • Bit 6: state_ret_save • Bit 5: state_ret_restore (active low) • Bit 4: power_en_ph_1 • Bit 3: power_en_ph_2 • Bit 2: power_isolation • Bit 1: power_por_reset_n • Bit 0: power_rstr_reset_n The default value of the bits in this register are such that the power island is not forced to be switched on when the power island controller output override enable bit in this register is enabled. When using this register to switch this power island on and off, the bits in this field are required to change in a specific order with individual register writes. The writes must happen in the order specified below. • 8'b00100100 : Power island is not being forced on • 8'b10100100 • 8'b10110100 • 8'b10111100 • 8'b10111111 • 8'b10111011 • 8'b00111011 : Power island is being forced on • 8'b10111011 • 8'b10111111 • 8'b11111111 • 8'b10111111 • 8'b10111110 • 8'b10100110 : Power island is being forced off, and in state retention • 8'b10110110 • 8'b10111110 • 8'b10111111 • 8'b10011111 • 8'b10111111 • 8'b10111011 • 8'b00111011 : Power island is being forced on, with state restored •

Register Name:			Common transceiver controller power island control status register
Register Mnemonic:			CMN_CDIAG_XCVRC_PWRI_STAT
Register Address:			00_0_0000 : 0010 : 01000
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/O	8'h24	**Power island controller output status:** This field indicates the current state of the output signals from the power island control state machine. The following are the power island control state machine output signals, and the bits in this field that they are associated with. • Bit 7: state_sandh • Bit 6: state_ret_save • Bit 5: state_ret_restore (active low) • Bit 4: power_en_ph_1 • Bit 3: power_en_ph_2 • Bit 2: power_isolation • Bit 1: power_por_reset_n • Bit 0: power_rstr_reset_n

Register Name:			Common diagnostic power island control override register **Note: This register must not be part of any register read write test where patterns other than those described in this register are written to this register.**
Register Mnemonic:			CMN_CDIAG_DIAG_PWRI_OVRD
Register Address:			00_0_0000 : 0010 : 01001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Power island controller input override enable**: When enabled, the power island control state machine input override bits in this register will drive the power island control state machine directly, and override any control from other state machines in the macro. **Note: The CDB power island must be triggered to be enabled, by accessing a register outside of the common control module control and diagnostic registers space, prior to enabling this function.** **Note: This bit must never change state in the same register write that any of the override bits it controls in this register are changed.** • 1'b0: Override disabled • 1'b1: Override enabled
14	R/W	1'h0	**Power island controller output override enable**: When enabled, the power island control state machine output override bits in this register will drive the power island control state machine outputs, and override any control from power island control state machine. **Note: This bit must never change state in the same register write that any of the override bits it controls in this register are changed.** • 1'b0: Override disabled • 1'b1: Override enabled
13:12	R/O	2'h0	Reserved
11	R/W	1'h0	**Power suspend request override**: When enabled, this bit will override the power_suspend_req input of the power island control state machine.
10	R/O	1'h1	**Power suspend acknowledge**: This is the current state of the power_suspend_ack output from the power island control state machine. **Note: This bit will always be set to 1'b1 after the power island is enabled, as a function of the power island controller input override enable bit in this register.**
9	R/W	1'h0	**Power recover request override**: When enabled, this bit will override the power_recover_req input of the power island control state machine.
8	R/O	1'h0	**Power recover acknowledge**: This is the current state of the power_recover_ack output from the power island control state machine. **Note: This bit will always be set to 1'b1 after the power island is enabled, as a function of the power island controller input override enable bit in this register.**

Register Name:			Common diagnostic power island control override register **Note: This register must not be part of any register read write test where patterns other than those described in this register are written to this register.**
Register Mnemonic:			CMN_CDIAG_DIAG_PWRI_OVRD
Register Address:			00_0_0000 : 0010 : 01001
Bits	**Type**	**Reset**	**Description**
7:0	R/W	8'h24	**Power island controller output override:** When enabled, the bits in this field will override the output signals from the power island control state machine. The following are the power island control state machine output signals, and the bits in this field that they are associated with. • Bit 7: state_sandh • Bit 6: state_ret_save • Bit 5: state_ret_restore (active low) • Bit 4: power_en_ph_1 • Bit 3: power_en_ph_2 • Bit 2: power_isolation • Bit 1: power_por_reset_n • Bit 0: power_rstr_reset_n The default value of the bits in this register are such that the power island is not forced to be switched on when the power island controller output override enable bit in this register is enabled. When using this register to switch this power island on and off, the bits in this field are required to change in a specific order with individual register writes. The writes must happen in the order specified below. • 8'b00100100 : Power island is not being forced on • 8'b10100100 • 8'b10110100 • 8'b10111100 • 8'b10111111 • 8'b10111011 • 8'b00111011 : Power island is being forced on • 8'b10111011 • 8'b10111111 • 8'b11111111 • 8'b10111111 • 8'b10111110 • 8'b10100110 : Power island is being forced off, and in state retention • 8'b10110110 • 8'b10111110 • 8'b10111111 • 8'b10011111 • 8'b10111111 • 8'b10111011 • 8'b00111011 : Power island is being forced on, with state restored •

Register Name:			Common diagnostic power island control status register
Register Mnemonic:			CMN_CDIAG_DIAG_PWRI_STAT
Register Address:			00_0_0000 : 0010 : 01010
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/O	8'hA4	**Power island controller output status:** This field indicates the current state of the output signals from the power island control state machine. The following are the power island control state machine output signals, and the bits in this field that they are associated with. • Bit 7: state_sandh • Bit 6: state_ret_save • Bit 5: state_ret_restore (active low) • Bit 4: power_en_ph_1 • Bit 3: power_en_ph_2 • Bit 2: power_isolation • Bit 1: power_por_reset_n • Bit 0: power_rstr_reset_n

Register Name:			PLL full rate and data rate clock control register
Register Mnemonic:			CMN_CDIAG_PRATECLK_CTRL
Register Address:			00_0_0000 : 0010 : 01011
Bits	**Type**	**Reset**	**Description**
15:2	R/O	14'h0	Reserved
1	R/W	1'h0	**Common PLL 1 full rate and data rate source clock select:** Selects the PLL source clock to use when generating the cmn_pll1_clk_fullrt, cmn_pll1_clk_datart0, and cmn_pll1_clk_datart1 clocks. • 1'b0 : cmnda_pll1_clk_0 • 1'b1 : cmnda_pll1_clk_1
0	R/W	1'h0	**Common PLL 0 full rate and data rate source clock select:** Selects the PLL source clock to use when generating the cmn_pll0_clk_fullrt, cmn_pll0_clk_datart0, and cmn_pll0_clk_datart1 clocks. • 1'b0 : cmnda_pll0_clk_0 • 1'b1 : cmnda_pll0_clk_1

Register Name:			Reference clock receiver override register **Note: If this register is required to be written in order to receiver a clean and valid reference clock, then this register must be the first register that is written in the PMA after releasing the CDB from reset.**
Register Mnemonic:			CMN_CDIAG_REFCLK_OVRD
Register Address:			00_0_0000 : 0010 : 01100
Bits	**Type**	**Reset**	**Description**
15:13	R/O	3'h0	Reserved
12	R/W	1'h0	**Derived reference clock source select**: Selects which PLL is the source for the derived reference clock, by driving the cmnda_ref_clk_der_src_sel signal to the analog. • 1'b0: PLL 0 • 1'b1: PLL 1
11:10	R/O	2'h0	Reserved
9:8	R/W	2'h1	**Digital reference clock receiver hysteresis adjust**: Control the amount of hysteresis used for the digital reference clock receiver, by driving the cmnda_ref_clk_dig_hyst_adj signal to the analog. • 2'b00: -1X nominal • 2'b01: Nominal • 2'b10: +1X nominal • 2'b11: +2X nominal
7	R/O	1'h0	Reserved
6	R/W	1'h0	**Analog reference clock enable override**: This bit can be used to force the cmnda_ref_clk_en signal going to the analog to the active state. • 1'b0 : No effect on the state of the cmnda_ref_clk_en signal. • 1'b1 : The cmnda_ref_clk_en is forced to the active state.
5	R/W	1'h0	**Reference clock AC coupling cap bypass**: Controls the bypassing of the AC coupling caps in the differential receiver. • 1'b0: AC coupling caps not bypassed. • 1'b1: AC coupling caps bypassed.
4	R/W	1'h0	**Derived reference clock enable**: Enables the derived reference clock function, by driving the cmnda_ref_clk_der_en signal to the analog. **Note: The option of using this clock is only intended to be for PCIe modes of operation.** • 1'b0: Derived clock function disabled. • 1'b1: Derived clock function enabled.
3:2	R/W	2'h2	**Reference clock high pass filter control**: Controls the cutoff frequency of the high pass filter in the reference clock receiver input. This controls this function by controlling the cmnda_ref_clk_filt_ctrl signal going into the analog. The following are the valid settings for this field and the corresponding cut off frequencies. • 2'b00 : 20 - 49 MHz • 2'b01 : 50 - 99 MHz • 2'b10 : 100 - 149 MHz • 2'b11 : 150 - 166 MHz
1	R/O	1'h0	Reserved
0	R/W	1'h0	**Reference clock receiver circuit clock control:** • 1'b0 : The analog reference clock receiver circuit for the digital drives the cmn_ref_clk_rcv pin. • 1'b1 : The analog reference clock receiver circuit for the PLL drives the cmn_ref_clk_rcv pin.

Register Name:			Reference clock receiver test register
Register Mnemonic:			CMN_CDIAG_REFCLK_TEST
Register Address:			00_0_0000 : 0010 : 01101
Bits	**Type**	**Reset**	**Description**
15:10	R/O	6'h0	Reserved
9:8	R/W	2'h0	Reserved - spare
7:6	R/O	2'h0	Reserved
5	R/W	1'h0	**Reference clock driver 0 test mode enable**: Enables the reference clock driver DC test mode, by controlling the cmnda_ref_clk0_drv_test_en signal going into the analog.
4	R/W	1'h0	**Reference clock driver 0 test mode value**: When enabled by the reference clock driver 0 test mode enable bit in this register, the value in this bit will be driven by the reference clock driver, by controlling the cmnda_ref_clk0_drv_test_val signal going into the analog.
3	R/O	1'h0	Reserved
2	R/W	1'h0	**Reference clock receiver test mode enable**: Enables the reference clock receiver DC test mode, by controlling the cmnda_ref_clk_rcv_test_en signal going into the analog.
1	R/O	1'h0	**Reference clock receiver test mode PLL value**: When enabled by the reference clock receiver test mode enable bit in this register, the value in this bit will be the value present on the PLL reference clock receiver. This bit is driven by thecmnda_ref_clk_rcv_test_pll_val signal driven from the analog.
0	R/O	1'h0	**Reference clock receiver test mode digital value**: When enabled by the reference clock receiver test mode enable bit in this register, the value in this bit will be the value present on the digital reference clock receiver. This bit is driven by thecmnda_ref_clk_rcv_test_dig_val signal driven from the analog.

Register Name:			Power state machine clock receiver control register
Register Mnemonic:			CMN_CDIAG_PSMCLK_CTRL
Register Address:			00_0_0000 : 0010 : 01110
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12'h0	Reserved
3:0	R/W	4'h5	**PSM clock divider value:** The value of this field is used to control the divider setting of the PSM clock divider. The following are the encoded value for this field. • 4'b0000 : Reserved • 4'b0001 : Divide by 1 • 4'b0010 : Divide by 2 • 4'b0011 : Divide by 3 • ... • 4'b1111 : Divide by 15

Register Name:			Signal detect clock oscillator control register
Register Mnemonic:			CMN_CDIAG_SDOSC_CTRL
Register Address:			00_0_0000 : 0010 : 01111
Bits	**Type**	**Reset**	**Description**
15:2	R/O	14'h0	Reserved
1	R/W	1'h0	**Oscillator Enable Override Enable**: This bit enables the oscillator enable override bit in this register to directly control the signal detect oscillator. • 1'b0 : Override disabled. • 1'b1 : Override enabled
0	R/W	1'h0	**Oscillator Enable Override**: When enabled by the oscillator enable override enable bit in this register, this bit can be used to directly control the enable of the signal detect oscillator. • 1'b0 : Disabled. • 1'b1 : Enabled

Register Name:			Reference clock bump driver 0 control register
Register Mnemonic:			CMN_CDIAG_REFCLK_DRV0_CTRL
Register Address:			00_0_0000 : 0010 : 10000
Bits	**Type**	**Reset**	**Description**
15:10	R/O	6'h0	Reserved
9:8	R/W	2'h2	**Clock driver drive current tune**: Controls the amplitude of the reference clock driver, by controlling the cmnda_ref_clk0_itune signal going to the analog. The following is the encoding of this field. • 2'b00: 300mV pk-pk differential • 2'b01: 350mV pk-pk differential • 2'b10: 400mV pk-pk differential 2'b11: 450mV pk-pk differential •
7	R/O	1'h0	Reserved
6	R/W	1'h1	**Reference clock driver high Z**: When the reference clock driver is disabled, this controls if the driver outputs are high Z or pulled low, by controlling the cmnda_ref_clk0_drv_highz signal going to the analog. • 1'b0: Driver outputs pulled low • 1'b1: Driver outputs high Z
5	R/W	1'h0	**Clock driver termination**: Enables the termination in the reference clock driver, by controlling the cmnda_ref_clk0_termination signal going to the analog. • 1'b0: Disabled • 1'b1: Enabled
4	R/W	1'h0	**Clock select**: Selects which reference clock that will be driven by the reference clock driver, by controlling the cmnda_ref_clk0_clk_select signal going to the analog. • 1'b0: Received reference clock • 1'b1: Derived reference clock
3	R/W	1'h0	**Clock gate enable override enable**: This bit enables the clock gate enable override bit in this register to override the clock gate of the reference clock driver.
2	R/W	1'h0	**Clock gate enable override**: When enabled by the clock gate enable override enable bit in this register, this bit can be used to directly control the clock gate enable of the reference clock driver by controlling the cmnda_ref_clk0_clk_gate_en signal going to the analog.
1	R/W	1'h1	**Driver enable override enable**: This bit enables the driver enable override bit in this register to override the enable of the reference clock driver. **Note: This driver must not be enabled when a reference clock signal is being driven to the reference clock 0 bumps.**
0	R/W	1'h0	**Driver enable override**: When enabled by the driver enable override enable bit in this register, this bit can be used to directly control the enable of the reference clock driver by controlling the cmnda_ref_clk0_drv_en signal going to the analog. **Note: This driver must not be enabled when a reference clock signal is being driven to the reference clock 0 bumps.**

Register Name:			Common control CDB diagnostic register
Register Mnemonic:			CMN_CDIAG_CDB_DIAG
Register Address:			00_0_0000 : 0010 : 11100
Bits	**Type**	**Reset**	**Description**
15:1	R/O	15'h0	Reserved
0	R/C	1'h0	**CDB bus error**: This bit will be set when the internal CDB watchdog timer expires. It is automatically cleared on a read of this register.

Register Name:			Common control reset diagnostic register
Register Mnemonic:			CMN_CDIAG_RST_DIAG
Register Address:			00_0_0000 : 0010 : 11101
Bits	**Type**	**Reset**	**Description**
15:2	R/O	14'h0	Reserved
1	R/O	1'hX	Current state of the cdb_isl_ctrl_sm_reset_n reset.
0	R/O	1'hX	Current state of the cmn_reset_sync_n reset.

Register Name:			Common control cover your alternatives register
Register Mnemonic:			CMN_CDIAG_DCYA
Register Address:			00_0_0000 : 0010 : 11111
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8’h0	Reserved
7:0	R/W	8’h0	Reserved - spare

10.2.7 Bandgap Calibration Controller Registers

Base address bits 15:5 :	00_0_0000 : 0011
Address bits 4:0	**Register name and location of detailed description**
00000	Bandgap calibration control register on page 150
00001	Bandgap calibration override register on page 151
00010	Bandgap calibration start register on page 151
00011	Bandgap calibration tune register on page 151
00100	Bandgap calibration initialization timer register on page 151
00101	Bandgap calibration iteration timer register on page 152
00110 - 11111	Reserved

Table 93: Bandgap calibration controller registers address space

Register Name:			Bandgap calibration control register
Register Mnemonic:			CMN_BGCAL_CTRL
Register Address:			00_0_0000 : 0011 : 00000
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Start bandgap calibration**: Activating (1'b1) this bit will start the bandgap calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1'b0) until the bandbap calibration process done bit in this register is cleared. **Note: This bit is intended to be for diagnostics purposes only. This calibration process is automatically activated internally by the startup state machine. When using this bit, the user must wait until after the internally activated process completes.**
14	R/O	1'h0	**Bandgap calibration process done**: This bit will be set to 1'b1 when the bandgap calibration process is complete. It will be cleared by cmn_reset_n, or by the deactivation of the start bandgap calibration bit in this register after calibration is complete. **Note: This bit is intended to be for diagnostics purposes only.** **Note: This bit is not likely to be observed as being set after internal automatic calibration is complete, because the internally generated run signal will be driven inactive immediately after the done signal is activated, and therefore the internal done signal will be cleared.** **Note: Three cmn_ref_clk cycles are required after the start of the calibration process to clear this signal.**
13	R/O	1'h0	**No analog calibration response** : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the analog has not responded indicating that a valid calibration value has been reached. Note: Due to the nature of the calibration process, it is not possible to determine if the analog responded or not for the lowest code of 5'b10001. The reason for this is, when sequencing to lower codes, the calibration function is looking for the analog response to transition from 1'b1 to 1'b0. If the lowest calibration code is the correct code, the analog will respond with 1'b1, and the algorithm can't sequence to a lower code to look for the transition from 1'b1 to 1'b0. Similarly, if the analog is not responding, it is not possible to sequence to a lower code to detect this. Therefore, even when the calibration function selects this calibration code as valid, this bit will be set.
12	R/O	1'h0	**Current analog comparator response**: This is the current state of the analog comparator response signal (cmnda_bias_bgcal_comp). This signal is not synchronized, and is provided for diagnostic purposes only.
11:6	R/O	6'h0	Reserved
5:0	R/O	6'h00	**Bandgap calibration code**: This is the calibration code that was determined by the bandgap calibration process. The following indicates how this encoding maps to the cmnda_bias_bgcal_up and cmnda_bias_bgcal_azsel signals going to the analog bandgap function. The 6 bit value on the left is the value of this field. The 1 bit value on the right corresponds to the value of cmnda_bias_bgcal_up. The 5 bit field corresponds to the value of cmnda_bias_bgcal_azsel. • 6'b011111 : 1'b1, 5'b11111 • 6'b011110 : 1'b1, 5'b11110 • ... • 6'b000010 : 1'b1, 5'b00010 • 6'b000001 : 1'b1, 5'b00001 • 6'b000000 : 1'b1, 5'b00000 • 6'b111111 : 1'b0, 5'b00001 • 6'b111110 : 1'b0, 5'b00010 • ... • 6'b100010 : 1'b0, 5'b11110 • 6'b100001 : 1'b0, 5'b11111

Register Name:			Bandgap calibration override register
Register Mnemonic:			CMN_BGCAL_OVRD
Register Address:			00_0_0000 : 0011 : 00001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Bandgap code override enable**: Activation (1'b1) of this register bit allows the bandgap codes determined during the automatic calibration process to be overridden. The override value is specified using the bandgap code override value field of this register. Note: The value of this field must not be changed while the calibration function is running.
14	R/W	1'h0	**Analog calibration enable override**: Activation (1'b1) of this register bit will force the analog calibration circuits to be enabled by activating the cmnda_bias_bgcal_en enable. Note: The value of this field must not be changed while the calibration function is running.
13:6	R/O	8'h0	Reserved
5:0	R/W	6'h00	**Bandgap code override value**: These bits are used to override the bandgap code determined during the automatic calibration process. The code written to these bits is valid when the bandgap code override enable bit in this register is active. The codes in this field correspond to those described in the bandgap calibration code field in the Bandgap calibration control register on page 150. Note: The value of this field must not be changed while the calibration function is running.

Register Name:			Bandgap calibration start register
Register Mnemonic:			CMN_BGCAL_START
Register Address:			00_0_0000 : 0011 : 00010
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Bandgap calibration direction**: This controls the direction that the automatic calibration process steps the calibration codes in. • 1'b0 : From 4'b1001 to 4'b0111. • 1'b1 : From 4'b0111 to 4'b0000.
14:6	R/O	9'h0	Reserved
5:0	R/W	6'h00	**Start bandgap calibration code**: This is the calibration code that the calibration process starts with when automatic calibration is run. The codes in this field correspond to those described in the bandgap calibration code field in the Bandgap calibration control register on page 150.

Register Name:			Bandgap calibration tune register
Register Mnemonic:			CMN_BGCAL_TUNE
Register Address:			00_0_0000 : 0011 : 00011
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h00	**Bandgap calibration tune value**: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note: This value is a twos complement value, so the calibrated code can be increased or decreased.

Register Name:			Bandgap calibration initialization timer register
Register Mnemonic:			CMN_BGCAL_INIT_TMR
Register Address:			00_0_0000 : 0011 : 00100
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h07D	**Initialization wait timer value**: This is the number of cmn_ref_clk clocks to wait between when the analog bandgap calibration circuits are enabled, and when the first values are placed on the bandgap calibration signals going to the analog. **Note: This results in the minimum number of cmn_ref_clk clocks required to wait 5 uSec.** **Note: This should never be set to a value of less than 1.**

Register Name:			Bandgap calibration iteration timer register
Register Mnemonic:			CMN_BGCAL_ITER_TMR
Register Address:			00_0_0000 : 0011 : 00101
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h07D	**Iteration wait timer value**: This is the number of cmn_ref_clk clocks to wait between when a value is placed on the bandgap calibration signals going to the analog, and when the comparator value coming from the analog circuits can be checked. **Note: This results in the minimum number of cmn_ref_clk clocks required to wait 5 uSec.** **Note: This should never be set to a value of less than 3.**

10.2.8 External Bias Current Calibration Controller Registers

Base address bits 15:5 :	00_0_0000 : 0011
Address bits 4:0	**Register name and location of detailed description**
10000	External bias current calibration control register on page 153
10001	External bias current calibration override register on page 154
10010	External bias current calibration start register on page 154
10011	External bias current calibration tune register on page 154
10100	External bias current calibration initialization timer register on page 155
10101	External bias current calibration iteration timer register on page 155
10110 - 11111	Reserved

Table 94: External bias current calibration controller registers address space

Register Name:			External bias current calibration control register
Register Mnemonic:			CMN_IBCAL_CTRL
Register Address:			00_0_0000 : 0011 : 10000
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Start calibration**: Activating (1'b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1'b0) until the calibration process done bit in this register is cleared. **Note: This signal is intended to be for diagnostics purposes only. This calibration process is automatically activated internally by the startup state machine. When using this bit, the user must wait until after the internally activated process completes.**
14	R/O	1'h0	**Calibration process done**: This bit will be set to 1'b1 when the calibration process is complete. It will be cleared by cmn_reset_n, or by the deactivation of the start calibration bit in this register after calibration is complete. **Note: This bit is intended to be for diagnostics purposes only.** **Note: This bit is not likely to be observed as being set after internal automatic calibration is complete, because the internally generated run signal will be driven inactive immediately after the done signal is activated, and therefore the internal done signal will be cleared.** **Note: Three cmn_ref_clk cycles are required after the start of the resistor calibration process to clear this signal.**
13	R/O	1'h0	**No analog calibration response** : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the analog has not responded indicating that a valid calibration value has been reached. Note: Due to the nature of the calibration process, it is not possible to determine if the analog responded or not for the lowest code of 7'b00000. The reason for this is, when sequencing to lower codes, the calibration function is looking for the analog response to transition from 1'b1 to 1'b0. If the lowest calibration code is the correct code, the analog will respond with 1'b1, and the algorithm can't sequence to a lower code to look for the transition from 1'b1 to 1'b0. Similarly, if the analog is not responding, it is not possible to sequence to a lower code to detect this. Therefore, even when the calibration function selects this calibration code as valid, this bit will be set.
12	R/O	1'h0	**Current analog comparator response**: This is the current state of the analog comparator response signal (cmnda_ibiascal_comp). This signal is not synchronized, and is provided for diagnostic purposes only.
11:7	R/O	5'h0	Reserved
6:0	R/O	7'h00	**Calibration code**: This is the calibration code that was determined by the calibration process. The following are the values for the code: • 7'b0000000: Minimum value. • 7'b0000001 • 7'b0000010 • 7'b0000011 • ... • 7'b0111110 • 7'b0111111: Maximum value. Note: The most significant bit is the calibration code sign bit, and is always 1'b0 in this application. **Note: The reset value for this field is with the common calibration power island switched off. In cases where this power island is switched on, the reset value be 7'b0011111.**

Register Name:			External bias current calibration override register
Register Mnemonic:			CMN_IBCAL_OVRD
Register Address:			00_0_0000 : 0011 : 10001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Calibration code override enable**: Activation (1'b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register. **Note:** The value of this field must not be changed while the calibration function is running.
14	R/W	1'h0	**Analog calibration enable override**: Activation (1'b1) of this register bit will force the analog calibration circuits to be enabled by activating the cmnda_ibiascal_en enable and the cmnda_ibiascal_clk clock. **Note:** The value of this field must not be changed while the calibration function is running.
13:7	R/O	7'h0	Reserved
6:0	R/W	7'h00	**Calibration code override value**: These bits are used to override the calibration code determined during the automatic resistor calibration process. The code written to these bits is valid when the calibration code override enable bit in this register is active. The following are the values for the code: • 7'b0000000: Minimum value. • 7'b0000001 • 7'b0000010 • 7'b0000011 • ... • 7'b0111110 • 7'b0111111: Maximum value. Note: The most significant bit is the calibration code sign bit, and is always 1'b0 in this application. **Note:** The value of this field must not be changed while the calibration function is running.

Register Name:			External bias current calibration start register
Register Mnemonic:			CMN_IBCAL_START
Register Address:			00_0_0000 : 0011 : 10010
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Calibration direction**: This controls the direction that the automatic calibration process steps the calibration codes in. • 1'b0 : From 7'b0000000 to 7'b0111111. • 1'b1 : From 7'b0111111 to 7'b0000000.
14:7	R/O	8'h0	Reserved
6:0	R/W	7'h01F	**Start resistor calibration code**: This is the calibration code that the resistor calibration process starts with when automatic calibration is run. The following are the values for the code. • 7'b0000000: Minimum value. • 7'b0000001 • 7'b0000010 • 7'b0000011 • ... • 7'b0111110 • 7'b0111111: Maximum value. Note: The most significant bit is the calibration code sign bit, and is always 1'b0 in this application.

Register Name:			External bias current calibration tune register
Register Mnemonic:			CMN_IBCAL_TUNE
Register Address:			00_0_0000 : 0011 : 10011
Bits	**Type**	**Reset**	**Description**
15:7	R/O	9'h0	Reserved
6:0	R/W	7'h00	**Calibration tune value**: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note: This value is a twos complement value, so the calibrated code can be increased or decreased.

Register Name:			External bias current calibration initialization timer register
Register Mnemonic:			CMN_IBCAL_INIT_TMR
Register Address:			00_0_0000 : 0011 : 10100
Bits	**Type**	**Reset**	**Description**
15:7	R/O	9'h0	Reserved
6:0	R/W	7'h19	**Initialization wait timer value**: This is the number of cmn_ref_clk clocks to wait between when the analog calibration circuits are enabled, and when the first calibration selection value is placed on the calibration code bus, going to the analog. **Note: This results in the minimum number of cmn_ref_clk clocks required to wait 1 uSec.** **Note: This should never be set to a value of less than 1.**

Register Name:			External bias current calibration iteration timer register
Register Mnemonic:			CMN_IBCAL_ITER_TMR
Register Address:			00_0_0000 : 0011 : 10101
Bits	**Type**	**Reset**	**Description**
15:7	R/O	9'h0	Reserved
6:0	R/W	7'h07	**Iteration wait timer value**: This is the number of cmn_ref_clk clocks to wait between when a value is placed on the calibration selection bus going to the analog, and when the comparator value coming from the analog circuits can be checked. **Note: This should never be set to a value of less than 3.**

10.2.9 PLL 0 Controller Registers

Base address bits 15:5 :	00_0_0000 : 0100
Address bits 4:0	**Register name and location of detailed description**
00000	PLL 0 VCO calibration control register on page 157
00001	PLL 0 VCO calibration start point register on page 157
00010	PLL 0 VCO calibration timer control register on page 158
00011	PLL 0 VCO calibration override register on page 158
00100	PLL 0 VCO calibration initialization timer register on page 158
00101	PLL 0 VCO calibration iteration timer register on page 158
00110	PLL 0 VCO calibration reference clock timer start value register on page 159
00111	Reserved
01000	PLL 0 VCO calibration PLL clock counter start value register on page 159
01001 - 01111	Reserved
10000	PLL 0 feedback divider integer register mode 0 on page 160
10001	PLL 0 feedback divider fractional low register mode 0 on page 160
10010	PLL 0 feedback divider fractional high register mode 0 on page 160
10011	PLL 0 feedback divider high threshold register mode 0 on page 160
10100	PLL 0 delta sigma modulator diagnostics register mode 0 on page 161
10101	PLL 0 delta sigma modulator feedback divider value high override register mode 0 on page 161
10110	PLL 0 delta sigma modulator feedback divider value low override register mode 0 on page 161
10111	Reserved
11000	PLL 0 spread spectrum control register 1 mode 0 on page 161
11001	PLL 0 spread spectrum control register 2 mode 0 on page 161
11010	PLL 0 spread spectrum control register 3 mode 0 on page 162
11011	PLL 0 spread spectrum control register 4 mode 0 on page 162
11100	PLL 0 lock reference counter start value register on page 159
11101	PLL 0 lock reference counter idle value register on page 159
11110	PLL 0 lock PLL counter start value register on page 160
11111	PLL 0 lock PLL counter threshold value register on page 160

Table 95: PLL 0 controller registers address space (slave 0)

Base address bits 15:5 :	00_0_0000 : 0101
Address bits 4:0	**Register name and location of detailed description**
00000	PLL 0 feedback divider integer register mode 1 on page 162
00001	PLL 0 feedback divider fractional low register mode 1 on page 162
00010	PLL 0 feedback divider fractional high register mode 1 on page 162
00011	PLL 0 feedback divider high threshold register mode 1 on page 162
00100	PLL 0 delta sigma modulator diagnostics register mode 1 on page 163
00101	PLL 0 delta sigma modulator feedback divider value high override register mode 1 on page 163
00110	PLL 0 delta sigma modulator feedback divider value low override register mode 1 on page 163
00111	Reserved
01000	PLL 0 spread spectrum control register 1 mode 1 on page 163
01001	PLL 0 spread spectrum control register 2 mode 1 on page 163
01010	PLL 0 spread spectrum control register 3 mode 1 on page 164
01011	PLL 0 spread spectrum control register 4 mode 1 on page 164
01100 - 11111	Reserved

Table 96: PLL 0 controller registers address space (slave 1)

Register Name:			PLL 0 VCO calibration control register
Register Mnemonic:			CMN_PLL0_VCOCAL_CTRL
Register Address:			00_0_0000 : 0100 : 00000
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Start VCO calibration**: Activating (1'b1) this bit will start a VCO calibration process. This bit must remain active until the VCO calibration process is complete (as indicated by the VCO calibration process done bit in this register). To start another VCO calibration process, the VCO calibration process done bit must have gone inactive from any prior calibration process. **Note: This bit is intended to be for diagnostics purposes only. This calibration process is automatically activated internally by the PLL control state machine. When using this bit, the user must wait until after the internally activated process completes.**
14	R/O	1'h0	**VCO calibration process done**: This bit will be set to 1'b1 when the VCO calibration process is complete. It will be cleared by the deactivation of the Start VCO calibration bit in this register. **Note: This bit is intended to be for diagnostics purposes only.** **Note: This bit is not likely to be observed as being set after internal automatic calibration is complete, because the internally generated run signal will be driven inactive immediately after the done signal is activated, and therefore the internal done signal will be cleared.**
13:8	R/O	6'h0	Reserved
7:0	R/O	8'h00	**VCO calibration code**: This is the calibration code that was determined by the VCO calibration process. This signal is valid when the VCO calibration process is complete. The values of this field correspond to different frequency bands the VCO will operate in. The frequency bands are controlled by the number of capacitors that are switched in the VCO analog circuit. This field specifies the number of capacitors that are switched in. The following are the values for this code: • 8'b00000000: 0 Capacitors activated • 8'b00000001: 1 Capacitor activated • 8'b00000010: 2 Capacitors activated • 8'b00000011: 3 Capacitors activated • ... • 8'b11111110: 254 Capacitors activated • 8'b11111111: 255 Capacitors activated **Note: This register is intended to be for diagnostics purposes only.** **Note: The reset value for this field is with the PLL controller power island switched off. In cases where this power island is switched on, the reset value be 8'h28.**

Register Name:			PLL 0 VCO calibration start point register
Register Mnemonic:			CMN_PLL0_VCOCAL_START
Register Address:			00_0_0000 : 0100 : 00001
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:12	R/W	3'h2	**VCO calibration initial step size control**: This field specifies the initial step size for the VCO calibration state machine. The following are the values that can be used in this field, and the corresponding step sizes. • 3'b000 : 1 • 3'b001 : 2 • 3'b010 : 4 • 3'b011 : 8 • 3'b100 : 16 • 3'b101 - 3'b111 : Reserved
11:8	R/O	4'h0	Reserved
7:0	R/W	8'h28	**VCO calibration code starting point value**: This field specifies the starting VCO code that is used by the VCO calibration state machine. The purpose of this value is such that the VCO calibration process starts at a point that is, on average, relatively close to the final calibration point. This allows the calibration time to be reduced, on average. The encoding of this field is the same as what is described in the VCO calibration control register. **Note: This field is intended to be for diagnostics purposes only.** **Note : This field must be set to a value that is always at least one step size away from the minimum and maximum calibration codes, and is a function of the step size specified in this register. For example, if the initial step size is 4, this value must be at least 4 greater than the minimum calibration code or 4 less than the maximum calibration code.**

Register Name:			PLL 0 VCO calibration timer control register
Register Mnemonic:			CMN_PLL0_VCOCAL_TCTRL
Register Address:			00_0_0000 : 0100 : 00010
Bits	**Type**	**Reset**	**Description**
15:3	R/O	13'h0	Reserved
2:0	R/W	3'h4	**VCO calibration initial time scale control**: This field specifies the calibration start time scaling factor applied to the VCO calibration when running the initial step size for the calibration code if not set to 1. Setting this value to a value other than 1 will reduce the calibration measurement time by the amount specified below. Then when the final calibration steps are made the full calibration time will be used to get the appropriate amount of resolution. • 3'b000 : Div 1 • 3'b001 : Div 2 • 3'b010 : Div 4 • 3'b011 : Div 8 • 3'b100 : Div 16 • 3'b101 : Div 32 • 3'b110 : Div 64 • 3'b111 : Div 128

Register Name:			PLL 0 VCO calibration override register
Register Mnemonic:			CMN_PLL0_VCOCAL_OVRD
Register Address:			00_0_0000 : 0100 : 00011
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**VCO calibration code override enable**: Activating (1'b1) this bit allows the VCO code determined during the automatic VCO calibration process to be overridden by the value driven by the VCO calibration code override value field in this register. **Note: This bit is intended to be for diagnostics purposes only.** **Note:** The value of this field must not be changed while the calibration function is running.
14:8	R/O	7'h0	Reserved
7:0	R/W	8'h00	**VCO calibration code override value**: This field is used to override the VCO code determined during the automatic VCO calibration process. The VCO code driven on this field is valid when the VCO calibration code override enable bit in this register is active. The encoding of this field is the same as what is described in the VCO calibration control register. **Note:** This field is intended to be for diagnostics purposes only. **Note:** The value of this field must not be changed while the calibration function is running.

Register Name:			PLL 0 VCO calibration initialization timer register
Register Mnemonic:			CMN_PLL0_VCOCAL_INIT_TMR
Register Address:			00_0_0000 : 0100 : 00100
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'h0	Reserved
13:0	R/W	14'h03E8	**Initialization wait timer value**: This is the number of clocks to wait between when the analog VCO calibration circuits are enabled, and when the first calibration code is driven to the analog. **Note: This results in the minimum number of cmn_ref_clk clocks required to wait 10 uSec.**

Register Name:			PLL 0 VCO calibration iteration timer register
Register Mnemonic:			CMN_PLL0_VCOCAL_ITER_TMR
Register Address:			00_0_0000 : 0100 : 00101
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'h0	Reserved
13:0	R/W	14'h0010	**Iteration wait timer value**: This is the number of clocks to wait between when a calibration code is driven to the analog, and when the clock rates are measured.

Register Name:			PLL 0 VCO calibration reference clock timer start value register
Register Mnemonic:			CMN_PLL0_VCOCAL_REFTIM_START
Register Address:			00_0_0000 : 0100 : 00110
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'h0	Reserved
13:0	R/W	14'h0C5F	**PLL VCO calibration reference clock timer start value** : This is the value that is loaded into the reference clock timer as the starting point for that timer, when running VCO calibration. The value in this field must correspond to a time that is approximately one full spread spectrum cycle long (31.67 uSec) **Note: The actual number of clocks counted is one larger than what is specified by this field.**

Register Name:			PLL 0 VCO calibration PLL clock counter start value register
Register Mnemonic:			CMN_PLL0_VCOCAL_PLLCNT_START
Register Address:			00_0_0000 : 0100 : 01000
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'h0	Reserved
13:0	R/W	14'h0C5F	**PLL VCO calibration PLL clock counter start value** : This is the value that is loaded into the PLL clock counter as the starting point for that counter, when running VCO calibration. When programming this register, it is important to understand how the frequency of the cmnda_pll0_fb_divider_clk is calculated relative to the reference clock. This is described in section 10.4 "Dual VCO PLL" on page 430. **Note: The actual number of clocks counted is one larger than what is specified by this field.**

Register Name:			PLL 0 lock reference counter start value register
Register Mnemonic:			CMN_PLL0_LOCK_REFCNT_START
Register Address:			00_0_0000 : 0100 : 11100
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:0	R/W	12'h0C8	**PLL lock reference counter start value** : This is the value that is loaded into the PLL lock detect reference counter as the starting point for that counter, when checking for PLL lock. Note: The value required here is a function of the PLL feedback divider clock mode and the reference clock frequency. Overview information in regards to this can be found in section 4.2 "Reference Clock Related Clocking and Reset" on page 297 and section 4.9 "Reference Clock Related Clock Rate Summary" on page 328. The following bullet points specify the required time the value in this register is required to generate. By default, the value in this register is set for FBR mode with a 100 MHz reference clock. • FBR mode and reference clock frequency greater than or equal to 100 MHz: 2 uSec • FBR mode and reference clock frequency less than 100 MHz: 8 uSec • DR mode: 800 nSec

Register Name:			PLL 0 lock reference counter idle value register
Register Mnemonic:			CMN_PLL0_LOCK_REFCNT_IDLE
Register Address:			00_0_0000 : 0100 : 11101
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:0	R/W	12'h004	**PLL lock reference counter idle value** : This is the value used by the PLL lock detection logic to specify the number of reference clocks between each phase of counting PLL clocks.

Register Name:			PLL 0 lock PLL counter start value register
Register Mnemonic:			CMN_PLL0_LOCK_PLLCNT_START
Register Address:			00_0_0000 : 0100 : 11110
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:0	R/W	12'h0C8	**PLL lock PLL counter start value** : This is the value that is loaded into the PLL lock detect PLL counter as the starting point for that counter, when checking for PLL lock. The value is the number of PLL feedback divider clock edges that are expected during the time specified by the PLL 0 lock reference counter start value register on page 159. When programming this register, it is important to understand how the frequency of the cmnda_pll0_fb_divider_clk is calculated relative to the reference clock. This is described in section 10.4 "Dual VCO PLL" on page 430.

Register Name:			PLL 0 lock PLL counter threshold value register
Register Mnemonic:			CMN_PLL0_LOCK_PLLCNT_THR
Register Address:			00_0_0000 : 0100 : 11111
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:0	R/W	12'h003	**PLL lock counter threshold value** : This is the value used by the PLL lock detection logic to determine if the PLL has locked. If the two counters in the PLL lock detection logic differ by less than this value, the PLL is considered locked.

Register Name:			PLL 0 feedback divider integer register mode 0
Register Mnemonic:			CMN_PLL0_INTDIV_M0
Register Address:			00_0_0000 : 0100 : 10000
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h000	**pll_fb_div_integer value**: Value of the pll_fb_div_integer signal.

Register Name:			PLL 0 feedback divider fractional low register mode 0
Register Mnemonic:			CMN_PLL0_FRACDIVL_M0
Register Address:			00_0_0000 : 0100 : 10001
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h0000	**pll_fb_div_fractional**: Value of the pll_fb_div_fractional[15:0] signal.

Register Name:			PLL 0 feedback divider fractional high register mode 0
Register Mnemonic:			CMN_PLL0_FRACDIVH_M0
Register Address:			00_0_0000 : 0100 : 10010
Bits	**Type**	**Reset**	**Description**
15:3	R/O	13'h0	Reserved
2:0	R/W	3'h0	**pll_fb_div_fractional**: Value of the pll_fb_div_fractional[18:16] signal.

Register Name:			PLL 0 feedback divider high threshold register mode 0
Register Mnemonic:			CMN_PLL0_HIGH_THR_M0
Register Address:			00_0_0000 : 0100 : 10011
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h000	**pll_fb_div_high_theshold**: Value of the pll_fb_div_high_threshold signal. The value of this register should be 2/3 the value of the combined integer and fractional divider values, and rounded up to the next even number.

Register Name:			PLL 0 delta sigma modulator diagnostics register mode 0
Register Mnemonic:			CMN_PLL0_DSM_DIAG_M0
Register Address:			00_0_0000 : 0100 : 10100
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Delta sigma bypass enable**: When set to 1'b1, the delta sigma modulator will be bypassed, and the output will be the value specified for the internal pll_fb_div_integer signal.
14	R/W	1'h1	**PLL feedback divider override enable** : When active (1'b1), the feedback divider low and high override values in the PLL 0 delta sigma modulator feedback divider value high override register mode 0 on page 161 and PLL 0 delta sigma modulator feedback divider value low override register mode 0 on page 161 registers will be used to override the feedback divider values driven to the analog.
13:4	R/O	10'h0	Reserved
3:0	R/W	4'h4	**PLL feedback divider latency adjustment**: This signal specifies a value to be subtracted from the feedback divider settings before they are output on the cmnda_pll0_fb_div_high and cmnda_pll0_fb_div_low signals.

Register Name:			PLL 0 delta sigma modulator feedback divider value high override register mode 0
Register Mnemonic:			CMN_PLL0_DSM_FBH_OVRD_M0
Register Address:			00_0_0000 : 0100 : 10101
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h01E	**PLL feedback divider high override value** : When enabled by the PLL feedback divider override enable bit in the PLL 0 delta sigma modulator diagnostics register mode 0 on page 161, the value in this field will be used to override the value on the cmnda_pll0_fb_div_high signal. Note: The value of this register should be 2/3 the value of the combined PLL feedback divider high and low values, and rounded up to the next even number.

Register Name:			PLL 0 delta sigma modulator feedback divider value low override register mode 0
Register Mnemonic:			CMN_PLL0_DSM_FBL_OVRD_M0
Register Address:			00_0_0000 : 0100 : 10110
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h00C	**PLL feedback divider low override value** : When enabled by the PLL feedback divider override enable bit in the PLL 0 delta sigma modulator diagnostics register mode 0 on page 161, the value in this field will be used to override the value on the cmnda_pll0_fb_div_low signal.

Register Name:			PLL 0 spread spectrum control register 1 mode 0
Register Mnemonic:			CMN_PLL0_SS_CTRL1_M0
Register Address:			00_0_0000 : 0100 : 11000
Bits	**Type**	**Reset**	**Description**
15:2	R/O	14'h0	Reserved
1	R/W	1'h1	**Spread spectrum waveform generator disable**: Setting this bit to a 1'b1 will disable the spread spectrum waveform generator.
0	R/W	1'h0	**Spread spectrum enable during VCO calibration** : Setting this bit to a 1'b1 will enable the spread spectrum function while VCO calibration is taking place.

Register Name:			PLL 0 spread spectrum control register 2 mode 0
Register Mnemonic:			CMN_PLL0_SS_CTRL2_M0
Register Address:			00_0_0000 : 0100 : 11001
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:0	R/W	14'h0000	**Amplitude step size**: Value of the amplitude_step_size pin on the spread spectrum waveform generator. Note: This field should not be set to 0 when spread spectrum is enabled.

Register Name:			PLL 0 spread spectrum control register 3 mode 0
Register Mnemonic:			CMN_PLL0_SS_CTRL3_M0
Register Address:			00_0_0000 : 0100 : 11010
Bits	**Type**	**Reset**	**Description**
15:7	R/O	9'h0	Reserved
6:0	R/W	7'h00	**Number of steps**: Value of the num_steps pin on the spread spectrum waveform generator. Note: This field should not be set to 0 when spread spectrum is enabled.

Register Name:			PLL 0 spread spectrum control register 4 mode 0
Register Mnemonic:			CMN_PLL0_SS_CTRL4_M0
Register Address:			00_0_0000 : 0100 : 11011
Bits	**Type**	**Reset**	**Description**
15:7	R/O	9'h0	Reserved
6:0	R/W	7'h00	**Time step size**: Value for the time_step_size pin on the spread spectrum waveform generator. Note: This field should not be set to 0 when spread spectrum is enabled.

Register Name:			PLL 0 feedback divider integer register mode 1
Register Mnemonic:			CMN_PLL0_INTDIV_M1
Register Address:			00_0_0000 : 0101 : 00000
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h000	**pll_fb_div_integer value**: Value of the pll_fb_div_integer signal.

Register Name:			PLL 0 feedback divider fractional low register mode 1
Register Mnemonic:			CMN_PLL0_FRACDIVL_M1
Register Address:			00_0_0000 : 0101 : 00001
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h0000	**pll_fb_div_fractional**: Value of the pll_fb_div_fractional[15:0] signal.

Register Name:			PLL 0 feedback divider fractional high register mode 1
Register Mnemonic:			CMN_PLL0_FRACDIVH_M1
Register Address:			00_0_0000 : 0101 : 00010
Bits	**Type**	**Reset**	**Description**
15:3	R/O	13'h0	Reserved
2:0	R/W	3'h0	**pll_fb_div_fractional**: Value of the pll_fb_div_fractional[18:16] signal.

Register Name:			PLL 0 feedback divider high threshold register mode 1
Register Mnemonic:			CMN_PLL0_HIGH_THR_M1
Register Address:			00_0_0000 : 0101 : 00011
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h000	**pll_fb_div_high_theshold**: Value of the pll_fb_div_high_threshold signal. The value of this register should be 2/3 the value of the combined integer and fractional divider values, and rounded up to the next even number.

Register Name:			PLL 0 delta sigma modulator diagnostics register mode 1
Register Mnemonic:			CMN_PLL0_DSM_DIAG_M1
Register Address:			00_0_0000 : 0101 : 00100
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Delta sigma bypass enable**: When set to 1'b1, the delta sigma modulator will be bypassed, and the output will be the value specified for the internal pll_fb_div_integer signal.
14	R/W	1'h1	**PLL feedback divider override enable** : When active (1'b1), the feedback divider low and high override values in the PLL 0 delta sigma modulator feedback divider value high override register mode 0 on page 161 and PLL 0 delta sigma modulator feedback divider value low override register mode 0 on page 161 registers will be used to override the feedback divider values driven to the analog.
13:4	R/O	10'h0	Reserved
3:0	R/W	4'h4	**PLL feedback divider latency adjustment**: This signal specifies a value to be subtracted from the feedback divider settings before they are output on the cmnda_pll0_fb_div_high and cmnda_pll0_fb_div_low signals.

Register Name:			PLL 0 delta sigma modulator feedback divider value high override register mode 1
Register Mnemonic:			CMN_PLL0_DSM_FBH_OVRD_M1
Register Address:			00_0_0000 : 0101 : 00101
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h010	**PLL feedback divider high override value** : When enabled by the PLL feedback divider override enable bit in the PLL 0 delta sigma modulator diagnostics register mode 0 on page 161, the value in this field will be used to override the value on the cmnda_pll0_fb_div_high signal. Note: The value of this register should be 2/3 the value of the combined PLL feedback divider high and low values, and rounded up to the next even number.

Register Name:			PLL 0 delta sigma modulator feedback divider value low override register mode 1
Register Mnemonic:			CMN_PLL0_DSM_FBL_OVRD_M1
Register Address:			00_0_0000 : 0101 : 00110
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h010	**PLL feedback divider low override value** : When enabled by the PLL feedback divider override enable bit in the PLL 0 delta sigma modulator diagnostics register mode 0 on page 161, the value in this field will be used to override the value on the cmnda_pll0_fb_div_low signal.

Register Name:			PLL 0 spread spectrum control register 1 mode 1
Register Mnemonic:			CMN_PLL0_SS_CTRL1_M1
Register Address:			00_0_0000 : 0101 : 01000
Bits	**Type**	**Reset**	**Description**
15:2	R/O	14'h0	Reserved
1	R/W	1'h1	**Spread spectrum waveform generator disable**: Setting this bit to a 1'b1 will disable the spread spectrum waveform generator.
0	R/W	1'h0	**Spread spectrum enable during VCO calibration** : Setting this bit to a 1'b1 will enable the spread spectrum function while VCO calibration is taking place.

Register Name:			PLL 0 spread spectrum control register 2 mode 1
Register Mnemonic:			CMN_PLL0_SS_CTRL2_M1
Register Address:			00_0_0000 : 0101 : 01001
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:0	R/W	14'h0000	**Amplitude step size**: Value of the amplitude_step_size pin on the spread spectrum waveform generator. Note: This field should not be set to 0 when spread spectrum is enabled.

Register Name:			PLL 0 spread spectrum control register 3 mode 1
Register Mnemonic:			CMN_PLL0_SS_CTRL3_M1
Register Address:			00_0_0000 : 0101 : 01010
Bits	**Type**	**Reset**	**Description**
15:7	R/O	9'h0	Reserved
6:0	R/W	7'h00	**Number of steps**: Value of the num_steps pin on the spread spectrum waveform generator. Note: This field should not be set to 0 when spread spectrum is enabled.

Register Name:			PLL 0 spread spectrum control register 4 mode 1
Register Mnemonic:			CMN_PLL0_SS_CTRL4_M1
Register Address:			00_0_0000 : 0101 : 01011
Bits	**Type**	**Reset**	**Description**
15:7	R/O	9'h0	Reserved
6:0	R/W	7'h00	**Time step size**: Value for the time_step_size pin on the spread spectrum waveform generator. Note: This field should not be set to 0 when spread spectrum is enabled.

10.2.10 PLL 1 Controller Registers

Base address bits 15:5 :	00_0_0000 : 0110
Address bits 4:0	**Register name and location of detailed description**
00000	PLL 1 VCO calibration control register on page 165
00001	PLL 1 VCO calibration start point register on page 165
00010	PLL 1 VCO calibration timer control register on page 166
00011	PLL 1 VCO calibration override register on page 166
00100	PLL 1 VCO calibration initialization timer register on page 166
00101	PLL 1 VCO calibration iteration timer register on page 166
00110	PLL 1 VCO calibration reference clock timer start value register on page 167
00111	Reserved
01000	PLL 1 VCO calibration PLL clock counter start value register on page 167
01001 - 01111	Reserved
10000	PLL 1 feedback divider integer register mode 0 on page 168
10001	PLL 1 feedback divider fractional low register mode 0 on page 168
10010	PLL 1 feedback divider fractional high register mode 0 on page 168
10011	PLL 1 feedback divider high threshold register mode 0 on page 168
10100	PLL 1 delta sigma modulator diagnostics register mode 0 on page 169
10101	PLL 1 delta sigma modulator feedback divider value high override register mode 0 on page 169
10110	PLL 1 delta sigma modulator feedback divider value low override register mode 0 on page 169
10111	Reserved
11000	PLL 1 spread spectrum control register 1 mode 0 on page 169
11001	PLL 1 spread spectrum control register 2 mode 0 on page 169
11010	PLL 1 spread spectrum control register 3 mode 0 on page 170
11011	PLL 1 spread spectrum control register 4 mode 0 on page 170
11100	PLL 1 lock reference counter start value register on page 167
11101	PLL 1 lock reference counter idle value register on page 167
11110	PLL 1 lock PLL counter start value register on page 168
11111	PLL 1 lock PLL counter threshold value register on page 168

Table 97: PLL 1 controller registers address space (slave 0)

Base address bits 15:5 :	00_0_0000 : 0111
Address bits 4:0	**Register name and location of detailed description**
00000 - 11111	Reserved

Table 98: PLL 1 controller registers address space (slave 1)

Register Name:			PLL 1 VCO calibration control register
Register Mnemonic:			CMN_PLL1_VCOCAL_CTRL
Register Address:			00_0_0000 : 0110 : 00000
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Start VCO calibration**: Activating (1'b1) this bit will start a VCO calibration process. This bit must remain active until the VCO calibration process is complete (as indicated by the VCO calibration process done bit in this register). To start another VCO calibration process, the VCO calibration process done bit must have gone inactive from any prior calibration process. **Note: This bit is intended to be for diagnostics purposes only. This calibration process is automatically activated internally by the PLL control state machine. When using this bit, the user must wait until after the internally activated process completes.**
14	R/O	1'h0	**VCO calibration process done**: This bit will be set to 1'b1 when the VCO calibration process is complete. It will be cleared by the deactivation of the Start VCO calibration bit in this register. **Note: This bit is intended to be for diagnostics purposes only.** **Note: This bit is not likely to be observed as being set after internal automatic calibration is complete, because the internally generated run signal will be driven inactive immediately after the done signal is activated, and therefore the internal done signal will be cleared.**
13:8	R/O	6'h0	Reserved
7:0	R/O	8'h00	**VCO calibration code**: This is the calibration code that was determined by the VCO calibration process. This signal is valid when the VCO calibration process is complete. The values of this field correspond to different frequency bands the VCO will operate in. The frequency bands are controlled by the number of capacitors that are switched in the VCO analog circuit. This field specifies the number of capacitors that are switched in. The following are the values for this code: • 8'b00000000: 0 Capacitors activated • 8'b00000001: 1 Capacitor activated • 8'b00000010: 2 Capacitors activated • 8'b00000011: 3 Capacitors activated • ... • 8'b11111110: 254 Capacitors activated • 8'b11111111: 255 Capacitors activated **Note: This register is intended to be for diagnostics purposes only.** **Note: The reset value for this field is with the PLL controller power island switched off. In cases where this power island is switched on, the reset value be 8'h28.**

Register Name:			PLL 1 VCO calibration start point register
Register Mnemonic:			CMN_PLL1_VCOCAL_START
Register Address:			00_0_0000 : 0110 : 00001
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:12	R/W	3'h2	**VCO calibration initial step size control**: This field specifies the initial step size for the VCO calibration state machine. The following are the values that can be used in this field, and the corresponding step sizes. • 3'b000 : 1 • 3'b001 : 2 • 3'b010 : 4 • 3'b011 : 8 • 3'b100 : 16 • 3'b101 - 3'b111 : Reserved **Note: This field is intended to be for diagnostics purposes only.**
11:8	R/O	4'h0	Reserved
7:0	R/W	8'h28	**VCO calibration code starting point value**: This field specifies the starting VCO code that is used by the VCO calibration state machine. The purpose of this value is such that the VCO calibration process starts at a point that is, on average, relatively close to the final calibration point. This allows the calibration time to be reduced, on average. The encoding of this field is the same as what is described in the VCO calibration control register. **Note: This field is intended to be for diagnostics purposes only.** **Note : This field must be set to a value that is always at least one step size away from the minimum and maximum calibration codes, and is a function of the step size specified in this register. For example, if the initial step size is 4, this value must be at least 4 greater than the minimum calibration code or 4 less than the maximum calibration code.**

Register Name:			PLL 1 VCO calibration timer control register
Register Mnemonic:			CMN_PLL1_VCOCAL_TCTRL
Register Address:			00_0_0000 : 0110 : 00010
Bits	**Type**	**Reset**	**Description**
15:3	R/O	13'h0	Reserved
2:0	R/W	3'h4	**VCO calibration initial time scale control**: This field specifies the calibration start time scaling factor applied to the VCO calibration when running the initial step size for the calibration code if not set to 1. Setting this value to a value other than 1 will reduce the calibration measurement time by the amount specified below. Then when the final calibration steps are made the full calibration time will be used to get the appropriate amount of resolution. • 3'b000 : Div 1 • 3'b001 : Div 2 • 3'b010 : Div 4 • 3'b011 : Div 8 • 3'b100 : Div 16 • 3'b101 : Div 32 • 3'b110 : Div 64 • 3'b111 : Div 128

Register Name:			PLL 1 VCO calibration override register
Register Mnemonic:			CMN_PLL1_VCOCAL_OVRD
Register Address:			00_0_0000 : 0110 : 00011
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**VCO calibration code override enable**: Activating (1'b1) this bit allows the VCO code determined during the automatic VCO calibration process to be overridden by the value driven by the VCO calibration code override value field in this register. **Note: This bit is intended to be for diagnostics purposes only.** **Note: This bit should not be activated while a calibration is currently in progress.**
14:8	R/O	7'h0	Reserved
7:0	R/W	8'h00	**VCO calibration code override value**: This field is used to override the VCO code determined during the automatic VCO calibration process. The VCO code driven on this field is valid when the VCO calibration code override enable bit in this register is active. The encoding of this field is the same as what is described in the VCO calibration control register. Note: This field is intended to be for diagnostics purposes only. **Note:** The value of this field must not be changed while the calibration function is running.

Register Name:			PLL 1 VCO calibration initialization timer register
Register Mnemonic:			CMN_PLL1_VCOCAL_INIT_TMR
Register Address:			00_0_0000 : 0110 : 00100
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'h0	Reserved
13:0	R/W	14'h03E8	**Initialization wait timer value**: This is the number of clocks to wait between when the analog VCO calibration circuits are enabled, and when the first calibration code is driven to the analog. Note: The reset value for this field corresponds to a time of 10 uSec.

Register Name:			PLL 1 VCO calibration iteration timer register
Register Mnemonic:			CMN_PLL1_VCOCAL_ITER_TMR
Register Address:			00_0_0000 : 0110 : 00101
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'h0	Reserved
13:0	R/W	14'h0010	**Iteration wait timer value**: This is the number of clocks to wait between when a calibration code is driven to the analog, and when the clock rates are measured.

Register Name:			PLL 1 VCO calibration reference clock timer start value register
Register Mnemonic:			CMN_PLL1_VCOCAL_REFTIM_START
Register Address:			00_0_0000 : 0110 : 00110
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'h0	Reserved
13:0	R/W	14'h0C5F	**PLL VCO calibration reference clock timer start value** : This is the value that is loaded into the reference clock timer as the starting point for that timer, when running VCO calibration. The value in this field must correspond to a time that is approximately one full spread spectrum cycle long (31.67 uSec) **Note: The actual number of clocks counted is one larger than what is specified by this field.**

Register Name:			PLL 1 VCO calibration PLL clock counter start value register
Register Mnemonic:			CMN_PLL1_VCOCAL_PLLCNT_START
Register Address:			00_0_0000 : 0110 : 01000
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'h0	Reserved
13:0	R/W	14'h0C5F	**PLL VCO calibration PLL clock counter start value** : This is the value that is loaded into the PLL clock counter as the starting point for that counter, when running VCO calibration. When programming this register, it is important to understand how the frequency of the cmnda_pll0_fb_divider_clk is calculated relative to the reference clock. This is described in section 10.4 "Dual VCO PLL" on page 430. **Note: The actual number of clocks counted is one larger than what is specified by this field.**

Register Name:			PLL 1 lock reference counter start value register
Register Mnemonic:			CMN_PLL1_LOCK_REFCNT_START
Register Address:			00_0_0000 : 0110 : 11100
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:0	R/W	12'h0C8	**PLL lock reference counter start value** : This is the value that is loaded into the PLL lock detect reference counter as the starting point for that counter, when checking for PLL lock. Note: The value required here is a function of the PLL feedback divider clock mode and the reference clock frequency. Overview information in regards to this can be found in section 4.2 "Reference Clock Related Clocking and Reset" on page 297 and section 4.9 "Reference Clock Related Clock Rate Summary" on page 328. The following bullet points specify the required time the value in this register is required to generate. By default, the value in this register is set for FBR mode with a 100 MHz reference clock. • FBR mode and reference clock frequency greater than or equal to 100 MHz: 2 uSec • FBR mode and reference clock frequency less than 100 MHz: 8 uSec • DR mode: 800 nSec

Register Name:			PLL 1 lock reference counter idle value register
Register Mnemonic:			CMN_PLL1_LOCK_REFCNT_IDLE
Register Address:			00_0_0000 : 0110 : 11101
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:0	R/W	12'h004	**PLL lock reference counter idle value** : This is the value used by the PLL lock detection logic to specify the number of reference clocks between each phase of counting PLL clocks.

Register Name:			PLL 1 lock PLL counter start value register
Register Mnemonic:			CMN_PLL1_LOCK_PLLCNT_START
Register Address:			00_0_0000 : 0110 : 11110
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:0	R/W	12'h0C8	**PLL lock PLL counter start value** : This is the value that is loaded into the PLL lock detect PLL counter as the starting point for that counter, when checking for PLL lock. The value is the number of PLL feedback divider clock edges that are expected during the time specified by the PLL 0 lock reference counter start value register on page 159. When programming this register, it is important to understand how the frequency of the cmnda_pll0_fb_divider_clk is calculated relative to the reference clock. This is described in section 10.4 "Dual VCO PLL" on page 430.

Register Name:			PLL 1 lock PLL counter threshold value register
Register Mnemonic:			CMN_PLL1_LOCK_PLLCNT_THR
Register Address:			00_0_0000 : 0110 : 11111
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:0	R/W	12'h003	**PLL lock counter threshold value** : This is the value used by the PLL lock detection logic to determine if the PLL has locked. If the two counters in the PLL lock detection logic differ by less than this value, the PLL is considered locked.

Register Name:			PLL 1 feedback divider integer register mode 0
Register Mnemonic:			CMN_PLL1_INTDIV_M0
Register Address:			00_0_0000 : 0110 : 10000
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h000	**pll_fb_div_integer value**: Value of the pll_fb_div_integer signal.

Register Name:			PLL 1 feedback divider fractional low register mode 0
Register Mnemonic:			CMN_PLL1_FRACDIVL_M0
Register Address:			00_0_0000 : 0110 : 10001
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h0000	**pll_fb_div_fractional**: Value of the pll_fb_div_fractional[15:0] signal.

Register Name:			PLL 1 feedback divider fractional high register mode 0
Register Mnemonic:			CMN_PLL1_FRACDIVH_M0
Register Address:			00_0_0000 : 0110 : 10010
Bits	**Type**	**Reset**	**Description**
15:3	R/O	13'h0	Reserved
2:0	R/W	3'h0	**pll_fb_div_fractional**: Value of the pll_fb_div_fractional[18:16] signal.

Register Name:			PLL 1 feedback divider high threshold register mode 0
Register Mnemonic:			CMN_PLL1_HIGH_THR_M0
Register Address:			00_0_0000 : 0110 : 10011
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h000	**pll_fb_div_high_theshold**: Value of the pll_fb_div_high_threshold signal. The value of this register should be 2/3 the value of the combined integer and fractional divider values, and rounded up to the next even number.

Register Name:			PLL 1 delta sigma modulator diagnostics register mode 0
Register Mnemonic:			CMN_PLL1_DSM_DIAG_M0
Register Address:			00_0_0000 : 0110 : 10100
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Delta sigma bypass enable**: When set to 1'b1, the delta sigma modulator will be bypassed, and the output will be the value specified for the internal pll_fb_div_integer signal.
14	R/W	1'h1	**PLL feedback divider override enable** : When active (1'b1), the feedback divider low and high override values in the PLL 1 delta sigma modulator feedback divider value high override register mode 0 on page 169 and PLL 1 delta sigma modulator feedback divider value low override register mode 0 on page 169 registers will be used to override the feedback divider values driven to the analog.
13:4	R/O	10'h0	Reserved
3:0	R/W	4'h4	**PLL feedback divider latency adjustment**: This signal specifies a value to be subtracted from the feedback divider settings before they are output on the cmnda_pll1_fb_div_high and cmnda_pll1_fb_div_low signals.

Register Name:			PLL 1 delta sigma modulator feedback divider value high override register mode 0
Register Mnemonic:			CMN_PLL1_DSM_FBH_OVRD_M0
Register Address:			00_0_0000 : 0110 : 10101
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h010	**PLL feedback divider high override value** : When enabled by the PLL feedback divider override enable bit in the PLL 1 delta sigma modulator diagnostics register mode 0 on page 169, the value in this field will be used to override the value on the cmnda_pll1_fb_div_high signal. Note: The value of this register should be 2/3 the value of the combined PLL feedback divider high and low values, and rounded up to the next even number.

Register Name:			PLL 1 delta sigma modulator feedback divider value low override register mode 0
Register Mnemonic:			CMN_PLL1_DSM_FBL_OVRD_M0
Register Address:			00_0_0000 : 0110 : 10110
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h010	**PLL feedback divider low override value** : When enabled by the PLL feedback divider override enable bit in the PLL 1 delta sigma modulator diagnostics register mode 0 on page 169, the value in this field will be used to override the value on the cmnda_pll1_fb_div_low signal.

Register Name:			PLL 1 spread spectrum control register 1 mode 0
Register Mnemonic:			CMN_PLL1_SS_CTRL1_M0
Register Address:			00_0_0000 : 0110 : 11000
Bits	**Type**	**Reset**	**Description**
15:2	R/O	14'h0	Reserved
1	R/W	1'h1	**Spread spectrum waveform generator disable**: Setting this bit to a 1'b1 will disable the spread spectrum waveform generator.
0	R/W	1'h0	**Spread spectrum enable during VCO calibration** : Setting this bit to a 1'b1 will enable the spread spectrum function while VCO calibration is taking place.

Register Name:			PLL 1 spread spectrum control register 2 mode 0
Register Mnemonic:			CMN_PLL1_SS_CTRL2_M0
Register Address:			00_0_0000 : 0110 : 11001
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:0	R/W	14'h0000	**Amplitude step size**: Value of the amplitude_step_size pin on the spread spectrum waveform generator. Note: This field should not be set to 0 when spread spectrum is enabled.

Register Name:			PLL 1 spread spectrum control register 3 mode 0
Register Mnemonic:			CMN_PLL1_SS_CTRL3_M0
Register Address:			00_0_0000 : 0110 : 11010
Bits	**Type**	**Reset**	**Description**
15:7	R/O	1'h0	Reserved
6:0	R/W	7'h00	**Number of steps**: Value of the num_steps pin on the spread spectrum waveform generator. Note: This field should not be set to 0 when spread spectrum is enabled.

Register Name:			PLL 1 spread spectrum control register 4 mode 0
Register Mnemonic:			CMN_PLL1_SS_CTRL4_M0
Register Address:			00_0_0000 : 0110 : 11011
Bits	**Type**	**Reset**	**Description**
15:7	R/O	9'h0	Reserved
6:0	R/W	7'h00	**Time step size**: Value for the time_step_size pin on the spread spectrum waveform generator. Note: This field should not be set to 0 when spread spectrum is enabled.

10.2.11 Transmitter Resistor Calibration Controller Registers

Base address bits 15:5 :	00_0_0000 : 1000
Address bits 4:0	**Register name and location of detailed description**
00000	TX pull-up resistor calibration control register on page 171
00001	TX pull-up resistor calibration override register on page 172
00010	TX pull-up resistor calibration start register on page 172
00011	TX pull-up resistor calibration tune register on page 172
00100	TX pull-up resistor calibration initialization timer register on page 173
00101	TX pull-up resistor calibration iteration timer register on page 173
00110 - 00111	Reserved

Table 99: Transmitter pull up resistor calibration controller registers address space

Base address bits 15:5 :	00_0_0000 : 1000
Address bits 4:0	**Register name and location of detailed description**
01000	TX pull-down resistor calibration control register on page 174
01001	TX pull-down resistor calibration override register on page 175
01010	TX pull-down resistor calibration start register on page 175
01011	TX pull-down resistor calibration tune register on page 175
01100	TX pull-down resistor calibration initialization timer register on page 176
01101	TX pull-down resistor calibration iteration timer register on page 176
01110 - 01111	Reserved

Table 100: Transmitter pull down resistor calibration controller registers address space

Register Name:			TX pull-up resistor calibration control register
Register Mnemonic:			CMN_TXPUCAL_CTRL
Register Address:			00_0_0000 : 1000 : 00000
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Start resistor calibration**: Activating (1'b1) this bit will start the resistor calibration process. This signal must remain active until the resistor calibration process is complete. To start another resistor calibration process, this register must first be set inactive (1'b0) until the resistor calibration process done bit in this register is cleared. **Note: This signal is intended to be for diagnostics purposes only. This calibration process is automatically activated internally by the startup state machine. When using this bit, the user must wait until after the internally activated process completes.**
14	R/O	1'h0	**Resistor calibration process done**: This bit will be set to 1'b1 when the resistor calibration process is complete. It will be cleared by cmn_reset_n, or by the deactivation of the start resistor calibration bit in this register after calibration is complete. **Note: This bit is intended to be for diagnostics purposes only.** **Note: This bit is not likely to be observed as being set after internal automatic calibration is complete, because the internally generated run signal will be driven inactive immediately after the done signal is activated, and therefore the internal done signal will be cleared.** **Note: Three cmn_ref_clk cycles are required after the start of the resistor calibration process to clear this signal.**
13	R/O	1'h0	**No analog calibration response** : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the analog has not responded indicating that a valid calibration value has been reached. Note: Due to the nature of the calibration process, it is not possible to determine if the analog responded or not for the lowest code of 7'b0000000. The reason for this is, when sequencing to lower codes, the calibration function is looking for the analog response to transition from 1'b1 to 1'b0. If the lowest calibration code is the correct code, the analog will respond with 1'b1, and the algorithm can't sequence to a lower code to look for the transition from 1'b1 to 1'b0. Similarly, if the analog is not responding, it is not possible to sequence to a lower code to detect this. Therefore, even when the calibration function selects this calibration code as valid, this bit will be set.
12	R/O	1'h0	**Current analog comparator response**: This is the current state of the analog comparator response signal (cmnda_rescal_comp_tx_useg). This signal is not synchronized, and is provided for diagnostic purposes only.
11:7	R/O	5'h0	Reserved
6:0	R/O	7'h00	**Resistor calibration code**: This is the calibration code that was determined by the resistor calibration process. The following are the values for the code. • 7'b0000000: No resistors active. • 7'b0000001: 1 resistor active. • 7'b0000010: 2 resistors active. • 7'b0000011: 3 resistors active. • 7'b0000100: 4 resistors active. • ... • 7'b0111100: 60 resistors active. Note: The most significant bit is the calibration code sign bit, and is always 1'b0 in this application. **Note: The reset value for this field is with the common calibration power island switched off. In cases where this power island is switched on, the reset value be 7'b0101101.**

Register Name:			TX pull-up resistor calibration override register
Register Mnemonic:			CMN_TXPUCAL_OVRD
Register Address:			00_0_0000 : 1000 : 00001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Resistor code override enable**: Activation (1'b1) of this register bit allows the resistor codes determined during the automatic resistor calibration process to be overridden. The override value is specified using the resistor code override value field of this register. **Note:** The value of this field must not be changed while the calibration function is running.
14	R/W	1'h0	**Analog calibration enable override**: Activation (1'b1) of this register bit will force the analog calibration circuits to be enabled by activating the cmnda_rescal_en_tx_useg enable and the cmnda_rescal_clk_tx-_useg clock. **Note:** The value of this field must not be changed while the calibration function is running.
13:7	R/O	7'h0	Reserved
6:0	R/W	7'h00	**Resistor code override value**: These bits are used to override the resistor code determined during the automatic resistor calibration process. The resistor code written to these bits is valid when the resistor code override enable bit in this register is active. The following are the values for the code. • 7'b0000000: No resistors active. • 7'b0000001: 1 resistor active. • 7'b0000010: 2 resistors active. • 7'b0000011: 3 resistors active. • 7'b0000100: 4 resistors active. • ... • 7'b0111100: 60 resistors active. Note: The most significant bit is the calibration code sign bit, and is always 1'b0 in this application. **Note:** The value of this field must not be changed while the calibration function is running.

Register Name:			TX pull-up resistor calibration start register
Register Mnemonic:			CMN_TXPUCAL_START
Register Address:			00_0_0000 : 1000 : 00010
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Resistor calibration direction**: This controls the direction that the automatic calibration process steps the calibration codes in. • 1'b0 : From 7'b0000000 to 7'b0111100. • 1'b1 : From 7'b0111100 to 7'b0000000.
14:7	R/O	8'h0	Reserved
6:0	R/W	7'h2D	**Start resistor calibration code**: This is the calibration code that the resistor calibration process starts with when automatic calibration is run. The following are the values for the code. • 7'b0000000: No resistors active. • 7'b0000001: 1 resistor active. • 7'b0000010: 2 resistors active. • 7'b0000011: 3 resistors active. • 7'b0000100: 4 resistors active. • ... • 7'b0111100: 60 resistors active. Note: The most significant bit is the calibration code sign bit, and is always 1'b0 in this application.

Register Name:			TX pull-up resistor calibration tune register
Register Mnemonic:			CMN_TXPUCAL_TUNE
Register Address:			00_0_0000 : 1000 : 00011
Bits	**Type**	**Reset**	**Description**
15:7	R/O	9'h0	Reserved
6:0	R/W	7'h00	**Resistor calibration tune value**: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note: This value is a twos complement value, so the calibrated code can be increased or decreased.

Register Name:			TX pull-up resistor calibration initialization timer register
Register Mnemonic:			CMN_TXPUCAL_INIT_TMR
Register Address:			00_0_0000 : 1000 : 00100
Bits	**Type**	**Reset**	**Description**
15:7	R/O	9'h0	Reserved
6:0	R/W	7'h1E	**Initialization wait timer value**: This is the number of cmn_ref_clk clocks to wait between when the analog resistor calibration circuits are enabled, and when the first resistor selection values are placed on the resistor selection bus, going to the analog. **Note: This results in the minimum number of cmn_ref_clk clocks required to wait 1.2 uSec.** **Note: This should never be set to a value of less than 1.**

Register Name:			TX pull-up resistor calibration iteration timer register
Register Mnemonic:			CMN_TXPUCAL_ITER_TMR
Register Address:			00_0_0000 : 1000 : 00101
Bits	**Type**	**Reset**	**Description**
15:7	R/O	9'h0	Reserved
6:0	R/W	7'h06	**Iteration wait timer value**: This is the number of cmn_ref_clk clocks to wait between when a value is placed on the resistor selection bus going to the analog, and when the comparator value coming from the analog circuits can be checked. **Note: This results in the minimum number of cmn_ref_clk clocks required to wait 240 nSec.** **Note: This should never be set to a value of less than 3.**

Register Name:			TX pull-down resistor calibration control register
Register Mnemonic:			CMN_TXPDCAL_CTRL
Register Address:			00_0_0000 : 1000 : 01000
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Start resistor calibration**: Activating (1'b1) this bit will start the resistor calibration process. This signal must remain active until the resistor calibration process is complete. To start another resistor calibration process, this register must first be set inactive (1'b0) until the resistor calibration process done bit in this register is cleared. **Note: This signal is intended to be for diagnostics purposes only. This calibration process is automatically activated internally by the startup state machine. When using this bit, the user must wait until after the internally activated process completes.**
14	R/O	1'h0	**Resistor calibration process done**: This bit will be set to 1'b1 when the resistor calibration process is complete. It will be cleared by cmn_reset_n, or by the deactivation of the start resistor calibration bit in this register after calibration is complete. **Note: This bit is intended to be for diagnostics purposes only.** **Note: This bit is not likely to be observed as being set after internal automatic calibration is complete, because the internally generated run signal will be driven inactive immediately after the done signal is activated, and therefore the internal done signal will be cleared.** **Note: Three cmn_ref_clk cycles are required after the start of the resistor calibration process to clear this signal.**
13	R/O	1'h0	**No analog calibration response** : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the analog has not responded indicating that a valid calibration value has been reached. Note: Due to the nature of the calibration process, it is not possible to determine if the analog responded or not for the lowest code of 7'b0000000. The reason for this is, when sequencing to lower codes, the calibration function is looking for the analog response to transition from 1'b1 to 1'b0. If the lowest calibration code is the correct code, the analog will respond with 1'b1, and the algorithm can't sequence to a lower code to look for the transition from 1'b1 to 1'b0. Similarly, if the analog is not responding, it is not possible to sequence to a lower code to detect this. Therefore, even when the calibration function selects this calibration code as valid, this bit will be set.
12	R/O	1'h0	**Current analog comparator response**: This is the current state of the analog comparator response signal (cmnda_rescal_comp_tx_dseg). This signal is not synchronized, and is provided for diagnostic purposes only.
11:7	R/O	5'h0	Reserved
6:0	R/O	7'h00	**Resistor calibration code**: This is the calibration code that was determined by the resistor calibration process. The following are the values for the code. • 7'b0000000: No resistors active. • 7'b0000001: 1 resistor active. • 7'b0000010: 2 resistors active. • 7'b0000011: 3 resistors active. • 7'b0000100: 4 resistors active. • ... • 7'b0111100: 60 resistors active. Note: The most significant bit is the calibration code sign bit, and is always 1'b0 in this application. **Note: The reset value for this field is with the common calibration power island switched off. In cases where this power island is switched on, the reset value be 7'b0101101.**

Register Name:			TX pull-down resistor calibration override register
Register Mnemonic:			CMN_TXPDCAL_OVRD
Register Address:			00_0_0000 : 1000 : 01001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Resistor code override enable**: Activation (1'b1) of this register bit allows the resistor codes determined during the automatic resistor calibration process to be overridden. The override value is specified using the resistor code override value field of this register. **Note:** The value of this field must not be changed while the calibration function is running.
14	R/W	1'h0	**Analog calibration enable override**: Activation (1'b1) of this register bit will force the analog calibration circuits to be enabled by activating the cmnda_rescal_en_tx_dseg enable and the cmnda_rescal_clk_tx-_dseg clock. **Note:** The value of this field must not be changed while the calibration function is running.
13:7	R/O	7'h0	Reserved
6:0	R/W	7'h00	**Resistor code override value**: These bits are used to override the resistor code determined during the automatic resistor calibration process. The resistor code written to these bits is valid when the resistor code override enable bit in this register is active. The following are the values for the code. • 7'b0000000: No resistors active. • 7'b0000001: 1 resistor active. • 7'b0000010: 2 resistors active. • 7'b0000011: 3 resistors active. • 7'b0000100: 4 resistors active. • ... • 7'b0111100: 60 resistors active. Note: The most significant bit is the calibration code sign bit, and is always 1'b0 in this application. **Note:** The value of this field must not be changed while the calibration function is running.

Register Name:			TX pull-down resistor calibration start register
Register Mnemonic:			CMN_TXPDCAL_START
Register Address:			00_0_0000 : 1000 : 01010
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h1	**Resistor calibration direction**: This controls the direction that the automatic calibration process steps the calibration codes in. • 1'b0 : From 7'b0000000 to 7'b0111100. • 1'b1 : From 7'b0111100 to 7'b0000000.
14:7	R/O	8'h0	Reserved
6:0	R/W	7'h2D	**Start resistor calibration code**: This is the calibration code that the resistor calibration process starts with when automatic calibration is run. The following are the values for the code. • 7'b0000000: No resistors active. • 7'b0000001: 1 resistor active. • 7'b0000010: 2 resistors active. • 7'b0000011: 3 resistors active. • 7'b0000100: 4 resistors active. • ... • 7'b0111100: 60 resistors active. Note: The most significant bit is the calibration code sign bit, and is always 1'b0 in this application.

Register Name:			TX pull-down resistor calibration tune register
Register Mnemonic:			CMN_TXPDCAL_TUNE
Register Address:			00_0_0000 : 1000 : 01011
Bits	**Type**	**Reset**	**Description**
15:7	R/O	9'h0	Reserved
6:0	R/W	7'h00	**Resistor calibration tune value**: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note: This value is a twos complement value, so the calibrated code can be increased or decreased.

Register Name:			TX pull-down resistor calibration initialization timer register
Register Mnemonic:			CMN_TXPDCAL_INIT_TMR
Register Address:			00_0_0000 : 1000 : 01100
Bits	**Type**	**Reset**	**Description**
15:7	R/O	9'h0	Reserved
6:0	R/W	7'h1E	**Initialization wait timer value**: This is the number of cmn_ref_clk clocks to wait between when the analog resistor calibration circuits are enabled, and when the first resistor selection values are placed on the resistor selection bus, going to the analog. **Note: This results in the minimum number of cmn_ref_clk clocks required to wait 1.2 uSec.** **Note: This should never be set to a value of less than 1.**

Register Name:			TX pull-down resistor calibration iteration timer register
Register Mnemonic:			CMN_TXPDCAL_ITER_TMR
Register Address:			00_0_0000 : 1000 : 01101
Bits	**Type**	**Reset**	**Description**
15:7	R/O	9'h0	Reserved
6:0	R/W	7'h06	**Iteration wait timer value**: This is the number of cmn_ref_clk clocks to wait between when a value is placed on the resistor selection bus going to the analog, and when the comparator value coming from the analog circuits can be checked. **Note: This results in the minimum number of cmn_ref_clk clocks required to wait 240 nSec.** **Note: This should never be set to a value of less than 3.**

10.2.12 Receiver Resistor Calibration Controller Registers

Base address bits 15:5 :	**00_0_0000 : 1000**
Address bits 4:0	**Register name and location of detailed description**
10000	RX resistor calibration control register on page 177
10001	RX resistor calibration override register on page 178
10010	RX resistor calibration start register on page 178
10011	RX resistor calibration tune register on page 178
10100	RX resistor calibration initialization timer register on page 179
10101	RX resistor calibration iteration timer register on page 179
10110 - 11111	Reserved

Table 101: Receiver resistor calibration controller registers address space

Register Name:			RX resistor calibration control register
Register Mnemonic:			CMN_RXCAL_CTRL
Register Address:			00_0_0000 : 1000 : 10000
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Start resistor calibration**: Activating (1'b1) this bit will start the resistor calibration process. This signal must remain active until the resistor calibration process is complete. To start another resistor calibration process, this register must first be set inactive (1'b0) until the resistor calibration process done bit in this register is cleared. **Note: This signal is intended to be for diagnostics purposes only. This calibration process is automatically activated internally by the startup state machine. When using this bit, the user must wait until after the internally activated process completes.**
14	R/O	1'h0	**Resistor calibration process done**: This bit will be set to 1'b1 when the resistor calibration process is complete. It will be cleared by cmn_reset_n, or by the deactivation of the start resistor calibration bit in this register after calibration is complete. **Note: This bit is intended to be for diagnostics purposes only.** **Note: This bit is not likely to be observed as being set after internal automatic calibration is complete, because the internally generated run signal will be driven inactive immediately after the done signal is activated, and therefore the internal done signal will be cleared.** **Note: Three cmn_ref_clk cycles are required after the start of the resistor calibration process to clear this signal.**
13	R/O	1'h0	**No analog calibration response** : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the analog has not responded indicating that a valid calibration value has been reached. Note: Due to the nature of the calibration process, it is not possible to determine if the analog responded or not for the lowest code of 5'b00000. The reason for this is, when sequencing to lower codes, the calibration function is looking for the analog response to transition from 1'b1 to 1'b0. If the lowest calibration code is the correct code, the analog will respond with 1'b1, and the algorithm can't sequence to a lower code to look for the transition from 1'b1 to 1'b0. Similarly, if the analog is not responding, it is not possible to sequence to a lower code to detect this. Therefore, even when the calibration function selects this calibration code as valid, this bit will be set.
12	R/O	1'h0	**Current analog comparator response**: This is the current state of the analog comparator response signal (cmnda_rescal_comp_rx). This signal is not synchronized, and is provided for diagnostic purposes only.
11:5	R/O	7'h0	Reserved
4:0	R/O	5'h00	**Resistor calibration code**: This is the calibration code that was determined by the resistor calibration process. The following are the values for the code: • 5'b00000: No resistors active • 5'b00001: 1 resistor active • 5'b00010: 2 resistors active • 5'b00011: 3 resistors active • ... • 5'b01110: 14 resistors active. • 5'b01111: 15 resistors active. Note: The most significant bit is the calibration code sign bit, and is always 1'b0 in this application.

Register Name:			RX resistor calibration override register
Register Mnemonic:			CMN_RXCAL_OVRD
Register Address:			00_0_0000 : 1000 : 10001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Resistor code override enable**: Activation (1'b1) of this register bit allows the resistor codes determined during the automatic resistor calibration process to be overridden. The override value is specified using the resistor code override value field of this register. **Note:** The value of this field must not be changed while the calibration function is running.
14	R/W	1'h0	**Analog calibration enable override**: Activation (1'b1) of this register bit will force the analog calibration circuits to be enabled by activating the cmnda_rescal_en_rx enable and the cmnda_rescal_clk_rx clock. **Note:** The value of this field must not be changed while the calibration function is running.
13:5	R/O	9'h0	Reserved
4:0	R/W	5'h00	**Resistor code override value**: These bits are used to override the resistor code determined during the automatic resistor calibration process. The resistor code written to these bits is valid when the resistor code override enable bit in this register is active. The following are the values for the code: • 5'b00000: No resistors active • 5'b00001: 1 resistor active • 5'b00010: 2 resistors active • 5'b00011: 3 resistors active • ... • 5'b01110: 14 resistors active. • 5'b01111: 15 resistors active. Note: The most significant bit is the calibration code sign bit, and is always 1'b0 in this application. **Note:** The value of this field must not be changed while the calibration function is running.

Register Name:			RX resistor calibration start register
Register Mnemonic:			CMN_RXCAL_START
Register Address:			00_0_0000 : 1000 : 10010
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Resistor calibration direction**: This controls the direction that the automatic calibration process steps the calibration codes in. • 1'b0 : From 4'b0000 to 4'b0111. • 1'b1 : From 4'b0111 to 4'b0000.
14:5	R/O	10'h0	Reserved
4:0	R/W	5'h08	**Start resistor calibration code**: This is the calibration code that the resistor calibration process starts with when automatic calibration is run. The following are the values for the code. • 5'b00000: No resistors active • 5'b00001: 1 resistor active • 5'b00010: 2 resistors active • 5'b00011: 3 resistors active • ... • 5'b01110: 14 resistors active. • 5'b01111: 15 resistors active. Note: The most significant bit is the calibration code sign bit, and is always 1'b0 in this application.

Register Name:			RX resistor calibration tune register
Register Mnemonic:			CMN_RXCAL_TUNE
Register Address:			00_0_0000 : 1000 : 10011
Bits	**Type**	**Reset**	**Description**
15:5	R/O	11'h0	Reserved
4:0	R/W	5'h00	**Resistor calibration tune value**: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note: This value is a twos complement value, so the calibrated code can be increased or decreased.

Register Name:			RX resistor calibration initialization timer register
Register Mnemonic:			CMN_RXCAL_INIT_TMR
Register Address:			00_0_0000 : 1000 : 10100
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:0	R/W	12'h2EE	**Initialization wait timer value**: This is the number of cmn_ref_clk clocks to wait between when the analog resistor calibration circuits are enabled, and when the first resistor selection values are placed on the resistor selection bus, going to the analog. **Note: This results in the minimum number of cmn_ref_clk clocks required to wait 30 uSec.** **Note: This should never be set to a value of less than 1.**

Register Name:			RX resistor calibration iteration timer register
Register Mnemonic:			CMN_RXCAL_ITER_TMR
Register Address:			00_0_0000 : 1000 : 10101
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:0	R/W	12'h006	**Iteration wait timer value**: This is the number of cmn_ref_clk clocks to wait between when a value is placed on the resistor selection bus going to the analog, and when the comparator value coming from the analog circuits can be checked. **Note: This results in the minimum number of cmn_ref_clk clocks required to wait 240 nSec.** **Note: This should never be set to a value of less than 3.**

10.2.13 Signal Detect Clock Calibration Registers

Base address bits 15:5 :	00_0_0000 : 1001
Address bits 4:0	**Register name and location of detailed description**
00000	Signal detect clock calibration control register on page 180
00001	Signal detect clock calibration start point register on page 181
00010	Signal detect clock calibration timer control register on page 181
00011	Signal detect clock calibration override register on page 182
00100	Signal detect clock calibration initialization timer register on page 182
00101	Signal detect clock calibration iteration timer register on page 182
00110	Signal detect clock calibration reference clock timer start value register on page 182
00111	Reserved
01000	Signal detect clock calibration PLL clock counter start value register on page 183
01001 - 11111	Reserved

Table 102: Signal detect clock calibration registers address space

Register Name:			Signal detect clock calibration control register
Register Mnemonic:			CMN_SD_CAL_CTRL
Register Address:			00_0_0000 : 1001 : 00000
Bits	**Type**	**Reset**	**Description**
15	R/W	1’h0	**Start calibration**: Activating (1’b1) this bit will start a calibration process. This bit must remain active until the calibration process is complete (as indicated by the calibration process done bit in this register). To start another calibration process, the calibration process done bit must have gone inactive from any prior calibration process. **Note: This bit is intended to be for diagnostics purposes only. This calibration process is automatically activated internally by the startup state machine. When using this bit, the user must wait until after the internally activated process completes.**
14	R/O	1’h0	**Calibration process done**: This bit will be set to 1’b1 when the calibration process is complete. It will be cleared by the deactivation of the Start calibration bit in this register. **Note: This bit is intended to be for diagnostics purposes only.** **Note: This bit is not likely to be observed as being set after internal automatic calibration is complete, because the internally generated run signal will be driven inactive immediately after the done signal is activated, and therefore the internal done signal will be cleared.**
13:5	R/O	9’h0	Reserved
4:0	R/O	5’h00	**Calibration code**: This is the calibration code that was determined by the calibration process. This signal is valid when the calibration process is complete. The values of this field correspond to different frequency bands the will operate in. The frequency bands are controlled by the number of capacitors that are switched in the analog circuit. This field specifies the number of capacitors that are switched in. The following are the values for this code: • 5’b00000: 0 Capacitors activated • 5’b00001: 1 Capacitor activated • 5’b00010: 2 Capacitors activated • 5’b00011: 3 Capacitors activated • ... • 5’b11110: 30 Capacitors activated • 5’b11111: 31 Capacitors activated **Note: This signal is intended to be for diagnostics purposes only.** **Note: The reset value for this field is with the common calibration power island switched off. In cases where this power island is switched on, the reset value be 5’b11110**

Register Name:			Signal detect clock calibration start point register
Register Mnemonic:			CMN_SD_CAL_START
Register Address:			00_0_0000 : 1001 : 00001
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:12	R/W	3'h1	**Calibration initial step size control**: This field specifies the initial step size for the calibration state machine. The following are the values that can be used in this field, and the corresponding step sizes. • 3'b000 : 1 • 3'b001 : 2 • 3'b010 : 4 • 3'b011 : 8 • 3'b100 : 16 • 3'b101 - 3'b111 : Reserved
11:5	R/O	7'h0	Reserved
4:0	R/W	5'h1E	**Calibration code starting point value**: This field specifies the starting code that is used by the calibration state machine. The purpose of this value is such that the calibration process starts at a point that is, on average, relatively close to the final calibration point. This allows the calibration time to be reduced, on average. The following are the values for this code: • 5'b00000: 0 Capacitors activated • 5'b00001: 1 Capacitor activated • 5'b00010: 2 Capacitors activated • 5'b00011: 3 Capacitors activated • ... • 5'b11110: 30 Capacitors activated • 5'b11111: 31 Capacitors activated **Note: This field is intended to be for diagnostics purposes only.** **Note : This field must be set to a value that is always at least one step size away from the minimum and maximum calibration codes, and is a function of the step size specified in this register. For example, if the initial step size is 4, this value must be at least 4 greater than the minimum calibration code or 4 less than the maximum calibration code.** **Note: The reset value for this field is with the common calibration power island switched off. In cases where this power island is switched on, the reset value be 5'b11110**

Register Name:			Signal detect clock calibration timer control register
Register Mnemonic:			CMN_SD_CAL_TCTRL
Register Address:			00_0_0000 : 1001 : 00010
Bits	**Type**	**Reset**	**Description**
15:3	R/O	13'h0	Reserved
2:0	R/W	3'h1	**Calibration initial time scale control**: This field specifies the calibration start time scaling factor applied to the calibration when running the initial step size for the calibration code is not set to 1. Setting this value to a value other than 1 will reduce the calibration measurement time by the amount specified below. Then when the final calibration steps are made the full calibration time will be used to get the appropriate amount of resolution. • 3'b000 : Div 1 • 3'b001 : Div 2 • 3'b010 : Div 4 • 3'b011 : Div 8 • 3'b100 : Div 16 • 3'b101 : Div 32 • 3'b110 : Div 64 • 3'b111 : Div 128

Register Name:			Signal detect clock calibration override register
Register Mnemonic:			CMN_SD_CAL_OVRD
Register Address:			00_0_0000 : 1001 : 00011
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Calibration code override enable**: Activating (1'b1) this bit allows the code determined during the automatic calibration process to be overridden by the value driven by the calibration code override value field in this register. **Note: This bit is intended to be for diagnostics purposes only.** **Note: This bit should not be activated while a calibration is currently in progress.**
14:5	R/O	10'h0	Reserved
4:0	R/W	5'h00	**Calibration code override value**: This field is used to override the code determined during the automatic calibration process. The code driven on this field is valid when the calibration code override enable bit in this register is active. The following are the values for this code: • 5'b00000: 0 Capacitors activated • 5'b00001: 1 Capacitor activated • 5'b00010: 2 Capacitors activated • 5'b00011: 3 Capacitors activated • ... • 5'b11110: 30 Capacitors activated • 5'b11111: 31 Capacitors activated **Note: This field is intended to be for diagnostics purposes only.** Note: The value of this field must not be changed while the calibration function is running.

Register Name:			Signal detect clock calibration initialization timer register
Register Mnemonic:			CMN_SD_CAL_INIT_TMR
Register Address:			00_0_0000 : 1001 : 00100
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h06	**Initialization wait timer value**: This is the number of clocks to wait between when the analog calibration circuits are enabled, and when the first calibration code is driven to the analog. **Note: This results in the minimum number of cmn_ref_clk clocks required to wait 80 nSec for the analog plus four reference clocks for the digital clock gating to be enabled.**

Register Name:			Signal detect clock calibration iteration timer register
Register Mnemonic:			CMN_SD_CAL_ITER_TMR
Register Address:			00_0_0000 : 1001 : 00101
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h02	**Iteration wait timer value**: This is the number of clocks to wait between when a calibration code is driven to the analog, and when the clock rates are measured. **Note: This results in the minimum number of cmn_ref_clk clocks required to wait 80 nSec.**

Register Name:			Signal detect clock calibration reference clock timer start value register
Register Mnemonic:			CMN_SD_CAL_REFTIM_START
Register Address:			00_0_0000 : 1001 : 00110
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h0E	**Calibration reference clock timer start value** : This is the value that is loaded into the reference clock timer as the starting point for that timer, when running calibration. **Note: The actual number of clocks counted is one larger than what is specified by this field.**

Register Name:			Signal detect clock calibration PLL clock counter start value register
Register Mnemonic:			CMN_SD_CAL_PLLCNT_START
Register Address:			00_0_0000 : 1001 : 01000
Bits	**Type**	**Reset**	**Description**
15:10	R/O	5'h0	Reserved
9:0	R/W	10'h12B	**Calibration feedback clock counter start value** : This is the value that is loaded into the PLL clock counter as the starting point for that counter, when running calibration. **Note: The actual number of clocks counted is one larger than what is specified by this field.**

10.2.14 Common Clock Measurement Module Registers

Base address bits 15:5 :	00_0_0000 : 1100
Address bits 4:0	**Register name and location of detailed description**
00000	Clock frequency measurement control register on page 183
00001	Test clock selection register on page 184
00010	Reference clock timer value register on page 184
00011	Test clock counter value register on page 184
00100 - 11111	Reserved

Table 103: Common clock measurement module registers address space

Register Name:			Clock frequency measurement control register
Register Mnemonic:			CMN_CMSMT_CLK_FREQ_MSMT_CTRL
Register Address:			00_0_0000 : 1100 : 00000
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Run test clock measurement**: Activating (1'b1) this bit will run the test clock measurement process. This bit must remain active until the test clock measurement process is complete, as indicated by the test clock measurement done bit in this register. To start another measurement process, this bit must first be driven inactive then driven active again. **Note: Both of the clocks used to generate the ref_clk and test_clk clocks must be active prior to setting or clearing this bit.** **Note: The values in the Test clock selection register and Reference clock timer value register must be set prior to activating this bit.**
14	R/O	1'h0	**Test clock measurement done**: This bit will be set to 1'b1 when the test clock measurement process is complete. It will be cleared by the deactivation of the start test clock measurement bit in this register.
13:0	R/O	14'h0	Reserved

Register Name:			Test clock selection register
Register Mnemonic:			CMN_CMSMT_TEST_CLK_SEL
Register Address:			00_0_0000 : 1100 : 00001
Bits	**Type**	**Reset**	**Description**
15:3	R/O	13'h0	Reserved
2:0	R/W	3'h0	**Test clock select**: This field drives the test_clk_select pin, in order to control an external MUX for selecting between multiple test clocks to measure. The following is the encoding for this field, and the clock each value selects. • 3'b000 : PLL 0 reference clock • 3'b001 : PLL 0 feedback divider clock • 3'b010 : PLL 0 full rate clock • 3'b011 : PLL 1 reference clock • 3'b100 : PLL 1 feedback divider clock • 3'b101 : PLL 1 full rate clock • 3'b110 : Process monitor clock • 3'b111 : Signal detect clock

Register Name:			Reference clock timer value register
Register Mnemonic:			CMN_CMSMT_REF_CLK_TMR_VALUE
Register Address:			00_0_0000 : 1100 : 00010
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:0	R/W	12'h000	**Reference clock timer value** : This specifies the amount of time, in reference clock cycles, to count test clock cycles. This value minus 1 is loaded into the reference clock timer. A value of 0 for this field is not valid when running this function.

Register Name:			Test clock counter value register
Register Mnemonic:			CMN_CMSMT_TEST_CLK_CNT_VALUE
Register Address:			00_0_0000 : 1100 : 00011
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:0	R/O	12'h000	**Test clock counter value**: When the test clock measurement process is complete, the value in this field specifies the number of test clock cycles that were counted in the time specified by the reference clock timer value. This field is only valid while the test clock measurement done bit in the Clock frequency measurement control register is active.

10.2.15 Common PLL 0 Control And Diagnostics Registers

Base address bits 15:5 :	**00_0_0000 : 1101**
Address bits 4:0	**Register name and location of detailed description**
00000	PLL 0 control register mode 0 on page 186
00001	PLL 0 clock select register mode 0 on page 187
00010	PLL 0 override register mode 0 on page 187
00011	PLL 0 current trim register mode 0 on page 188
00100	PLL 0 charge pump proportional path adjust register mode 0 on page 188
00101	PLL 0 charge pump integral path adjust register mode 0 on page 188
00110	PLL 0 proportional path filter adjust register mode 0 on page 188
00111	PLL 0 charge pump tune register mode 0 on page 188
01000 - 01111	Reserved
10000	PLL 0 control register mode 1 on page 189
10001	PLL 0 clock select register mode 1 on page 190
10010	PLL 0 override register mode 1 on page 190
10011	PLL 0 current trim register mode 1 on page 191

Table 104: Common PLL 0 control and diagnostic registers address space

Base address bits 15:5 :	00_0_0000 : 1101
Address bits 4:0	**Register name and location of detailed description**
10100	PLL 0 charge pump proportional path adjust register mode 1 on page 191
10101	PLL 0 charge pump integral path adjust register mode 1 on page 191
10110	PLL 0 proportional path filter adjust register mode 1 on page 191
10111	PLL 0 charge pump tune register mode 1 on page 191
11000 - 11111	Reserved

Table 104: Common PLL 0 control and diagnostic registers address space

Register Name:			PLL 0 control register mode 0
Register Mnemonic:			CMN_PDIAG_PLL0_CTRL_M0
Register Address:			00_0_0000 : 1101 : 00000
Bits	**Type**	**Reset**	**Description**
15:12	R/W	4'h1	**PLL ring oscillator VCO temperature drift control**: This field controls the Ring VCO Frequency drift with temperature. It controls the mix of vtref bias and external bias to keep the temperature drift for the ring low, by driving the cmnda_pll0_vco_ring_cmos_sel signal going into the common analog. • 4'b0001: Minimum • ... • 4'b1111: Maximum
11:9	R/O	3'h0	Reserved
8	R/W	1'h0	**PLL VCO select**: Selects the VCO mode of operation, by driving the cmnda_pll0_vco_sel signal going into the common analog. • 1'b0: LC tank mode • 1'b1: Ring oscillator mode
7:6	R/O	2'h0	Reserved
5	R/W	1'h0	**PLL feedback divider clock select**: This signal selects which internal PLL clock will be used to drive the cmnda_pll0_fb_divider_clk, by driving the cmnda_pll0_fb_divider_clk_sel signal going into the common analog. • 1'b0: Feedback divider clock • 1'b1: PLL digital rate clock
4	R/W	1'h1	**PLL feedback divider pre-scale**: controls the feedback divider pre-scale, by driving the cmnd-a_pll0_div24_sel signal going into the common analog. One should read the description of the PLL in section 10.4 "Dual VCO PLL" on page 430 for detailed information about the function of this divider control. • 1'b0: Divide by 2 - When using the delta sigma modulator, this setting results in lower noise on the PLL output clock, but higher power. • 1'b1: Divide by 4 - When using the delta sigma modulator, this setting results in higher noise on the PLL output clock, but lower power. This setting is ideal for cases when the delta sigma modulator is not used, because is results in both lower noise and lower power.
3:2	R/O	2'h0	Reserved
1:0	R/W	2'h2	**PLL PFD reset delay**: Controls the minimum reset pulse width for the PFD. This drives the cmnd-a_pll0_pfd_rst_dly signal going into the common analog. The following lists the reset pulse width values for typical conditions for each of the possible values for this field. • 2'b00 : Delay = 257.5 pSec (minimum) • 2'b01 : Delay = 337.4 pSec • 2'b10 : Delay = 415.2 pSec • 2'b11 : Delay = 493.2 pSec (maximum)

Register Name:			PLL 0 clock select register mode 0
Register Mnemonic:			CMN_PDIAG_PLL0_CLK_SEL_M0
Register Address:			00_0_0000 : 1101 : 00001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**PLL 0 clock 1 divider enable**: This bit enables the divider used to generate the cmnda_pll0_clk_1, from the PLL high speed clock.
14:12	R/W	3'h0	**PLL 0 clock 1 divider select**: This field selects the divider value used to generate the cmnda_pll0_clk_1, from the PLL high speed clock, by driving the cmnda_pll0_clk_1_div_sel signal to the analog. The encoding of this signal is as follows. • All codes currently result in divide by 16.
11:8	R/W	4'h6	**PLL 0 clock 0 and derived reference clock divider select**: This field selects the divider value used to generate the cmnda_pll0_clk_0 and derived reference clock, from the PLL high speed clock, by driving the cmnda_pll0_clk_0_div_sel signal to the analog. The encoding of this signal is as follows. Note: There are different divider values per bit for associated with each clock.

Bits	**3**	**2**	**1**	**0**	**Total PLL clock division**	**Total derived reference clock division**
PLL clk div per bit	**1'b0 = /2 1'b1 = /3**	**1'b0 = /1 1'b1 = /2**	**1'b0 = /4 1'b1 = /5**	**1'b0 = /1 1'b1 = /2**		
Der ref clk div per bit	**1'b0 = /2 1'b1 = /3**	**1'b0 = /1 1'b1 = /2**	**1'b0 = /20 1'b1 = /25**	**1'b0 = /1 1'b1 = /1**		
Bit Values	1'b0	1'b0	1'b0	1'b0	8	40
	1'b0	1'b0	1'b0	1'b1	16	40
	1'b0	1'b0	1'b1	1'b0	10	50
	1'b0	1'b0	1'b1	1'b1	20	50
	1'b0	1'b1	1'b0	1'b0	16	80
	1'b0	1'b1	1'b0	1'b1	32	80
	1'b0	1'b1	1'b1	1'b0	20	100
	1'b0	1'b1	1'b1	1'b1	40	100
	1'b1	1'b0	1'b0	1'b0	12	60
	1'b1	1'b0	1'b0	1'b1	24	60
	1'b1	1'b0	1'b1	1'b0	15	75
	1'b1	1'b0	1'b1	1'b1	30	75
	1'b1	1'b1	1'b0	1'b0	24	120
	1'b1	1'b1	1'b0	1'b1	48	120
	1'b1	1'b1	1'b1	1'b0	30	150
	1'b1	1'b1	1'b1	1'b1	60	150

7:2	R/O	6'h0	Reserved
1:0	R/W	2'h1	**PLL 0 clock select**: This field selects one of 3 possible high speed output clocks from PLL 0, to drive on the high speed analog clock 0, by driving the cmnda_pll0_clk_sel signal to the analog. The encoding for this is as follows. • 2'b00: Divide by 1 • 2'b01: Divide by 2 • 2'b10: Divide by 4 • 2'b11: Reserved

Register Name:			PLL 0 override register mode 0
Register Mnemonic:			CMN_PDIAG_PLL0_OVRD_M0
Register Address:			00_0_0000 : 1101 : 00010
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12'h0	Reserved
3	R/W	1'h0	**PLL VCO calibration enable override enable**: When active (1'b1), the PLL VCO calibration enable override bit in this register, can be used to directly control the enable of the VCO calibration function in the PLL (instead of the VCO calibration module).
2	R/W	1'h0	**PLL VCO calibration enable override**: When enabled by the PLL VCO calibration enable override enable bit in this register, this bit will directly control the enable of the VCO calibration function in the PLL.
1	R/W	1'h0	**PLL phase lock detect enable** : Enables the diagnostic PLL phase lock detect function in the analog. This activates the cmnda_pll0_ph_lock_en signal going into the analog.
0	R/O	1'h0	**PLL phase lock detected** : When enabled by the PLL phase lock detect enable bit in this register, this bit indicates that a PLL phase lock has been detected. This is the current value of the cmnda_pll0_ph_lock_-detect signal coming from the analog.

Register Name:			PLL 0 current trim register mode 0
Register Mnemonic:			CMN_PDIAG_PLL0_ITRIM_M0
Register Address:			00_0_0000 : 1101 : 00011
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h0F	**PLL VCO bias current trim code**: Controls the tank currents in the PLL LC tank circuit. This field drives the cmnda_pll0_vco_bias_current_trim signal going to the analog.

Register Name:			PLL 0 charge pump proportional path adjust register mode 0
Register Mnemonic:			CMN_PDIAG_PLL0_CP_PADJ_M0
Register Address:			00_0_0000 : 1101 : 00100
Bits	**Type**	**Reset**	**Description**
15:8	R/W	8'h00	**PLL charge pump proportional path capacitance adjust**: Adjusts the charge pump proportional path capacitance, by driving the cmnda_pll0_cp_prop_cap_adj signal going to the analog.
7:0	R/W	8'h28	**PLL charge pump proportional path current adjust**: Adjusts the charge pump proportional path current, by driving the cmnda_pll0_cp_prop_cur_adj signal going to the analog.

Register Name:			PLL 0 charge pump integral path adjust register mode 0
Register Mnemonic:			CMN_PDIAG_PLL0_CP_IADJ_M0
Register Address:			00_0_0000 : 1101 : 00101
Bits	**Type**	**Reset**	**Description**
15:8	R/W	8'h08	**PLL charge pump integral path capacitance adjust**: Adjusts the charge pump integral path capacitance, by driving the cmnda_pll0_cp_int_cap_adj signal going to the analog.
7:0	R/W	8'h08	**PLL charge pump integral path current adjust**: Adjusts the charge pump integral path current, by driving the cmnda_pll0_cp_int_cur_adj signal going to the analog.

Register Name:			PLL 0 proportional path filter adjust register mode 0
Register Mnemonic:			CMN_PDIAG_PLL0_FILT_PADJ_M0
Register Address:			00_0_0000 : 1101 : 00110
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:8	R/W	4'h0	**PLL proportional path filter capacitance adjust**: Adjusts the proportional path filter capacitance, by driving the cmnda_pll0_filt_c_adj signal going to the analog.
7:4	R/O	4'h0	Reserved
3:0	R/W	4'h0	**PLL proportional path filter resistance adjust**: Adjusts the proportional path filter resistance, by driving the cmnda_pll0_filt_r_adj signal going to the analog.

Register Name:			PLL 0 charge pump tune register mode 0
Register Mnemonic:			CMN_PDIAG_PLL0_CP_TUNE_M0
Register Address:			00_0_0000 : 1101 : 00111
Bits	**Type**	**Reset**	**Description**
15:2	R/O	14'h0	Reserved
1:0	R/W	2'h1	**PLL charge pump calibration reference voltage tune**: Adjusts the charge pump calibration reference voltage, by driving the cmnda_pll0_cp_vref_tune signal going to the analog. • 2'b00 : minimum • 2'b01 • 2'b10 • 2'b11 maximum

Register Name:			PLL 0 control register mode 1
Register Mnemonic:			CMN_PDIAG_PLL0_CTRL_M1
Register Address:			00_0_0000 : 1101 : 10000
Bits	**Type**	**Reset**	**Description**
15:12	R/W	4'h1	**PLL ring oscillator VCO level shifter enable**: This field controls the Ring VCO Frequency drift with temperature. It controls the mix of vtref bias and external bias to keep the temperature drift for the ring low, by driving the cmnda_pll0_vco_ring_cmos_sel signal going into the common analog. • 4'b0001: Minimum • ... • 4'b1111: Maximum
11:9	R/O	3'h0	Reserved
8	R/W	1'h0	**PLL VCO select**: Selects the VCO mode of operation, by driving the cmnda_pll0_vco_sel signal going into the common analog. • 1'b0: LC tank mode • 1'b1: Ring oscillator mode
7:6	R/O	2'h0	Reserved
5	R/W	1'h0	**PLL feedback divider clock select**: This signal selects which internal PLL clock will be used to drive the cmnda_pll0_fb_divider_clk,driving the cmnda_pll0_fb_divider_clk_sel signal going into the common analog. • 1'b0: Feedback divider clock • 1'b1: PLL digital rate clock
4	R/W	1'h1	**PLL feedback divider pre-scale**: controls the feedback divider pre-scale, by driving the cmnda_pll0_div24_sel signal going into the common analog. One should read the description of the PLL in section 10.4 "Dual VCO PLL" on page 430 for detailed information about the function of this divider control. • 1'b0: Divide by 2 - When using the delta sigma modulator, this setting results in lower noise on the PLL output clock, but higher power. • 1'b1: Divide by 4 - When using the delta sigma modulator, this setting results in higher noise on the PLL output clock, but lower power. This setting is ideal for cases when the delta sigma modulator is not used, because is results in both lower noise and lower power.
3:2	R/O	2'h0	Reserved
1:0	R/W	2'h2	**PLL PFD reset delay**: Controls the minimum reset pulse width for the PFD. This drives the cmnda_pll0_pfd_rst_dly signal going into the common analog. The following lists the reset pulse width values for typical conditions for each of the possible values for this field. • 2'b00 : Delay = 257.5 pSec (minimum) • 2'b01 : Delay = 337.4 pSec • 2'b10 : Delay = 415.2 pSec • 2'b11 : Delay = 493.2 pSec (maximum)

Register Name:			PLL 0 clock select register mode 1
Register Mnemonic:			CMN_PDIAG_PLL0_CLK_SEL_M1
Register Address:			00_0_0000 : 1101 : 10001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**PLL 0 clock 1 divider enable**: This bit enables the divider used to generate the cmnda_pll0_clk_1, from the PLL high speed clock.
14:12	R/W	3'h0	**PLL 0 clock 1 divider select**: This field selects the divider value used to generate the cmnda_pll0_clk_1, from the PLL high speed clock, by driving the cmnda_pll0_clk_1_div_sel signal to the analog. The encoding of this signal is as follows. • All codes currently result in divide by 16.
11:8	R/W	4'h4	**PLL 0 clock 0 and derived reference clock divider select**: This field selects the divider value used to generate the cmnda_pll0_clk_0 and derived reference clock, from the PLL high speed clock, by driving the cmnda_pll0_clk_0_div_sel signal to the analog. The encoding of this signal is as follows. Note: There are different divider values per bit for associated with each clock.

Bits	**3**	**2**	**1**	**0**	**Total PLL clock division**	**Total derived reference clock division**
PLL clk div per bit	**1'b0 = /2 1'b1 = /3**	**1'b0 = /1 1'b1 = /2**	**1'b0 = /4 1'b1 = /5**	**1'b0 = /1 1'b1 = /2**		
Der ref clk div per bit	**1'b0 = /2 1'b1 = /3**	**1'b0 = /1 1'b1 = /2**	**1'b0 = /20 1'b1 = /25**	**1'b0 = /1 1'b1 = /1**		
Bit Values	1'b0	1'b0	1'b0	1'b0	8	40
	1'b0	1'b0	1'b0	1'b1	16	40
	1'b0	1'b0	1'b1	1'b0	10	50
	1'b0	1'b0	1'b1	1'b1	20	50
	1'b0	1'b1	1'b0	1'b0	16	80
	1'b0	1'b1	1'b0	1'b1	32	80
	1'b0	1'b1	1'b1	1'b0	20	100
	1'b0	1'b1	1'b1	1'b1	40	100
	1'b1	1'b0	1'b0	1'b0	12	60
	1'b1	1'b0	1'b0	1'b1	24	60
	1'b1	1'b0	1'b1	1'b0	15	75
	1'b1	1'b0	1'b1	1'b1	30	75
	1'b1	1'b1	1'b0	1'b0	24	120
	1'b1	1'b1	1'b0	1'b1	48	120
	1'b1	1'b1	1'b1	1'b0	30	150
	1'b1	1'b1	1'b1	1'b1	60	150

7:2	R/O	6'h0	Reserved
1:0	R/W	2'h0	**PLL 0 clock select**: This field selects one of 3 possible high speed output clocks from PLL 0, to drive on the high speed analog clock 0, by driving the cmnda_pll0_clk_sel signal to the analog. The encoding for this is as follows. • 2'b00: Divide by 1 • 2'b01: Divide by 2 • 2'b10: Divide by 4 • 2'b11: Reserved

Register Name:			PLL 0 override register mode 1
Register Mnemonic:			CMN_PDIAG_PLL0_OVRD_M1
Register Address:			00_0_0000 : 1101 : 10010
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12'h0	Reserved
3	R/W	1'h0	**PLL VCO calibration enable override enable**: When active (1'b1), the PLL VCO calibration enable override bit in this register, can be used to directly control the enable of the VCO calibration function in the PLL (instead of the VCO calibration module).
2	R/W	1'h0	**PLL VCO calibration enable override**: When enabled by the PLL VCO calibration enable override enable bit in this register, this bit will directly control the enable of the VCO calibration function in the PLL.
1	R/W	1'h0	**PLL phase lock detect enable** : Enables the diagnostic PLL phase lock detect function in the analog. This activates the cmnda_pll0_ph_lock_en signal going into the analog.
0	R/O	1'h0	**PLL phase lock detected** : When enabled by the PLL phase lock detect enable bit in this register, this bit indicates that a PLL phase lock has been detected. This is the current value of the cmnda_pll0_ph_lock_-detect signal coming from the analog.

Register Name:			PLL 0 current trim register mode 1
Register Mnemonic:			CMN_PDIAG_PLL0_ITRIM_M1
Register Address:			00_0_0000 : 1101 : 10011
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h0F	**PLL VCO bias current trim code**: Controls the tank currents in the PLL LC tank circuit. This field drives the cmnda_pll0_vco_bias_current_trim signal going to the analog.

Register Name:			PLL 0 charge pump proportional path adjust register mode 1
Register Mnemonic:			CMN_PDIAG_PLL0_CP_PADJ_M1
Register Address:			00_0_0000 : 1101 : 10100
Bits	**Type**	**Reset**	**Description**
15:8	R/W	8'h00	**PLL charge pump proportional path capacitance adjust**: Adjusts the charge pump proportional path capacitance, by driving the cmnda_pll0_cp_prop_cap_adj signal going to the analog.
7:0	R/W	8'h28	**PLL charge pump proportional path current adjust**: Adjusts the charge pump proportional path current, by driving the cmnda_pll0_cp_prop_cur_adj signal going to the analog.

Register Name:			PLL 0 charge pump integral path adjust register mode 1
Register Mnemonic:			CMN_PDIAG_PLL0_CP_IADJ_M1
Register Address:			00_0_0000 : 1101 : 10101
Bits	**Type**	**Reset**	**Description**
15:8	R/W	8'h08	**PLL charge pump integral path capacitance adjust**: Adjusts the charge pump integral path capacitance, by driving the cmnda_pll0_cp_int_cap_adj signal going to the analog.
7:0	R/W	8'h08	**PLL charge pump integral path current adjust**: Adjusts the charge pump integral path current, by driving the cmnda_pll0_cp_int_cur_adj signal going to the analog.

Register Name:			PLL 0 proportional path filter adjust register mode 1
Register Mnemonic:			CMN_PDIAG_PLL0_FILT_PADJ_M1
Register Address:			00_0_0000 : 1101 : 10110
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:8	R/W	4'h0	**PLL proportional path filter capacitance adjust**: Adjusts the proportional path filter capacitance, by driving the cmnda_pll0_filt_c_adj signal going to the analog.
7:4	R/O	4'h0	Reserved
3:0	R/W	4'h0	**PLL proportional path filter resistance adjust**: Adjusts the proportional path filter resistance, by driving the cmnda_pll0_filt_r_adj signal going to the analog.

Register Name:			PLL 0 charge pump tune register mode 1
Register Mnemonic:			CMN_PDIAG_PLL0_CP_TUNE_M1
Register Address:			00_0_0000 : 1101 : 10111
Bits	**Type**	**Reset**	**Description**
15:2	R/O	14'h0	Reserved
1:0	R/W	2'h1	**PLL charge pump calibration reference voltage tune**: Adjusts the charge pump calibration reference voltage, by driving the cmnda_pll0_cp_vref_tune signal going to the analog. • 2'b00 : minimum • 2'b01 • 2'b10 • 2'b11 maximum •

10.2.16 Common PLL 1 Control And Diagnostics Registers

Base address bits 15:5 :	00_0_0000 : 1110
Address bits 4:0	**Register name and location of detailed description**
00000	PLL 1 control register mode 0 on page 193
00001	PLL 1 clock select register mode 0 on page 194
00010	PLL 1 override register mode 0 on page 194
00011	PLL 1 current trim register mode 0 on page 195
00100	PLL 1 charge pump proportional path adjust register mode 0 on page 195
00101	PLL 1 charge pump integral path adjust register mode 0 on page 195
00110	PLL 1 proportional path filter adjust register mode 0 on page 195
00111	PLL 1 charge pump tune register mode 0 on page 195
01000 - 11111	Reserved

Table 105: Common PLL 1 control and diagnostic registers address space

Register Name:			PLL 1 control register mode 0
Register Mnemonic:			CMN_PDIAG_PLL1_CTRL_M0
Register Address:			00_0_0000 : 1110 : 00000
Bits	**Type**	**Reset**	**Description**
15:12	R/W	4'h1	**PLL ring oscillator VCO level shifter enable**: This field controls the Ring VCO Frequency drift with temperature. It controls the mix of vtref bias and external bias to keep the temperature drift for the ring low, by driving the cmnda_pll1_vco_ring_cmos_sel signal going into the common analog. 4'b0001: Minimum • ... • 4'b1111: Maximum
11:9	R/O	3'h0	Reserved
8	R/W	1'h0	**PLL VCO select**: Selects the VCO mode of operation, by driving the cmnda_pll1_vco_sel signal going into the common analog. • 1'b0: LC tank mode • 1'b1: Ring oscillator mode
7:6	R/O	2'h0	Reserved
5	R/W	1'h0	**PLL feedback divider clock select**: This signal selects which internal PLL clock will be used to drive the cmnda_pll1_fb_divider_clk,driving the cmnda_pll1_fb_divider_clk_sel signal going into the common analog. • 1'b0: Feedback divider clock • 1'b1: PLL digital rate clock
4	R/W	1'h1	**PLL feedback divider pre-scale**: controls the feedback divider pre-scale, by driving the cmnda_pll1_div24_sel signal going into the common analog. One should read the description of the PLL in section 10.4 "Dual VCO PLL" on page 430 for detailed information about the function of this divider control. • 1'b0: Divide by 2 - When using the delta sigma modulator, this setting results in lower noise on the PLL output clock, but higher power. • 1'b1: Divide by 4 - When using the delta sigma modulator, this setting results in higher noise on the PLL output clock, but lower power. This setting is ideal for cases when the delta sigma modulator is not used, because is results in both lower noise and lower power.
3:2	R/O	2'h0	Reserved
1:0	R/W	2'h2	**PLL PFD reset delay**: Controls the minimum reset pulse width for the PFD. This drives the cmnda_pll1_pfd_rst_dly signal going into the common analog. The following lists the reset pulse width values for typical conditions for each of the possible values for this field. • 2'b00 : Delay = 257.5 pSec (minimum) • 2'b01 : Delay = 337.4 pSec • 2'b10 : Delay = 415.2 pSec • 2'b11 : Delay = 493.2 pSec (maximum)

Register Name:			PLL 1 clock select register mode 0
Register Mnemonic:			CMN_PDIAG_PLL1_CLK_SEL_M0
Register Address:			00_0_0000 : 1110 : 00001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**PLL 1 clock 1 divider enable**: This bit enables the divider used to generate the cmnda_pll1_clk_1, from the PLL high speed clock.
14:12	R/W	3'h0	**PLL 1 clock 1 divider select**: This field selects the divider value used to generate the cmnda_pll1_clk_1, from the PLL high speed clock, by driving the cmnda_pll1_clk_1_div_sel signal to the analog. The encoding of this signal is as follows. • All codes currently result in divide by 16.
11:8	R/W	4'h4	**PLL 1 clock 0 and derived reference clock divider select**: This field selects the divider value used to generate the cmnda_pll1_clk_0 and derived reference clock, from the PLL high speed clock, by driving the cmnda_pll1_clk_0_div_sel signal to the analog. The encoding of this signal is as follows. Note: There are different divider values per bit for associated with each clock.

Bits	**3**	**2**	**1**	**0**	**Total PLL clock division**	**Total derived reference clock division**
PLL clk div per bit	**1'b0 = /2 1'b1 = /3**	**1'b0 = /1 1'b1 = /2**	**1'b0 = /4 1'b1 = /5**	**1'b0 = /1 1'b1 = /2**		
Der ref clk div per bit	**1'b0 = /2 1'b1 = /3**	**1'b0 = /1 1'b1 = /2**	**1'b0 = /20 1'b1 = /25**	**1'b0 = /1 1'b1 = /1**		
Bit Values	1'b0	1'b0	1'b0	1'b0	8	40
	1'b0	1'b0	1'b0	1'b1	16	40
	1'b0	1'b0	1'b1	1'b0	10	50
	1'b0	1'b0	1'b1	1'b1	20	50
	1'b0	1'b1	1'b0	1'b0	16	80
	1'b0	1'b1	1'b0	1'b1	32	80
	1'b0	1'b1	1'b1	1'b0	20	100
	1'b0	1'b1	1'b1	1'b1	40	100
	1'b1	1'b0	1'b0	1'b0	12	60
	1'b1	1'b0	1'b0	1'b1	24	60
	1'b1	1'b0	1'b1	1'b0	15	75
	1'b1	1'b0	1'b1	1'b1	30	75
	1'b1	1'b1	1'b0	1'b0	24	120
	1'b1	1'b1	1'b0	1'b1	48	120
	1'b1	1'b1	1'b1	1'b0	30	150
	1'b1	1'b1	1'b1	1'b1	60	150

7:2	R/O	6'h0	Reserved
1:0	R/W	2'h0	**PLL 1 clock select**: This field selects one of 3 possible high speed output clocks from PLL 1, to drive on the high speed analog clock 1, by driving the cmnda_pll1_clk_sel signal to the analog. The encoding for this is as follows. • 2'b00: Divide by 1 • 2'b01: Divide by 2 • 2'b10: Divide by 4 • 2'b11: Reserved

Register Name:			PLL 1 override register mode 0
Register Mnemonic:			CMN_PDIAG_PLL1_OVRD_M0
Register Address:			00_0_0000 : 1110 : 00010
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12'h0	Reserved
3	R/W	1'h0	**PLL VCO calibration enable override enable**: When active (1'b1), the PLL VCO calibration enable override bit in this register, can be used to directly control the enable of the VCO calibration function in the PLL (instead of the VCO calibration module).
2	R/W	1'h0	**PLL VCO calibration enable override**: When enabled by the PLL VCO calibration enable override enable bit in this register, this bit will directly control the enable of the VCO calibration function in the PLL.
1	R/W	1'h0	**PLL phase lock detect enable** : Enables the diagnostic PLL phase lock detect function in the analog. This activates the cmnda_pll1_ph_lock_en signal going into the analog.
0	R/O	1'h0	**PLL phase lock detected** : When enabled by the PLL phase lock detect enable bit in this register, this bit indicates that a PLL phase lock has been detected. This is the current value of the cmnda_pll1_ph_lock_-detect signal coming from the analog.

Register Name:			PLL 1 current trim register mode 0
Register Mnemonic:			CMN_PDIAG_PLL1_ITRIM_M0
Register Address:			00_0_0000 : 1110 : 00011
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h0F	**PLL VCO bias current trim code**: Controls the tank currents in the PLL LC tank circuit. This field drives the cmnda_pll1_vco_bias_current_trim signal going to the analog.

Register Name:			PLL 1 charge pump proportional path adjust register mode 0
Register Mnemonic:			CMN_PDIAG_PLL1_CP_PADJ_M0
Register Address:			00_0_0000 : 1110 : 00100
Bits	**Type**	**Reset**	**Description**
15:8	R/W	8'h00	**PLL charge pump proportional path capacitance adjust**: Adjusts the charge pump proportional path capacitance, by driving the cmnda_pll1_cp_prop_cap_adj signal going to the analog.
7:0	R/W	8'h28	**PLL charge pump proportional path current adjust**: Adjusts the charge pump proportional path current, by driving the cmnda_pll0_cp_prop_cur_adj signal going to the analog.

Register Name:			PLL 1 charge pump integral path adjust register mode 0
Register Mnemonic:			CMN_PDIAG_PLL1_CP_IADJ_M0
Register Address:			00_0_0000 : 1110 : 00101
Bits	**Type**	**Reset**	**Description**
15:8	R/W	8'h08	**PLL charge pump integral path capacitance adjust**: Adjusts the charge pump integral path capacitance, by driving the cmnda_pll1_cp_int_cap_adj signal going to the analog.
7:0	R/W	8'h08	**PLL charge pump integral path current adjust**: Adjusts the charge pump integral path current, by driving the cmnda_pll1_cp_int_cur_adj signal going to the analog.

Register Name:			PLL 1 proportional path filter adjust register mode 0
Register Mnemonic:			CMN_PDIAG_PLL1_FILT_PADJ_M0
Register Address:			00_0_0000 : 1110 : 00110
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:8	R/W	4'h0	**PLL proportional path filter capacitance adjust**: Adjusts the proportional path filter capacitance, by driving the cmnda_pll1_filt_c_adj signal going to the analog.
7:4	R/O	4'h0	Reserved
3:0	R/W	4'h0	**PLL proportional path filter resistance adjust**: Adjusts the proportional path filter resistance, by driving the cmnda_pll1_filt_r_adj signal going to the analog.

Register Name:			PLL 1 charge pump tune register mode 0
Register Mnemonic:			CMN_PDIAG_PLL1_CP_TUNE_M0
Register Address:			00_0_0000 : 1110 : 00111
Bits	**Type**	**Reset**	**Description**
15:2	R/O	14'h0	Reserved
1:0	R/W	2'h1	**PLL charge pump calibration reference voltage tune**: Adjusts the charge pump calibration reference voltage, by driving the cmnda_pll1_cp_vref_tune signal going to the analog. • 2'b00 : minimum • 2'b01 • 2'b10 • 2'b11 maximum •

10.2.17 Common Functions Control And Diagnostics Registers

Base address bits 15:5 :	00_0_0000 : 1111
Address bits 4:0	**Register name and location of detailed description**
00000	Bandgap override register on page 196
00001	Bias override register 1 on page 197
00010	Bias override register 2 on page 197
00011	Voltage regulator control register on page 198
00100	Common process monitor control register on page 198
00101	Sample and hold bandgap calibration code register on page 198
00110	Sample and hold resistor calibration code register on page 198
00111	Sample and hold signal detect clock calibration code register on page 199
01000	ATB control register 1 on page 199
01001	ATB control register 2 on page 199
01010	ATB ADC control register 0 on page 199
01011	ATB ADC control register 1 on page 200
01100	Common high speed reset release state machine control register on page 201
01101	Common functions reset diagnostic register on page 201
01110	Common digital functions cover your alternatives register on page 201
01111	Common analog cover your alternatives register on page 201
10000 - 11111	Reserved

Table 106: Common functions control and diagnostic registers address space

Register Name:			Bandgap override register
Register Mnemonic:			CMN_DIAG_BANDGAP_OVRD
Register Address:			00_0_0000 : 1111 : 00000
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4’h0	Reserved
11:8	R/O	4’hX	**Bandgap startup circuit startup count** : Identifies the status of the bandgap startup counter. This bit is driven by the cmnda_bias_bg_start_count signal coming from the analog.
7:5	R/O	3’h0	Reserved
4	R/W	1’h0	**Bandgap startup circuit select** : Selects the startup circuit to be used for the bias / bandgap circuits, by driving the cmnda_bias_bg_start_sel signal going to the analog. • 1’b0: Analog comparator startup circuit • 1’b1: Ring oscillator startup circuit
3:2	R/W	2’h1	**Bandgap startup circuit sense voltage adjust** : This field is used to adjust the bandgap startup circuit sense voltage, by driving the cmnda_bias_bg_start_adj signal to the analog. The following list shows the encoding of this field. • 2’b00 : 543.0 mV • 2’b01 : 558.5 mV • 2’b10 : 574.0 mV • 2’b11 : Reserved •
1:0	R/W	2’h1	**Bandgap voltage adjust** : This field is used to adjust the bandgap voltage, by driving the cmnda_bias_b-g_adj signal to the analog. The following list shows the encoding of this field. • 2’b00 : 414.7 mV • 2’b01 : 440.7 mV • 2’b10 : 446.7 mV • 2’b11 : 446.7 mV

Register Name:			Bias override register 1
Register Mnemonic:			CMN_DIAG_BIAS_OVRD1
Register Address:			00_0_0000 : 1111 : 00001
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:12	R/W	3'h3	**Receiver resistor calibration current adjust**: This field is used to adjust the receiver resistor calibration bias current. It drives the cmnda_bias_rx_rescal_adj signal going to the analog. The following list shows the encoding of this field. • 3'b000 : 110.72 uA • 3'b001 : 112.50 uA • 3'b010 : 114.29 uA • 3'b011 : 116.07 uA • 3'b100 : 117.86 uA • 3'b101 : 119.64 uA • 3'b110 : 121.43 uA • 3'b111 : 123.22 uA
11	R/O	1'h0	Reserved
10:8	R/W	3'h6	**Transmitter resistor calibration current adjust**: This field is used to adjust the transmitter resistor calibration bias current. It drives the cmnda_bias_tx_rescal_adj signal going to the analog. The following list shows the encoding of this field. • 3'b000 : 110.72 uA • 3'b001 : 112.50 uA • 3'b010 : 114.29 uA • 3'b011 : 116.07 uA • 3'b100 : 117.86 uA • 3'b101 : 119.64 uA • 3'b110 : 121.43 uA • 3'b111 : 123.22 uA
7:4	R/W	4'h0	Reserved - spare
3:1	R/O	3'h0	Reserved
0	R/W	1'h0	Reserved - spare

Register Name:			Bias override register 2
Register Mnemonic:			CMN_DIAG_BIAS_OVRD2
Register Address:			00_0_0000 : 1111 : 00010
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5	R/W	1'h0	**Bias filter bypass enable override enable**: When active (1'b1), the bias filter bypass enable override bit in this register, can be used to directly control the bias filter bypass enable function.
4	R/W	1'h0	**Bias filter bypass enable override**: When enabled by the bias filter bypass enable override enable bit in this register, this bit will directly control the bias filter bypass enable function.
3:2	R/O	2'h0	Reserved
1:0	R/W	2'h0	**Regulator bandgap reference voltage adjust**: This field is used to adjust the regulator bandgap reference voltage, by driving the cmnda_bias_vreg_adj signal to the analog. The encoding of this field is as follows. • 2'b00 : 850 mV • 2'b01 : 830 mV • 2'b10 : 810 mV • 2'b11 : Reserved •

Register Name:			Voltage regulator control register
Register Mnemonic:			CMN_DIAG_VREG_CTRL
Register Address:			00_0_0000 : 1111 : 00011
Bits	**Type**	**Reset**	**Description**
15:1	R/O	15'h0	Reserved
0	R/W	1'h0	**Voltage regulator reference voltage select**: Selects the reference voltage used for the voltage regulator in common, by driving the cmnda_vreg_ref_sel signal to the analog. • 1'b0: Bandgap supply • 1'b1: Core supply

Register Name:			Common process monitor control register
Register Mnemonic:			CMN_DIAG_PM_CTRL
Register Address:			00_0_0000 : 1111 : 00100
Bits	**Type**	**Reset**	**Description**
15:5	R/O	11'h0	Reserved
4	R/W	1'h0	**Process monitor enable**: Enables the analog process monitor, by driving the cmnda_pcm_en signal to the analog. • 1'b0: Disabled • 1'b1: Enabled
3	R/O	1'h0	Reserved
2:0	R/W	3'h0	**Process monitor mode select**: Selects the mode of the analog process monitor, by driving the cmnda_pcm_sel signal to the analog. • 000 : SVT 11nm oscillator, • 001 : LVT 11nm oscillator, • 010 : ULVT 11nm oscillator, • 011 : IO oscillator, • 100 : SVT 36nm oscillator, • 101 : LVT 36nm oscillator, • 110 : ULVT 36nm oscillator, • 111 : Reserved

Register Name:			Sample and hold bandgap calibration code register
Register Mnemonic:			CMN_DIAG_SH_BANDGAP
Register Address:			00_0_0000 : 1111 : 00101
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5	R/O	1'h0	**Bandgap up value**: Bandgap calibration up signal value, as it is currently captured in the sample and hold latches.
4:0	R/O	5'h00	**Bandgap auto zero select value**: Bandgap calibration auto zero select signal value, as it is currently captured in the sample and hold latches.

Register Name:			Sample and hold resistor calibration code register
Register Mnemonic:			CMN_DIAG_SH_RESISTOR
Register Address:			00_0_0000 : 1111 : 00110
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'h0	Reserved
13:8	R/O	6'h00	**TX resistor code**: TX resistor calibration code signal value, as it is currently captured in the sample and hold latches.
7:4	R/O	4'h0	Reserved
3:0	R/O	4'h0	**RX resistor code**: RX resistor calibration code signal value, as it is currently captured in the sample and hold latches.

Register Name:			Sample and hold signal detect clock calibration code register
Register Mnemonic:			CMN_DIAG_SH_SDCLK
Register Address:			00_0_0000 : 1111 : 00111
Bits	**Type**	**Reset**	**Description**
15:5	R/O	11'h0	Reserved
4:0	R/O	5'h00	**Signal detect clock code**: Signal detect clock calibration code signal value, as it is currently captured in the sample and hold latches.

Register Name:			ATB control register 1
Register Mnemonic:			CMN_DIAG_ATB_CTRL1
Register Address:			00_0_0000 : 1111 : 01000
Bits	**Type**	**Reset**	**Description**
15:2	R/O	14'h0	Reserved
1	R/W	1'h0	**Core side ATB enable**: When active (1'b) and the ATB enable bit in this register is also active, the ATB signals will be driven on the core side ATB signals.
0	R/W	1'h0	**ATB enable**: When active (1'b1), the ATB test function is enabled.

Register Name:			ATB control register 2
Register Mnemonic:			CMN_DIAG_ATB_CTRL2
Register Address:			00_0_0000 : 1111 : 01001
Bits	**Type**	**Reset**	**Description**
15:13	R/O	3'h0	Reserved
12:11	R/W	2'h0	**ATB component type select**: These bits specify which component type is currently selected by the ATB, as specified below. • 2'b00: Common • 2'b01: Transmitter • 2'b10: Receiver • 2'b11: Reserved
10:6	R/W	5'h00	**ATB component sub address**: Specifies the sub address of the component being selected. In this design, the sub address must be 0 when accessing the common, and the lane number when accessing a transmitter or receiver.
5:0	R/W	6'h00	**ATB test point address**: Specifies the exact point in the selected analog component to be observed.

Register Name:			ATB ADC control register 0
Register Mnemonic:			CMN_DIAG_ATB_ADC_CTRL0
Register Address:			00_0_0000 : 1111 : 01010
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**ATB analog ADC enable**: This enables the analog ADC function, by driving the cmnda_atba2d_en signal to the analog. This bit must be set to the enabled state before starting an ATB ADC process using the other bits in this register. • 1'b0: ADC disabled • 1'b1: ADC enabled
14	R/W	1'h0	**Start ATB ADC process**: Activating (1'b1) this bit will start the ATB ADC process. This signal must remain active until the ATB ADC process is complete, as indicated by the ATB ADC process done bit in this register. To start another ATB ADC process, this bit must first be deactivated until the ATB ADC process done bit in this register is also cleared. This bit drives the cmnda_atba2d_start signal to the analog.
13	R/O	1'h0	**ATB ADC process done**: This bit will be set to 1'b1 when the ATB ADC process is complete, and the data in the ATB ADC code field of this register is considered valid. Once set to 1'b1, it will remain set until the Start ATB ADC process bit in this register is set to 1'b0. This bit is driven by the cmnda_atba2d_dvalid signal from the analog.
12:8	R/O	5'h0	Reserved
7:0	R/O	8'h00	**ATB ADC data code**: This is the digital code representing the level of the analog ATB signal that was digitized by the analog ADC. This code is only valid when the ATB ADC process done bit in this register is active (1'b1). The voltage represented by this value is equal to code/256*0.94.

Register Name:			ATB ADC control register 1
Register Mnemonic:			CMN_DIAG_ATB_ADC_CTRL1
Register Address:			00_0_0000 : 1111 : 01011
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'h0	Reserved
13	R/W	1'h0	**ATB ADC offset correction enable** : Enables internal auto generated offset correction mode, by driving the cmnda_atba2d_en_off_cor signal to the analog. • 1'b0: Disabled • 1'b1: Enabled
12	R/W	1'h0	**ATB ADC force cap values :** Forces a positive or negative voltage on the internal cap, by driving the cmnda_atba2d_frc_val signal to the analog. • 1'b0: Forces a negative voltage on the cap. • 1'b1: Forces a positive voltage on the cap.
11	R/W	1'h0	**ATB ADC enable manual offset correction :** When this signal is active, the value in the ATB ADC manual offset correction value field of this register is used to manually control the offset correction, by driving the cmnda_atba2d_off_byp_en signal to the analog. • 1'b0: Disabled • 1'b1: Enabled
10:9	R/O	2'h0	Reserved
8:4	R/W	5'h00	**ATB ADC manual offset correction value :** When the ATB ADC enable manual offset correction bit in this register is active, this field is used to manually control the offset correction, by driving the cmnda_atba2d_off_adj_byp signal to the analog.
3:0	R/W	4'h0	**ATB ADC mode** : This field indicates the mode the analog to digital converter is in. This field directly controls the cmnda_atba2d_adcmode signal going to the analog. For current mode measurements (ATB analog to digital converter mode == 3'b101), the current is equal to (Code/256)*0.6901/10.5K uA. The following list describes each of the modes, and the value used to select it. • 4'b0000: Hi-Z Mode • 4'b0001: atb0 and local gnd • 4'b0010: atb0 assumed remote gnd • 4'b0011: atb1 and local gnd • 4'b0100: atb1 assumed remote gnd • 4'b0101: 10K between atb0 and agnd • 4'b0110: Offset Measure Mode volt offset • 4'b0111: Offset Measure Mode current offset • 4'b1000: BIST mode - force cap 1 polarity based on the ADC force cap values bit in this register. • 4'b1001: BIST mode - force cap 2 polarity based on the ADC force cap values bit in this register. • 4'b1010: BIST mode - force cap 3 polarity based on the ADC force cap values bit in this register. • 4'b1011: BIST mode - force cap 4 polarity based on the ADC force cap values bit in this register. • 4'b1100: BIST mode - force cap 5 polarity based on the ADC force cap values bit in this register. • 4'b1101: BIST mode - force cap 6 polarity based on the ADC force cap values bit in this register. • 4'b1110: BIST mode - force cap 7 polarity based on the ADC force cap values bit in this register. • 4'b1111: BIST mode - force cap 8 polarity based on the ADC force cap values bit in this register. BIST mode procedure: 1. Calibrate the signal detect clock oscillator. 2. Select the BIST mode for the cap being measured. 3. Set the ADC force cap values bit in this register to 1'b0. 4. Measure the output. 5. Set the ADC force cap values bit in this register to 1'b1. 6. Measure the output. 7. Calculate the difference between the two measured outputs. The difference should be 32 for caps 1-7 and 15 for cap 8, +/- 1. This procedure should be run 4 times and average the results, to reduce the affects of noise.

Register Name:			Common high speed reset release state machine control register
Register Mnemonic:			CMN_DIAG_HSRRSM_CTRL
Register Address:			00_0_0000 : 1111 : 01100
Bits	**Type**	**Reset**	**Description**
15:7	R/O	9'h0	Reserved
6:4	R/W	3'h1	**Transceiver reset delay** : Species the number of PSM clock cycles the transceiver common high speed reset state machine stays in the delay state.
3	R/O	1'h0	Reserved
2:0	R/W	3'h1	**Transmitter reset delay** : Species the number of PSM clock cycles the transmitter common high speed reset state machine stays in the delay state.

Register Name:			Common functions reset diagnostic register
Register Mnemonic:			CMN_DIAG_RST_DIAG
Register Address:			00_0_0000 : 1111 : 01101
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11	R/O	1'hX	Current state of the cmn_sd_clk_cal_fb_clk_reset_n reset.
10	R/O	1'hX	Current state of the cmn_sd_clk_cal_ref_clk_reset_n reset.
9	R/O	1'hX	Current state of the cmn_pll1_dsm_reset_n reset.
8	R/O	1'hX	Current state of the cmn_pll0_dsm_reset_n reset.
7	R/O	1'hX	Current state of the cmn_pll1_vco_cal_fbdiv_clk_reset_n reset.
6	R/O	1'hX	Current state of the cmn_pll1_lock_det_fbdiv_clk_reset_n reset.
5	R/O	1'hX	Current state of the cmn_pll1_vco_cal_ref_clk_reset_n reset.
4	R/O	1'hX	Current state of the cmn_pll1_lock_det_ref_clk_reset_n reset.
3	R/O	1'hX	Current state of the cmn_pll0_vco_cal_fbdiv_clk_reset_n reset.
2	R/O	1'hX	Current state of the cmn_pll0_lock_det_fbdiv_clk_reset_n reset.
1	R/O	1'hX	Current state of the cmn_pll0_vco_cal_ref_clk_reset_n reset.
0	R/O	1'hX	Current state of the cmn_pll0_lock_det_ref_clk_reset_n reset.

Register Name:			Common digital functions cover your alternatives register
Register Mnemonic:			CMN_DIAG_DCYA
Register Address:			00_0_0000 : 1111 : 01110
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h0	Reserved - spare

Register Name:			Common analog cover your alternatives register
Register Mnemonic:			CMN_DIAG_ACYA
Register Address:			00_0_0000 : 1111 : 01111
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:4	R/W	4'h0	Reserved - spare
3:0	R/W	4'h0	PLL charge pump proportional gain adjust: Adjusts the charge pump gain for the PLLs to help manage bandwidth. The following is the encoding of this field. • 4'b0000: vctrip + 0 mV • 4'b0001: vctrip + 15 mV • 4'b0011: vctrip + 30 mV • 4'b0111: vctrip + 45 mV • 4'b1111: vctrip + 60 mV

10.2.18 Power State Machine Registers

Base address bits 15:5 :	01_m_nnnn : 0000
Address bits 4:0	**Register name and location of detailed description**
00000	Power state machine control register on page 203
00001	Power state machine reset control register on page 204
00010	PSM calibration in delay timer register on page 205
00011	A0 in delay timer register on page 205
00100	A0 in bypass timer register on page 205
00101	A1 in delay timer register on page 205
00110	A2 in delay timer register on page 206
00111	A3 in delay timer register on page 206
01000	A4 in delay timer register on page 206
01001	A5 in delay timer register on page 206
01010	PSM calibration out delay timer register on page 206
01011	A0 out delay timer register on page 207
01100	A1 out delay timer register on page 207
01101	A2 out delay timer register on page 207
01110	A3 out delay timer register on page 207
01111	A4 out delay timer register on page 207
10000	A5 out delay timer register on page 208
10001	Ready delay timer register on page 208
10010	Power state machine diagnostic register on page 208
10011	PSM current state register 0 on page 209
10100	PSM current state register 1 on page 209
10101 - 11110	Reserved
11111	Power state machine user defined control register on page 209

Table 107: Powers state machine registers address space

Register Name:			Power state machine control register
Register Mnemonic:			XCVR_PSM_CTRL
Register Address:			01_m_nnnn : 0000 : 00000
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14	R/W	1'h0	**Bypass A0 in delay from PSM ready** : When this bit is active (1'b1), the A0 input delay is bypassed when transitioning from the PSM ready state to the A0 power state. The result of this is the amount of time spent in the A0 in delay state is the value specified in the A0 in bypass timer register. The intention of this bit is to bypass the delay for cases when the required analog components being controlled are already enabled in the ready power state, and no additional delay is required when transitioning to the A0 power state.
13	R/W	1'h0	**Bypass A0 in delay from A5** : When this bit is active (1'b1), the A0 input delay is bypassed when transitioning from the A5 to the A0 power state. The result of this is the amount of time spent in the A0 in delay state is the value specified in the A0 in bypass timer register. The intention of this bit is to bypass the delay for cases when the required analog components being controlled are already enabled in the A5 power state, and no additional delay is required when transitioning to the A0 power state.
12	R/W	1'h0	**Bypass A0 in delay from A4** : When this bit is active (1'b1), the A0 input delay is bypassed when transitioning from the A4 to the A0 power state. The result of this is the amount of time spent in the A0 in delay state is the value specified in the A0 in bypass timer register. The intention of this bit is to bypass the delay for cases when the required analog components being controlled are already enabled in the A4 power state, and no additional delay is required when transitioning to the A0 power state.
11	R/W	1'h0	**Bypass A0 in delay from A3** : When this bit is active (1'b1), the A0 input delay is bypassed when transitioning from the A3 to the A0 power state. The result of this is the amount of time spent in the A0 in delay state is the value specified in the A0 in bypass timer register. The intention of this bit is to bypass the delay for cases when the required analog components being controlled are already enabled in the A3 power state, and no additional delay is required when transitioning to the A0 power state.
10	R/W	1'h0	**Bypass A0 in delay from A2** : When this bit is active (1'b1), the A0 input delay is bypassed when transitioning from the A2 to the A0 power state. The result of this is the amount of time spent in the A0 in delay state is the value specified in the A0 in bypass timer register. The intention of this bit is to bypass the delay for cases when the required analog components being controlled are already enabled in the A2 power state, and no additional delay is required when transitioning to the A0 power state.
9	R/W	1'h1	**Bypass A0 in delay from A1** : When this bit is active (1'b1), the A0 input delay is bypassed when transitioning from the A1 to the A0 power state. The result of this is the amount of time spent in the A0 in delay state is the value specified in the A0 in bypass timer register. The intention of this bit is to bypass the delay for cases when the required analog components being controlled are already enabled in the A1 power state, and no additional delay is required when transitioning to the A0 power state.
8	R/O	1'h0	Reserved
7:1	R/O	7'h0	Reserved
0	R/W	1'h1	Reserved - spare (must remain set to 1'b1).

Register Name:			Power state machine reset control register
Register Mnemonic:			XCVR_PSM_RCTRL
Register Address:			01_m_nnnn : 0000 : 00001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h1	**RX reset active ready** : Controls the state the receiver reset is changed to when in the ready power state. 1'b0: Reset not active 1'b1: Reset active
14	R/W	1'h0	**RX reset active calibration** : Controls the state the receiver reset is changed to when in the calibration power state. 1'b0: Reset not active 1'b1: Reset active
13	R/W	1'h1	**RX reset active A5** : Controls the state the receiver reset is changed to when in the A5 entry power state. 1'b0: Reset not active 1'b1: Reset active
12	R/W	1'h1	**RX reset active A4** : Controls the state the receiver reset is changed to when in the A4 entry power state. 1'b0: Reset not active 1'b1: Reset active
11	R/W	1'h1	**RX reset active A3** : Controls the state the receiver reset is changed to when in the A3 entry power state. 1'b0: Reset not active 1'b1: Reset active
10	R/W	1'h1	**RX reset active A2** : Controls the state the receiver reset is changed to when in the A2 entry power state. 1'b0: Reset not active 1'b1: Reset active
9	R/W	1'h0	**RX reset active A1** : Controls the state the receiver reset is changed to when in the A1 entry power state. 1'b0: Reset not active 1'b1: Reset active
8	R/W	1'h0	**RX reset active A0** : Controls the state the receiver reset is changed to when in the A0 entry power state. 1'b0: Reset not active 1'b1: Reset active
7	R/W	1'h1	**TX reset active ready** : Controls the state the transmitter reset is changed to when in the ready power state. 1'b0: Reset not active 1'b1: Reset active
6	R/W	1'h1	**TX reset active calibration** : Controls the state the transmitter reset is changed to when in the calibration power state. 1'b0: Reset not active 1'b1: Reset active
5	R/W	1'h1	**TX reset active A5** : Controls the state the transmitter reset is changed to when in the A5 entry power state. 1'b0: Reset not active 1'b1: Reset active
4	R/W	1'h1	**TX reset active A4** : Controls the state the transmitter reset is changed to when in the A4 entry power state. 1'b0: Reset not active 1'b1: Reset active
3	R/W	1'h1	**TX reset active A3** : Controls the state the transmitter reset is changed to when in the A3 entry power state. 1'b0: Reset not active 1'b1: Reset active
2	R/W	1'h1	**TX reset active A2** : Controls the state the transmitter reset is changed to when in the A2 entry power state. 1'b0: Reset not active 1'b1: Reset active
1	R/W	1'h0	**TX reset active A1** : Controls the state the transmitter reset is changed to when in the A1 entry power state. 1'b0: Reset not active 1'b1: Reset active
0	R/W	1'h0	**TX reset active A0** : Controls the state the transmitter reset is changed to when in the A0 entry power state. 1'b0: Reset not active 1'b1: Reset active

Register Name:			PSM calibration in delay timer register
Register Mnemonic:			XCVR_PSM_CALIN_TMR
Register Address:			01_m_nnnn : 0000 : 00010
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:0	R/W	12'h096	**PSM calibration in delay state timer value** : Value used for the timer when the power state machine is in the PSM calibration in delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for $t_{psm_lane_cal_in_del.}$ Note, the smallest value this field is allowed to be is 12'h001. With that said, setting this field to a value smaller than the reset value will likely result in incorrect function.

Register Name:			A0 in delay timer register
Register Mnemonic:			XCVR_PSM_A0IN_TMR
Register Address:			01_m_nnnn : 0000 : 00011
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:0	R/W	12'h096	**A0 in delay state timer value** : Value used for the timer when the power state machine is in the A0 in delay state, unless the timer is bypassed under the control of the bypass A0 bits in the Power state machine control register. This timer delay is specified as the number of PSM clocks to count to implement the time required for $t_{psm_A0_in_del.}$ Note, the smallest value this field is allowed to be is 12'h001. With that said, setting this field to a value smaller than the reset value will likely result in incorrect function.

Register Name:			A0 in bypass timer register
Register Mnemonic:			XCVR_PSM_A0BYP_TMR
Register Address:			01_m_nnnn : 0000 : 00100
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h0A	**A0 in delay state bypass timer value** : Value used for the timer when the power state machine is in the A0 in delay state and the timer is bypassed under the control of the bypass A0 bits in the Power state machine control register. This timer delay is specified as the number of PSM clocks to count to implement the time required for $t_{psm_A0_in_byp.}$ Note, the smallest value this field is allowed to be is 6'h01. With that said, setting this field to a value smaller than the reset value will likely result in incorrect function.

Register Name:			A1 in delay timer register
Register Mnemonic:			XCVR_PSM_A1IN_TMR
Register Address:			01_m_nnnn : 0000 : 00101
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h10	**A1 in delay state timer value** : Value used for the timer when the power state machine is in the A1 in delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for $t_{psm_A1_in_del.}$ Note, the smallest value this field is allowed to be is 6'h01. With that said, setting this field to a value smaller than the reset value will likely result in incorrect function.

Register Name:			A2 in delay timer register
Register Mnemonic:			XCVR_PSM_A2IN_TMR
Register Address:			01_m_nnnn : 0000 : 00110
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h10	**A2 in delay state timer value** : Value used for the timer when the power state machine is in the A2 in delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for $t_{psm_A2_in_del.}$ Note, the smallest value this field is allowed to be is 6'h01. With that said, setting this field to a value smaller than the reset value will likely result in incorrect function.

Register Name:			A3 in delay timer register
Register Mnemonic:			XCVR_PSM_A3IN_TMR
Register Address:			01_m_nnnn : 0000 : 00111
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h10	**A3 in delay state timer value** : Value used for the timer when the power state machine is in the A3 in delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for $t_{psm_A3_in_del.}$ Note, the smallest value this field is allowed to be is 6'h01. With that said, setting this field to a value smaller than the reset value will likely result in incorrect function.

Register Name:			A4 in delay timer register
Register Mnemonic:			XCVR_PSM_A4IN_TMR
Register Address:			01_m_nnnn : 0000 : 01000
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h10	**A4 in delay state timer value** : Value used for the timer when the power state machine is in the A4 in delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for $t_{psm_A4_in_del.}$ Note, the smallest value this field is allowed to be is 6'h01. With that said, setting this field to a value smaller than the reset value will likely result in incorrect function.

Register Name:			A5 in delay timer register
Register Mnemonic:			XCVR_PSM_A5IN_TMR
Register Address:			01_m_nnnn : 0000 : 01001
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h10	**A5 in delay state timer value** : Value used for the timer when the power state machine is in the A5 in delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for $t_{psm_A5_in_del.}$ Note, the smallest value this field is allowed to be is 6'h01. With that said, setting this field to a value smaller than the reset value will likely result in incorrect function.

Register Name:			PSM calibration out delay timer register
Register Mnemonic:			XCVR_PSM_CALOUT_TMR
Register Address:			01_m_nnnn : 0000 : 01010
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h01	**PSM calibration out delay state timer value** : Value used for the timer when the power state machine is in the PSM calibration out delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for $t_{psm_lane_cal_out_del.}$ Note, the smallest value this field is allowed to be is 6'h01. With that said, setting this field to a value smaller than the reset value will likely result in incorrect function.

Register Name:			A0 out delay timer register
Register Mnemonic:			XCVR_PSM_A0OUT_TMR
Register Address:			01_m_nnnn : 0000 : 01011
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h01	**A0 out delay state timer value** : Value used for the timer when the power state machine is in the A0 out delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for $t_{psm_A0_out_del.}$ Note, the smallest value this field is allowed to be is 6'h01. With that said, setting this field to a value smaller than the reset value will likely result in incorrect function.

Register Name:			A1 out delay timer register
Register Mnemonic:			XCVR_PSM_A1OUT_TMR
Register Address:			01_m_nnnn : 0000 : 01100
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h01	**A1 out delay state timer value** : Value used for the timer when the power state machine is in the A1 out delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for $t_{psm_A1_out_del.}$ Note, the smallest value this field is allowed to be is 6'h01. With that said, setting this field to a value smaller than the reset value will likely result in incorrect function.

Register Name:			A2 out delay timer register
Register Mnemonic:			XCVR_PSM_A2OUT_TMR
Register Address:			01_m_nnnn : 0000 : 01101
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h01	**A2 out delay state timer value** : Value used for the timer when the power state machine is in the A2 out delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for $t_{psm_A2_out_del.}$ Note, the smallest value this field is allowed to be is 6'h01. With that said, setting this field to a value smaller than the reset value will likely result in incorrect function.

Register Name:			A3 out delay timer register
Register Mnemonic:			XCVR_PSM_A3OUT_TMR
Register Address:			01_m_nnnn : 0000 : 01110
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h01	**A3 out delay state timer value** : Value used for the timer when the power state machine is in the A3 out delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for $t_{psm_A3_out_del.}$ Note, the smallest value this field is allowed to be is 6'h01. With that said, setting this field to a value smaller than the reset value will likely result in incorrect function.

Register Name:			A4 out delay timer register
Register Mnemonic:			XCVR_PSM_A4OUT_TMR
Register Address:			01_m_nnnn : 0000 : 01111
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h01	**A4 out delay state timer value** : Value used for the timer when the power state machine is in the A4 out delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for $t_{psm_A4_out_del.}$ Note, the smallest value this field is allowed to be is 6'h01. With that said, setting this field to a value smaller than the reset value will likely result in incorrect function.

Register Name:			A5 out delay timer register
Register Mnemonic:			XCVR_PSM_A5OUT_TMR
Register Address:			01_m_nnnn : 0000 : 10000
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h01	**A5 out delay state timer value** : Value used for the timer when the power state machine is in the A5 out delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for $t_{psm_A5_out_del.}$ Note, the smallest value this field is allowed to be is 6'h01. With that said, setting this field to a value smaller than the reset value will likely result in incorrect function.

Register Name:			Ready delay timer register
Register Mnemonic:			XCVR_PSM_RDY_TMR
Register Address:			01_m_nnnn : 0000 : 10001
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h10	**Ready delay state timer value** : Value used for the timer when the power state machine is in the ready state. This timer delay is specified as the number of PSM clocks to count to implement the minimum time ($t_{psm_rdy_out_del.)}$ required to remain in this state. Note, the smallest value this field is allowed to be is 6'h01. With that said, setting this field to a value smaller than the reset value will likely result in incorrect function.

Register Name:			Power state machine diagnostic register
Register Mnemonic:			XCVR_PSM_DIAG
Register Address:			01_m_nnnn : 0000 : 10010
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14	R/W	1'hx	**Force calibration exit acknowledge** : Setting this bit to 1'b1 forces the psm_cal_exit_ack pin of the power state machine active.
13	R/W	1'h0	**Force A5 exit acknowledge** : Setting this bit to 1'b1 forces the psm_a5_exit_ack pin of the power state machine active.
12	R/W	1'h0	**Force A4 exit acknowledge** : Setting this bit to 1'b1 forces the psm_a4_exit_ack pin of the power state machine active.
11	R/W	1'h0	**Force A3 exit acknowledge** : Setting this bit to 1'b1 forces the psm_a3_exit_ack pin of the power state machine active.
10	R/W	1'h0	**Force A2 exit acknowledge** : Setting this bit to 1'b1 forces the psm_a2_exit_ack pin of the power state machine active.
9	R/W	1'h0	**Force A1 exit acknowledge** : Setting this bit to 1'b1 forces the psm_a1_exit_ack pin of the power state machine active.
8	R/W	1'h0	**Force A0 exit acknowledge** : Setting this bit to 1'b1 forces the psm_a0_exit_ack pin of the power state machine active.
7	R/O	1'h0	Reserved
6	R/W	1'hx	**Force calibration entry acknowledge** : Setting this bit to 1'b1 forces the psm_cal_entry_ack pin of the power state machine active.
5	R/W	1'h0	**Force A5 entry acknowledge** : Setting this bit to 1'b1 forces the psm_a5_entry_ack pin of the power state machine active.
4	R/W	1'h0	**Force A4 entry acknowledge** : Setting this bit to 1'b1 forces the psm_a4_entry_ack pin of the power state machine active.
3	R/W	1'h0	**Force A3 entry acknowledge** : Setting this bit to 1'b1 forces the psm_a3_entry_ack pin of the power state machine active.
2	R/W	1'h0	**Force A2 entry acknowledge** : Setting this bit to 1'b1 forces the psm_a2_entry_ack pin of the power state machine active.
1	R/W	1'h0	**Force A1 entry acknowledge** : Setting this bit to 1'b1 forces the psm_a1_entry_ack pin of the power state machine active.
0	R/W	1'h0	**Force A0 entry acknowledge** : Setting this bit to 1'b1 forces the psm_a0_entry_ack pin of the power state machine active.

Register Name:			PSM current state register 0
Register Mnemonic:			XCVR_PSM_ST_0
Register Address:			01_m_nnnn : 0000 : 10011
Bits	**Type**	**Reset**	**Description**
15:0	R/O	16'h0000	PSM current state[15:0] : Indicates bits 15:0 of the current state of the power state machine.

Register Name:			PSM current state register 1
Register Mnemonic:			XCVR_PSM_ST_1
Register Address:			01_m_nnnn : 0000 : 10100
Bits	**Type**	**Reset**	**Description**
15:10	R/O	6'h0	Reserved
9:0	R/O	10'h000	PSM current state[25:16] : Indicates bits 25:16 of the current state of the power state machine.

Register Name:			Power state machine user defined control register
Register Mnemonic:			XCVR_PSM_USER_DEF_CTRL
Register Address:			01_m_nnnn : 0000 : 11111
Bits	**Type**	**Reset**	**Description**
15:5	R/W	11'h0	Reserved - spare
4	R/W	1'h0	**Force PSM gated clock on**: Setting this bit to 1'b1 will force the PSM gated clock on, independent of the internal PSM state machine clock gate controls.
3:0	R/W	4'h0	Reserved - spare

10.2.19 TX Coefficient Calculator Registers

Base address bits 15:5 :	01_m_nnnn : 0010
Address bits 4:0	**Register name and location of detailed description**
00000	TX coefficient controller control register on page 212
00001	TX pre-cursor override register on page 213
00010	TX main-cursor override register on page 213
00011	TX post-cursor override register on page 213
00100	TX pre-cursor current value register on page 214
00101	TX main-cursor current value register on page 214
00110	TX post-cursor current value register on page 214
00111	LF multiplier value register on page 215
01000	Calculated pre emphasis multiplier value 00 register on page 215
01001	Calculated pre emphasis multiplier value 01 register on page 215
01010	Calculated pre emphasis multiplier value 10 register on page 216
01011	Calculated pre emphasis multiplier value 11 register on page 216
01100	Calculated post emphasis multiplier value 00 register on page 216
01101	Calculated post emphasis multiplier value 01 register on page 217
01110	Calculated post emphasis multiplier value 10 register on page 217
01111	Calculated post emphasis multiplier value 11 register on page 217
10000	Margin full swing multiplier value 000 register on page 218
10001	Margin full swing multiplier value 001 register on page 218
10010	Margin full swing multiplier value 010 register on page 219
10011	Margin full swing multiplier value 011 register on page 219
10100	Margin full swing multiplier value 100 register on page 220
10101	Margin full swing multiplier value 101 register on page 220
10110	Margin full swing multiplier value 110 register on page 221
10111	Margin full swing multiplier value 111 register on page 221
11000	Margin half swing multiplier value 000 register on page 222
11001	Margin half swing multiplier value 001 register on page 222
11010	Margin half swing multiplier value 010 register on page 223
11011	Margin half swing multiplier value 011 register on page 223
11100	Margin half swing multiplier value 100 register on page 224
11101	Margin half swing multiplier value 101 register on page 224
11110	Margin half swing multiplier value 110 register on page 225
11111	Margin half swing multiplier value 111 register on page 225

Table 108: TX coefficient calculator registers address space (slave 0)

Base address bits 15:5 :	01_m_nnnn : 0011
Address bits 4:0	**Register name and location of detailed description**
00000	Preset 0 pre emphasis coefficient multiplier value register on page 226
00001	Preset 1 pre emphasis coefficient multiplier value register on page 226
00010	Preset 2 pre emphasis coefficient multiplier value register on page 226
00011	Preset 3 pre emphasis coefficient multiplier value register on page 227
00100	Preset 4 pre emphasis coefficient multiplier value register on page 227
00101	Preset 5 pre emphasis coefficient multiplier value register on page 227
00110	Preset 6 pre emphasis coefficient multiplier value register on page 228
00111	Preset 7 pre emphasis coefficient multiplier value register on page 228
01000	Preset 8 pre emphasis coefficient multiplier value register on page 228
01001	Preset 9 pre emphasis coefficient multiplier value register on page 229
01010 - 01111	Reserved
10000	Preset 0 post emphasis coefficient multiplier value register on page 229
10001	Preset 1 post emphasis coefficient multiplier value register on page 229
10010	Preset 2 post emphasis coefficient multiplier value register on page 230
10011	Preset 3 post emphasis coefficient multiplier value register on page 230
10100	Preset 4 post emphasis coefficient multiplier value register on page 230
10101	Preset 5 post emphasis coefficient multiplier value register on page 231
10110	Preset 6 post emphasis coefficient multiplier value register on page 231
10111	Preset 7 post emphasis coefficient multiplier value register on page 231
11000	Preset 8 post emphasis coefficient multiplier value register on page 232
11001	Preset 9 post emphasis coefficient multiplier value register on page 232
11010 - 11111	Reserved

Table 109: TX coefficient calculator registers address space (slave 1)

Register Name:			TX coefficient controller control register
Register Mnemonic:			TX_TXCC_CTRL
Register Address:			01_m_nnnn : 0010 : 00000
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'h0	Reserved
13:12	R/W	2'h2	**Margin multiplier rounding control**: This field controls the rounding function on the margin multiplier. • 2'b00 : Round to nearest integer • 2'b01 : Floor • 2'b10 : Ceiling
11:10	R/W	2'h2	**LF value multiplier rounding control**: This field controls the rounding function on the LF value multiplier. • 2'b00 : Round to nearest integer • 2'b01 : Floor • 2'b10 : Ceiling
9:8	R/W	2'h2	**Calculated post-emphasis multiplier rounding control**: This field controls the rounding function on the calculated post-emphasis multiplier. • 2'b00 : Round to nearest integer • 2'b01 : Floor • 2'b10 : Ceiling
7:6	R/W	2'h2	**Calculated pre-emphasis multiplier rounding control**: This field controls the rounding function on the pre-emphasis multiplier. • 2'b00 : Round to nearest integer • 2'b01 : Floor • 2'b10 : Ceiling
5:4	R/W	2'h0	**Coefficient calculator multiplier rounding control**: This field controls the rounding function on the coefficient calculator multiplier. • 2'b00 : Round to nearest integer • 2'b01 : Floor • 2'b10 : Ceiling
3	R/W	1'h0	**De-emphasis control standard mode 3 value**: This bit controls the de-emphasis mode when the xcvr_-standard_mode is set to 2'b11. • 1'b0 : Use the calculated de-emphasis values based on the 2 least significant bits of the top level tx_de-emphasis signal. • 1'b1 : Use the coefficient values on all 18 bits of the top level tx_deemphasis signal.
2	R/W	1'h1	**De-emphasis control standard mode 2 value**: This bit controls the de-emphasis mode when the xcvr_-standard_mode is set to 2'b10. • 1'b0 : Use the calculated de-emphasis values based on the 2 least significant bits of the top level tx_de-emphasis signal. • 1'b1 : Use the coefficient values on all 18 bits of the top level tx_deemphasis signal.
1	R/W	1'h0	**De-emphasis control standard mode 1 value**: This bit controls the de-emphasis mode when the xcvr_-standard_mode is set to 2'b01. • 1'b0 : Use the calculated de-emphasis values based on the 2 least significant bits of the top level tx_de-emphasis signal. • 1'b1 : Use the coefficient values on all 18 bits of the top level tx_deemphasis signal.
0	R/W	1'h0	**De-emphasis control standard mode 0 value**: This bit controls the de-emphasis mode when the xcvr_-standard_mode is set to 2'b00. • 1'b0 : Use the calculated de-emphasis values based on the 2 least significant bits of the top level tx_de-emphasis signal. • 1'b1 : Use the coefficient values on all 18 bits of the top level tx_deemphasis signal.

Register Name:			TX pre-cursor override register
Register Mnemonic:			TX_TXCC_PRE_OVRD
Register Address:			01_m_nnnn : 0010 : 00001
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8	R/W	1'h0	**Pre-cursor override enable**: When enabled, the pre-cursor field in this register is used to override the pre-cursor value.
7:6	R/O	2'h0	Reserved
5:0	R/W	6'h00	**Pre-cursor override value**: When enabled by the pre-cursor override enable bit in this register, the value in this field is used to override the pre-cursor value. Note: This field is 6 bits wide, to match the pre-cursor data input width on the tx_deemphasis input pin, only the least significant 5 bits are used to actually drive the H bridge driver controller. In the calculated modes, this will override the calculated pre-cursor value. **Note : This register is for diagnostic purposes only. When this field is used, one must make sure that the value does not result in the H bridge driver controller input pin restrictions being violated. These restrictions are documented in the pin list of H bridge driver controller specification.**

Register Name:			TX main-cursor override register
Register Mnemonic:			TX_TXCC_MAIN_OVRD
Register Address:			01_m_nnnn : 0010 : 00010
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8	R/W	1'h0	**Main-cursor override enable**: When enabled, the main-cursor field in this register is used to override the main-cursor value.
7:6	R/O	2'h0	Reserved
5:0	R/W	6'h00	**Main-cursor override value**: When enabled by the main-cursor override enable bit in this register, the value in this field is used to override the main-cursor value. Note: This field is 6 bits wide, to match the main-cursor data input width on the tx_deemphasis input pin, only the least significant 5 bits are used to actually drive the H bridge driver controller. In the calculated modes, this will override the calculated main-cursor value. **Note : This register is for diagnostic purposes only. When this field is used, one must make sure that the value does not result in the H bridge driver controller input pin restrictions being violated. These restrictions are documented in the pin list of H bridge driver controller specification.**

Register Name:			TX post-cursor override register
Register Mnemonic:			TX_TXCC_POST_OVRD
Register Address:			01_m_nnnn : 0010 : 00011
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8	R/W	1'h0	**Post-cursor override enable**: When enabled, the post-cursor field in this register is used to override the post-cursor value.
7:6	R/O	2'h0	Reserved
5:0	R/W	6'h00	**Post-cursor override value**: When enabled by the post-cursor override enable bit in this register, the value in this field is used to override the post-cursor value. Note: This field is 6 bits wide, to match the post-cursor data input width on the tx_deemphasis input pin, only the least significant 5 bits are used to actually drive the H bridge driver controller. In the calculated modes, this will override the calculated post-cursor value. **Note : This register is for diagnostic purposes only. When this field is used, one must make sure that the value does not result in the H bridge driver controller input pin restrictions being violated. These restrictions are documented in the pin list of H bridge driver controller specification.**

Register Name:			TX pre-cursor current value register
Register Mnemonic:			TX_TXCC_PRE_CVAL
Register Address:			01_m_nnnn : 0010 : 00100
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/O	6'hXX	**Pre-cursor value**: The value in this field indicates the current value of the pre-cursor (C_{-1}) coefficient. The value of this field can be any of the following, depending on the current mode of operation. • The override value when override is enabled, under the control of the TX pre-cursor override register on page 213 • The pre-cursor data on the tx_deemphasis input pin when the calculated de-emphasis values are disabled, under the control of the TX coefficient controller control register on page 212. • The internally calculated pre-cursor value when the calculated de-emphasis values are enabled, under the control of the TX coefficient controller control register on page 212. **Note: The value of this is a function of input values and internal logic. Therefore, there is not a fixed reset value.**

Register Name:			TX main-cursor current value register
Register Mnemonic:			TX_TXCC_MAIN_CVAL
Register Address:			01_m_nnnn : 0010 : 00101
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/O	6'hXX	**Main-cursor value**: The value in this field indicates the current value of the main-cursor (C_0) coefficient. The value of this field can be any of the following, depending on the current mode of operation. • The override value when override is enabled, under the control of the TX main-cursor override register on page 213 • The main-cursor data on the tx_deemphasis input pin when the calculated de-emphasis values are disabled, under the control of the TX coefficient controller control register on page 212. • All 0s when the calculated de-emphasis values are enabled, under the control of the TX coefficient controller control register on page 212. **Note: The value of this is a function of input values and internal logic. Therefore, there is not a fixed reset value.**

Register Name:			TX post-cursor current value register
Register Mnemonic:			TX_TXCC_POST_CVAL
Register Address:			01_m_nnnn : 0010 : 00110
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/O	6'hXX	**Post-cursor value**: The value in this field indicates the current value of the post-cursor (C_{+1}) coefficient. The value of this field can be any of the following, depending on the current mode of operation. • The override value when override is enabled, under the control of the TX post-cursor override register on page 213 • The post-cursor data on the tx_deemphasis input pin when the calculated de-emphasis values are disabled, under the control of the TX coefficient controller control register on page 212. • The internally calculated post-cursor value when the calculated de-emphasis values are enabled, under the control of the TX coefficient controller control register on page 212. **Note: The value of this is a function of input values and internal logic. Therefore, there is not a fixed reset value.**

Register Name:			LF multiplier value register
Register Mnemonic:			TX_TXCC_LF_MULT
Register Address:			01_m_nnnn : 0010 : 00111
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h2A	**LF multiplier value**: The value in this field specifies the multiplier value used to generate the LF value from the FS value. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Calculated pre emphasis multiplier value 00 register
Register Mnemonic:			TX_TXCC_CPRE_MULT_00
Register Address:			01_m_nnnn : 0010 : 01000
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h00	**Calculated pre emphasis multiplier value 00**: The value in this field specifies the multiplier value used to generate the calculated pre emphasis value from the FS value when tx_deemphasis[1:0] = 2'b00. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Calculated pre emphasis multiplier value 01 register
Register Mnemonic:			TX_TXCC_CPRE_MULT_01
Register Address:			01_m_nnnn : 0010 : 01001
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h00	**Calculated pre emphasis multiplier value 01**: The value in this field specifies the multiplier value used to generate the calculated pre emphasis value from the FS value when tx_deemphasis[1:0] = 2'b01. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Calculated pre emphasis multiplier value 10 register
Register Mnemonic:			TX_TXCC_CPRE_MULT_10
Register Address:			01_m_nnnn : 0010 : 01010
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h00	**Calculated pre emphasis multiplier value 10**: The value in this field specifies the multiplier value used to generate the calculated pre emphasis value from the FS value when tx_deemphasis[1:0] = 2'b10. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Calculated pre emphasis multiplier value 11 register
Register Mnemonic:			TX_TXCC_CPRE_MULT_11
Register Address:			01_m_nnnn : 0010 : 01011
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h00	**Calculated pre emphasis multiplier value 11**: The value in this field specifies the multiplier value used to generate the calculated pre emphasis value from the FS value when tx_deemphasis[1:0] = 2'b11. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Calculated post emphasis multiplier value 00 register
Register Mnemonic:			TX_TXCC_CPOST_MULT_00
Register Address:			01_m_nnnn : 0010 : 01100
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h1C	**Calculated post emphasis multiplier value 00**: The value in this field specifies the multiplier value used to generate the calculated post emphasis value from the FS value when tx_deemphasis[1:0] = 2'b00. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Calculated post emphasis multiplier value 01 register
Register Mnemonic:			TX_TXCC_CPOST_MULT_01
Register Address:			01_m_nnnn : 0010 : 01101
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h11	**Calculated post emphasis multiplier value 01**: The value in this field specifies the multiplier value used to generate the calculated post emphasis value from the FS value when tx_deemphasis[1:0] = 2'b01. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Calculated post emphasis multiplier value 10 register
Register Mnemonic:			TX_TXCC_CPOST_MULT_10
Register Address:			01_m_nnnn : 0010 : 01110
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h00	**Calculated post emphasis multiplier value 10**: The value in this field specifies the multiplier value used to generate the calculated post emphasis value from the FS value when tx_deemphasis[1:0] = 2'b10. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Calculated post emphasis multiplier value 11 register
Register Mnemonic:			TX_TXCC_CPOST_MULT_11
Register Address:			01_m_nnnn : 0010 : 01111
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h00	**Calculated post emphasis multiplier value 11**: The value in this field specifies the multiplier value used to generate the calculated post emphasis value from the FS value when tx_deemphasis[1:0] = 2'b11. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Margin full swing multiplier value 000 register
Register Mnemonic:			TX_TXCC_MGNFS_MULT_000
Register Address:			01_m_nnnn : 0010 : 10000
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h00	**Margin full swing multiplier value 000**: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when tx_vmargin = 3'b000 and tx_low_power_swing_en = 1'b0. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The largest value this is allowed to be set to is 54/128. Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Margin full swing multiplier value 001 register
Register Mnemonic:			TX_TXCC_MGNFS_MULT_001
Register Address:			01_m_nnnn : 0010 : 10001
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h00	**Margin full swing multiplier value 001**: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when tx_vmargin = 3'b001 and tx_low_power_swing_en = 1'b0. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The largest value this is allowed to be set to is 54/128. Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Margin full swing multiplier value 010 register
Register Mnemonic:			TX_TXCC_MGNFS_MULT_010
Register Address:			01_m_nnnn : 0010 : 10010
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h07	**Margin full swing multiplier value 010**: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when tx_vmargin = 3'b010 and tx_low_power_swing_en = 1'b0. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The largest value this is allowed to be set to is 54/128. Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Margin full swing multiplier value 011 register
Register Mnemonic:			TX_TXCC_MGNFS_MULT_011
Register Address:			01_m_nnnn : 0010 : 10011
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h0F	**Margin full swing multiplier value 011**: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when tx_vmargin = 3'b011 and tx_low_power_swing_en = 1'b0. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The largest value this is allowed to be set to is 54/128. Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Margin full swing multiplier value 100 register
Register Mnemonic:			TX_TXCC_MGNFS_MULT_100
Register Address:			01_m_nnnn : 0010 : 10100
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h15	**Margin full swing multiplier value 100**: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when tx_vmargin = 3'b100 and tx_low_power_swing_en = 1'b0. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The largest value this is allowed to be set to is 54/128. Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Margin full swing multiplier value 101 register
Register Mnemonic:			TX_TXCC_MGNFS_MULT_101
Register Address:			01_m_nnnn : 0010 : 10101
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h1C	**Margin full swing multiplier value 101**: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when tx_vmargin = 3'b101 and tx_low_power_swing_en = 1'b0. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The largest value this is allowed to be set to is 54/128. Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Margin full swing multiplier value 110 register
Register Mnemonic:			TX_TXCC_MGNFS_MULT_110
Register Address:			01_m_nnnn : 0010 : 10110
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h23	**Margin full swing multiplier value 110**: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when tx_vmargin = 3'b110 and tx_low_power_swing_en = 1'b0. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The largest value this is allowed to be set to is 54/128. Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Margin full swing multiplier value 111 register
Register Mnemonic:			TX_TXCC_MGNFS_MULT_111
Register Address:			01_m_nnnn : 0010 : 10111
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h2A	**Margin full swing multiplier value 111**: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when tx_vmargin = 3'b111 and tx_low_power_swing_en = 1'b0. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The largest value this is allowed to be set to is 54/128. Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Margin half swing multiplier value 000 register
Register Mnemonic:			TX_TXCC_MGNHS_MULT_000
Register Address:			01_m_nnnn : 0010 : 11000
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h1C	**Margin half swing multiplier value 000**: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when tx_vmargin = 3'b000 and tx_low_power_swing_en = 1'b1. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The largest value this is allowed to be set to is 54/128. Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Margin half swing multiplier value 001 register
Register Mnemonic:			TX_TXCC_MGNHS_MULT_001
Register Address:			01_m_nnnn : 0010 : 11001
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h1C	**Margin half swing multiplier value 001**: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when tx_vmargin = 3'b001 and tx_low_power_swing_en = 1'b1. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The largest value this is allowed to be set to is 54/128. Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Margin half swing multiplier value 010 register
Register Mnemonic:			TX_TXCC_MGNHS_MULT_010
Register Address:			01_m_nnnn : 0010 : 11010
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h20	**Margin half swing multiplier value 010**: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when tx_vmargin = 3'b010 and tx_low_power_swing_en = 1'b1. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The largest value this is allowed to be set to is 54/128. Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Margin half swing multiplier value 011 register
Register Mnemonic:			TX_TXCC_MGNHS_MULT_011
Register Address:			01_m_nnnn : 0010 : 11011
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h24	**Margin half swing multiplier value 011**: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when tx_vmargin = 3'b011 and tx_low_power_swing_en = 1'b1. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The largest value this is allowed to be set to is 54/128. Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Margin half swing multiplier value 100 register
Register Mnemonic:			TX_TXCC_MGNHS_MULT_100
Register Address:			01_m_nnnn : 0010 : 11100
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h28	**Margin half swing multiplier value 100**: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when tx_vmargin = 3'b100 and tx_low_power_swing_en = 1'b1. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The largest value this is allowed to be set to is 54/128. Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Margin half swing multiplier value 101 register
Register Mnemonic:			TX_TXCC_MGNHS_MULT_101
Register Address:			01_m_nnnn : 0010 : 11101
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h2C	**Margin half swing multiplier value 101**: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when tx_vmargin = 3'b101 and tx_low_power_swing_en = 1'b1. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The largest value this is allowed to be set to is 54/128. Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Margin half swing multiplier value 110 register
Register Mnemonic:			TX_TXCC_MGNHS_MULT_110
Register Address:			01_m_nnnn : 0010 : 11110
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h33	**Margin half swing multiplier value 110**: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when tx_vmargin = 3'b110 and tx_low_power_swing_en = 1'b1. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The largest value this is allowed to be set to is 54/128. Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Margin half swing multiplier value 111 register
Register Mnemonic:			TX_TXCC_MGNHS_MULT_111
Register Address:			01_m_nnnn : 0010 : 11111
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h36	**Margin half swing multiplier value 111**: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when tx_vmargin = 3'b111 and tx_low_power_swing_en = 1'b1. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The largest value this is allowed to be set to is 54/128. Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Preset 0 pre emphasis coefficient multiplier value register
Register Mnemonic:			TX_TXCC_P0PRE_COEF_MULT
Register Address:			01_m_nnnn : 0011 : 00000
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h00	**Preset 0 pre emphasis coefficient multiplier value** : The value in this field specifies the multiplier value used to generate the preset 0 pre emphasis coefficient value from the FS value. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Preset 1 pre emphasis coefficient multiplier value register
Register Mnemonic:			TX_TXCC_P1PRE_COEF_MULT
Register Address:			01_m_nnnn : 0011 : 00001
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h00	**Preset 1 pre emphasis coefficient multiplier value** : The value in this field specifies the multiplier value used to generate the preset 1 pre emphasis coefficient value from the FS value. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Preset 2 pre emphasis coefficient multiplier value register
Register Mnemonic:			TX_TXCC_P2PRE_COEF_MULT
Register Address:			01_m_nnnn : 0011 : 00010
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h00	**Preset 2 pre emphasis coefficient multiplier value** : The value in this field specifies the multiplier value used to generate the preset 2 pre emphasis coefficient value from the FS value. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Preset 3 pre emphasis coefficient multiplier value register
Register Mnemonic:			TX_TXCC_P3PRE_COEF_MULT
Register Address:			01_m_nnnn : 0011 : 00011
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h00	**Preset 3 pre emphasis coefficient multiplier value** : The value in this field specifies the multiplier value used to generate the preset 3 pre emphasis coefficient value from the FS value. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Preset 4 pre emphasis coefficient multiplier value register
Register Mnemonic:			TX_TXCC_P4PRE_COEF_MULT
Register Address:			01_m_nnnn : 0011 : 00100
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h00	**Preset 4 pre emphasis coefficient multiplier value** : The value in this field specifies the multiplier value used to generate the preset 4 pre emphasis coefficient value from the FS value. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Preset 5 pre emphasis coefficient multiplier value register
Register Mnemonic:			TX_TXCC_P5PRE_COEF_MULT
Register Address:			01_m_nnnn : 0011 : 00101
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h0D	**Preset 5 pre emphasis coefficient multiplier value** : The value in this field specifies the multiplier value used to generate the preset 5 pre emphasis coefficient value from the FS value. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Preset 6 pre emphasis coefficient multiplier value register
Register Mnemonic:			TX_TXCC_P6PRE_COEF_MULT
Register Address:			01_m_nnnn : 0011 : 00110
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h10	**Preset 6 pre emphasis coefficient multiplier value** : The value in this field specifies the multiplier value used to generate the preset 6 pre emphasis coefficient value from the FS value. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Preset 7 pre emphasis coefficient multiplier value register
Register Mnemonic:			TX_TXCC_P7PRE_COEF_MULT
Register Address:			01_m_nnnn : 0011 : 00111
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h0D	**Preset 7 pre emphasis coefficient multiplier value** : The value in this field specifies the multiplier value used to generate the preset 7 pre emphasis coefficient value from the FS value. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Preset 8 pre emphasis coefficient multiplier value register
Register Mnemonic:			TX_TXCC_P8PRE_COEF_MULT
Register Address:			01_m_nnnn : 0011 : 01000
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h10	**Preset 8 pre emphasis coefficient multiplier value** : The value in this field specifies the multiplier value used to generate the preset 8 pre emphasis coefficient value from the FS value. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Preset 9 pre emphasis coefficient multiplier value register
Register Mnemonic:			TX_TXCC_P9PRE_COEF_MULT
Register Address:			01_m_nnnn : 0011 : 01001
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h16	**Preset 9 pre emphasis coefficient multiplier value** : The value in this field specifies the multiplier value used to generate the preset 9 pre emphasis coefficient value from the FS value. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

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Register Name:			Preset 0 post emphasis coefficient multiplier value register
Register Mnemonic:			TX_TXCC_P0POST_COEF_MULT
Register Address:			01_m_nnnn : 0011 : 10000
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h22	**Preset 0 post emphasis coefficient multiplier value** : The value in this field specifies the multiplier value used to generate the preset 0 post emphasis coefficient value from the FS value. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Preset 1 post emphasis coefficient multiplier value register
Register Mnemonic:			TX_TXCC_P1POST_COEF_MULT
Register Address:			01_m_nnnn : 0011 : 10001
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h16	**Preset 1 post emphasis coefficient multiplier value** : The value in this field specifies the multiplier value used to generate the preset 1 post emphasis coefficient value from the FS value. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Preset 2 post emphasis coefficient multiplier value register
Register Mnemonic:			TX_TXCC_P2POST_COEF_MULT
Register Address:			01_m_nnnn : 0011 : 10010
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h1B	**Preset 2 post emphasis coefficient multiplier value** : The value in this field specifies the multiplier value used to generate the preset 2 post emphasis coefficient value from the FS value. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Preset 3 post emphasis coefficient multiplier value register
Register Mnemonic:			TX_TXCC_P3POST_COEF_MULT
Register Address:			01_m_nnnn : 0011 : 10011
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h10	**Preset 3 post emphasis coefficient multiplier value** : The value in this field specifies the multiplier value used to generate the preset 3 post emphasis coefficient value from the FS value. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Preset 4 post emphasis coefficient multiplier value register
Register Mnemonic:			TX_TXCC_P4POST_COEF_MULT
Register Address:			01_m_nnnn : 0011 : 10100
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h00	**Preset 4 post emphasis coefficient multiplier value** : The value in this field specifies the multiplier value used to generate the preset 4 post emphasis coefficient value from the FS value. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Preset 5 post emphasis coefficient multiplier value register
Register Mnemonic:			TX_TXCC_P5POST_COEF_MULT
Register Address:			01_m_nnnn : 0011 : 10101
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h00	**Preset 5 post emphasis coefficient multiplier value** : The value in this field specifies the multiplier value used to generate the preset 5 post emphasis coefficient value from the FS value. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Preset 6 post emphasis coefficient multiplier value register
Register Mnemonic:			TX_TXCC_P6POST_COEF_MULT
Register Address:			01_m_nnnn : 0011 : 10110
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h00	**Preset 6 post emphasis coefficient multiplier value** : The value in this field specifies the multiplier value used to generate the preset 6 post emphasis coefficient value from the FS value. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Preset 7 post emphasis coefficient multiplier value register
Register Mnemonic:			TX_TXCC_P7POST_COEF_MULT
Register Address:			01_m_nnnn : 0011 : 10111
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h1B	**Preset 7 post emphasis coefficient multiplier value** : The value in this field specifies the multiplier value used to generate the preset 7 post emphasis coefficient value from the FS value. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Preset 8 post emphasis coefficient multiplier value register
Register Mnemonic:			TX_TXCC_P8POST_COEF_MULT
Register Address:			01_m_nnnn : 0011 : 11000
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h10	**Preset 8 post emphasis coefficient multiplier value** : The value in this field specifies the multiplier value used to generate the preset 8 post emphasis coefficient value from the FS value. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

Register Name:			Preset 9 post emphasis coefficient multiplier value register
Register Mnemonic:			TX_TXCC_P9POST_COEF_MULT
Register Address:			01_m_nnnn : 0011 : 11001
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h00	**Preset 9 post emphasis coefficient multiplier value** : The value in this field specifies the multiplier value used to generate the preset 9 post emphasis coefficient value from the FS value. The following describes the multiplier value each bit in this field corresponds to. • Bit 7 : 128/128 = 1.000000 • Bit 6 : 64/128 = 0.500000 • Bit 5 : 32/128 = 0.250000 • Bit 4 : 16/128 = 0.125000 • Bit 3 : 8/128 = 0.062500 • Bit 2 : 4/128 = 0.031250 • Bit 1 : 2/128 = 0.015625 • Bit 0 : 1/128 = 0.0078125 Note: The output of the multiplier is 6 bits wide. It is possible to set the multiplier value here to a sufficiently large value to cause the multiplier result to be such that it can't be represented by a 6 bit value. In such a case, the multiplier output will be driven to 6'b111111.

10.2.20 Transceiver Control And Diagnostics Registers

Base address bits 15:5 :	01_m_nnnn : 0111
Address bits 4:0	**Register name and location of detailed description**
00000	Transceiver power island control timer register on page 233
00001	Transceiver transceiver calibration power island control override register on page 234
00010	Transceiver transceiver calibration power island control status register on page 235
00011	Transceiver transceiver data path power island control override register on page 236
00100	Transceiver transceiver data path power island control status register on page 237
00101	Transceiver PLL data rate clock control register on page 238
00110	Transceiver high speed clock select register on page 239
00111	Transceiver high speed clock divider register on page 240
01000	TX clock control register on page 240
01001	RX clock control register on page 241
01010	Transceiver bidirectional control register on page 242
01011	Power state controller override register on page 243
01100	Transceiver control reset diagnostic register on page 243
01101	Transceiver clock reset control register on page 243
01110 - 01110	Reserved
01111	Transceiver digital cover your alternatives register on page 243
10000 - 11111	Reserved

Table 110: Transceiver control and diagnostic registers address space

Register Name:			Transceiver power island control timer register
Register Mnemonic:			XCVR_DIAG_PWRI_TMR
Register Address:			01_m_nnnn : 0111 : 00000
Bits	**Type**	**Reset**	**Description**
15:13	R/O	3’h0	Reserved
12:8	R/W	5’h05	**Power enable phase 2 timer value**: This specifies the number of PSM clock cycles the power island control state machines in the transceiver will wait in the power phase 2 enable states, in order to allow enough time for the second phase of the switched domain to power up, before deactivating the isolation functions. The default value of this register corresponds to a delay of 40 nSec. Note: A value of 0 in this field is not valid, and will result in a delay of 1 clock cycle. Note: The value of this field must be the same for all lanes that make up a link.
7:5	R/O	3’h0	Reserved
4:0	R/W	5’h05	**Power enable phase 1 timer value**: This specifies the number of PSM clock cycles the power island control state machines in the transceiver will wait in the power phase 1 enable states, in order to allow enough time for the first phase of the switched domain to power up, before enabling the second phase of the power up. The default value of this register corresponds to a delay of 40 nSec. Note: A value of 0 in this field is not valid, and will result in a delay of 1 clock cycle. Note: The value of this field must be the same for all lanes that make up a link.

Register Name:			Transceiver transceiver calibration power island control override register **Note: This register must not be part of any register read write test where patterns other than those described in this register are written to this register.**
Register Mnemonic:			XCVR_DIAG_XCAL_PWRI_OVRD
Register Address:			01_m_nnnn : 0111 : 00001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Power island controller input override enable**: When enabled, the power island control state machine input override bits in this register will drive the power island control state machine directly, and override any control from other state machines in the macro. **Note: The CDB power island must be triggered to be enabled, by accessing a register outside of the common control module control and diagnostic registers space, prior to enabling this function.** **Note: This bit must never change state in the same register write that any of the override bits it controls in this register are changed.** **Note: The value of this field must be the same for all lanes that make up a link.** • 1'b0: Override disabled • 1'b1: Override enabled
14	R/W	1'h0	**Power island controller output override enable**: When enabled, the power island control state machine output override bits in this register will drive the power island control state machine outputs, and override any control from power island control state machine. **Note: This bit must never change state in the same register write that any of the override bits it controls in this register are changed.** **Note: The value of this field must be the same for all lanes that make up a link.** • 1'b0: Override disabled • 1'b1: Override enabled
13:12	R/O	2'h0	Reserved
11	R/W	1'h0	**Power suspend request override**: When enabled, this bit will override the power_suspend_req input of the power island control state machine. **Note: The value of this field must be the same for all lanes that make up a link.**
10	R/O	1'h1	**Power suspend acknowledge**: This is the current state of the power_suspend_ack output from the power island control state machine. **Note: This bit will always be set to 1'b1 after the power island is enabled, as a function of the power island controller input override enable bit in this register.**
9	R/W	1'h0	**Power recover request override**: When enabled, this bit will override the power_recover_req input of the power island control state machine. **Note: The value of this field must be the same for all lanes that make up a link.**
8	R/O	1'h0	**Power recover acknowledge**: This is the current state of the power_recover_ack output from the power island control state machine. **Note: This bit will always be set to 1'b1 after the power island is enabled, as a function of the power island controller input override enable bit in this register.**

Register Name:			Transceiver transceiver calibration power island control override register **Note: This register must not be part of any register read write test where patterns other than those described in this register are written to this register.**
Register Mnemonic:			XCVR_DIAG_XCAL_PWRI_OVRD
Register Address:			01_m_nnnn : 0111 : 00001
Bits	**Type**	**Reset**	**Description**
7:0	R/W	8'h24	**Power island controller output override:** When enabled, the bits in this field will override the output signals from the power island control state machine. The following are the power island control state machine output signals, and the bits in this field that they are associated with. • Bit 7: state_sandh • Bit 6: state_ret_save • Bit 5: state_ret_restore (active low) • Bit 4: power_en_ph_1 • Bit 3: power_en_ph_2 • Bit 2: power_isolation • Bit 1: power_por_reset_n • Bit 0: power_rstr_reset_n The default value of the bits in this register are such that the power island is not forced to be switched on when the power island controller output override enable bit in this register is enabled. When using this register to switch this power island on and off, the bits in this field are required to change in a specific order with individual register writes. The writes must happen in the order specified below. • 8'b00100100 : Power island is not being forced on • 8'b10100100 • 8'b10110100 • 8'b10111100 • 8'b10111111 • 8'b10111011 • 8'b00111011 : Power island is being forced on • 8'b10111011 • 8'b10111111 • 8'b11111111 • 8'b10111111 • 8'b10111110 • 8'b10100110 : Power island is being forced off, and in state retention • 8'b10110110 • 8'b10111110 • 8'b10111111 • 8'b10011111 • 8'b10111111 • 8'b10111011 • 8'b00111011 : Power island is being forced on, with state restored **Note: The value of this field must be the same for all lanes that make up a link.**

Register Name:			Transceiver transceiver calibration power island control status register
Register Mnemonic:			XCVR_DIAG_XCAL_PWRI_STAT
Register Address:			01_m_nnnn : 0111 : 00010
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/O	8'h24	**Power island controller output status:** This field indicates the current state of the output signals from the power island control state machine. The following are the power island control state machine output signals, and the bits in this field that they are associated with. • Bit 7: state_sandh • Bit 6: state_ret_save • Bit 5: state_ret_restore (active low) • Bit 4: power_en_ph_1 • Bit 3: power_en_ph_2 • Bit 2: power_isolation • Bit 1: power_por_reset_n • Bit 0: power_rstr_reset_n

Register Name:			Transceiver transceiver data path power island control override register **Note: This register must not be part of any register read write test where patterns other than those described in this register are written to this register.**
Register Mnemonic:			XCVR_DIAG_XDP_PWRI_OVRD
Register Address:			01_m_nnnn : 0111 : 00011
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Power island controller input override enable**: When enabled, the power island control state machine input override bits in this register will drive the power island control state machine directly, and override any control from other state machines in the macro. **Note: The CDB power island must be triggered to be enabled, by accessing a register outside of the common control module control and diagnostic registers space, prior to enabling this function.** **Note: This bit must never change state in the same register write that any of the override bits it controls in this register are changed.** **Note: The value of this field must be the same for all lanes that make up a link.** • 1'b0: Override disabled • 1'b1: Override enabled
14	R/W	1'h0	**Power island controller output override enable**: When enabled, the power island control state machine output override bits in this register will drive the power island control state machine outputs, and override any control from power island control state machine. **Note: This bit must never change state in the same register write that any of the override bits it controls in this register are changed.** **Note: The value of this field must be the same for all lanes that make up a link.** • 1'b0: Override disabled • 1'b1: Override enabled
13:12	R/O	2'h0	Reserved
11	R/W	1'h0	**Power suspend request override**: When enabled, this bit will override the power_suspend_req input of the power island control state machine. **Note: The value of this field must be the same for all lanes that make up a link.**
10	R/O	1'h1	**Power suspend acknowledge**: This is the current state of the power_suspend_ack output from the power island control state machine. **Note: This bit will always be set to 1'b1 after the power island is enabled, as a function of the power island controller input override enable bit in this register.**
9	R/W	1'h0	**Power recover request override**: When enabled, this bit will override the power_recover_req input of the power island control state machine. **Note: The value of this field must be the same for all lanes that make up a link.**
8	R/O	1'h0	**Power recover acknowledge**: This is the current state of the power_recover_ack output from the power island control state machine. **Note: This bit will always be set to 1'b1 when the power island controller input override enable bit in this register.**

Register Name:			Transceiver transceiver data path power island control override register **Note: This register must not be part of any register read write test where patterns other than those described in this register are written to this register.**
Register Mnemonic:			XCVR_DIAG_XDP_PWRI_OVRD
Register Address:			01_m_nnnn : 0111 : 00011
Bits	**Type**	**Reset**	**Description**
7:0	R/W	8'h24	**Power island controller output override:** When enabled, the bits in this field will override the output signals from the power island control state machine. The following are the power island control state machine output signals, and the bits in this field that they are associated with. • Bit 7: state_sandh • Bit 6: state_ret_save • Bit 5: state_ret_restore (active low) • Bit 4: power_en_ph_1 • Bit 3: power_en_ph_2 • Bit 2: power_isolation • Bit 1: power_por_reset_n • Bit 0: power_rstr_reset_n The default value of the bits in this register are such that the power island is not forced to be switched on when the power island controller output override enable bit in this register is enabled. When using this register to switch this power island on and off, the bits in this field are required to change in a specific order with individual register writes. The writes must happen in the order specified below. • 8'b00100100 : Power island is not being forced on • 8'b10100100 • 8'b10110100 • 8'b10111100 • 8'b10111111 • 8'b10111011 • 8'b00111011 : Power island is being forced on • 8'b10111011 • 8'b10111111 • 8'b11111111 • 8'b10111111 • 8'b10111110 • 8'b10100110 : Power island is being forced off, and in state retention • 8'b10110110 • 8'b10111110 • 8'b10111111 • 8'b10011111 • 8'b10111111 • 8'b10111011 • 8'b00111011 : Power island is being forced on, with state restored **Note: The value of this field must be the same for all lanes that make up a link.**

Register Name:			Transceiver transceiver data path power island control status register
Register Mnemonic:			XCVR_DIAG_XDP_PWRI_STAT
Register Address:			01_m_nnnn : 0111 : 00100
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/O	8'h24	**Power island controller output status:** This field indicates the current state of the output signals from the power island control state machine. The following are the power island control state machine output signals, and the bits in this field that they are associated with. • Bit 7: state_sandh • Bit 6: state_ret_save • Bit 5: state_ret_restore (active low) • Bit 4: power_en_ph_1 • Bit 3: power_en_ph_2 • Bit 2: power_isolation • Bit 1: power_por_reset_n • Bit 0: power_rstr_reset_n

Register Name:			Transceiver PLL data rate clock control register
Register Mnemonic:			XCVR_DIAG_PLLDRC_CTRL
Register Address:			01_m_nnnn : 0111 : 00101
Bits	**Type**	**Reset**	**Description**
15:14	R/W	2'h0	**Digital PLL clock select standard mode 3**: This bit controls which full rate PLL clock is selected, when xcvr_standard_mode is set to 2'b11. • 2'b00 : PLL clock 0 from the PLL 0. • 2'b01 : PLL clock 1 from the PLL 0. • 2'b10 : PLL clock 0 from the PLL 1. • 2'b11 : PLL clock 1 from the PLL 1.
13:12	R/W	2'h0	**Digital PLL data rate divider standard mode 3 value**: This field will directly control the xcvr_pll_clk_-datart_div signal, which controls which divided clock is selected when generating the xcvr_pll_clk_datart clock, when xcvr_standard_mode is set to 2'b11. • 2'b00 : Divide by 1. • 2'b01 : Divide by 2. • 2'b10 : Divide by 4. • 2'b11 : Divide by 8.
11:10	R/W	2'h0	**Digital PLL clock select standard mode 2**: This bit controls which full rate PLL clock is selected, when xcvr_standard_mode is set to 2'b10. • 2'b00 : PLL clock 0 from the PLL 0. • 2'b01 : PLL clock 1 from the PLL 0. • 2'b10 : PLL clock 0 from the PLL 1. • 2'b11 : PLL clock 1 from the PLL 1.
9:8	R/W	2'h0	**Digital PLL data rate divider standard mode 2 value**: This field will directly control the xcvr_pll_clk_-datart_div signal, which controls which divided clock is selected when generating the xcvr_pll_clk_datart clock, when xcvr_standard_mode is set to 2'b10. • 2'b00 : Divide by 1. • 2'b01 : Divide by 2. • 2'b10 : Divide by 4. • 2'b11 : Divide by 8.
7:6	R/W	2'h0	**Digital PLL clock select standard mode 1**: This bit controls which full rate PLL clock is selected, when xcvr_standard_mode is set to 2'b01. • 2'b00 : PLL clock 0 from the PLL 0. • 2'b01 : PLL clock 1 from the PLL 0. • 2'b10 : PLL clock 0 from the PLL 1. • 2'b11 : PLL clock 1 from the PLL 1.
5:4	R/W	2'h1	**Digital PLL data rate divider standard mode 1 value**: This field will directly control the xcvr_pll_clk_-datart_div signal, which controls which divided clock is selected when generating the xcvr_pll_clk_datart clock, when xcvr_standard_mode is set to 2'b01. • 2'b00 : Divide by 1. • 2'b01 : Divide by 2. • 2'b10 : Divide by 4. • 2'b11 : Divide by 8.
3:2	R/W	2'h0	**Digital PLL clock select standard mode 0**: This bit controls which full rate PLL clock is selected, when xcvr_standard_mode is set to 2'b00. • 2'b00 : PLL clock 0 from the PLL 0. • 2'b01 : PLL clock 1 from the PLL 0. • 2'b10 : PLL clock 0 from the PLL 1. • 2'b11 : PLL clock 1 from the PLL 1.
1:0	R/W	2'h2	**Digital PLL data rate divider standard mode 0 value**: This field will directly control the xcvr_pll_clk_-datart_div signal, which controls which divided clock is selected when generating the xcvr_pll_clk_datart clock, when xcvr_standard_mode is set to 2'b00. • 2'b00 : Divide by 1. • 2'b01 : Divide by 2. • 2'b10 : Divide by 4. • 2'b11 : Divide by 8.

Register Name:			Transceiver high speed clock select register
Register Mnemonic:			XCVR_DIAG_HSCLK_SEL
Register Address:			01_m_nnnn : 0111 : 00110
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'h0	Reserved
13:12	R/W	2'h0	**High speed clock select standard mode 3**: This specifies which analog high speed clock is selected when xcvr_standard_mode is set to 2'b11, by driving the xcvrda_clk_sel signal to the analog, as specified below. • 2'b00 : clock 0. • 2'b01 : clock 1. • 2'b10 : Reserved. • 2'b11 : Reserved.
11:10	R/O	2'h0	Reserved
9:8	R/W	2'h0	**High speed clock select standard mode 2**: This specifies which analog high speed clock is selected when xcvr_standard_mode is set to 2'b10, by driving the xcvrda_clk_sel signal to the analog, as specified below. • 2'b00 : clock 0. • 2'b01 : clock 1. • 2'b10 : Reserved. • 2'b11 : Reserved.
7:6	R/O	2'h0	Reserved
5:4	R/W	2'h0	**High speed clock select standard mode 1**: This specifies which analog high speed clock is selected when xcvr_standard_mode is set to 2'b01, by driving the xcvrda_clk_sel signal to the analog, as specified below. • 2'b00 : clock 0. • 2'b01 : clock 1. • 2'b10 : Reserved. • 2'b11 : Reserved.
3:2	R/O	2'h0	Reserved
1:0	R/W	2'h0	**High speed clock select standard mode 0**: This specifies which analog high speed clock is selected when xcvr_standard_mode is set to 2'b00, by driving the xcvrda_clk_sel signal to the analog, as specified below. • 2'b00 : clock 0. • 2'b01 : clock 1. • 2'b10 : Reserved. • 2'b11 : Reserved.

Register Name:			Transceiver high speed clock divider register
Register Mnemonic:			XCVR_DIAG_HSCLK_DIV
Register Address:			01_m_nnnn : 0111 : 00111
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:12	R/W	3'h0	**Transceiver clock divider select standard mode 3**: This field selects the divider value used to divide the selected analog high speed clock by when generating the transmit and receive clocks, when xcvr_standard_mode is set to 2'b11, by driving the xcvrda_xcvr_clk_div_sel signal to the analog, as specified below. • 3'b000: Div 1 • 3'b001: Div 2 • 3'b010: Div 3 • 3'b011: Div 4 • 3'b100: Div 5 • 3'b101: Div 6 • 3'b110: Div 7 • 3'b111: Div 8
11	R/O	1'h0	Reserved
10:8	R/W	3'h0	**Transceiver clock divider select standard mode 2**: This field selects the divider value used to divide the selected analog high speed clock by when generating the transmit and receive clocks, when xcvr_standard_mode is set to 2'b10, by driving the xcvrda_xcvr_clk_div_sel signal to the analog, as specified below. • 3'b000: Div 1 • 3'b001: Div 2 • 3'b010: Div 3 • 3'b011: Div 4 • 3'b100: Div 5 • 3'b101: Div 6 • 3'b110: Div 7 • 3'b111: Div 8
7	R/O	1'h0	Reserved
6:4	R/W	3'h0	**Transceiver clock divider select standard mode 1**: This field selects the divider value used to divide the selected analog high speed clock by when generating the transmit and receive clocks, when xcvr_standard_mode is set to 2'b01, by driving the xcvrda_xcvr_clk_div_sel signal to the analog, as specified below. • 3'b000: Div 1 • 3'b001: Div 2 • 3'b010: Div 3 • 3'b011: Div 4 • 3'b100: Div 5 • 3'b101: Div 6 • 3'b110: Div 7 • 3'b111: Div 8
3	R/O	1'h0	Reserved
2:0	R/W	3'h1	**Transceiver clock divider select standard mode 0**: This field selects the divider value used to divide the selected analog high speed clock by when generating the transmit and receive clocks, when xcvr_standard_mode is set to 2'b00, by driving the xcvrda_xcvr_clk_div_sel signal to the analog, as specified below. • 3'b000: Div 1 • 3'b001: Div 2 • 3'b010: Div 3 • 3'b011: Div 4 • 3'b100: Div 5 • 3'b101: Div 6 • 3'b110: Div 7 • 3'b111: Div 8

Register Name:			TX clock control register
Register Mnemonic:			XCVR_DIAG_TXCLK_CTRL
Register Address:			01_m_nnnn : 0111 : 01000
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**TX serializer clock invert**: This bit is used to optionally invert the serializer clock (txda_ser_clk) for diagnostic purposes.
14:0	R/O	15'h0	Reserved

Register Name:			RX clock control register
Register Mnemonic:			XCVR_DIAG_RXCLK_CTRL
Register Address:			01_m_nnnn : 0111 : 01001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**RX deserializer clock invert**: This bit is used to optionally invert the deserializer clock (rxda_des_clk) for diagnostic purposes.
14	R/W	1'h0	**RX 2x clock enable:** This bit enables the receiver 2x clock function, by driving the rxda_des_clk_2x_en signal going to the analog. **Note: This clock is only designed to be used in modes where the data width is 20 bits.** 1'b0 : Disabled 1'b1 : Enabled
13	R/W	1'h0	**RX PI E path clock select**: Controls which PI clock drives the E path clocks, by driving the rxda_pi_i_drv_e_en signal going to the analog. • 1'b0: E PI clock • 1'b1: IQ PI clock
12:8	R/O	5'h0	Reserved
7	R/W	1'h0	**PI output clock divider enable standard mode 3**: This bit controls the rxda_pi_out_clk_div_en to enable the PI output clock divider, when xcvr_standard_mode is set to 2'b11. 1'b0 : Disabled 1'b1 : Enabled
6	R/W	1'h0	**PI output clock divider enable standard mode 3**: This bit controls the rxda_pi_out_clk_div_en to enable the PI output clock divider, when xcvr_standard_mode is set to 2'b11. 1'b0 : Disabled 1'b1 : Enabled
5	R/W	1'h0	**PI output clock divider enable standard mode 3**: This bit controls the rxda_pi_out_clk_div_en to enable the PI output clock divider, when xcvr_standard_mode is set to 2'b11. 1'b0 : Disabled 1'b1 : Enabled
4	R/W	1'h0	**PI output clock divider enable standard mode 3**: This bit controls the rxda_pi_out_clk_div_en to enable the PI output clock divider, when xcvr_standard_mode is set to 2'b11. 1'b0 : Disabled 1'b1 : Enabled
3:0	R/O	4'h0	Reserved

Register Name:			Transceiver bidirectional control register
Register Mnemonic:			XCVR_DIAG_BIDI_CTRL
Register Address:			01_m_nnnn : 0111 : 01010
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7	R/W	1'h1	**Receiver enable standard mode 3**: When bidirectional bumps are implemented in the transceiver, this bit is a global enable for the receiver function, when xcvr_standard_mode is set to 2'b11. When bidirectional bumps are not implemented, this bit will be reset to 1'b1, and should not be changed. When bidirectional bumps are implemented in the transceiver, this bit and the corresponding transmitter enable bit must not be set to 1'b1 at the same time. • 1'b0 : RX disabled • 1'b1 : RX enabled
6	R/W	1'h1	**Receiver enable standard mode 2**: When bidirectional bumps are implemented in the transceiver, this bit is a global enable for the receiver function, when xcvr_standard_mode is set to 2'b10. When bidirectional bumps are not implemented, this bit will be reset to 1'b1, and should not be changed. When bidirectional bumps are implemented in the transceiver, this bit and the corresponding transmitter enable bit must not be set to 1'b1 at the same time. • 1'b0 : RX disabled • 1'b1 : RX enabled
5	R/W	1'h1	**Receiver enable standard mode 1**: When bidirectional bumps are implemented in the transceiver, this bit is a global enable for the receiver function, when xcvr_standard_mode is set to 2'b01. When bidirectional bumps are not implemented, this bit will be reset to 1'b1, and should not be changed. When bidirectional bumps are implemented in the transceiver, this bit and the corresponding transmitter enable bit must not be set to 1'b1 at the same time. • 1'b0 : RX disabled • 1'b1 : RX enabled
4	R/W	1'h1	**Receiver enable standard mode 0**: When bidirectional bumps are implemented in the transceiver, this bit is a global enable for the receiver function, when xcvr_standard_mode is set to 2'b00. When bidirectional bumps are not implemented, this bit will be reset to 1'b1, and should not be changed. When bidirectional bumps are implemented in the transceiver, this bit and the corresponding transmitter enable bit must not be set to 1'b1 at the same time. • 1'b0 : RX disabled • 1'b1 : RX enabled
3	R/W	1'h1	**Transmitter enable standard mode 3**: When bidirectional bumps are implemented in the transceiver, this bit is a global enable for the transmitter function, when xcvr_standard_mode is set to 2'b11. When bidirectional bumps are not implemented, this bit will be reset to 1'b1, and should not be changed. When bidirectional bumps are implemented in the transceiver, this bit and the corresponding receiver enable bit must not be set to 1'b1 at the same time. • 1'b0 : TX disabled • 1'b1 : TX enabled
2	R/W	1'h1	**Transmitter enable standard mode 2**: When bidirectional bumps are implemented in the transceiver, this bit is a global enable for the transmitter function, when xcvr_standard_mode is set to 2'b10. When bidirectional bumps are not implemented, this bit will be reset to 1'b1, and should not be changed. When bidirectional bumps are implemented in the transceiver, this bit and the corresponding receiver enable bit must not be set to 1'b1 at the same time. • 1'b0 : TX disabled • 1'b1 : TX enabled
1	R/W	1'h1	**Transmitter enable standard mode 1**: When bidirectional bumps are implemented in the transceiver, this bit is a global enable for the transmitter function, when xcvr_standard_mode is set to 2'b01. When bidirectional bumps are not implemented, this bit will be reset to 1'b1, and should not be changed. When bidirectional bumps are implemented in the transceiver, this bit and the corresponding receiver enable bit must not be set to 1'b1 at the same time. • 1'b0 : TX disabled • 1'b1 : TX enabled
0	R/W	1'h1	**Transmitter enable standard mode 0**: When bidirectional bumps are implemented in the transceiver, this bit is a global enable for the transmitter function, when xcvr_standard_mode is set to 2'b00. When bidirectional bumps are not implemented, this bit will be reset to 1'b1, and should not be changed. When bidirectional bumps are implemented in the transceiver, this bit and the corresponding receiver enable bit must not be set to 1'b1 at the same time. • 1'b0 : TX disabled • 1'b1 : TX enabled

Register Name:			Power state controller override register
Register Mnemonic:			XCVR_DIAG_PSC_OVRD
Register Address:			01_m_nnnn : 0111 : 01011
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12'h0	Reserved
3	R/W	1'h0	**Receiver DFE enable mask value standard mode 3**: This bit will controls the rx_dfe_eq_enable_std_mask digital signal, which can be used to mask off the rxda_dfe_eq_enable signal coming from the power state controller and going into the analog, when xcvr_standard_mode is set to 2'b11. • 1'b0 : The signal will be masked off. 1'b1 : The signal to pass through unchanged.
2	R/W	1'h1	**Receiver DFE enable mask value standard mode 2**: This bit will controls the rx_dfe_eq_enable_std_mask digital signal, which can be used to mask off the rxda_dfe_eq_enable signal coming from the power state controller and going into the analog, when xcvr_standard_mode is set to 2'b10. • 1'b0 : The signal will be masked off. 1'b1 : The signal to pass through unchanged.
1	R/W	1'h1	**Receiver DFE enable mask value standard mode 1**: This bit will controls the rx_dfe_eq_enable_std_mask digital signal, which can be used to mask off the rxda_dfe_eq_enable signal coming from the power state controller and going into the analog, when xcvr_standard_mode is set to 2'b01. • 1'b0 : The signal will be masked off. 1'b1 : The signal to pass through unchanged.
0	R/W	1'h0	**Receiver DFE enable mask value standard mode 0**: This bit will controls the rx_dfe_eq_enable_std_mask digital signal, which can be used to mask off the rxda_dfe_eq_enable signal coming from the power state controller and going into the analog, when xcvr_standard_mode is set to 2'b00. • 1'b0 : The signal will be masked off. 1'b1 : The signal to pass through unchanged.

Register Name:			Transceiver control reset diagnostic register
Register Mnemonic:			XCVR_DIAG_RST_DIAG
Register Address:			01_m_nnnn : 0111 : 01100
Bits	**Type**	**Reset**	**Description**
15:3	R/O	13'h0	Reserved
2	R/O	1'h0	Current state of the tx_coef_calc_reset_n reset.
1	R/O	1'h0	Current state of the xcvr_psm_reset_n reset.
0	R/O	1'h0	Current state of the xcvr_ref_clk_reset_n reset.

Register Name:			Transceiver clock reset control register
Register Mnemonic:			XCVR_DIAG_XCVR_CLK_CTRL
Register Address:			01_m_nnnn : 0111 : 01101
Bits	**Type**	**Reset**	**Description**
15:6	R/O	11'h0	Reserved
5:0	R/W	6'h0D	**Transceiver clock enable delay timer value**: This specifies the number of xcvr_psm_clk clock cycles that the transceiver high speed clock reset release state machine will wait between when it drives the analog transceiver clock enable signal active, and when it will initiate the process to release the reset for this clock. The default value of this register corresponds to a delay of at least 100 nSec.

Register Name:			Transceiver digital cover your alternatives register
Register Mnemonic:			XCVR_DIAG_DCYA
Register Address:			01_m_nnnn : 0111 : 01111
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h0	Reserved - spare

10.2.21 Driver Controller Control And Diagnostics Registers

Base address bits 15:5 :	01_m_nnnn : 0110
Address bits 4:0	**Register name and location of detailed description**
00000	Lane fast common mode enable timeout register on page 244
00001	Lane fast common mode enable sample wait timer register on page 244
00010	Lane fast common mode enable margin timer register on page 244
00011	Lane fast common mode enable tuning register on page 245
00100	Transmitter LFPS control register on page 245
00101	TX receiver detect tuning register on page 245
00110	TX driver diagnostic register on page 246
00111 - 11111	Reserved

Table 111: Driver controller control and diagnostic registers address space

Register Name:			Lane fast common mode enable timeout register
Register Mnemonic:			DRV_DIAG_LANE_FCM_EN_TO
Register Address:			01_m_nnnn : 0110 : 00000
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Bypass fast establishment of common mode enable**: When enabled, the fast establishment of common mode function will be bypassed. When bypassed, the lane fast common mode enable timeout value field in this register can be used to implement a programmable delay between when fast establishment of common mode is normally initiated, and when it will complete.
14:12	R/O	3'h0	Reserved
11:0	R/W	12'h1F4	**Lane fast common mode enable timeout value**: The usage of the value of this field is a function of the state of the bypass fast establishment of common mode enable bit in this register. • Bypass disabled (normal operation): This specifies the maximum number of reference clock cycles the fast establishment of common mode process will wait for the tx_{p,m} signals to be reported as above the reference voltage, before timing out. The default value of this register corresponds to a delay of 20 uSec. • Bypass enabled: This can be used to specify the number of reference clock cycles between when fast establishment of common mode would normally be initiated, and when it is acknowledged to have been completed.

Register Name:			Lane fast common mode enable sample wait timer register
Register Mnemonic:			DRV_DIAG_LANE_FCM_EN_SWAIT_TMR
Register Address:			01_m_nnnn : 0110 : 00001
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12'h0	Reserved
3:0	R/W	4'h6	**Lane fast common mode enable sample wait timer value**: This specifies the number of reference clock cycles the fast establishment of common mode process will wait between changing the state of the signals controlling the analog common mode sense circuits, and testing the results of the new state.

Register Name:			Lane fast common mode enable margin timer register
Register Mnemonic:			DRV_DIAG_LANE_FCM_EN_MGN_TMR
Register Address:			01_m_nnnn : 0110 : 00010
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:0	R/W	12'h096	**Lane fast common mode enable margin timer value**: This specifies the number of reference clock cycles the fast establishment of common mode process will enable all the margin segments for. The default value of this register corresponds to a delay of 6 uSec.

Register Name:			Lane fast common mode enable tuning register
Register Mnemonic:			DRV_DIAG_LANE_FCM_EN_TUNE
Register Address:			01_m_nnnn : 0110 : 00011
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:8	R/W	4'h8	**Common mode sense reference DAC voltage initial test**: This field sets the common mode detect reference voltage for the common mode detect comparator, when the fast establishment of common mode function is checking the initial state of the txda_cm_sense_comp_out signal. This field drives the txda_cm_sense_vref_dac signal going to the analog, and it is only driven when the fast establishment of common mode function is enabled. • 4'b0000 : Minimum • ... • 4'b1111 : Maximum
7:4	R/W	4'hC	**Common mode sense reference DAC voltage high test**: This field sets the common mode detect reference voltage for the common mode detect comparator, when the fast establishment of common mode function is waiting for the txda_cm_sense_comp_out signal from the analog to be driven high. This field drives the txda_cm_sense_vref_dac signal going to the analog, and it is only driven when the fast establishment of common mode function is enabled. • 4'b0000 : Minimum • ... • 4'b1111 : Maximum
3:0	R/W	4'h4	**Common mode sense reference DAC voltage low test**: This field sets the common mode detect reference voltage for the common mode detect comparator, when the fast establishment of common mode function is waiting for the txda_cm_sense_comp_out signal from the analog to be driven low. This field drives the txda_cm_sense_vref_dac signal going to the analog, and it is only driven when the fast establishment of common mode function is enabled. • 4'b0000 : Minimum • ... • 4'b1111 : Maximum

Register Name:			Transmitter LFPS control register
Register Mnemonic:			DRV_DIAG_LFPS_CTRL
Register Address:			01_m_nnnn : 0110 : 00100
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:0	R/W	8'h0F	**LFPS half period clocks**: Specifies the number of clock cycles required to implement one half of a LFPS period, by the transmitter LFPS controller.

Register Name:			TX receiver detect tuning register
Register Mnemonic:			DRV_DIAG_RCVDET_TUNE
Register Address:			01_m_nnnn : 0110 : 00101
Bits	**Type**	**Reset**	**Description**
15:5	R/O	11'h0	Reserved
4	R/W	1'h0	**Receiver detect comparators output and or control**: This bit controls how the receiver detect comparators are used to drive the txda_rcvdet_detected_n signal to the digital. This bit drives the txda_rcvdet_and_orb signal going to the analog. • 1'b0: Comparator outputs are ORed to drive txda_rcvdet_detected_n. • 1'b1: Comparator outputs are ANDed to drive txda_rcvdet_detected_n.
3:0	R/W	4'hC	**Receiver detect reference DAC voltage**: This field sets the receiver detect reference voltage for the receiver detect comparator. This field drives the txda_rcvdet_vref_dac signal going to the analog. • 4'b0000 : Minimum • ... • 4'b1111 : Maximum

Register Name:			TX driver diagnostic register
Register Mnemonic:			DRV_DIAG_TX_DRV
Register Address:			01_m_nnnn : 0110 : 00110
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7	R/W	1'h1	**TX boost enable**: Increases the transmitter amplitude for fast data transitions, by controlling the txda_drv_boost_en signal going to the analog.
6	R/O	1'h0	Reserved
5:4	R/W	2'h2	**TX boost tune**: Controls the transmitter boost amplitude when the transmitter boost function is enabled using the TX boost enable bit in this register, by controlling the txda_drv_boost_tune signal going to the analog. The following specifies the encoding of this field. • 2'b00: Minimum Boost • 2'b01: • 2'b10: • 2'b11: Maximum Boost
3:2	R/O	2'h0	Reserved
1	R/W	1'h1	**TX pre-driver pull up control**: When the pre-driver is disabled, this bit controls the state of the pre-driver output, by controlling the txda_drv_predrv_pullup signal going to the analog. • 1'b0 : Pulled low • 1'b1 : Pulled high
0	R/W	1'h1	**TX driver margin type**: Selects the margining type the driver will operate in, by controlling the txda_drv_-margin_type signal going to the analog. • 1'b0 : Classical margining - High power margining that complies with the driver differential and common-mode output impedance specs. • 1'b1 : Low-Power margining - Low power margining that complies with the driver differential output impedance spec. However, this violates the driver common-mode output impedance spec to save power whenever a reduced swing driver configuration is used. Note: When fast establishment of common mode is active, the analog signal that this drives will be forced to 1'b0.

10.2.22 Transmitter Power State Controller Registers

Base address bits 15:5 :	01_m_nnnn : 1000
Address bits 4:0	**Register name and location of detailed description**
00000	Transmitter A0 power state definition register on page 247
00001	Transmitter A1 power state definition register on page 248
00010	Transmitter A2 power state definition register on page 248
00011	Transmitter A3 power state definition register on page 249
00100	Transmitter A4 power state definition register on page 249
00101	Transmitter A5 power state definition register on page 250
00110	Transmitter calibration power state definition register on page 250
00111	Transmitter ready power state definition register on page 251
01000 - 11111	Reserved

Table 112: Transmitter power state controller registers address space

Register Name:			Transmitter A0 power state definition register
Register Mnemonic:			TX_PSC_A0
Register Address:			01_m_nnnn : 1000 : 00000
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4’h0	Reserved
11	R/W	1’h0	**TX driver common mode enable extend control**: Specifies which power states the tx_cmn_-mode_en_ext signal is considered valid in and can be used to force the driver to continue to be in the common mode state.
10	R/W	1’h0	**Force txda_lfps_sel active**: Setting this bit forces the txda_lfps_sel signal going to the analog to be driven active. Note: This function is implemented for additional power savings. In the analog, this signal controls a MUX select associated with the driver. Setting this bit to 1’b1 will force the MUX to select the LFPS data path. By selecting this path in low power states, instead of the serializer path, the driver inputs will be driven to 1’b0, instead of having the potential to toggle.
9	R/W	1’h0	**LFPS clock gate enable**: Enables the LFPS clock gate when an LFPS clock is required.
8	R/W	1’h0	Reserved - spare
7	R/W	1’h1	**Transmitter low current mode**: Enables a low current consumption mode within the common mode voltage circuit in the driver, via the txda_drv_idle_lowi_en signal going to the analog.
6	R/W	1’h1	**Transmitter mission mode enable**: Enables the analog circuits in the driver required to run in mission mode, via the txda_drv_mission_en signal going to the analog.
5	R/W	1’h1	**TX driver common mode enable**: Enables the common mode voltage circuits in the driver.
4	R/W	1’h1	**TX driver enable**: Enables the transmitter driver, via the H bridge driver controller.
3	R/W	1’h1	**TX post-emphasis enable (C_{+1})**: Enables the transmitter circuits related to the post-emphasis function.
2	R/W	1’h1	**TX pre-emphasis enable (C_{-1})**: Enables the transmitter circuits related to the pre-emphasis function.
1	R/W	1’h1	**TX pre-driver enable**: Enables the transmitter pre-driver, driver data selection MUX, and receiver detect.
0	R/W	1’h1	**TX serializer enable**: Enables the serializer and related clock divider circuits.

Register Name:			Transmitter A1 power state definition register
Register Mnemonic:			TX_PSC_A1
Register Address:			01_m_nnnn : 1000 : 00001
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11	R/W	1'h0	**TX driver common mode enable extend control**
10	R/W	1'h1	**Force txda_lfps_sel active**
9	R/W	1'h0	**LFPS clock gate enable**
8	R/W	1'h0	Reserved - spare
7	R/W	1'h1	**Transmitter low current mode**
6	R/W	1'h0	**Transmitter mission mode enable**
5	R/W	1'h1	**TX driver common mode enable**
4	R/W	1'h0	**TX driver enable**
3	R/W	1'h1	**TX post-emphasis enable (C_{+1})**
2	R/W	1'h1	**TX pre-emphasis enable (C_{-1})**
1	R/W	1'h1	**TX pre-driver enable**
0	R/W	1'h1	**TX serializer enable**

Register Name:			Transmitter A2 power state definition register
Register Mnemonic:			TX_PSC_A2
Register Address:			01_m_nnnn : 1000 : 00010
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11	R/W	1'h0	**TX driver common mode enable extend control**
10	R/W	1'h1	**Force txda_lfps_sel active**
9	R/W	1'h0	**LFPS clock gate enable**
8	R/W	1'h0	Reserved - spare
7	R/W	1'h1	**Transmitter low current mode**
6	R/W	1'h0	**Transmitter mission mode enable**
5	R/W	1'h1	**TX driver common mode enable**
4	R/W	1'h0	**TX driver enable**
3	R/W	1'h1	**TX post-emphasis enable (C_{+1})**
2	R/W	1'h1	**TX pre-emphasis enable (C_{-1})**
1	R/W	1'h1	**TX pre-driver enable**
0	R/W	1'h0	**TX serializer enable**

Register Name:			Transmitter A3 power state definition register
Register Mnemonic:			TX_PSC_A3
Register Address:			01_m_nnnn : 1000 : 00011
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11	R/W	1'h0	**TX driver common mode enable extend control**
10	R/W	1'h1	**Force txda_lfps_sel active**
9	R/W	1'h0	**LFPS clock gate enable**
8	R/W	1'h0	Reserved - spare
7	R/W	1'h1	**Transmitter low current mode**
6	R/W	1'h0	**Transmitter mission mode enable**
5	R/W	1'h1	**TX driver common mode enable**
4	R/W	1'h0	**TX driver enable**
3	R/W	1'h1	**TX post-emphasis enable (C_{+1})**
2	R/W	1'h1	**TX pre-emphasis enable (C_{-1})**
1	R/W	1'h1	**TX pre-driver enable**
0	R/W	1'h0	**TX serializer enable**

Register Name:			Transmitter A4 power state definition register
Register Mnemonic:			TX_PSC_A4
Register Address:			01_m_nnnn : 1000 : 00100
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11	R/W	1'h1	**TX driver common mode enable extend control**
10	R/W	1'h0	**Force txda_lfps_sel active**
9	R/W	1'h0	**LFPS clock gate enable**
8	R/W	1'h0	Reserved - spare
7	R/W	1'h1	**Transmitter low current mode**
6	R/W	1'h0	**Transmitter mission mode enable**
5	R/W	1'h0	**TX driver common mode enable**
4	R/W	1'h0	**TX driver enable**
3	R/W	1'h0	**TX post-emphasis enable (C_{+1})**
2	R/W	1'h0	**TX pre-emphasis enable (C_{-1})**
1	R/W	1'h0	**TX pre-driver enable**
0	R/W	1'h0	**TX serializer enable**

Register Name:			Transmitter A5 power state definition register
Register Mnemonic:			TX_PSC_A5
Register Address:			01_m_nnnn : 1000 : 00101
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11	R/W	1'h0	**TX driver common mode enable extend control**
10	R/W	1'h0	**Force txda_lfps_sel active**
9	R/W	1'h0	**LFPS clock gate enable**
8	R/W	1'h0	Reserved - spare
7	R/W	1'h0	**Transmitter low current mode**
6	R/W	1'h0	**Transmitter mission mode enable**
5	R/W	1'h0	**TX driver common mode enable**
4	R/W	1'h0	**TX driver enable**
3	R/W	1'h0	**TX post-emphasis enable (C_{+1})**
2	R/W	1'h0	**TX pre-emphasis enable (C_{-1})**
1	R/W	1'h0	**TX pre-driver enable**
0	R/W	1'h0	**TX serializer enable**

Register Name:			Transmitter calibration power state definition register
Register Mnemonic:			TX_PSC_CAL
Register Address:			01_m_nnnn : 1000 : 00110
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11	R/W	1'h0	**TX driver common mode enable extend control**
10	R/W	1'h0	**Force txda_lfps_sel active**
9	R/W	1'h0	**LFPS clock gate enable**
8	R/W	1'h0	Reserved - spare
7	R/W	1'h1	**Transmitter low current mode**
6	R/W	1'h0	**Transmitter mission mode enable**
5	R/W	1'h1	**TX driver common mode enable**
4	R/W	1'h0	**TX driver enable**
3	R/W	1'h0	**TX post-emphasis enable (C_{+1})**
2	R/W	1'h0	**TX pre-emphasis enable (C_{-1})**
1	R/W	1'h0	**TX pre-driver enable**
0	R/W	1'h0	**TX serializer enable**

Register Name:			Transmitter ready power state definition register
Register Mnemonic:			TX_PSC_RDY
Register Address:			01_m_nnnn : 1000 : 00111
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11	R/W	1'h0	**TX driver common mode enable extend control**
10	R/W	1'h0	**Force txda_lfps_sel active**
9	R/W	1'h0	**LFPS clock gate enable**
8	R/W	1'h0	Reserved - spare
7	R/W	1'h1	**Transmitter low current mode**
6	R/W	1'h0	**Transmitter mission mode enable**
5	R/W	1'h1	**TX driver common mode enable**
4	R/W	1'h0	**TX driver enable**
3	R/W	1'h0	**TX post-emphasis enable (C_{+1})**
2	R/W	1'h0	**TX pre-emphasis enable (C_{-1})**
1	R/W	1'h0	**TX pre-driver enable**
0	R/W	1'h0	**TX serializer enable**

10.2.23 Receiver Detect Registers

Base address bits 15:5 :	**01_m_nnnn : 1001**
Address bits 4:0	**Register name and location of detailed description**
00000	Transmit receiver detect control register on page 251
00001	Transmit receiver detect override register on page 252
00010	Transmit receiver detect enable timer register on page 252
00011	Transmit receiver detect start timer register on page 252
00100 - 11111	Reserved

Table 113: Receiver detect registers address space

Register Name:			Transmit receiver detect control register
Register Mnemonic:			TX_RCVDET_CTRL
Register Address:			01_m_nnnn : 1001 : 00000
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Start receiver detect**: Activating (1'b1) this bit will start the receiver detect process. This bit must remain active until the receiver detect process is complete, as indicated by the receiver detect process done bit in this register. To start another receiver detect process, this register must first be set inactive (1'b0) until the receiver detect process done bit in this register is cleared. **Note: This bit is intended to be for diagnostics purposes only. The receiver detect process must not be under the control of the top level tx_rcv_detect_en pin when attempting to initiate the process using this register.**
14	R/O	1'h0	**Receiver detect process done**: This bit will be set to 1'b1 when the receiver detect process is complete. It will be cleared by cmn_reset_n, or by the deactivation of the start receiver detect bit in this register. **Note: This bit is intended to be for diagnostics purposes only.**
13	R/O	1'h0	**Receiver detected**: When the receiver detect process is complete, this register bit will indicate the current state of the tx_rcv_detected pin.
12:0	R/O	13'h0	Reserved

Register Name:			Transmit receiver detect override register
Register Mnemonic:			TX_RCVDET_OVRD
Register Address:			01_m_nnnn : 1001 : 00001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Receiver detect override enable**: Activation (1'b1) of this register bit enables the tx_rcv_detected output from the receiver detect state machine to be driven directly by the receiver detect override bit in this register.
14	R/W	1'h0	**Receiver detect override**: When the receiver detect override enable bit in this register is active (1'b1), this bit will directly control the tx_rcv_detected output from the receiver detect state machine.
13:0	R/O	14'h0	Reserved

Register Name:			Transmit receiver detect enable timer register
Register Mnemonic:			TX_RCVDET_EN_TMR
Register Address:			01_m_nnnn : 1001 : 00010
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h0000	**Enable wait time value**: This is the number of clocks the receiver detect state machine waits between driving the txda_rcvdet_en signal active and driving the txda_rcvdet_start signal active going to the analog.

Register Name:			Transmit receiver detect start timer register
Register Mnemonic:			TX_RCVDET_ST_TMR
Register Address:			01_m_nnnn : 1001 : 00011
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h09C4	**Start wait time value**: This is the number of clocks the receiver detect state machine waits between driving the txda_rcvdet_start signal active and checking the results on the txda_rcvdet_detected_n signal coming from the analog. The default required wait time is 100us.

10.2.24 Transmitter BIST Controller Registers

One should note that the BIST registers should not be accessed when the diagnostic power island is switched off. Additionally, these registers do not retain their state when the diagnostic power island is switched off and are reset when the power island is switched back on.

Base address bits 15:5 :	**01_m_nnnn : 1010**
Address bits 4:0	**Register name and location of detailed description**
00000	Transmit BIST control register on page 253
00001	Transmit BIST user defined data write register on page 253
00010	Transmit BIST PRBS seed 0 register on page 253
00011	Transmit BIST PRBS seed 1 register on page 254
00010 - 11111	Reserved

Table 114: Transmitter BIST controller registers address space

Register Name:			Transmit BIST control register **Note: This register is on the diagnostic power island, and cannot be written to when this power island is switched off.**
Register Mnemonic:			TX_BIST_CTRL
Register Address:			01_m_nnnn : 1010 : 00000
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:8	R/W	4'h0	**Transmitter BIST mode**: Controls which mode the BIST will operate in. The value of this field must match the corresponding field for the receive BIST controller. The following are the values used for this field, and what BIST mode they correspond to. • 4'b0000 : User defined data FIFO • 4'b0001 - 4'b0111 : Reserved • 4'b1000 : PRBS7 : 7 bit LFSR • 4'b1001 : PRBS15 : 15 bit LFSR • 4'b1010 : PRBS23 : 23 bit LFSR • 4'b1011 : PRBS31 : 31 bit LFSR • 4'b1100 - 4'b1111 : Reserved
7:5	R/O	3'h0	Reserved
4	R/W	1'h0	**Transmitter BIST force error**: When this bit transitions from 1'b0 to 1'b1, the transmit BIST controller will force an error to be transmitted from the BIST logic, by inverting one of the parallel data bits.
3:2	R/O	2'h0	Reserved
1	W/O	1'h0	**Transmitter BIST user defined data FIFO clear**: Writing a 1'b1 to this bit will clear the transmitter BIST user defined data FIFO. Note : This bit is automatically cleared after it is written to. Note : This clear function simply resets the FIFO pointers. It does not clear the contents of the FIFO.
0	R/W	1'h0	**Transmitter BIST enable**: This bit enables the transmitter BIST function. Note : The remaining bits in this register must be stable when changing this bit. Therefore, it is best to enable and disable this function using a read / modify / write of this bit.

Register Name:			Transmit BIST user defined data write register **Note: This register is on the diagnostic power island, and cannot be written to when this power island is switched off.**
Register Mnemonic:			TX_BIST_UDDWR
Register Address:			01_m_nnnn : 1010 : 00001
Bits	**Type**	**Reset**	**Description**
15:10	R/O	6'h0	Reserved
9:0	W/O	10'h000	**Transmitter BIST user defined data**: Writing a data word to this field will result in that data word being placed in the next available position in the transmitter BIST user defined data FIFO. Note, when in 20 bit mode, all 10 of these bits are used. When in 16 bit mode, only the least significant 8 bits are used. Note: The FIFO in this implementation is 18 words deep. It is up to the user to not write more than 18 words of data into this FIFO. Exceeding this amount of data will generate unpredictable results.

Register Name:			Transmit BIST PRBS seed 0 register **Note: This register is on the diagnostic power island, and cannot be written to when this power island is switched off.**
Register Mnemonic:			TX_BIST_SEED0
Register Address:			01_m_nnnn : 1010 : 00010
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h0001	**Transmitter BIST PRBS seed (15:0)**: When the BIST is in PRBS mode, this field provides a seed for the PRBS, such that different lanes can have different BIST patterns. Note: This field contains the least significant 16 bits of the full 31 bit seed value. Note: When combined, this register and the Transmit BIST PRBS seed 1 register on page 254 must never be all 0s.

Register Name:			Transmit BIST PRBS seed 1 register **Note: This register is on the diagnostic power island, and cannot be written to when this power island is switched off.**
Register Mnemonic:			TX_BIST_SEED1
Register Address:			01_m_nnnn : 1010 : 00011
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:0	R/W	15'h0000	**Transmitter BIST PRBS seed (30:16)**: When the BIST is in PRBS mode, this field provides a seed for the PRBS, such that different lanes can have different BIST patterns. Note: This field contains the most significant 15 bits of the full 31 bit seed value. Note: When combined, this register and the Transmit BIST PRBS seed 0 register on page 253 must never be all 0s.

10.2.25 Transmitter Control And Diagnostics Registers

Base address bits 15:5 :	**01_m_nnnn : 1111**
Address bits 4:0	**Register name and location of detailed description**
00000	TX sync FIFO diagnostic control register on page 254
00001	TX sync FIFO diagnostic timer register on page 255
00010	TX electrical idle diagnostic register on page 255
00011	Reserved
00100	Reserved
00101	Transmitter control reset diagnostic register on page 256
00110	Transmitter digital cover your alternatives register on page 256
00111	Transmitter analog cover your alternatives register on page 256
11000 - 11111	Reserved

Table 115: Transmitter control and diagnostics registers address space

Register Name:			TX sync FIFO diagnostic control register
Register Mnemonic:			TX_DIAG_SFIFO_CTRL
Register Address:			01_m_nnnn : 1111 : 00000
Bits	**Type**	**Reset**	**Description**
15:5	R/O	11'h0	Reserved
4	R/W	1'h0	**FIFO enqueue pointer bump**: This bit can be used to decrement the enqueue pointer relative to the dequeue pointer, for diagnostic purposes. Changing this bit from a value of 1'b0 to 1'b1 will trigger a single decrement of the enqueue pointer.
3	R/O	1'h0	**FIFO pointers overlapping**: This bit indicates that the current enqueue and dequeue pointers have been detected as overlapping. This is not necessarily an error condition. For example, the pointers could be overlapping but the enqueue pointer could be skewed later in time relative to the dequeue pointer. In this scenario, the actual enqueue would take place after the dequeue. The intention of this bit is to be used with the FIFO enqueue pointer bump bit in this register to move the pointers around and determine their relative positions. **Note: The reset value for this field is with the transceiver data path power island switched off. In cases where this power island is switched on, the reset value be 1'b1**
2	R/O	1'h0	**FIFO alignment acknowledge**: This bit indicates that the FIFO alignment process is complete, as initiated either automatically by the hardware of the FIFO alignment enable override bits in this register. This bit is driven directly by the fifo_align_ack pin of the FIFO.
1	R/W	1'h0	**FIFO alignment enable override enable**: This bit enables the FIFO alignment enable override register to drive the fifo_align_en pin of the FIFO directly for diagnostic purposes. • 1'b0: Override disabled • 1'b1: Override enabled.
0	R/W	1'h0	**FIFO alignment enable override**: When enabled by the FIFO alignment enable override enable bit in this register, this bit directly controls the fifo_align_en pin of the FIFO to provide a means of running the FIFO alignment function for diagnostic purposes.

Register Name:			TX sync FIFO diagnostic timer register
Register Mnemonic:			TX_DIAG_SFIFO_TMR
Register Address:			01_m_nnnn : 1111 : 00001
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'h0	Reserved
13:8	R/W	6'h06	**FIFO alignment settle delay**: This field specifies the number of clocks to wait for a prior change to the enqueue pointer to complete before initiating the check phase of the alignment procedure in the sync FIFO. It drives the fifo_align_settle_del pin of the FIFO.
7:6	R/O	2'h0	Reserved
5:0	R/W	6'h0C	**FIFO alignment detect delay**: This field specifies the number of clocks to wait in the delay state for each phase of the alignment procedure in the sync FIFO. It drives the fifo_align_detect_del pin of the FIFO.

Register Name:			TX electrical idle diagnostic register
Register Mnemonic:			TX_DIAG_ELEC_IDLE
Register Address:			01_m_nnnn : 1111 : 00010
Bits	**Type**	**Reset**	**Description**
15:12	R/W	4'h4	**TX electrical idle exit delay 16 bit data width** : This field controls the amount of additional delay added to the electrical idle signal after the sync FIFO, when exiting the electrical idle state for 16 bit data width modes. The valid values that can be used for this field are the same as those in the TX electrical idle entry delay 20 bit data width field of this register.
11:8	R/W	4'h4	**TX electrical idle entry delay 16 bit data width** : This field controls the amount of additional delay added to the electrical idle signal after the sync FIFO, when entering the electrical idle state for 16 bit data width modes. The valid values that can be used for this field are the same as those in the TX electrical idle entry delay 20 bit data width field of this register.
7:4	R/W	4'h3	**TX electrical idle exit delay 20 bit data width** : This field controls the amount of additional delay added to the electrical idle signal after the sync FIFO, when exiting the electrical idle state for 20 bit data width modes. The valid values that can be used for this field are the same as those in the TX electrical idle entry delay 20 bit data width field of this register.
3:0	R/W	4'h3	**TX electrical idle entry delay 20 bit data width** : This field controls the amount of additional delay added to the electrical idle signal after the sync FIFO, when entering the electrical idle state for 20 bit data width modes. The following are the valid values that can be used for this field, and the corresponding delays. Note: In the cases of 1/2 clock cycle delays, this is implemented on the falling edge of the clock, so the result is a function of the duty cycle of the clock. • 4'b0000 : Reserved. • 4'b0001 : Reserved. • 4'b0010 : 1.0 clock delays. • 4'b0011 : 1.5 clock delays. • 4'b0100 : 2.0 clock delays. • 4'b0101 : 2.5 clock delays. • 4'b0110 : 3.0 clock delays. • 4'b0111 : 3.5 clock delays. • 4'b1000 : 4.0 clock delays. • 4'b1001 : 4.5 clock delays. • 4'b1010 : 5.0 clock delays. • 4'b1011 : 5.5 clock delays. • 4'b1100 : 6.0 clock delays. • 4'b1101 : 6.5 clock delays. • 4'b1110 : 7.0 clock delays. • 4'b1111 : 7.5 clock delays.

Register Name:			Transmitter control reset diagnostic register
Register Mnemonic:			TX_DIAG_RST_DIAG
Register Address:			01_m_nnnn : 1111 : 00101
Bits	**Type**	**Reset**	**Description**
15:5	R/O	11'h0	Reserved
4	R/O	1'h0	Current state of the txda_tx_clk_reset_n reset.
3	R/O	1'h0	Current state of the tx_dig_reset_n reset.
2	R/O	1'h0	Current state of the tx_sync_fifo_deq_rst_n reset.
1	R/O	1'h0	Current state of the tx_sync_fifo_enq_rst_n reset.
0	R/O	1'h0	Current state of the tx_lfps_reset_n reset.

Register Name:			Transmitter digital cover your alternatives register
Register Mnemonic:			TX_DIAG_DCYA
Register Address:			01_m_nnnn : 1111 : 00110
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h0	Reserved - spare

Register Name:			Transmitter analog cover your alternatives register
Register Mnemonic:			TX_DIAG_ACYA
Register Address:			01_m_nnnn : 1111 : 00111
Bits	**Type**	**Reset**	**Description**
15:1	R/W	15'h0	Reserved - spare
0	R/W	1'h0	**HBDC latch control**: Controls the state of the latches associated with the H bridge driver controller related signals in the H bridge driver encoder logic digital in the transmitter analog, as well as the boost enable and level control signals. These latches can be closed to hold the current state of these signals. While closed, the signals that drive the latch inputs can be changed by writing the registers that control them, and not effect the state of the driver. Once such register writes are complete, the latches can be made transparent again to apply the changes. • 1'b0: Latches transparent • 1'b1: Latches gated

10.2.26 Receiver Power State Controller Registers

Base address bits 15:5 :	**10_m_nnnn : 0000**
Address bits 4:0	**Register name and location of detailed description**
00000	Receiver A0 power state definition register (!) on page 257
00001	Receiver A1 power state definition register (!) on page 257
00010	Receiver A2 power state definition register (!) on page 258
00011	Receiver A3 power state definition register (!) on page 258
00100	Receiver A4 power state definition register (!) on page 258
00101	Receiver A5 power state definition register (!) on page 259
00110	Receiver calibration power state definition register (!) on page 259
00111	Receiver ready power state definition register (!) on page 259
01000 - 11111	Reserved

Table 116: Receiver power state controller registers address space

Register Name:			Receiver A0 power state definition register (!)
Register Mnemonic:			RX_PSC_A0
Register Address:			10_m_nnnn : 0000 : 00000
Bits	**Type**	**Reset**	**Description**
15:13	R/O	3'h0	Reserved
12	R/W	1'h0	**RX signal detect enable extend control**: Specifies which power states the rx_sig_det_en_ext signal is considered valid in and can be used to force the signal detect functions to remain on.
11	R/W	1'h1	**RX signal detect filter enable**: Enables the receiver signal detect filter function in the digital receiver controller.
10	R/W	1'h0	**RX LFPS detect filter enable**: Enables the receiver LFPS detect filter function in the digital receiver controller.
9	R/W	1'h0	Reserved - spare
8	R/W	1'h1	**RX signal detect enable**: Enables the receiver signal detect function. This drives the rxda_sd_en signal going to the analog. Note: This bit needs to be active for LFPS detect functions to work, because these functions share the same analog circuits.
7:5	R/W	3'h0	Reserved - spare
4	R/W	1'h1	**RX equalizer engine enable**: Specifies which power state the REE runs in. • 1'b0: Disabled • 1'b1: Enabled
3	R/W	1'h1	**RX DFE equalization enable**: Enables the receiver DFE equalization circuits, via the rxda_dfe_eq_enable signal.
2	R/W	1'h1	**RX PI enable**: Enables the receiver circuits related to the PI and associated clocking components.
1	R/W	1'h0	**RX e path enable (calibration and eye surf only)** : Enables the receiver circuits related to the eye plot PI and e path deserializer for calibration and eye surf.
0	R/W	1'h1	**RX enable**: Enables the receiver circuits related to the CDRLF, Sampler, FE, and Deserializer.

Register Name:			Receiver A1 power state definition register (!)
Register Mnemonic:			RX_PSC_A1
Register Address:			10_m_nnnn : 0000 : 00001
Bits	**Type**	**Reset**	**Description**
15:13	R/O	3'h0	Reserved
12	R/W	1'h0	**RX signal detect enable extend control**
11	R/W	1'h1	**RX signal detect filter enable**
10	R/W	1'h0	**RX LFPS detect filter enable**
9	R/W	1'h0	Reserved - spare
8	R/W	1'h1	**RX signal detect enable**
7:5	R/W	3'h0	Reserved - spare
4	R/W	1'h1	**RX equalizer engine enable**
3	R/W	1'h1	**RX DFE equalization enable**.
2	R/W	1'h1	**RX PI enable**
1	R/W	1'h0	**RX e path enable (calibration and eye surf only)**
0	R/W	1'h1	**RX enable**

Register Name:			Receiver A2 power state definition register (!)
Register Mnemonic:			RX_PSC_A2
Register Address:			10_m_nnnn : 0000 : 00010
Bits	**Type**	**Reset**	**Description**
15:13	R/O	3'h0	Reserved
12	R/W	1'h0	**RX signal detect enable extend control**
11	R/W	1'h1	**RX signal detect filter enable**
10	R/W	1'h0	**RX LFPS detect filter enable**
9	R/W	1'h0	Reserved - spare
8	R/W	1'h1	**RX signal detect enable**
7:5	R/W	3'h0	Reserved - spare
4	R/W	1'h0	**RX equalizer engine enable**
3	R/W	1'h0	**RX DFE equalization enable**.
2	R/W	1'h0	**RX PI enable**
1	R/W	1'h0	**RX e path enable (calibration and eye surf only)**
0	R/W	1'h0	**RX enable**

Register Name:			Receiver A3 power state definition register (!)
Register Mnemonic:			RX_PSC_A3
Register Address:			10_m_nnnn : 0000 : 00011
Bits	**Type**	**Reset**	**Description**
15:13	R/O	3'h0	Reserved
12	R/W	1'h0	**RX signal detect enable extend control**
11	R/W	1'h0	**RX signal detect filter enable**
10	R/W	1'h0	**RX LFPS detect filter enable**
9	R/W	1'h0	Reserved - spare
8	R/W	1'h0	**RX signal detect enable**
7:5	R/W	3'h0	Reserved - spare
4	R/W	1'h0	**RX equalizer engine enable**
3	R/W	1'h0	**RX DFE equalization enable**.
2	R/W	1'h0	**RX PI enable**
1	R/W	1'h0	**RX e path enable (calibration and eye surf only)**
0	R/W	1'h0	**RX enable**

Register Name:			Receiver A4 power state definition register (!)
Register Mnemonic:			RX_PSC_A4
Register Address:			10_m_nnnn : 0000 : 00100
Bits	**Type**	**Reset**	**Description**
15:13	R/O	3'h0	Reserved
12	R/W	1'h1	**RX signal detect enable extend control**
11	R/W	1'h0	**RX signal detect filter enable**
10	R/W	1'h0	**RX LFPS detect filter enable**
9	R/W	1'h0	Reserved - spare
8	R/W	1'h0	**RX signal detect enable**
7:5	R/W	3'h0	Reserved - spare
4	R/W	1'h0	**RX equalizer engine enable**
3	R/W	1'h0	**RX DFE equalization enable**.
2	R/W	1'h0	**RX PI enable**
1	R/W	1'h0	**RX e path enable (calibration and eye surf only)**
0	R/W	1'h0	**RX enable**

Register Name:			Receiver A5 power state definition register (!)
Register Mnemonic:			RX_PSC_A5
Register Address:			10_m_nnnn : 0000 : 00101
Bits	**Type**	**Reset**	**Description**
15:13	R/O	3'h0	Reserved
12	R/W	1'h0	**RX signal detect enable extend control**
11	R/W	1'h0	**RX signal detect filter enable**
10	R/W	1'h0	**RX LFPS detect filter enable**
9	R/W	1'h0	Reserved - spare
8	R/W	1'h0	**RX signal detect enable**
7:5	R/W	3'h0	Reserved - spare
4	R/W	1'h0	**RX equalizer engine enable**
3	R/W	1'h0	**RX DFE equalization enable**.
2	R/W	1'h0	**RX PI enable**
1	R/W	1'h0	**RX e path enable (calibration and eye surf only)**
0	R/W	1'h0	**RX enable**

Register Name:			Receiver calibration power state definition register (!)
Register Mnemonic:			RX_PSC_CAL
Register Address:			10_m_nnnn : 0000 : 00110
Bits	**Type**	**Reset**	**Description**
15:13	R/O	3'h0	Reserved
12	R/W	1'h0	**RX signal detect enable extend control**
11	R/W	1'h0	**RX signal detect filter enable**
10	R/W	1'h0	**RX LFPS detect filter enable**
9	R/W	1'h0	Reserved - spare
8	R/W	1'h1	**RX signal detect enable**
7:5	R/W	3'h0	Reserved - spare
4	R/W	1'h0	**RX equalizer engine enable**
3	R/W	1'h1	**RX DFE equalization enable**.
2	R/W	1'h1	**RX PI enable**
1	R/W	1'h1	**RX e path enable (calibration and eye surf only)**
0	R/W	1'h1	**RX enable**

Register Name:			Receiver ready power state definition register (!)
Register Mnemonic:			RX_PSC_RDY
Register Address:			10_m_nnnn : 0000 : 00111
Bits	**Type**	**Reset**	**Description**
15:13	R/O	3'h0	Reserved
12	R/W	1'h0	**RX signal detect enable extend control**
11	R/W	1'h0	**RX signal detect filter enable**
10	R/W	1'h0	**RX LFPS detect filter enable**
9	R/W	1'h0	Reserved - spare
8	R/W	1'h0	**RX signal detect enable**
7:5	R/W	3'h0	Reserved - spare
4	R/W	1'h0	**RX equalizer engine enable**
3	R/W	1'h0	**RX DFE equalization enable**.
2	R/W	1'h0	**RX PI enable**
1	R/W	1'h0	**RX e path enable (calibration and eye surf only)**
0	R/W	1'h0	**RX enable**

10.2.27 Signal Detect Calibration 0 Controller Registers

Base address bits 15:5 :	10_m_nnnn : 0010
Address bits 4:0	**Register name and location of detailed description**
00000	Signal detect calibration 0 control register (!) on page 260
00001	Signal detect calibration 0 override register (!) on page 261
00010	Signal detect calibration 0 start register (!) on page 261
00011	Signal detect calibration 0 tune register (!) on page 261
00100	Signal detect calibration 0 initialization timer register (!) on page 261
00101	Signal detect calibration 0 iteration timer register (!) on page 262
00110 - 00111	Reserved

Table 117: Signal detect calibration 0 registers address space

Register Name:			Signal detect calibration 0 control register (!)
Register Mnemonic:			RX_SDCAL0_CTRL
Register Address:			10_m_nnnn : 0010 : 00000
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Start calibration**: Activating (1'b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1'b0) until the bandbap calibration process done bit in this register is cleared. **Note: This bit is intended to be for diagnostics purposes only. This calibration process is automatically activated internally by the power state machine. When using this bit, the user must wait until after the internally activated process completes.**
14	R/O	1'h0	**Calibration process done**: This bit will be set to 1'b1 when the calibration process is complete. It will be cleared by cmn_reset_n, or by the deactivation of the start calibration bit in this register after calibration is complete. **Note: This bit is intended to be for diagnostics purposes only.** **Note: This bit is not likely to be observed as being set after internal automatic calibration is complete, because the internally generated run signal will be driven inactive immediately after the done signal is activated, and therefore the internal done signal will be cleared.** **Note: Three cmn_ref_clk cycles are required after the start of the calibration process to clear this signal.**
13	R/O	1'h0	**No analog calibration response** : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the analog has not responded indicating that a valid calibration value has been reached. Note: Due to the nature of the calibration process, it is not possible to determine if the analog responded or not for the lowest code of 4'b1001. The reason for this is, when sequencing to lower codes, the calibration function is looking for the analog response to transition from 1'b1 to 1'b0. If the lowest calibration code is the correct code, the analog will respond with 1'b1, and the algorithm can't sequence to a lower code to look for the transition from 1'b1 to 1'b0. Similarly, if the analog is not responding, it is not possible to sequence to a lower code to detect this. Therefore, even when the calibration function selects this calibration code as valid, this bit will be set.
12	R/O	1'h0	**Current analog comparator response**: This is the current state of the analog comparator response signal (rxda_sd_cal_0_comp). This signal is not synchronized, and is provided for diagnostic purposes only.
11:5	R/O	7'h0	Reserved
4:0	R/O	5'h00	**Calibration code**: This is the calibration code that was determined by the calibration process. The following indicates how this encoding maps to the rxda_sd_cal_0_up and rxda_sd_cal_0_code signals going to the analog function. The five bit value on the left is the value of this field. The one bit value on the right corresponds to the value of rxda_sd_cal_0_up. The four bit field corresponds to the value of rxda_sd_cal_0_code. • 5'b01111 : 1'b1, 4'b1111 • 5'b01110 : 1'b1, 4'b1110 • ... • 5'b00010 : 1'b1, 4'b0010 • 5'b00001 : 1'b1, 4'b0001 • 5'b00000 : 1'b1, 4'b0000 • 5'b11111 : 1'b0, 4'b0001 • 5'b11110 : 1'b0, 4'b0010 • ... • 5'b10010 : 1'b0, 4'b1110 • 5'b10001 : 1'b0, 4'b1111

Register Name:			Signal detect calibration 0 override register (!)
Register Mnemonic:			RX_SDCAL0_OVRD
Register Address:			10_m_nnnn : 0010 : 00001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Calibration code override enable**: Activation (1'b1) of this register bit allows the codes determined during the automatic calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14	R/W	1'h0	**Analog calibration enable override**: Activation (1'b1) of this register bit will force the analog calibration circuits to be enabled by activating the rxda_sd_cal_0_en enable and the rxda_sd_cal_0_clk clock.
13:5	R/O	9'h0	Reserved
4:0	R/W	5'h00	**Calibration code override value**: These bits are used to override the calibration code determined during the automatic calibration process. The code written to these bits is valid when the calibration code override enable bit in this register is active. The codes in this field correspond to those described in the calibration code field in the Signal detect calibration 0 control register (!) on page 260.

Register Name:			Signal detect calibration 0 start register (!)
Register Mnemonic:			RX_SDCAL0_START
Register Address:			10_m_nnnn : 0010 : 00010
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Calibration direction**: This controls the direction that the automatic calibration process steps the calibration codes in. • 1'b0 : From 5'b10001 to 5'b01111. • 1'b1 : From 5'b01111 to 5'b00000.
14:5	R/O	10'h0	Reserved
4:0	R/W	5'h00	**Start calibration code**: This is the calibration code that the calibration process starts with when automatic calibration is run. The codes in this field correspond to those described in the calibration code field in the Signal detect calibration 0 control register (!) on page 260.

Register Name:			Signal detect calibration 0 tune register (!)
Register Mnemonic:			RX_SDCAL0_TUNE
Register Address:			10_m_nnnn : 0010 : 00011
Bits	**Type**	**Reset**	**Description**
15:5	R/O	11'h0	Reserved
4:0	R/W	5'h00	**Calibration tune value**: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note: This value is a twos complement value, so the calibrated code can be increased or decreased.

Register Name:			Signal detect calibration 0 initialization timer register (!)
Register Mnemonic:			RX_SDCAL0_INIT_TMR
Register Address:			10_m_nnnn : 0010 : 00100
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h019	**Initialization wait timer value**: This is the number of cmn_ref_clk clocks to wait between when the analog calibration circuits are enabled, and when the first values are placed on the calibration code signals going to the analog. **Note: This results in the minimum number of cmn_ref_clk clocks required to wait 1 uSec.** **Note: This should never be set to a value of less than 1.**

Register Name:			Signal detect calibration 0 iteration timer register (!)
Register Mnemonic:			RX_SDCAL0_ITER_TMR
Register Address:			10_m_nnnn : 0010 : 00101
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h07D	**Iteration wait timer value**: This is the number of cmn_ref_clk clocks to wait between when a value is placed on the calibration code signals going to the analog, and when the comparator value coming from the analog circuits can be checked. **Note: This results in the minimum number of cmn_ref_clk clocks required to wait 5 uSec.** **Note: This should never be set to a value of less than 3.**

10.2.28 Signal Detect Calibration 1 Controller Registers

Base address bits 15:5 :	10_m_nnnn : 0010
Address bits 4:0	**Register name and location of detailed description**
01000	Signal detect calibration 1 control register (!) on page 263
01001	Signal detect calibration 1 override register (!) on page 264
01010	Signal detect calibration 1 start register (!) on page 264
01011	Signal detect calibration 1 tune register (!) on page 264
01100	Signal detect calibration 1 initialization timer register (!) on page 264
01101	Signal detect calibration 1 iteration timer register (!) on page 265
01110 - 01111	Reserved

Table 118: Signal detect calibration 1 registers address space

Register Name:			Signal detect calibration 1 control register (!)
Register Mnemonic:			RX_SDCAL1_CTRL
Register Address:			10_m_nnnn : 0010 : 01000
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Start calibration**: Activating (1'b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1'b0) until the bandbap calibration process done bit in this register is cleared. **Note: This bit is intended to be for diagnostics purposes only. This calibration process is automatically activated internally by the power state machine. When using this bit, the user must wait until after the internally activated process completes.**
14	R/O	1'h0	**Calibration process done**: This bit will be set to 1'b1 when the calibration process is complete. It will be cleared by cmn_reset_n, or by the deactivation of the start calibration bit in this register after calibration is complete. **Note: This bit is intended to be for diagnostics purposes only.** **Note: This bit is not likely to be observed as being set after internal automatic calibration is complete, because the internally generated run signal will be driven inactive immediately after the done signal is activated, and therefore the internal done signal will be cleared.** **Note: Three cmn_ref_clk cycles are required after the start of the calibration process to clear this signal.**
13	R/O	1'h0	**No analog calibration response** : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the analog has not responded indicating that a valid calibration value has been reached. Note: Due to the nature of the calibration process, it is not possible to determine if the analog responded or not for the lowest code of 4'b1001. The reason for this is, when sequencing to lower codes, the calibration function is looking for the analog response to transition from 1'b1 to 1'b0. If the lowest calibration code is the correct code, the analog will respond with 1'b1, and the algorithm can't sequence to a lower code to look for the transition from 1'b1 to 1'b0. Similarly, if the analog is not responding, it is not possible to sequence to a lower code to detect this. Therefore, even when the calibration function selects this calibration code as valid, this bit will be set.
12	R/O	1'h0	**Current analog comparator response**: This is the current state of the analog comparator response signal (rxda_sd_cal_0_comp). This signal is not synchronized, and is provided for diagnostic purposes only.
11:5	R/O	7'h0	Reserved
4:0	R/O	5'h00	**Calibration code**: This is the calibration code that was determined by the calibration process. The following indicates how this encoding maps to the rxda_sd_cal_1_up and rxda_sd_cal_1_code signals going to the analog calibration function. The five bit value on the left is the value of this field. The one bit value on the right corresponds to the value of rxda_sd_cal_1_up. The four bit field corresponds to the value of rxda_sd_cal_1_code. • 5'b01111 : 1'b1, 4'b1111 • 5'b01110 : 1'b1, 4'b1110 • ... • 5'b00010 : 1'b1, 4'b0010 • 5'b00001 : 1'b1, 4'b0001 • 5'b00000 : 1'b1, 4'b0000 • 5'b11111 : 1'b0, 4'b0001 • 5'b11110 : 1'b0, 4'b0010 • ... • 5'b10010 : 1'b0, 4'b1110 • 5'b10001 : 1'b0, 4'b1111

Register Name:			Signal detect calibration 1 override register (!)
Register Mnemonic:			RX_SDCAL1_OVRD
Register Address:			10_m_nnnn : 0010 : 01001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Calibration code override enable**: Activation (1'b1) of this register bit allows the calibration code determined during the automatic calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14	R/W	1'h0	**Analog calibration enable override**: Activation (1'b1) of this register bit will force the analog calibration circuits to be enabled by activating the rxda_sd_cal_1_en enable and the rxda_sd_cal_1_clk clock.
13:5	R/O	9'h0	Reserved
4:0	R/W	5'h00	**Calibration code override value**: These bits are used to override the calibration code determined during the automatic calibration process. The code written to these bits is valid when the calibration code override enable bit in this register is active. The codes in this field correspond to those described in the calibration code field in the Signal detect calibration 1 control register (!) on page 263.

Register Name:			Signal detect calibration 1 start register (!)
Register Mnemonic:			RX_SDCAL1_START
Register Address:			10_m_nnnn : 0010 : 01010
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Calibration direction**: This controls the direction that the automatic calibration process steps the calibration codes in. • 1'b0 : From 5'b10001 to 5'b01111. • 1'b1 : From 5'b01111 to 5'b00000.
14:5	R/O	10'h0	Reserved
4:0	R/W	5'h00	**Start calibration code**: This is the calibration code that the calibration process starts with when automatic calibration is run. The codes in this field correspond to those described in the calibration code field in the Signal detect calibration 1 control register (!) on page 263.

Register Name:			Signal detect calibration 1 tune register (!)
Register Mnemonic:			RX_SDCAL1_TUNE
Register Address:			10_m_nnnn : 0010 : 01011
Bits	**Type**	**Reset**	**Description**
15:5	R/O	11'h0	Reserved
4:0	R/W	5'h00	**Calibration tune value**: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note: This value is a twos complement value, so the calibrated code can be increased or decreased.

Register Name:			Signal detect calibration 1 initialization timer register (!)
Register Mnemonic:			RX_SDCAL1_INIT_TMR
Register Address:			10_m_nnnn : 0010 : 01100
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h019	**Initialization wait timer value**: This is the number of cmn_ref_clk clocks to wait between when the analog calibration circuits are enabled, and when the first values are placed on the calibration code signals going to the analog. **Note: This results in the minimum number of cmn_ref_clk clocks required to wait 1 uSec.** **Note: This should never be set to a value of less than 1.**

Register Name:			Signal detect calibration 1 iteration timer register (!)
Register Mnemonic:			RX_SDCAL1_ITER_TMR
Register Address:			10_m_nnnn : 0010 : 01101
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h07D	**Iteration wait timer value**: This is the number of cmn_ref_clk clocks to wait between when a value is placed on the calibration code signals going to the analog, and when the comparator value coming from the analog circuits can be checked. **Note: This results in the minimum number of cmn_ref_clk clocks required to wait 5 uSec.** **Note: This should never be set to a value of less than 3.**

10.2.29 Sampler Error DAC Control Registers

Base address bits 15:5 :	**10_m_nnnn : 0010**
Address bits 4:0	**Register name and location of detailed description**
11000	Sampler error DAC control register (!) on page 265
11001 - 11111	Reserved

Table 119: Sampler error DAC control registers address space

Register Name:			Sampler error DAC control register (!)
Register Mnemonic:			RX_SAMP_DAC_CTRL
Register Address:			10_m_nnnn : 0010 : 11000
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h14	**Sampler error DAC value**: Specifies the input value to the sampler error DAC. This value is a twos complement binary number, with the range specified below. • 6'b011111: Maximum. • 6'b000000: Center point of the data eye. • 6'b100001: Minimum. **Note : Even though this register can be set to as low as 6'b100001, it should never be set below the value of 6'b000000 when the VGA gain target adjustment algorithm is enabled in the REE. Doing so can result in an underflow condition.**

10.2.30 Receiver Sampler Latch Calibration Registers

Base address bits 15:5 :	10_m_nnnn : 0011
Address bits 4:0	**Register name and location of detailed description**
00000	RX sampler latch calibration control register (!) on page 266
00001	RX sampler latch calibration I predictive positive status register (!) on page 267
00010	RX sampler latch calibration I predictive positive override register (!) on page 267
00011	RX sampler latch calibration I predictive negative status register (!) on page 267
00100	RX sampler latch calibration I predictive negative override register (!) on page 268
00101	RX sampler latch calibration Q predictive positive status register (!) on page 268
00110	RX sampler latch calibration Q predictive positive override register (!) on page 268
00111	RX sampler latch calibration Q predictive negative status register (!) on page 269
01000	RX sampler latch calibration Q predictive negative override register (!) on page 269
01001	RX sampler latch calibration e predictive positive status register (!) on page 269
01010	RX sampler latch calibration e predictive positive override register (!) on page 270
01011	RX sampler latch calibration e predictive negative status register (!) on page 270
01100	RX sampler latch calibration e predictive negative override register (!) on page 270
01101	RX sampler latch calibration initialization timer value register (!) on page 271
01110	RX sampler latch calibration run timer value register (!) on page 271
01111	RX sampler latch calibration diagnostic control register (!) on page 272
10000	RX sampler latch calibration disable register (!) on page 273
10001 - 11111	Reserved

Table 120: Receiver sampler latch calibration registers address space

Register Name:			RX sampler latch calibration control register (!)
Register Mnemonic:			RX_SLC_CTRL
Register Address:			10_m_nnnn : 0011 : 00000
Bits	**Type**	**Reset**	**Description**
15	R/W	1’h0	**Start RX sampler latch calibration**: Activating (1’b1) this bit will start the RX sampler latch calibration process. This bit should remain active until the RX sampler latch calibration process is complete. To start another RX sampler latch calibration process, this bit must first be inactive (1’b0) until the RX sampler latch calibration done bit goes inactive. **Note: This bit is intended to be for diagnostics purposes only. This calibration process is automatically activated internally, by the power state machine. When using this bit, the user must wait until after the internally activated process completes.**
14	R/O	1’h0	**RX sampler latch calibration process done**: This bit will be set to 1’b1 when the RX sampler latch calibration process is complete. It will be cleared by cmn_reset_n, or by clearing the start RX sampler latch process bit. **Note: This bit is intended to be for diagnostics purposes only.** **Note: This bit is not likely to be observed as being set after internal automatic calibration is complete, because the internally generated run signal will be driven inactive immediately after the done signal is activated, and therefore the internal done signal will be cleared.**
13:2	R/O	12’h0	Reserved
1:0	R/W	2’h1	**RX sampler latch calibration scaler**: This field specifies the scaler value used for the input data accumulator. The following is the encoding used for this signal. • 2’b00: x1 • 2’b01: x2 • 2’b10: x4 • 2’b11: x8

Register Name:			RX sampler latch calibration I predictive positive status register (!)
Register Mnemonic:			RX_SLC_IPP_STAT
Register Address:			10_m_nnnn : 0011 : 00001
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:8	R/O	7'h00	**I even latch receiver sampler calibration DAC value**: This field contains the current value that the RX sampler latch calibration module has calculated to drive to the DAC that controls the offset for the I even latch in the analog sampler component.
7	R/O	1'h0	Reserved
6:0	R/O	7'h00	**I odd latch receiver sampler calibration DAC value**: This field contains the current value that the RX sampler latch calibration module has calculated to drive to the DAC that controls the offset for the I odd latch in the analog sampler component.

Register Name:			RX sampler latch calibration I predictive positive override register (!)
Register Mnemonic:			RX_SLC_IPP_OVRD
Register Address:			10_m_nnnn : 0011 : 00010
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**I even latch receiver sampler calibration DAC override enable**: When this bit is active (1'b1), the value in the I even latch receiver sampler calibration DAC override value field of this register will override the calibrated value for the I even sampler latch.
14:8	R/W	7'h00	**I even latch receiver sampler calibration DAC override value**: This field contains the value that can be used to override the calibrated value for the I even sampler latch. Note: The value of this field is a signed magnitude number with a range of 5'b11111 (min) to 5'b01111 (max). A value of 5'b10000 (-0) is not valid.
7	R/W	1'h0	**I odd latch receiver sampler calibration DAC override enable**: When this bit is active (1'b1), the value in the I odd latch receiver sampler calibration DAC override value field of this register will override the calibrated value for the I odd sampler latch.
6:0	R/W	7'h00	**I odd latch receiver sampler calibration DAC override value**: This field contains the value that can be used to override the calibrated value for the I odd sampler latch. Note: The value of this field is a signed magnitude number with a range of 5'b11111 (min) to 5'b01111 (max). A value of 5'b10000 (-0) is not valid.

Register Name:			RX sampler latch calibration I predictive negative status register (!)
Register Mnemonic:			RX_SLC_IPM_STAT
Register Address:			10_m_nnnn : 0011 : 00011
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:8	R/O	7'h00	**I even latch receiver sampler calibration DAC value**: This field contains the current value that the RX sampler latch calibration module has calculated to drive to the DAC that controls the offset for the I even latch in the analog sampler component.
7	R/O	1'h0	Reserved
6:0	R/O	7'h00	**I odd latch receiver sampler calibration DAC value**: This field contains the current value that the RX sampler latch calibration module has calculated to drive to the DAC that controls the offset for the I odd latch in the analog sampler component.

Register Name:			RX sampler latch calibration I predictive negative override register (!)
Register Mnemonic:			RX_SLC_IPM_OVRD
Register Address:			10_m_nnnn : 0011 : 00100
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**I eve latch receiver sampler calibration DAC override enable**: When this bit is active (1'b1), the value in the I even latch receiver sampler calibration DAC override value field of this register will override the calibrated value for the I even sampler latch.
14:8	R/W	7'h00	**I even latch receiver sampler calibration DAC override value**: This field contains the value that can be used to override the calibrated value for the I even sampler latch. Note: The value of this field is a signed magnitude number with a range of 5'b11111 (min) to 5'b01111 (max). A value of 5'b10000 (-0) is not valid.
7	R/W	1'h0	**I odd latch receiver sampler calibration DAC override enable**: When this bit is active (1'b1), the value in the I odd latch receiver sampler calibration DAC override value field of this register will override the calibrated value for the I odd sampler latch.
6:0	R/W	7'h00	**I odd latch receiver sampler calibration DAC override value**: This field contains the value that can be used to override the calibrated value for the I odd sampler latch. Note: The value of this field is a signed magnitude number with a range of 5'b11111 (min) to 5'b01111 (max). A value of 5'b10000 (-0) is not valid.

Register Name:			RX sampler latch calibration Q predictive positive status register (!)
Register Mnemonic:			RX_SLC_QPP_STAT
Register Address:			10_m_nnnn : 0011 : 00101
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:8	R/O	7'h00	**Q even latch receiver sampler calibration DAC value**: This field contains the current value that the RX sampler latch calibration module has calculated to drive to the DAC that controls the offset for the Q even latch in the analog sampler component.
7	R/O	1'h0	Reserved
6:0	R/O	7'h00	**Q odd latch receiver sampler calibration DAC value**: This field contains the current value that the RX sampler latch calibration module has calculated to drive to the DAC that controls the offset for the Q odd latch in the analog sampler component.

Register Name:			RX sampler latch calibration Q predictive positive override register (!)
Register Mnemonic:			RX_SLC_QPP_OVRD
Register Address:			10_m_nnnn : 0011 : 00110
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Q even latch receiver sampler calibration DAC override enable**: When this bit is active (1'b1), the value in the Q even latch receiver sampler calibration DAC override value field of this register will override the calibrated value for the Q even sampler latch.
14:8	R/W	7'h00	**Q even latch receiver sampler calibration DAC override value**: This field contains the value that can be used to override the calibrated value for the Q even sampler latch. Note: The value of this field is a signed magnitude number with a range of 5'b11111 (min) to 5'b01111 (max). A value of 5'b10000 (-0) is not valid.
7	R/W	1'h0	**Q odd latch receiver sampler calibration DAC override enable**: When this bit is active (1'b1), the value in the Q odd latch receiver sampler calibration DAC override value field of this register will override the calibrated value for the Q odd sampler latch.
6:0	R/W	7'h00	**Q odd latch receiver sampler calibration DAC override value**: This field contains the value that can be used to override the calibrated value for the Q odd sampler latch. Note: The value of this field is a signed magnitude number with a range of 5'b11111 (min) to 5'b01111 (max). A value of 5'b10000 (-0) is not valid.

Register Name:			RX sampler latch calibration Q predictive negative status register (!)
Register Mnemonic:			RX_SLC_QPM_STAT
Register Address:			10_m_nnnn : 0011 : 00111
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:8	R/O	7'h00	**Q even latch receiver sampler calibration DAC value**: This field contains the current value that the RX sampler latch calibration module has calculated to drive to the DAC that controls the offset for the Q even latch in the analog sampler component.
7	R/O	1'h0	Reserved
6:0	R/O	7'h00	**Q odd latch receiver sampler calibration DAC value**: This field contains the current value that the RX sampler latch calibration module has calculated to drive to the DAC that controls the offset for the Q odd latch in the analog sampler component.

Register Name:			RX sampler latch calibration Q predictive negative override register (!)
Register Mnemonic:			RX_SLC_QPM_OVRD
Register Address:			10_m_nnnn : 0011 : 01000
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Q even latch receiver sampler calibration DAC override enable**: When this bit is active (1'b1), the value in the Q even latch receiver sampler calibration DAC override value field of this register will override the calibrated value for the Q even sampler latch.
14:8	R/W	7'h00	**Q even latch receiver sampler calibration DAC override value**: This field contains the value that can be used to override the calibrated value for the Q even sampler latch. Note: The value of this field is a signed magnitude number with a range of 5'b11111 (min) to 5'b01111 (max). A value of 5'b10000 (-0) is not valid.
7	R/W	1'h0	**Q odd latch receiver sampler calibration DAC override enable**: When this bit is active (1'b1), the value in the Q odd latch receiver sampler calibration DAC override value field of this register will override the calibrated value for the Q odd sampler latch.
6:0	R/W	7'h00	**Q odd latch receiver sampler calibration DAC override value**: This field contains the value that can be used to override the calibrated value for the Q odd sampler latch. Note: The value of this field is a signed magnitude number with a range of 5'b11111 (min) to 5'b01111 (max). A value of 5'b10000 (-0) is not valid.

Register Name:			RX sampler latch calibration e predictive positive status register (!)
Register Mnemonic:			RX_SLC_EPP_STAT
Register Address:			10_m_nnnn : 0011 : 01001
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:8	R/O	7'h00	**e even latch receiver sampler calibration DAC value**: This field contains the current value that the RX sampler latch calibration module has calculated to drive to the DAC that controls the offset for the e even latch in the analog sampler component.
7	R/O	1'h0	Reserved
6:0	R/O	7'h00	**e odd latch receiver sampler calibration DAC value**: This field contains the current value that the RX sampler latch calibration module has calculated to drive to the DAC that controls the offset for the e odd latch in the analog sampler component.

Register Name:			RX sampler latch calibration e predictive positive override register (!)
Register Mnemonic:			RX_SLC_EPP_OVRD
Register Address:			10_m_nnnn : 0011 : 01010
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**e even latch receiver sampler calibration DAC override enable**: When this bit is active (1'b1), the value in the e even latch receiver sampler calibration DAC override value field of this register will override the calibrated value for the e even sampler latch.
14:8	R/W	7'h00	**e even latch receiver sampler calibration DAC override value**: This field contains the value that can be used to override the calibrated value for the e even sampler latch. Note: The value of this field is a signed magnitude number with a range of 5'b11111 (min) to 5'b01111 (max). A value of 5'b10000 (-0) is not valid.
7	R/W	1'h0	**e odd latch receiver sampler calibration DAC override enable**: When this bit is active (1'b1), the value in the e odd latch receiver sampler calibration DAC override value field of this register will override the calibrated value for the e odd sampler latch.
6:0	R/W	7'h00	**e odd latch receiver sampler calibration DAC override value**: This field contains the value that can be used to override the calibrated value for the e odd sampler latch. Note: The value of this field is a signed magnitude number with a range of 5'b11111 (min) to 5'b01111 (max). A value of 5'b10000 (-0) is not valid.

Register Name:			RX sampler latch calibration e predictive negative status register (!)
Register Mnemonic:			RX_SLC_EPM_STAT
Register Address:			10_m_nnnn : 0011 : 01011
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:8	R/O	7'h00	**e even latch receiver sampler calibration DAC value**: This field contains the current value that the RX sampler latch calibration module has calculated to drive to the DAC that controls the offset for the e even latch in the analog sampler component.
7	R/O	1'h0	Reserved
6:0	R/O	7'h00	**e odd latch receiver sampler calibration DAC value**: This field contains the current value that the RX sampler latch calibration module has calculated to drive to the DAC that controls the offset for the e odd latch in the analog sampler component.

Register Name:			RX sampler latch calibration e predictive negative override register (!)
Register Mnemonic:			RX_SLC_EPM_OVRD
Register Address:			10_m_nnnn : 0011 : 01100
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**e even latch receiver sampler calibration DAC override enable**: When this bit is active (1'b1), the value in the e even latch receiver sampler calibration DAC override value field of this register will override the calibrated value for the e even sampler latch.
14:8	R/W	7'h00	**e even latch receiver sampler calibration DAC override value**: This field contains the value that can be used to override the calibrated value for the e even sampler latch. Note: The value of this field is a signed magnitude number with a range of 5'b11111 (min) to 5'b01111 (max). A value of 5'b10000 (-0) is not valid.
7	R/W	1'h0	**e odd latch receiver sampler calibration DAC override enable**: When this bit is active (1'b1), the value in the e odd latch receiver sampler calibration DAC override value field of this register will override the calibrated value for the e odd sampler latch.
6:0	R/W	7'h00	**e odd latch receiver sampler calibration DAC override value**: This field contains the value that can be used to override the calibrated value for the e odd sampler latch. Note: The value of this field is a signed magnitude number with a range of 5'b11111 (min) to 5'b01111 (max). A value of 5'b10000 (-0) is not valid.

Register Name:			RX sampler latch calibration initialization timer value register (!)
Register Mnemonic:			RX_SLC_INIT_TMR
Register Address:			10_m_nnnn : 0011 : 01101
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h0010	**RX sampler latch calibration initialization timer value** : This is the value that will be used for the RX sampler latch calibration initialization timer, which controls the time the rxda_sampler_latch_cal_en is held active prior to the individual calibration processes being kicked off. Note: This register should not be set to 16'h0000.

Register Name:			RX sampler latch calibration run timer value register (!)
Register Mnemonic:			RX_SLC_RUN_TMR
Register Address:			10_m_nnnn : 0011 : 01110
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'hA000	**RX sampler latch calibration run timer value** : This is the value that will be used for the RX sampler latch calibration run timer, which controls the run time for each calibration process. Note: This register should not be set to 16'h0000.

Register Name:			RX sampler latch calibration diagnostic control register (!)
Register Mnemonic:			RX_SLC_DIAG_CTRL
Register Address:			10_m_nnnn : 0011 : 01111
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Diagnostic control enable** : This bit enables the selected RX sampler latch calibration data sub module for diagnostic purposes. **Note: The diagnostic control override select field must be set to select the desired RX sampler latch calibration data sub module when this bit is set. This bit MUST also be cleared before changing the value of the diagnostic control override select field.** **Note: When these diagnostic features are used, the user must disable the respective automatic calibration functions in the RX sampler latch calibration disable register (!) on page 273. Due to the way this is implemented, if a given i sampler latch calibration function is being controlled by the diagnostic features here, all of the i sampler calibration functions must be disabled in the RX sampler latch calibration disable register (!) on page 273. The same is required for both the e and q sampler latch calibration functions.**
14:7	R/O	8'h0	Reserved
6	R/W	1'h0	**Voter override neg** : When enabled using the voter override enable bit in this register, writing a 1'b1 in this register bit will force the voter in the selected RX sampler latch calibration data sub module to activate the voter neg signal for a single clock cycle. **Note : This bit must be cleared after each time it is enabled, in order to perform follow on voter override functions with this bit.** **Note : It is not valid to activate both the voter override neg and voter override pos at the same time.**
5	R/W	1'h0	**Voter override pos** : When enabled using the voter override enable bit in this register, writing a 1'b1 in this register bit will force the voter in the selected RX sampler latch calibration data sub module to activate the voter pos signal for a single clock cycle. **Note : This bit must be cleared after each time it is enabled, in order to perform follow on voter override functions with this bit.** **Note : It is not valid to activate both the voter override neg and voter override pos at the same time.**
4	R/W	1'h0	**Voter override enable** : Setting this bit to a 1'b1 will enable the voter override function in the selected RX sampler latch calibration data sub module. **Note : This function is intended to be for diagnostic and verification purposes only.** **Note : Both the voter override neg and voter override pos bits in this register must be set to 1'b0 when this bit is initially set to 1'b1 when enabling this function.**
3:0	R/W	4'h0	**Diagnostic control override select** : Selects the RX sampler latch calibration data sub module to enable for diagnostic and verification purposes. This field also selects which sub module's rx_sampler_latch_cal_diag output port will be routed to the top level rx_sampler_latch_cal_diag output port. The following are the values used to select the required RX sampler latch calibration data sub module. • 4'b0000 : i odd positive coefficient. • 4'b0001 : q odd positive coefficient. • 4'b0010 : e odd positive coefficient. • 4'b0011 : Reserved. • 4'b0100 : i odd negative coefficient. • 4'b0101 : q odd negative coefficient. • 4'b0110 : e odd negative coefficient. • 4'b0111 : Reserved. • 4'b1000 : i even positive coefficient. • 4'b1001 : q even positive coefficient. • 4'b1010 : e even positive coefficient. • 4'b1011 : Reserved. • 4'b1100 : i even negative coefficient. • 4'b1101 : q even negative coefficient. • 4'b1110 : e even negative coefficient. • 4'b1111 : Reserved.

Register Name:			RX sampler latch calibration disable register (!)
Register Mnemonic:			RX_SLC_DIS
Register Address:			10_m_nnnn : 0011 : 10000
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14	R/W	1'h0	**e even negative coefficient disable :** Writing a 1'b1 to this bit will disable auto calibration for the e even negative coefficient RX sampler latch calibration data sub module.
13	R/W	1'h0	**q even negative coefficient disable :** Writing a 1'b1 to this bit will disable auto calibration for the q even negative coefficient RX sampler latch calibration data sub module.
12	R/W	1'h0	**i even negative coefficient disable :** Writing a 1'b1 to this bit will disable auto calibration for the i even negative coefficient RX sampler latch calibration data sub module.
11	R/O	1'h0	Reserved
10	R/W	1'h0	**e even positive coefficient disable :** Writing a 1'b1 to this bit will disable auto calibration for the e even positive coefficient RX sampler latch calibration data sub module.
9	R/W	1'h0	**q even positive coefficient disable :** Writing a 1'b1 to this bit will disable auto calibration for the q even positive coefficient RX sampler latch calibration data sub module.
8	R/W	1'h0	**i even positive coefficient disable :** Writing a 1'b1 to this bit will disable auto calibration for the i even positive coefficient RX sampler latch calibration data sub module.
7	R/O	1'h0	Reserved
6	R/W	1'h0	**e odd negative coefficient disable :** Writing a 1'b1 to this bit will disable auto calibration for the e odd negative coefficient RX sampler latch calibration data sub module.
5	R/W	1'h0	**q odd negative coefficient disable :** Writing a 1'b1 to this bit will disable auto calibration for the q odd negative coefficient RX sampler latch calibration data sub module.
4	R/W	1'h0	**i odd negative coefficient disable :** Writing a 1'b1 to this bit will disable auto calibration for the i odd negative coefficient RX sampler latch calibration data sub module.
3	R/O	1'h0	Reserved
2	R/W	1'h0	**e odd positive coefficient disable :** Writing a 1'b1 to this bit will disable auto calibration for the e odd positive coefficient RX sampler latch calibration data sub module.
1	R/W	1'h0	**q odd positive coefficient disable :** Writing a 1'b1 to this bit will disable auto calibration for the q odd positive coefficient RX sampler latch calibration data sub module.
0	R/W	1'h0	**i odd positive coefficient disable :** Writing a 1'b1 to this bit will disable auto calibration for the i odd positive coefficient RX sampler latch calibration data sub module.

10.2.31 Receiver Equalizer Engine (REE) Registers

Base address bits 15:5 :	**10_m_nnnn : 1000**
Address bits 4:0	**Register name and location of detailed description**
PCIe TX equalization state machine registers	
00000	REE PCIe TX equalization control state machine control register (!) on page 277
00001	REE PCIe TX equalization control state machine equalization enable mask for PCIe evaluation register (!) on page 278
00010	REE PCIe TX equalization control state machine equalization enable mask for PCIe post evaluation equalization register (!) on page 279
00011	REE PCIe TX equalization control state machine run post evaluation timer value register (!) on page 279
00100	REE PCIe TX equalization control state machine time-out timer value register (!) on page 279
00101	REE PCIe TX equalization control state machine max incremental evaluation value register (!) on page 280
00110	Reserved
00111	Reserved
General control state machine 1 registers	

Table 121: Receiver equalizer engine registers address space (slave 0)

Base address bits 15:5 :	10_m_nnnn : 1000
Address bits 4:0	**Register name and location of detailed description**
01000	REE general control state machine 1 control register (!) on page 280
01001	REE general control state machine 1 phase 1 equalization enable mask register (!) on page 281
01010	REE general control state machine 1 phase 2 equalization enable mask register (!) on page 282
01011	REE general control state machine 1 start timer value register (!) on page 282
01100	REE general control state machine 1 run phase 1 timer value register (!) on page 282
01101	REE general control state machine 1 run phase 2 timer value register (!) on page 283
01110	Reserved
01111	Reserved
General control state machine 2 registers	
10000	REE general control state machine 2 control register (!) on page 283
10001	REE general control state machine 2 phase 1 equalization enable mask register (!) on page 284
10010	REE general control state machine 2 phase 2 equalization enable mask register (!) on page 285
10011	REE general control state machine 2 start timer value register (!) on page 285
10100	REE general control state machine 2 run phase 1 timer value register (!) on page 285
10101	REE general control state machine 2 run phase 2 timer value register (!) on page 286
10110	Reserved
10111	Reserved
Periodic general control state machine registers	
11000	REE periodic general control state machine control register (!) on page 286
11001	REE periodic general control state machine phase 1 equalization enable mask register (!) on page 287
11010	REE periodic general control state machine phase 2 equalization enable mask register (!) on page 288
11011	REE periodic general control state machine start timer value register (!) on page 288
11100	REE periodic general control state machine run phase 1 timer value register (!) on page 288
11101	REE periodic general control state machine run phase 2 timer value register (!) on page 289
11110	Reserved
11111	Reserved

Table 121: Receiver equalizer engine registers address space (slave 0)

Base address bits 15:5 :	10_m_nnnn : 1001
Address bits 4:0	**Register name and location of detailed description**
USB 3 general control state machine registers	
00000	REE USB 3 general control state machine control register (!) on page 289
00001	REE USB 3 general control state machine phase 1 equalization enable mask register (!) on page 290
00010	REE USB 3 general control state machine phase 2 equalization enable mask register (!) on page 291
00011	REE USB 3 general control state machine start timer value register (!) on page 291
00100	REE USB 3 general control state machine run phase 1 timer value register (!) on page 291
00101	REE USB 3 general control state machine run phase 2 timer value register (!) on page 292
00110	Reserved
00111	Reserved
E path enable state machine registers	
01000	REE analog enable control state machine delay timer value register (!) on page 292
01001	Reserved
01010	Reserved
01011	Reserved
Reserved for future functions	
01100	Reserved
01101	Reserved
01110	Reserved
01111	Reserved
TX post emphasis registers (Peaking amp function)	
10000	REE TX post cursor control register (!) on page 292
10001	REE TX post cursor code control register (!) on page 293
10010	REE TX post cursor upper threshold register (!) on page 293
10011	REE TX post cursor lower threshold register (!) on page 293
10100	REE TX post cursor input override register (!) on page 293
10101	REE TX post cursor code override 0 register (!) on page 293
10110	REE TX post cursor code override 1 register (!) on page 294
10111	REE TX post cursor diagnostics register (!) on page 294
TX pre emphasis registers (Tap function)	
11000	REE TX pre cursor control register (!) on page 295
11001	REE TX pre cursor override register (!) on page 295
11010	REE TX pre cursor diagnostics register (!) on page 296
11011	Reserved
Reserved for future functions	
11100	Reserved
11101	Reserved
11110	Reserved
11111	Reserved

Table 122: Receiver equalizer engine registers address space (slave 1)

Base address bits 15:5 :	10_m_nnnn : 1010
Address bits 4:0	**Register name and location of detailed description**
Peaking amp function registers	
00000	REE peaking amp control register (!) on page 297
00001	REE peaking amp code control register (!) on page 297
00010	REE peaking amp upper threshold register (!) on page 298
00011	REE peaking amp lower threshold register (!) on page 298
00100	REE peaking amp input override register (!) on page 298
00101	REE peaking amp code override 0 register (!) on page 298
00110	REE peaking amp code override 1 register (!) on page 299
00111	REE peaking amp diagnostics register (!) on page 299
Attenuation function registers	
01000	REE attenuation control register (!) on page 299
01001	REE attenuation threshold register (!) on page 300
01010	REE attenuation counter register (!) on page 300
01011	REE attenuation override register (!) on page 300
01100	REE attenuation diagnostics register (!) on page 300
01101	Reserved
01110	Reserved
01111	Reserved
Tap 1 function registers	
10000	REE tap 1 control register (!) on page 301
10001	REE tap 1 override register (!) on page 301
10010	REE tap 1 diagnostics register (!) on page 302
10011	Reserved
Tap 2 function registers	
10100	REE tap 2 control register (!) on page 303
10101	REE tap 2 override register (!) on page 303
10110	REE tap 2 diagnostics register (!) on page 304
10111	Reserved
Tap 3 function registers	
11000	REE tap 3 control register (!) on page 305
11001	REE tap 3 override register (!) on page 305
11010	REE tap 3 diagnostics register (!) on page 306
11011	Reserved
Low frequency equalizer adaptive control registers	
11100	REE low frequency equalizer control register (!) on page 306
11101	REE low frequency equalizer override register (!) on page 307
11110	REE low frequency equalizer diagnostics register (!) on page 307
11111	Reserved

Table 123: Receiver equalizer engine registers address space (slave 2)

Base address bits 15:5 :	10_m_nnnn : 1011
Address bits 4:0	**Register name and location of detailed description**
VGA gain registers	
00000	REE VGA gain control register (!) on page 308
00001	REE VGA gain override register (!) on page 308
00010	REE VGA gain diagnostics register (!) on page 309
00011	REE VGA gain target adjust diagnostics register (!) on page 309
Offset correction registers	
00100	REE offset correction control register (!) on page 310
00101	REE offset correction override register (!) on page 310
00110	REE offset correction diagnostics register (!) on page 311
00111	Reserved
Short channel correction registers	
01000	REE short channel correction valid word counter register (!) on page 311
01001	REE short channel correction threshold counter register (!) on page 311
01010	Reserved
01011	Reserved
Reserved for future functions	
01100	Reserved
01101	Reserved
01110	Reserved
01111	Reserved
General REE control and diagnostic registers	
10000	REE adder configuration register (!) on page 312
10001	REE tap 1 clip control register (!) on page 312
10010	REE taps 2 and 3 clip control register (!) on page 313
10011	REE control data mask register (!) on page 314
10100	REE coefficient FIFO diagnostic register (!) on page 315
10101	REE diagnostic control register (!) on page 316
10110	REE TX equalization evaluator control register (!) on page 316
10111	REE control state machine gen mode control register 1 (!) on page 317
11000	REE control state machine gen mode control register 2 (!) on page 319
11001	REE TX equalization evaluator pre-emphasis register (!) on page 320
11010	REE TX equalization evaluator post-emphasis register (!) on page 320
11011	Reserved
11100	Reserved
11101	Reserved
11110	Reserved
11111	Reserved

Table 124: Receiver equalizer engine registers address space (slave 3)

Register Name:			REE PCIe TX equalization control state machine control register (!)
Register Mnemonic:			RX_REE_PTXEQSM_CTRL
Register Address:			10_m_nnnn : 1000 : 00000
Bits	**Type**	**Reset**	**Description**
15:0	R/O	16'h0	Reserved

Register Name:			REE PCIe TX equalization control state machine equalization enable mask for PCIe evaluation register (!)
Register Mnemonic:			RX_REE_PTXEQSM_EQENM_EVAL
Register Address:			10_m_nnnn : 1000 : 00001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	Reserved - spare
14	R/W	1'h0	**Ignore 1010 controller** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
13	R/W	1'h1	**TX equalization evaluator** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
12	R/W	1'h1	**TX post cursor control** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
11	R/W	1'h1	**TX pre cursor control** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
10	R/W	1'h0	**Short channel correction** : When set to 1'b1, this function is enabled when the TX equalization general control state machine is controlling the REE.
9	R/W	1'h1	**RX attenuation** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
8	R/W	1'h1	**RX VGA gain** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
7	R/W	1'h1	**RX offset correction coefficient** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
6	R/W	1'h1	**RX peaking amp gain** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
5	R/W	1'h0	**RX low frequency equalizer adaptive control** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
4	R/W	1'h0	Reserved - spare
3	R/W	1'h0	Reserved - spare
2	R/W	1'h1	**RX tap 3** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
1	R/W	1'h1	**RX tap 2** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
0	R/W	1'h1	**RX tap 1** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.

Register Name:			REE PCIe TX equalization control state machine equalization enable mask for PCIe post evaluation equalization register (!)
Register Mnemonic:			RX_REE_PTXEQSM_EQENM_PEVAL
Register Address:			10_m_nnnn : 1000 : 00010
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	Reserved - spare
14	R/W	1'h0	**Ignore 1010 controller** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
13	R/W	1'h0	**TX equalization evaluator** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
12	R/W	1'h0	**TX post cursor control** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
11	R/W	1'h0	**TX pre cursor control** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
10	R/W	1'h0	**Short channel correction** : When set to 1'b1, this function is enabled when the TX equalization general control state machine is controlling the REE.
9	R/W	1'h0	**RX attenuation** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
8	R/W	1'h0	**RX VGA gain** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
7	R/W	1'h0	**RX offset correction coefficient** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
6	R/W	1'h0	**RX peaking amp gain** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
5	R/W	1'h0	**RX low frequency equalizer adaptive control** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
4	R/W	1'h0	Reserved - spare
3	R/W	1'h0	Reserved - spare
2	R/W	1'h0	**RX tap 3** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
1	R/W	1'h0	**RX tap 2** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.
0	R/W	1'h0	**RX tap 1** : When set to 1'b1, this function is enabled when the PCIe TX equalization control state machine is controlling the REE.

Register Name:			REE PCIe TX equalization control state machine run post evaluation timer value register (!)
Register Mnemonic:			RX_REE_PTXEQSM_PEVAL_TMR
Register Address:			10_m_nnnn : 1000 : 00011
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h0000	**Run post evaluation equalization timer value** : This specifies number of clock cycles the state machine will wait in the Post Evaluation Equalization state.

Register Name:			REE PCIe TX equalization control state machine time-out timer value register (!)
Register Mnemonic:			RX_REE_PTXEQSM_TIMEOUT_TMR
Register Address:			10_m_nnnn : 1000 : 00100
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h30D4	**Time-out timer load value**: This specifies the number of clocks to run a PCIe evaluation before a time-out is indicated. If a time-out takes place, the state machine will indicate that no further equalization processes are required. Note: This value is shifted left 4 bits when loading the timer, the actual time is 16X this value.

Register Name:			REE PCIe TX equalization control state machine max incremental evaluation value register (!)
Register Mnemonic:			RX_REE_PTXEQSM_MAX_EVAL_CNT
Register Address:			10_m_nnnn : 1000 : 00101
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h3F	**Incremental evaluation counter load value**: This is the maximum number of incremental evaluations that will be performed plus one. When the number of incremental evaluations specified here minus one are performed, the state machine will indicate that no further equalization processes are required.

Register Name:			REE general control state machine 1 control register (!)
Register Mnemonic:			RX_REE_GCSM1_CTRL
Register Address:			10_m_nnnn : 1000 : 01000
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12'h0	Reserved
3	R/W	1'h1	**Equalization function reset enable**: Enables the reset of the functions controlled by this state machine, using the rx_ree_fcn_reset_n signal, when the equalization mode changes. • 1'b1 : Enabled • 1'b0 : Disabled
2	R/W	1'h0	**Loop enable**: Controls when the equalization functions in this state machine are run one time or loop continuously.
1	R/W	1'h0	**Force run equalization**: Setting this bit to a 1'b1, will force the general control state machine to run, independent of the macro functions that normally run the equalization.
0	R/W	1'h1	**Enable**: This bit enables the general control state machine function. • 1'b1 : Enabled • 1'b0 : Disabled

Register Name:			REE general control state machine 1 phase 1 equalization enable mask register (!)
Register Mnemonic:			RX_REE_GCSM1_EQENM_PH1
Register Address:			10_m_nnnn : 1000 : 01001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	Reserved - spare
14	R/W	1'h0	**Ignore 1010 controller** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
13	R/W	1'h0	**TX equalization evaluator** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
12	R/W	1'h0	**TX post cursor control** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
11	R/W	1'h0	**TX pre cursor control** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
10	R/W	1'h0	**Short channel correction** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
9	R/W	1'h1	**RX attenuation** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
8	R/W	1'h1	**RX VGA gain** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
7	R/W	1'h1	**RX offset correction coefficient** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
6	R/W	1'h1	**RX peaking amp gain** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
5	R/W	1'h1	**RX low frequency equalizer adaptive control** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
4	R/W	1'h0	Reserved - spare
3	R/W	1'h0	Reserved - spare
2	R/W	1'h1	**RX tap 3** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
1	R/W	1'h1	**RX tap 2** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
0	R/W	1'h1	**RX tap 1** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.

Register Name:			REE general control state machine 1 phase 2 equalization enable mask register (!)
Register Mnemonic:			RX_REE_GCSM1_EQENM_PH2
Register Address:			10_m_nnnn : 1000 : 01010
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	Reserved - spare
14	R/W	1'h0	**Ignore 1010 controller** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
13	R/W	1'h0	**TX equalization evaluator** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
12	R/W	1'h0	**TX post cursor control** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
11	R/W	1'h0	**TX pre cursor control** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
10	R/W	1'h0	**Short channel correction** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
9	R/W	1'h0	**RX attenuation** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
8	R/W	1'h1	**RX VGA gain** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
7	R/W	1'h1	**RX offset correction coefficient** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
6	R/W	1'h1	**RX peaking amp gain** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
5	R/W	1'h1	**RX low frequency equalizer adaptive control** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
4	R/W	1'h0	Reserved - spare
3	R/W	1'h0	Reserved - spare
2	R/W	1'h1	**RX tap 3** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
1	R/W	1'h1	**RX tap 2** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.
0	R/W	1'h1	**RX tap 1** : When set to 1'b1, this function is enabled when the general control state machine 1 is controlling the REE.

Register Name:			REE general control state machine 1 start timer value register (!)
Register Mnemonic:			RX_REE_GCSM1_START_TMR
Register Address:			10_m_nnnn : 1000 : 01011
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h0000	**Start timer value** : The number of clock cycles the state machine will wait in the Start Delay state. Note: This value is shifted left 8 bits when loading the timer, the actual time is 256X this value.

Register Name:			REE general control state machine 1 run phase 1 timer value register (!)
Register Mnemonic:			RX_REE_GCSM1_RUN_PH1_TMR
Register Address:			10_m_nnnn : 1000 : 01100
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h0F43	**Run phase 1 timer value** : This specifies the number of clock cycles the state machine will wait in the Run Equalization Phase 1 state. If this register is set to all 1s, the run in the Run Equalization Phase 1 state indefinitely. Note: This value is shifted left 8 bits when loading the timer, the actual time is 256X this value.

Register Name:			REE general control state machine 1 run phase 2 timer value register (!)
Register Mnemonic:			RX_REE_GCSM1_RUN_PH2_TMR
Register Address:			10_m_nnnn : 1000 : 01101
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h009D	**Run phase 2 timer value** : This specifies the number of clock cycles the state machine will wait in the Run Equalization Phase 2 state. Note: This value is shifted left 8 bits when loading the timer, the actual time is 256X this value.

Register Name:			REE general control state machine 2 control register (!)
Register Mnemonic:			RX_REE_GCSM2_CTRL
Register Address:			10_m_nnnn : 1000 : 10000
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12'h0	Reserved
3	R/W	1'h1	**Equalization function reset enable**: Enables the reset of the functions controlled by this state machine, using the rx_ree_fcn_reset_n signal, when the equalization mode changes. • 1'b1 : Enabled • 1'b0 : Disabled
2	R/W	1'h0	**Loop enable**: Controls when the equalization functions in this state machine are run one time or loop continuously.
1	R/W	1'h0	**Force run equalization**: Setting this bit to a 1'b1, will force the general control state machine to run, independent of the macro functions that normally run the equalization.
0	R/W	1'h1	**Enable**: This bit enables the general control state machine function. • 1'b1 : Enabled • 1'b0 : Disabled

Register Name:			REE general control state machine 2 phase 1 equalization enable mask register (!)
Register Mnemonic:			RX_REE_GCSM2_EQENM_PH1
Register Address:			10_m_nnnn : 1000 : 10001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	Reserved - spare
14	R/W	1'h0	**Ignore 1010 controller** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
13	R/W	1'h0	**TX equalization evaluator** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
12	R/W	1'h0	**TX post cursor control** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
11	R/W	1'h0	**TX pre cursor control** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
10	R/W	1'h0	**Short channel correction** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
9	R/W	1'h0	**RX attenuation** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
8	R/W	1'h0	**RX VGA gain** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
7	R/W	1'h1	**RX offset correction coefficient** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
6	R/W	1'h0	**RX peaking amp gain** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
5	R/W	1'h0	**RX low frequency equalizer adaptive control** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
4	R/W	1'h0	Reserved - spare
3	R/W	1'h0	Reserved - spare
2	R/W	1'h0	**RX tap 3** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
1	R/W	1'h0	**RX tap 2** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
0	R/W	1'h0	**RX tap 1** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.

Register Name:			REE general control state machine 2 phase 2 equalization enable mask register (!)
Register Mnemonic:			RX_REE_GCSM2_EQENM_PH2
Register Address:			10_m_nnnn : 1000 : 10010
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	Reserved - spare
14	R/W	1'h0	**Ignore 1010 controller** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
13	R/W	1'h0	**TX equalization evaluator** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
12	R/W	1'h0	**TX post cursor control** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
11	R/W	1'h0	**TX pre cursor control** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
10	R/W	1'h0	**Short channel correction** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
9	R/W	1'h0	**RX attenuation** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
8	R/W	1'h0	**RX VGA gain** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
7	R/W	1'h0	**RX offset correction coefficient** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
6	R/W	1'h0	**RX peaking amp gain** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
5	R/W	1'h0	**RX low frequency equalizer adaptive control** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
4	R/W	1'h0	Reserved - spare
3	R/W	1'h0	Reserved - spare
2	R/W	1'h0	**RX tap 3** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
1	R/W	1'h0	**RX tap 2** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.
0	R/W	1'h0	**RX tap 1** : When set to 1'b1, this function is enabled when the general control state machine 2 is controlling the REE.

Register Name:			REE general control state machine 2 start timer value register (!)
Register Mnemonic:			RX_REE_GCSM2_START_TMR
Register Address:			10_m_nnnn : 1000 : 10011
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h0000	**Start timer value** : The number of clock cycles the state machine will wait in the Start Delay state. Note: This value is shifted left 8 bits when loading the timer, the actual time is 256X this value.

Register Name:			REE general control state machine 2 run phase 1 timer value register (!)
Register Mnemonic:			RX_REE_GCSM2_RUN_PH1_TMR
Register Address:			10_m_nnnn : 1000 : 10100
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'hFFFF	**Run phase 1 timer value** : This specifies the number of clock cycles the state machine will wait in the Run Equalization Phase 1 state. If this register is set to all 1s, the run in the Run Equalization Phase 1 state indefinitely. Note: This value is shifted left 8 bits when loading the timer, the actual time is 256X this value.

Register Name:			REE general control state machine 2 run phase 2 timer value register (!)
Register Mnemonic:			RX_REE_GCSM2_RUN_PH2_TMR
Register Address:			10_m_nnnn : 1000 : 10101
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h0000	**Run phase 2 timer value** : This specifies the number of clock cycles the state machine will wait in the Run Equalization Phase 2 state. Note: This value is shifted left 8 bits when loading the timer, the actual time is 256X this value.

Register Name:			REE periodic general control state machine control register (!)
Register Mnemonic:			RX_REE_PERGCSM_CTRL
Register Address:			10_m_nnnn : 1000 : 11000
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12'h0	Reserved
3	R/W	1'h1	**Equalization function reset enable**: Enables the reset of the functions controlled by this state machine, using the rx_ree_fcn_reset_n pin, when the equalization mode changes. • 1'b1 : Enabled • 1'b0 : Disabled
2	R/W	1'h1	**Loop enable**: Controls when the equalization functions in this state machine are run one time or loop continuously.
1	R/W	1'h0	**Force run equalization**: Setting this bit to a 1'b1, will force the general control state machine to run, independent of the macro functions that normally run the equalization.
0	R/W	1'h1	**Enable**: This bit enables the general control state machine function. • 1'b1 : Enabled • 1'b0 : Disabled

Register Name:			REE periodic general control state machine phase 1 equalization enable mask register (!)
Register Mnemonic:			RX_REE_PERGCSM_EQENM_PH1
Register Address:			10_m_nnnn : 1000 : 11001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	Reserved - spare
14	R/W	1'h0	**Ignore 1010 controller** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
13	R/W	1'h0	**TX equalization evaluator** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
12	R/W	1'h0	**TX post cursor control** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
11	R/W	1'h0	**TX pre cursor control** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
10	R/W	1'h0	**Short channel correction** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
9	R/W	1'h1	**RX attenuation** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
8	R/W	1'h1	**RX VGA gain** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
7	R/W	1'h1	**RX offset correction coefficient** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
6	R/W	1'h1	**RX peaking amp gain** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
5	R/W	1'h0	**RX low frequency equalizer adaptive control** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
4	R/W	1'h0	Reserved - spare
3	R/W	1'h0	Reserved - spare
2	R/W	1'h1	**RX tap 3** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
1	R/W	1'h1	**RX tap 2** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
0	R/W	1'h1	**RX tap 1** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.

Register Name:			REE periodic general control state machine phase 2 equalization enable mask register (!)
Register Mnemonic:			RX_REE_PERGCSM_EQENM_PH2
Register Address:			10_m_nnnn : 1000 : 11010
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	Reserved - spare
14	R/W	1'h0	**Ignore 1010 controller** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
13	R/W	1'h0	**TX equalization evaluator** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
12	R/W	1'h0	**TX post cursor control** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
11	R/W	1'h0	**TX pre cursor control** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
10	R/W	1'h0	**Short channel correction** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
9	R/W	1'h0	**RX attenuation** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
8	R/W	1'h1	**RX VGA gain** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
7	R/W	1'h1	**RX offset correction coefficient** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
6	R/W	1'h1	**RX peaking amp gain** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
5	R/W	1'h0	**RX low frequency equalizer adaptive control** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
4	R/W	1'h0	Reserved - spare
3	R/W	1'h0	Reserved - spare
2	R/W	1'h1	**RX tap 3** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
1	R/W	1'h1	**RX tap 2** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.
0	R/W	1'h1	**RX tap 1** : When set to 1'b1, this function is enabled when the periodic general control state machine is controlling the REE.

Register Name:			REE periodic general control state machine start timer value register (!)
Register Mnemonic:			RX_REE_PERGCSM_START_TMR
Register Address:			10_m_nnnn : 1000 : 11011
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h1E85	**Start timer value** : The number of clock cycles the state machine will wait in the Start Delay state. Note: This value is shifted left 8 bits when loading the timer, the actual time is 256X this value.

Register Name:			REE periodic general control state machine run phase 1 timer value register (!)
Register Mnemonic:			RX_REE_PERGCSM_RUN_PH1_TMR
Register Address:			10_m_nnnn : 1000 : 11100
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h09D	**Run phase 1 timer value** : This specifies the number of clock cycles the state machine will wait in the Run Equalization Phase 1 state. If this register is set to all 1s, the run in the Run Equalization Phase 1 state indefinitely. Note: This value is shifted left 8 bits when loading the timer, the actual time is 256X this value.

Register Name:			REE periodic general control state machine run phase 2 timer value register (!)
Register Mnemonic:			RX_REE_PERGCSM_RUN_PH2_TMR
Register Address:			10_m_nnnn : 1000 : 11101
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h009D	**Run phase 2 timer value** : This specifies the number of clock cycles the state machine will wait in the Run Equalization Phase 2 state. Note: This value is shifted left 8 bits when loading the timer, the actual time is 256X this value.

Register Name:			REE USB 3 general control state machine control register (!)
Register Mnemonic:			RX_REE_U3GCSM_CTRL
Register Address:			10_m_nnnn : 1001 : 00000
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12'h0	Reserved
3	R/W	1'h1	**Equalization function reset enable**: Enables the reset of the functions controlled by this state machine, using the rx_ree_fcn_reset_n pin, when the equalization mode changes. • 1'b1 : Enabled • 1'b0 : Disabled
2	R/W	1'h0	**Loop enable**: Controls when the equalization functions in this state machine are run one time or loop continuously.
1	R/W	1'h0	**Force run equalization**: Setting this bit to a 1'b1, will force the general control state machine to run, independent of the macro functions that normally run the equalization.
0	R/W	1'h1	**Enable**: This bit enables the general control state machine function. • 1'b1 : Enabled • 1'b0 : Disabled

Register Name:			REE USB 3 general control state machine phase 1 equalization enable mask register (!)
Register Mnemonic:			RX_REE_U3GCSM_EQENM_PH1
Register Address:			10_m_nnnn : 1001 : 00001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	Reserved - spare
14	R/W	1'h0	**Ignore 1010 controller** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
13	R/W	1'h0	**TX equalization evaluator** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
12	R/W	1'h0	**TX post cursor control** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
11	R/W	1'h0	**TX pre cursor control** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
10	R/W	1'h0	**Short channel correction** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
9	R/W	1'h1	**RX attenuation** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
8	R/W	1'h1	**RX VGA gain** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
7	R/W	1'h1	**RX offset correction coefficient** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
6	R/W	1'h1	**RX peaking amp gain** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
5	R/W	1'h0	**RX low frequency equalizer adaptive control** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
4	R/W	1'h0	Reserved - spare
3	R/W	1'h0	Reserved - spare
2	R/W	1'h1	**RX tap 3** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
1	R/W	1'h1	**RX tap 2** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
0	R/W	1'h1	**RX tap 1** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.

Register Name:			REE USB 3 general control state machine phase 2 equalization enable mask register (!)
Register Mnemonic:			RX_REE_U3GCSM_EQENM_PH2
Register Address:			10_m_nnnn : 1001 : 00010
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	Reserved - spare
14	R/W	1'h0	**Ignore 1010 controller** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
13	R/W	1'h0	**TX equalization evaluator** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
12	R/W	1'h0	**TX post cursor control** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
11	R/W	1'h0	**TX pre cursor control** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
10	R/W	1'h0	**Short channel correction** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
9	R/W	1'h0	**RX attenuation** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
8	R/W	1'h1	**RX VGA gain** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
7	R/W	1'h1	**RX offset correction coefficient** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
6	R/W	1'h1	**RX peaking amp gain** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
5	R/W	1'h0	**RX low frequency equalizer adaptive control** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
4	R/W	1'h0	Reserved - spare
3	R/W	1'h0	Reserved - spare
2	R/W	1'h1	**RX tap 3** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
1	R/W	1'h1	**RX tap 2** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
0	R/W	1'h1	**RX tap 1** : When set to 1'b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.

Register Name:			REE USB 3 general control state machine start timer value register (!)
Register Mnemonic:			RX_REE_U3GCSM_START_TMR
Register Address:			10_m_nnnn : 1001 : 00011
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h0125	**Start timer value** : The number of clock cycles the state machine will wait in the Start Delay state. Note: This value is shifted left 8 bits when loading the timer, the actual time is 256X this value.

Register Name:			REE USB 3 general control state machine run phase 1 timer value register (!)
Register Mnemonic:			RX_REE_U3GCSM_RUN_PH1_TMR
Register Address:			10_m_nnnn : 1001 : 00100
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h07A2	**Run phase 1 timer value** : This specifies the number of clock cycles the state machine will wait in the Run Equalization Phase 1 state. If this register is set to all 1s, the run in the Run Equalization Phase 1 state indefinitely. Note: This value is shifted left 8 bits when loading the timer, the actual time is 256X this value.

Register Name:			REE USB 3 general control state machine run phase 2 timer value register (!)
Register Mnemonic:			RX_REE_U3GCSM_RUN_PH2_TMR
Register Address:			10_m_nnnn : 1001 : 00101
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h009D	**Run phase 2 timer value** : This specifies the number of clock cycles the state machine will wait in the Run Equalization Phase 2 state. Note: This value is shifted left 8 bits when loading the timer, the actual time is 256X this value.

Register Name:			REE analog enable control state machine delay timer value register (!)
Register Mnemonic:			RX_REE_ANAENSM_DEL_TMR
Register Address:			10_m_nnnn : 1001 : 01000
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h0064	**Analog enable delay timer value** : The number of clock cycles the state machine will wait in the Analog Enable Delay state. The time specified here is the number of clock cycles to wait between when the analog enable signal (ana_en) is initially driven active, until the ana_en_ack signal is driven active. The intention of this timer is to provide for the necessary amount of time between when the analog enable is driven active until when data controlled by the analog enable is considered valid.

Register Name:			REE TX post cursor control register (!)
Register Mnemonic:			RX_REE_TXPOST_CTRL
Register Address:			10_m_nnnn : 1001 : 10000
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11	R/W	1'h1	**Peaking amp feedback path enable**: Enables the peaking amp feedback path.
10:8	R/W	3'h0	**Peaking amp feedback scaler value**: Specifies the amount to scale the peaking amp feedback by. The following are the valid settings for this signal: • 3'b110 : /4 • 3'b111 : /2 • 3'b000 : x1 • 3'b001 : x2 • 3'b010 : x4
7	R/O	1'h0	Reserved
6:4	R/W	3'h0	**Peaking amp integrator accumulator scaler value**: Specifies the amount to scale the input to the peaking amp integrator accumulator by. The following are the valid settings for this field: • 3'b000: x1 • 3'b001: x2 • 3'b010: x4 • 3'b011: x8 • 3'b100: x16 • 3'b101 - 3'b111: Reserved Note: This function is unused in the TX post-cursor control.
3:0	R/W	4'h0	**Peaking amp sigma delta accumulator scaler value**: Specifies the amount to scale the input to the peaking amp sigma delta accumulator by. The following are the valid settings for this field: • 4'b0000: x1 • 4'b0001: x2 • 4'b0010: x4 • 4'b0011: x8 • 4'b0100: x16 • 4'b0101: x 32 • 4'b0110: x 64 • 4'b0111: x128 • 4'b1000: x256 • 4'b1001: x512 • 4'b1010: x1024 • 4'b1011: x2048 • 4'b1100: x 4096 • 4'b1101 x 8192 • 4'b1110 x 16384 • 4'b1111 x 32768

Register Name:			REE TX post cursor code control register (!)
Register Mnemonic:			RX_REE_TXPOST_CODE_CTRL
Register Address:			10_m_nnnn : 1001 : 10001
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'h0	Reserved
13:8	R/W	6'h00	**Peaking amp code maximum value**: This is the maximum value that the peaking amp code will be allowed to increase to. Note: This function is unused in the TX post-cursor control.
7:6	R/O	2'h0	Reserved
5:0	R/W	6'h00	**Peaking amp initial code**: Initial value the peaking amp code is set to when training starts. Note: This function is unused in the TX post-cursor control.

Register Name:			REE TX post cursor upper threshold register (!)
Register Mnemonic:			RX_REE_TXPOST_UTHR
Register Address:			10_m_nnnn : 1001 : 10010
Bits	Type	Reset	Description
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h008	**Peaking amp algorithm upper threshold**: This is the upper threshold value used in the peaking amp algorithm.

Register Name:			REE TX post cursor lower threshold register (!)
Register Mnemonic:			RX_REE_TXPOST_LTHR
Register Address:			10_m_nnnn : 1001 : 10011
Bits	Type	Reset	Description
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h006	**Peaking amp algorithm lower threshold**: This is the lower threshold value used in the peaking amp algorithm.

Register Name:			REE TX post cursor input override register (!)
Register Mnemonic:			RX_REE_TXPOST_IOVRD
Register Address:			10_m_nnnn : 1001 : 10100
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Peaking amp tap accumulator input override enable**: Setting this bit to a 1'b1 will allow the tap accumulator input in the peaking amp gain algorithm to be overridden by the peaking amp tap accumulator input override field in this register. Note: This function is unused in the TX post-cursor control.
14:8	R/O	7'h0	Reserved
7:0	R/W	8'h00	**Peaking amp tap accumulator input override** : Value that will override the tap accumulator input in the peaking amp gain algorithm, when the Peaking amp tap accumulator input override enable bit is active. Note: This function is unused in the TX post-cursor control.

Register Name:			REE TX post cursor code override 0 register (!)
Register Mnemonic:			RX_REE_TXPOST_COVRD0
Register Address:			10_m_nnnn : 1001 : 10101
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'h0	Reserved
13:8	R/W	6'h00	**Peaking amp code override value mode 1**: Value that will override the peaking amp code when in standard mode 1 when the peaking amp code override enable bit in the REE TX post cursor diagnostics register (!) on page 294 is active.
7:6	R/O	2'h0	Reserved
5:0	R/W	6'h00	**Peaking amp code override value mode 0**: Value that will override the peaking amp code when in standard mode 0 when the peaking amp code override enable bit in the REE TX post cursor diagnostics register (!) on page 294 is active.

Register Name:			REE TX post cursor code override 1 register (!)
Register Mnemonic:			RX_REE_TXPOST_COVRD1
Register Address:			10_m_nnnn : 1001 : 10110
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'h0	Reserved
13:8	R/W	6'h00	**Peaking amp code override value mode 3**: Value that will override the peaking amp code when in standard mode 3 when the peaking amp code override enable bit in the REE TX post cursor diagnostics register (!) on page 294 is active.
7:6	R/O	2'h0	Reserved
5:0	R/W	6'h00	**Peaking amp code override value mode 2**: Value that will override the peaking amp code when in standard mode 2 when the peaking amp code override enable bit in the REE TX post cursor diagnostics register (!) on page 294 is active.

Register Name:			REE TX post cursor diagnostics register (!)
Register Mnemonic:			RX_REE_TXPOST_DIAG
Register Address:			10_m_nnnn : 1001 : 10111
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Peaking amp code override enable**: Setting this bit to a 1'b1 will allow the peaking amp code to be overridden by the peaking amp code override value fields in the REE TX post cursor code override 0 register (!) on page 293 and REE TX post cursor code override 1 register (!) on page 294.
14	R/W	1'h0	**Voter override neg** : Writing a 1'b1 in this register bit will force the peaking amp voter function to activate the voter neg signal for a single clock cycle. **Note : This bit must be cleared after each time it is enabled, in order to perform follow on voter override functions with this bit.** **Note : It is not valid to activate both the voter override neg and voter override pos at the same time.**
13	R/W	1'h0	**Voter override pos** : Writing a 1'b1 in this register bit will force the peaking amp voter function to activate the voter pos signal for a single clock cycle. **Note : This bit must be cleared after each time it is enabled, in order to perform follow on voter override functions with this bit.** **Note : It is not valid to activate both the voter override neg and voter override pos at the same time.**
12	R/W	1'h0	**Voter override enable** : Setting this bit to a 1'b1 will allow only the voter override pos and voter override neg bits in this register to control the voter in the peaking amp. **Note : This function is intended to be for diagnostic and verification purposes only.** **Note : Both the voter override neg and voter override pos bits in this register must be set to 1'b0 when this bit is initially set to 1'b1 when enabling this function.**
11:8	R/O	4'h0	Reserved
7:6	R/O	2'h0	Reserved
5:0	R/O	6'h00	**Current peaking amp integrator accumulator**: Current value of the tap integrator accumulator, without the unused sign bit.

Register Name:			REE TX pre cursor control register (!)
Register Mnemonic:			RX_REE_TXPRE_CTRL
Register Address:			10_m_nnnn : 1001 : 11000
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11	R/W	1'h1	**Tap coefficient combinational logic zero crossing enable**: • 1'b0: Zero crossing combinational logic input not enabled. • 1'b1: Zero crossing combinational logic input enabled.
10	R/W	1'h1	**Tap coefficient combinational logic non zero crossing enable**: • 1'b0: non zero crossing combinational logic input not enabled. • 1'b1: non zero crossing combinational logic input enabled.
9	R/W	1'h0	**Tap coefficient combinational logic bit 0 only enable**: • 1'b0: All enabled combinational logic input modules will be used. • 1'b1: Only the enabled combinational logic input modules associated with bit 0 will be used.
8	R/W	1'h1	**Receiver DFE tap coefficient disable**: This bit disables the rxda_dfe_tap_coef output signal. • 1'b0 : rxda_dfe_tap_coef output enabled. • 1'b1 : rxda_dfe_tap_coef output disabled (all 0s). Note: This function is unused in the TX pre-cursor control.
7	R/O	1'h0	Reserved
6:4	R/W	3'h0	**Tap integrator accumulator scaler value**: Specifies the amount to scale the input to the tap integrator accumulator by. The following are the valid settings for this field: • 3'b000: x1 • 3'b001 - 3'b111: Reserved Note: This function is unused in the TX pre-cursor control.
3:0	R/W	4'h0	**Tap sigma delta accumulator scaler value**: Specifies the amount to scale the input to the tap sigma delta accumulator by. The following are the valid settings for this field: • 4'b0000: x1 • 4'b0001: x2 • 4'b0010: x4 • 4'b0011: x8 • 4'b0100: x16 • 4'b0101: x 32 • 4'b0110: x 64 • 4'b0111: x128 • 4'b1000: x256 • 4'b1001: x512 • 4'b1010: x1024 • 4'b1011: x2048 • 4'b1100: x 4096 • 4'b1101 x 8192 • 4'b1110 x 16384 • 4'b1111 x 32768

Register Name:			REE TX pre cursor override register (!)
Register Mnemonic:			RX_REE_TXPRE_OVRD
Register Address:			10_m_nnnn : 1001 : 11001
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7	R/W	1'h0	**Tap override enable**: Setting this bit to a 1'b1 will enable the tap override field in this register to override the tap integrator accumulator functions. Note: This function is unused in the TX pre-cursor control.
6	R/O	1'h0	Reserved
5:0	R/W	6'h00	**Tap override value**: When the tap override enable bit in this register is active, the value in this field will override the integrator accumulator value, as well as the input to the binary to thermometer encoder. Note: This function is unused in the TX pre-cursor control.

Register Name:			REE TX pre cursor diagnostics register (!)
Register Mnemonic:			RX_REE_TXPRE_DIAG
Register Address:			10_m_nnnn : 1001 : 11010
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14	R/W	1'h0	**Voter override neg** : Writing a 1'b1 in this register bit will force the tap voter function to activate the voter neg signal for a single clock cycle. **Note : This bit must be cleared after each time it is enabled, in order to perform follow on voter override functions with this bit.** **Note : It is not valid to activate both the voter override neg and voter override pos at the same time.**
13	R/W	1'h0	**Voter override pos** : Writing a 1'b1 in this register bit will force the tap voter function to activate the voter pos signal for a single clock cycle. **Note : This bit must be cleared after each time it is enabled, in order to perform follow on voter override functions with this bit.** **Note : It is not valid to activate both the voter override neg and voter override pos at the same time.**
12	R/W	1'h0	**Voter override enable** : Setting this bit to a 1'b1 will allow only the voter override pos and voter override neg bits in this register to control the voter in the tap. **Note : This function is intended to be for diagnostic and verification purposes only.** **Note : Both the voter override neg and voter override pos bits in this register must be set to 1'b0 when this bit is initially set to 1'b1 when enabling this function.**
11:6	R/O	6'h0	Reserved
5:0	R/O	6'h00	**Current tap integrator accumulator**: Current value of the tap integrator accumulator.

Register Name:			REE peaking amp control register (!)
Register Mnemonic:			RX_REE_PEAK_CTRL
Register Address:			10_m_nnnn : 1010 : 00000
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11	R/W	1'h1	**Peaking amp feedback path enable**: Enables the peaking amp feedback path.
10:8	R/W	3'h7	**Peaking amp feedback scaler value**: Specifies the amount to scale the peaking amp feedback by. The following are the valid settings for this signal: • 3'b110 : /4 • 3'b111 : /2 • 3'b000 : x1 • 3'b001 : x2 • 3'b010 : x4
7	R/O	1'h0	Reserved
6:4	R/W	3'h0	**Peaking amp integrator accumulator scaler value**: Specifies the amount to scale the input to the peaking amp integrator accumulator by. The following are the valid settings for this field: • 3'b000: x1 • 3'b001: x2 • 3'b010: x4 • 3'b011: x8 • 3'b100: x16 • 3'b101 - 3'b111: Reserved
3:0	R/W	4'h1	**Peaking amp sigma delta accumulator scaler value**: Specifies the amount to scale the input to the peaking amp sigma delta accumulator by. The following are the valid settings for this field: • 4'b0000: x1 • 4'b0001: x2 • 4'b0010: x4 • 4'b0011: x8 • 4'b0100: x16 • 4'b0101: x 32 • 4'b0110: x 64 • 4'b0111: x128 • 4'b1000: x256 • 4'b1001: x512 • 4'b1010: x1024 • 4'b1011: x2048 • 4'b1100: x 4096 • 4'b1101 x 8192 • 4'b1110 x 16384 • 4'b1111 x 32768

Register Name:			REE peaking amp code control register (!)
Register Mnemonic:			RX_REE_PEAK_CODE_CTRL
Register Address:			10_m_nnnn : 1010 : 00001
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'h0	Reserved
13:8	R/W	6'h2A	**Peaking amp code maximum value**: This is the maximum value that the peaking amp code will be allowed to increase to. Note: In this application the width of the integrator accumulator is 6, which is one larger that the number of bits required to generate a full 32 bit thermometer code. Therefore, as noted in the description of the REE gain integrator function, the thermometer code is generated using the most significant 5 bits of the integrator accumulator. As a result, the maximum value here needs to be 2X that of the maximum value of the thermometer encoded output.
7:6	R/O	2'h0	Reserved
5:0	R/W	6'h20	**Peaking amp initial code**: Initial value the peaking amp code is set to when training starts. Note: In this application the width of the integrator accumulator is 6, which is one larger that the number of bits required to generate a full 32 bit thermometer code. Therefore, as noted in the description of the REE gain integrator function, the thermometer code is generated using the most significant 5 bits of the integrator accumulator. As a result, the starting value here needs to be 2X that of the desired starting value of the thermometer encoded output.

Register Name:			REE peaking amp upper threshold register (!)
Register Mnemonic:			RX_REE_PEAK_UTHR
Register Address:			10_m_nnnn : 1010 : 00010
Bits	Type	Reset	Description
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h004	**Peaking amp algorithm upper threshold**: This is the upper threshold value used in the peaking amp algorithm.

Register Name:			REE peaking amp lower threshold register (!)
Register Mnemonic:			RX_REE_PEAK_LTHR
Register Address:			10_m_nnnn : 1010 : 00011
Bits	Type	Reset	Description
15:9	R/O	7'h0	Reserved
8:0	R/W	9'h1FA	**Peaking amp algorithm lower threshold**: This is the lower threshold value used in the peaking amp algorithm.

Register Name:			REE peaking amp input override register (!)
Register Mnemonic:			RX_REE_PEAK_IOVRD
Register Address:			10_m_nnnn : 1010 : 00100
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Peaking amp tap accumulator input override enable**: Setting this bit to a 1'b1 will allow the tap accumulator input in the peaking amp gain algorithm to be overridden by the peaking amp tap accumulator input override field in this register.
14:8	R/O	7'h0	Reserved
7:0	R/W	8'h00	**Peaking amp tap accumulator input override** : Value that will override the tap accumulator input in the peaking amp gain algorithm, when the Peaking amp tap accumulator input override enable bit is active.

Register Name:			REE peaking amp code override 0 register (!)
Register Mnemonic:			RX_REE_PEAK_COVRD0
Register Address:			10_m_nnnn : 1010 : 00101
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'h0	Reserved
13:8	R/W	6'h00	**Peaking amp code override value mode 1**: Value that will override the peaking amp code when in standard mode 1 when the peaking amp code override enable bit in the REE peaking amp diagnostics register (!) on page 299 is active. The value of this override field will ultimately drive the thermometer encoded value on the rxda_dfe_peaking_amp_gain signal driven to the analog.
7:6	R/O	2'h0	Reserved
5:0	R/W	6'h00	**Peaking amp code override value mode 0**: Value that will override the peaking amp code when in standard mode 0 when the peaking amp code override enable bit in the REE peaking amp diagnostics register (!) on page 299 is active. The value of this override field will ultimately drive the thermometer encoded value on the rxda_dfe_peaking_amp_gain signal driven to the analog.

Register Name:			REE peaking amp code override 1 register (!)
Register Mnemonic:			RX_REE_PEAK_COVRD1
Register Address:			10_m_nnnn : 1010 : 00110
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'h0	Reserved
13:8	R/W	6'h00	**Peaking amp code override value mode 3**: Value that will override the peaking amp code when in standard mode 3 when the peaking amp code override enable bit in the REE peaking amp diagnostics register (!) on page 299 is active. The value of this override field will ultimately drive the thermometer encoded value on the rxda_dfe_peaking_amp_gain signal driven to the analog.
7:6	R/O	2'h0	Reserved
5:0	R/W	6'h00	**Peaking amp code override value mode 2**: Value that will override the peaking amp code when in standard mode 2 when the peaking amp code override enable bit in the REE peaking amp diagnostics register (!) on page 299 is active. The value of this override field will ultimately drive the thermometer encoded value on the rxda_dfe_peaking_amp_gain signal driven to the analog.

Register Name:			REE peaking amp diagnostics register (!)
Register Mnemonic:			RX_REE_PEAK_DIAG
Register Address:			10_m_nnnn : 1010 : 00111
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Peaking amp code override enable**: Setting this bit to a 1'b1 will allow the peaking amp code to be overridden by the peaking amp code override value fields in the REE peaking amp code override 0 register (!) on page 298 and REE peaking amp code override 1 register (!) on page 299.
14	R/W	1'h0	**Voter override neg** : Writing a 1'b1 in this register bit will force the peaking amp voter function to activate the voter neg signal for a single clock cycle. **Note : This bit must be cleared after each time it is enabled, in order to perform follow on voter override functions with this bit.** **Note : It is not valid to activate both the voter override neg and voter override pos at the same time.**
13	R/W	1'h0	**Voter override pos** : Writing a 1'b1 in this register bit will force the peaking amp voter function to activate the voter pos signal for a single clock cycle. **Note : This bit must be cleared after each time it is enabled, in order to perform follow on voter override functions with this bit.** **Note : It is not valid to activate both the voter override neg and voter override pos at the same time.**
12	R/W	1'h0	**Voter override enable** : Setting this bit to a 1'b1 will allow only the voter override pos and voter override neg bits in this register to control the voter in the peaking amp. **Note : This function is intended to be for diagnostic and verification purposes only.** **Note : Both the voter override neg and voter override pos bits in this register must be set to 1'b0 when this bit is initially set to 1'b1 when enabling this function.**
11:8	R/O	4'h0	Reserved
7:6	R/O	2'h0	Reserved
5:0	R/O	6'h00	**Current peaking amp integrator accumulator**: Current value of the tap integrator accumulator, without the unused sign bit. **Note: The reset value for this field is with the transceiver data path power island switched off. In cases where this power island is switched on, the reset value be 6'b100000**

Register Name:			REE attenuation control register (!)
Register Mnemonic:			RX_REE_ATTEN_CTRL
Register Address:			10_m_nnnn : 1010 : 01000
Bits	**Type**	**Reset**	**Description**
15:5	R/O	11'h0	Reserved
4:0	R/W	5'h05	**Receiver DFE attenuation maximum value**: The maximum value the rxda_dfe_attenuation_bin will increase to. Note: The value on this field is not a twos complement value (There is no sign bit and it is always a positive number).

Register Name:			REE attenuation threshold register (!)
Register Mnemonic:			RX_REE_ATTEN_THR
Register Address:			10_m_nnnn : 1010 : 01001
Bits	**Type**	**Reset**	**Description**
15:13	R/O	3'h0	Reserved
12:8	R/W	5'h0C	**Attenuation high threshold value**: High threshold value to compare against the VGA gain accumulator value. Note: The value on this field is not a twos complement value (There is no sign bit and it is always a positive number).
7:5	R/O	3'h0	Reserved
4:0	R/W	5'h02	**Attenuation low threshold value**: Low threshold value to compare against the VGA gain accumulator value. Note: The value on this field is not a twos complement value (There is no sign bit and it is always a positive number).

Register Name:			REE attenuation counter register (!)
Register Mnemonic:			RX_REE_ATTEN_CNT
Register Address:			10_m_nnnn : 1010 : 01010
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h2100	**Attenuation counter max**: Value used to specify the maximum number of consecutive words above or below the specified thresholds which will result in triggering an increase or decrease in the rxda_dfe_attenuation_bin signal.

Register Name:			REE attenuation override register (!)
Register Mnemonic:			RX_REE_ATTEN_OVRD
Register Address:			10_m_nnnn : 1010 : 01011
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8	R/W	1'h0	**Attenuation override enable**: Setting this bit to a 1'b1 will allow the rxda_dfe_attenuation_bin signal to be overridden by the attenuation override value signal in this register.
7:5	R/O	3'h0	Reserved
4:0	R/W	5'h00	**Attenuation override value**: When enabled by the attenuation override enable bit in this register, this value will override the current attenuation value on the rxda_dfe_attenuation_bin output pin, and also force a corresponding change on the rxda_dfe_attenuation_therm pin. The following specifies the values that can be written to this register. • 6'b000000: Min • 6'b000001 • 6'b000011 • 6'b000111 • 6'b001111 • 6'b011111 • 6'b111111: Max

Register Name:			REE attenuation diagnostics register (!)
Register Mnemonic:			RX_REE_ATTEN_DIAG
Register Address:			10_m_nnnn : 1010 : 01100
Bits	**Type**	**Reset**	**Description**
15:5	R/O	11'h0	Reserved
4:0	R/O	5'h00	**Current attenuation value**: Current value of the attenuation.

Register Name:			REE tap 1 control register (!)
Register Mnemonic:			RX_REE_TAP1_CTRL
Register Address:			10_m_nnnn : 1010 : 10000
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11	R/W	1'h0	**Tap coefficient combinational logic zero crossing enable**: • 1'b0: Zero crossing combinational logic input not enabled. • 1'b1: Zero crossing combinational logic input enabled.
10	R/W	1'h1	**Tap coefficient combinational logic non zero crossing enable**: • 1'b0: non zero crossing combinational logic input not enabled. • 1'b1: non zero crossing combinational logic input enabled.
9	R/W	1'h0	**Tap coefficient combinational logic bit 0 only enable**: • 1'b0: All enabled combinational logic input modules will be used. • 1'b1: Only the enabled combinational logic input modules associated with bit 0 will be used.
8	R/W	1'h0	**Receiver DFE tap coefficient disable**: This bit disables the rxda_dfe_tap_coef output signal. • 1'b0 : rxda_dfe_tap_coef output enabled. • 1'b1 : rxda_dfe_tap_coef output disabled (all 0s).
7	R/O	1'h0	Reserved
6:4	R/W	3'h0	**Tap integrator accumulator scaler value**: Specifies the amount to scale the input to the tap integrator accumulator by. The following are the valid settings for this field: • 3'b000: x1 • 3'b001 - 3'b111: Reserved
3:0	R/W	4'h0	**Tap sigma delta accumulator scaler value**: Specifies the amount to scale the input to the tap sigma delta accumulator by. The following are the valid settings for this field: • 4'b0000: x1 • 4'b0001: x2 • 4'b0010: x4 • 4'b0011: x8 • 4'b0100: x16 • 4'b0101: x 32 • 4'b0110: x 64 • 4'b0111: x128 • 4'b1000: x256 • 4'b1001: x512 • 4'b1010: x1024 • 4'b1011: x2048 • 4'b1100: x 4096 • 4'b1101 x 8192 • 4'b1110 x 16384 • 4'b1111 x 32768

Register Name:			REE tap 1 override register (!)
Register Mnemonic:			RX_REE_TAP1_OVRD
Register Address:			10_m_nnnn : 1010 : 10001
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7	R/W	1'h0	**Tap override enable**: Setting this bit to a 1'b1 will enable the tap override field in this register to override the tap integrator accumulator functions.
6	R/O	1'h0	Reserved
5:0	R/W	6'h00	**Tap override value**: When the tap override enable bit in this register is active, the value in this field will override the integrator accumulator value, as well as the input to the binary to thermometer encoder.

Register Name:			REE tap 1 diagnostics register (!)
Register Mnemonic:			RX_REE_TAP1_DIAG
Register Address:			10_m_nnnn : 1010 : 10010
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14	R/W	1'h0	**Voter override neg** : Writing a 1'b1 in this register bit will force the tap voter function to activate the voter neg signal for a single clock cycle. **Note : This bit must be cleared after each time it is enabled, in order to perform follow on voter override functions with this bit.** **Note : It is not valid to activate both the voter override neg and voter override pos at the same time.**
13	R/W	1'h0	**Voter override pos** : Writing a 1'b1 in this register bit will force the tap voter function to activate the voter pos signal for a single clock cycle. **Note : This bit must be cleared after each time it is enabled, in order to perform follow on voter override functions with this bit.** **Note : It is not valid to activate both the voter override neg and voter override pos at the same time.**
12	R/W	1'h0	**Voter override enable** : Setting this bit to a 1'b1 will allow only the voter override pos and voter override neg bits in this register to control the voter in the tap. **Note : This function is intended to be for diagnostic and verification purposes only.** **Note : Both the voter override neg and voter override pos bits in this register must be set to 1'b0 when this bit is initially set to 1'b1 when enabling this function.**
11:6	R/O	6'h0	Reserved
5:0	R/O	6'h00	**Current tap integrator accumulator**: Current value of the tap integrator accumulator.

Register Name:			REE tap 2 control register (!)
Register Mnemonic:			RX_REE_TAP2_CTRL
Register Address:			10_m_nnnn : 1010 : 10100
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11	R/W	1'h0	**Tap coefficient combinational logic zero crossing enable**: • 1'b0: Zero crossing combinational logic input not enabled. • 1'b1: Zero crossing combinational logic input enabled. (This mode is currently not supported)
10	R/W	1'h1	**Tap coefficient combinational logic non zero crossing enable**: • 1'b0: non zero crossing combinational logic input not enabled. • 1'b1: non zero crossing combinational logic input enabled.
9	R/W	1'h0	**Tap coefficient combinational logic bit 0 only enable**: • 1'b0: All enabled combinational logic input modules will be used. • 1'b1: Only the enabled combinational logic input modules associated with bit 0 will be used.
8	R/W	1'h0	**Receiver DFE tap coefficient disable**: This bit disables the rxda_dfe_tap_coef output signal. • 1'b0 : rxda_dfe_tap_coef output enabled. • 1'b1 : rxda_dfe_tap_coef output disabled (all 0s).
7	R/O	1'h0	Reserved
6:4	R/W	3'h0	**Tap integrator accumulator scaler value**: Specifies the amount to scale the input to the tap integrator accumulator by. The following are the valid settings for this field: • 3'b000: x1 • 3'b001 - 3'b111: Reserved
3:0	R/W	4'h0	**Tap sigma delta accumulator scaler value**: Specifies the amount to scale the input to the tap sigma delta accumulator by. The following are the valid settings for this field: • 4'b0000: x1 • 4'b0001: x2 • 4'b0010: x4 • 4'b0011: x8 • 4'b0100: x16 • 4'b0101: x 32 • 4'b0110: x 64 • 4'b0111: x128 • 4'b1000: x256 • 4'b1001: x512 • 4'b1010: x1024 • 4'b1011: x2048 • 4'b1100: x 4096 • 4'b1101 x 8192 • 4'b1110 x 16384 • 4'b1111 x 32768

Register Name:			REE tap 2 override register (!)
Register Mnemonic:			RX_REE_TAP2_OVRD
Register Address:			10_m_nnnn : 1010 : 10101
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7	R/W	1'h0	**Tap override enable**: Setting this bit to a 1'b1 will enable the tap override field in this register to override the tap integrator accumulator functions.
6	R/O	1'h0	Reserved
5:0	R/W	6'h00	**Tap override value**: When the tap override enable bit in this register is active, the value in this field will override the integrator accumulator value, as well as the input to the binary to thermometer encoder.

Register Name:			REE tap 2 diagnostics register (!)
Register Mnemonic:			RX_REE_TAP2_DIAG
Register Address:			10_m_nnnn : 1010 : 10110
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14	R/W	1'h0	**Voter override neg** : Writing a 1'b1 in this register bit will force the tap voter function to activate the voter neg signal for a single clock cycle. **Note : This bit must be cleared after each time it is enabled, in order to perform follow on voter override functions with this bit.** **Note : It is not valid to activate both the voter override neg and voter override pos at the same time.**
13	R/W	1'h0	**Voter override pos** : Writing a 1'b1 in this register bit will force the tap voter function to activate the voter pos signal for a single clock cycle. **Note : This bit must be cleared after each time it is enabled, in order to perform follow on voter override functions with this bit.** **Note : It is not valid to activate both the voter override neg and voter override pos at the same time.**
12	R/W	1'h0	**Voter override enable** : Setting this bit to a 1'b1 will allow only the voter override pos and voter override neg bits in this register to control the voter in the tap. **Note : This function is intended to be for diagnostic and verification purposes only.** **Note : Both the voter override neg and voter override pos bits in this register must be set to 1'b0 when this bit is initially set to 1'b1 when enabling this function.**
11:6	R/O	6'h0	Reserved
5:0	R/O	6'h00	**Current tap integrator accumulator**: Current value of the tap integrator accumulator.

Register Name:			REE tap 3 control register (!)
Register Mnemonic:			RX_REE_TAP3_CTRL
Register Address:			10_m_nnnn : 1010 : 11000
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11	R/W	1'h0	**Tap coefficient combinational logic zero crossing enable**: • 1'b0: Zero crossing combinational logic input not enabled. • 1'b1: Zero crossing combinational logic input enabled. (This mode is currently not supported)
10	R/W	1'h1	**Tap coefficient combinational logic non zero crossing enable**: • 1'b0: non zero crossing combinational logic input not enabled. • 1'b1: non zero crossing combinational logic input enabled.
9	R/W	1'h0	**Tap coefficient combinational logic bit 0 only enable**: • 1'b0: All enabled combinational logic input modules will be used. • 1'b1: Only the enabled combinational logic input modules associated with bit 0 will be used.
8	R/W	1'h0	**Receiver DFE tap coefficient disable**: This bit disables the rxda_dfe_tap_coef output signal. • 1'b0 : rxda_dfe_tap_coef output enabled. • 1'b1 : rxda_dfe_tap_coef output disabled (all 0s).
7	R/O	1'h0	Reserved
6:4	R/W	3'h0	**Tap integrator accumulator scaler value**: Specifies the amount to scale the input to the tap integrator accumulator by. The following are the valid settings for this field: • 3'b000: x1 • 3'b001 - 3'b111: Reserved
3:0	R/W	4'h0	**Tap sigma delta accumulator scaler value**: Specifies the amount to scale the input to the tap sigma delta accumulator by. The following are the valid settings for this field: • 4'b0000: x1 • 4'b0001: x2 • 4'b0010: x4 • 4'b0011: x8 • 4'b0100: x16 • 4'b0101: x 32 • 4'b0110: x 64 • 4'b0111: x128 • 4'b1000: x256 • 4'b1001: x512 • 4'b1010: x1024 • 4'b1011: x2048 • 4'b1100: x 4096 • 4'b1101 x 8192 • 4'b1110 x 16384 • 4'b1111 x 32768

Register Name:			REE tap 3 override register (!)
Register Mnemonic:			RX_REE_TAP3_OVRD
Register Address:			10_m_nnnn : 1010 : 11001
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7	R/W	1'h0	**Tap override enable**: Setting this bit to a 1'b1 will enable the tap override field in this register to override the tap integrator accumulator functions.
6	R/O	1'h0	Reserved
5:0	R/W	6'h00	**Tap override value**: When the tap override enable bit in this register is active, the value in this field will override the integrator accumulator value, as well as the input to the binary to thermometer encoder.

Register Name:			REE tap 3 diagnostics register (!)
Register Mnemonic:			RX_REE_TAP3_DIAG
Register Address:			10_m_nnnn : 1010 : 11010
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14	R/W	1'h0	**Voter override neg** : Writing a 1'b1 in this register bit will force the tap voter function to activate the voter neg signal for a single clock cycle. **Note : This bit must be cleared after each time it is enabled, in order to perform follow on voter override functions with this bit.** **Note : It is not valid to activate both the voter override neg and voter override pos at the same time.**
13	R/W	1'h0	**Voter override pos** : Writing a 1'b1 in this register bit will force the tap voter function to activate the voter pos signal for a single clock cycle. **Note : This bit must be cleared after each time it is enabled, in order to perform follow on voter override functions with this bit.** **Note : It is not valid to activate both the voter override neg and voter override pos at the same time.**
12	R/W	1'h0	**Voter override enable** : Setting this bit to a 1'b1 will allow only the voter override pos and voter override neg bits in this register to control the voter in the tap. **Note : This function is intended to be for diagnostic and verification purposes only.** **Note : Both the voter override neg and voter override pos bits in this register must be set to 1'b0 when this bit is initially set to 1'b1 when enabling this function.**
11:6	R/O	6'h0	Reserved
5:0	R/O	6'h00	**Current tap integrator accumulator**: Current value of the tap integrator accumulator.

Register Name:			REE low frequency equalizer control register (!)
Register Mnemonic:			RX_REE_LFEQ_CTRL
Register Address:			10_m_nnnn : 1010 : 11100
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8	R/W	1'h0	**Receiver DFE coefficient disable**: This bit disables the rxda_dfe_coef output signal. • 1'b0 : rxda_dfe_coef output enabled. • 1'b1 : rxda_dfe_coef output disabled (all 0s).
7	R/O	1'h0	Reserved
6:4	R/W	3'h0	**Integrator accumulator scaler value**: Specifies the amount to scale the input to the integrator accumulator by. The following are the valid settings for this field: • 3'b000: x1 • 3'b001: x2 • 3'b010: x4 • 3'b011: x8 • 3'b100: x16 • 3'b101 - 3'b111: Reserved
3:0	R/W	4'h0	**Sigma delta accumulator scaler value**: Specifies the amount to scale the input to the sigma delta accumulator by. The following are the valid settings for this field: • 4'b0000: x1 • 4'b0001: x2 • 4'b0010: x4 • 4'b0011: x8 • 4'b0100: x16 • 4'b0101: x 32 • 4'b0110: x 64 • 4'b0111: x128 • 4'b1000: x256 • 4'b1001: x512 • 4'b1010: x1024 • 4'b1011: x2048 • 4'b1100: x 4096 • 4'b1101 x 8192 • 4'b1110 x 16384 • 4'b1111 x 32768

Register Name:			REE low frequency equalizer override register (!)
Register Mnemonic:			RX_REE_LFEQ_OVRD
Register Address:			10_m_nnnn : 1010 : 11101
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7	R/W	1'h0	**Override enable**: Setting this bit to a 1'b1 will enable the override field in this register to override the integrator accumulator functions.
6	R/O	1'h0	Reserved
5:0	R/W	6'h00	**Override value**: When the override enable bit in this register is active, the value in this field will override the integrator accumulator value, as well as the input to the binary to thermometer encoder.

Register Name:			REE low frequency equalizer diagnostics register (!)
Register Mnemonic:			RX_REE_LFEQ_DIAG
Register Address:			10_m_nnnn : 1010 : 11110
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14	R/W	1'h0	**Voter override neg** : Writing a 1'b1 in this register bit will force the voter function to activate the voter neg signal for a single clock cycle. **Note : This bit must be cleared after each time it is enabled, in order to perform follow on voter override functions with this bit.** **Note : It is not valid to activate both the voter override neg and voter override pos at the same time.**
13	R/W	1'h0	**Voter override pos** : Writing a 1'b1 in this register bit will force the voter function to activate the voter pos signal for a single cycle. **Note : This bit must be cleared after each time it is enabled, in order to perform follow on voter override functions with this bit.** **Note : It is not valid to activate both the voter override neg and voter override pos at the same time.**
12	R/W	1'h0	**Voter override enable** : Setting this bit to a 1'b1 will allow only the voter override pos and voter override neg bits in this register to control the voter. **Note : This function is intended to be for diagnostic and verification purposes only.** **Note : Both the voter override neg and voter override pos bits in this register must be set to 1'b0 when this bit is initially set to 1'b1 when enabling this function.**
11:6	R/O	6'h0	Reserved
5:0	R/O	6'h00	**Current integrator accumulator**: Current value of the integrator accumulator.

Register Name:			REE VGA gain control register (!)
Register Mnemonic:			RX_REE_VGA_GAIN_CTRL
Register Address:			10_m_nnnn : 1011 : 00000
Bits	**Type**	**Reset**	**Description**
15:13	R/O	3'h0	Reserved
12:8	R/W	5'h17	**VGA gain max**: Specifies the maximum value of the VGA gain integrator accumulator, and therefore also the maximum number of bits in the rxda_dfe_vga_gain thermometer code that will be set.
7	R/O	1'h0	Reserved
6:4	R/W	3'h0	**VGA gain integrator accumulator scaler value**: Specifies the amount to scale the input to the VGA gain integrator accumulator by. The following are the valid settings for this field: • 3'b000: x1 • 3'b001: x2 • 3'b010: x4 • 3'b011: x8 • 3'b100: x16 • 3'b101 - 3'b111: Reserved
3:0	R/W	4'h1	**VGA gain sigma delta accumulator scaler value**: Specifies the amount to scale the input to the VGA gain sigma delta accumulator by. The following are the valid settings for this field: • 4'b0000: x1 • 4'b0001: x2 • 4'b0010: x4 • 4'b0011: x8 • 4'b0100: x16 • 4'b0101: x 32 • 4'b0110: x 64 • 4'b0111: x128 • 4'b1000: x256 • 4'b1001: x512 • 4'b1010: x1024 • 4'b1011: x2048 • 4'b1100: x 4096 • 4'b1101 x 8192 • 4'b1110 x 16384 • 4'b1111 x 32768

Register Name:			REE VGA gain override register (!)
Register Mnemonic:			RX_REE_VGA_GAIN_OVRD
Register Address:			10_m_nnnn : 1011 : 00001
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**VGA gain target adjust override enable**: Setting this bit to a 1'b1 will enable the VGA gain target adjust override field in this register to override the VGA gain target adjust accumulator functions.
14:13	R/O	2'h0	Reserved
12:8	R/W	5'h00	**VGA gain target adjust override value**: When the VGA gain target adjust override enable bit in this register is active, the value in this field will override the accumulator value used to drive the vga_gain_tgt_adj signal.
7	R/W	1'h0	**VGA gain override enable**: Setting this bit to a 1'b1 will enable the VGA gain override field in this register to override the VGA gain integrator accumulator functions.
6:5	R/O	2'h0	Reserved
4:0	R/W	5'h00	**VGA gain override value**: When the VGA gain override enable bit in this register is active, the value in this field will override the integrator accumulator value, as well as the input to the binary to thermometer encoder.

Register Name:			REE VGA gain diagnostics register (!)
Register Mnemonic:			RX_REE_VGA_GAIN_DIAG
Register Address:			10_m_nnnn : 1011 : 00010
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14	R/W	1'h0	**Voter override neg** : Writing a 1'b1 in this register bit will force the VGA gain voter function to activate the voter neg signal for a single clock cycle. **Note : This bit must be cleared after each time it is enabled, in order to perform follow on voter override functions with this bit.** **Note : It is not valid to activate both the voter override neg and voter override pos at the same time.**
13	R/W	1'h0	**Voter override pos** : Writing a 1'b1 in this register bit will force the VGA gain voter function to activate the voter pos signal for a single clock cycle. **Note : This bit must be cleared after each time it is enabled, in order to perform follow on voter override functions with this bit.** **Note : It is not valid to activate both the voter override neg and voter override pos at the same time.**
12	R/W	1'h0	**Voter override enable** : Setting this bit to a 1'b1 will allow only the voter override pos and voter override neg bits in this register to control the voter in the VGA gain. **Note : This function is intended to be for diagnostic and verification purposes only.** **Note : Both the voter override neg and voter override pos bits in this register must be set to 1'b0 when this bit is initially set to 1'b1 when enabling this function.**
11:6	R/O	6'h0	Reserved
5:0	R/O	6'h00	**Current VGA gain integrator accumulator**: Current value of the VGA gain integrator accumulator. **Note: The reset value for this field is with the transceiver data path power island switched off. In cases where this power island is switched on, the reset value be 6'b001010**

Register Name:			REE VGA gain target adjust diagnostics register (!)
Register Mnemonic:			RX_REE_VGA_GAIN_TGT_DIAG
Register Address:			10_m_nnnn : 1011 : 00011
Bits	**Type**	**Reset**	**Description**
15:5	R/O	11'h0	Reserved
4:0	R/O	5'h00	**VGA gain target adjust value**: Amount to adjust the sampler reference DAC by.

Register Name:			REE offset correction control register (!)
Register Mnemonic:			RX_REE_OFF_COR_CTRL
Register Address:			10_m_nnnn : 1011 : 00100
Bits	**Type**	**Reset**	**Description**
15:7	R/O	9'h0	Reserved
6:4	R/W	3'h0	**Offset correction integrator accumulator scaler value**: Specifies the amount to scale the input to the offset correction integrator accumulator by. The following are the valid settings for this field: • 3'b000: x1 • 3'b001: x2 • 3'b010: x4 • 3'b011: x8 • 3'b100: x16 • 3'b101 - 3'b111: Reserved
3:0	R/W	4'h1	**Offset correction sigma delta accumulator scaler value**: Specifies the amount to scale the input to the offset correction sigma delta accumulator by. The following are the valid settings for this field: • 4'b0000: x1 • 4'b0001: x2 • 4'b0010: x4 • 4'b0011: x8 • 4'b0100: x16 • 4'b0101: x 32 • 4'b0110: x 64 • 4'b0111: x128 • 4'b1000: x256 • 4'b1001: x512 • 4'b1010: x1024 • 4'b1011: x2048 • 4'b1100: x 4096 • 4'b1101 x 8192 • 4'b1110 x 16384 • 4'b1111 x 32768

Register Name:			REE offset correction override register (!)
Register Mnemonic:			RX_REE_OFF_COR_OVRD
Register Address:			10_m_nnnn : 1011 : 00101
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7	R/W	1'h0	**Offset correction override enable**: Setting this bit to a 1'b1 will enable the offset correction override field in this register to override the offset correction integrator accumulator functions.
6	R/O	1'h0	Reserved
5:0	R/W	6'h00	**Offset correction override value**: When the offset correction override enable bit in this register is active, the value in this field will override the integrator accumulator value, as well as the input to the binary to thermometer encoder.

Register Name:			REE offset correction diagnostics register (!)
Register Mnemonic:			RX_REE_OFF_COR_DIAG
Register Address:			10_m_nnnn : 1011 : 00110
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14	R/W	1'h0	**Voter override neg** : Writing a 1'b1 in this register bit will force the offset correction voter function to activate the voter neg signal for a single clock cycle. **Note : This bit must be cleared after each time it is enabled, in order to perform follow on voter override functions with this bit.** **Note : It is not valid to activate both the voter override neg and voter override pos at the same time.**
13	R/W	1'h0	**Voter override pos** : Writing a 1'b1 in this register bit will force the offset correction voter function to activate the voter pos signal for a single clock cycle. **Note : This bit must be cleared after each time it is enabled, in order to perform follow on voter override functions with this bit.** **Note : It is not valid to activate both the voter override neg and voter override pos at the same time.**
12	R/W	1'h0	**Voter override enable** : Setting this bit to a 1'b1 will allow only the voter override pos and voter override neg bits in this register to control the voter in the offset correction. **Note : This function is intended to be for diagnostic and verification purposes only.** **Note : Both the voter override neg and voter override pos bits in this register must be set to 1'b0 when this bit is initially set to 1'b1 when enabling this function.**
11:6	R/O	6'h0	Reserved
5:0	R/O	6'h00	**Current offset correction integrator accumulator**: Current value of the offset correction integrator accumulator.

Register Name:			REE short channel correction valid word counter register (!)
Register Mnemonic:			RX_REE_SC_COR_WCNT
Register Address:			10_m_nnnn : 1011 : 01000
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h0A	**Valid word counter start value** : Value used for the starting value when counting the number of valid words.

Register Name:			REE short channel correction threshold counter register (!)
Register Mnemonic:			RX_REE_SC_COR_TCNT
Register Address:			10_m_nnnn : 1011 : 01001
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h04	**Threshold counter start value** : Value used for the starting value when counting the number of bits that are below the error threshold.

Register Name:			REE adder configuration register (!)
Register Mnemonic:			RX_REE_ADDR_CFG
Register Address:			10_m_nnnn : 1011 : 10000
Bits	**Type**	**Reset**	**Description**
15:11	R/O	5'h0	Reserved
10	R/W	1'h0	**TX post cursor tap 3 adder enable**: Setting this bit to 1'b1, enables the results of tap 3 to be added to the TX post cursor controller input.
9	R/W	1'h0	**TX post cursor tap 2 adder enable**: Setting this bit to 1'b1, enables the results of tap 2 to be added to the TX post cursor controller input.
8	R/W	1'h1	**TX post cursor tap 1 adder enable**: Setting this bit to 1'b1, enables the results of tap 1 to be added to the TX post cursor controller input.
7:3	R/O	5'h0	Reserved
2	R/W	1'h0	**RX peaking tap 3 adder enable**: Setting this bit to 1'b1, enables the results of tap 3 to be added to the RX peaking amp gain input.
1	R/W	1'h0	**RX peaking tap 2 adder enable**: Setting this bit to 1'b1, enables the results of tap 2 to be added to the RX peaking amp gain input.
0	R/W	1'h1	**RX peaking tap 1 adder enable**: Setting this bit to 1'b1, enables the results of tap 1 to be added to the RX peaking amp gain input.

Register Name:			REE tap 1 clip control register (!)
Register Mnemonic:			RX_REE_TAP1_CLIP
Register Address:			10_m_nnnn : 1011 : 10001
Bits	**Type**	**Reset**	**Description**
15:11	R/O	5'h0	Reserved
10:8	R/W	3'h5	**VGA target gain adjust multiplier**: Controls how much to multiply the VGA target gain adjust by, when calculating the tap threshold. The following are valid values, and the corresponding multiplier values. • 3'b000 : x 0.00 • 3'b001 : x 0.25 • 3'b010 : x 0.50 • 3'b011 : x 0.75 • 3'b100 : x 1.00 • 3'b101 : x 1.25 • 3'b110 : x 1.50 • 3'b111 : x 1.75 **Note : By setting this field to 3'b000, the threshold value of the tap can be directly controlled by the threshold adjust field in this register.** **Note : The VGA gain adjust signal is always a negative number.**
7:5	R/O	3'h0	Reserved
4:0	R/W	5'h19	**Threshold adjust**: Controls how much the threshold can be adjusted by, after multiplying the VGA target gain adjust multiplier with the VGA target gain adjust. Note: This field is a positive number (not twos complement). **Note : This field must not be set to a value of 0.** **Note : The combination of the multiplier and threshold fields in this register controls the positive and negative thresholds for the taps, according to the following equation.** **Note: The logic associated with this limits the range of the magnitude to 0 to 31, before creating the positive and negative thresholds.** **+/- ((VGA target gain adjust * VGA target gain adjust multiplier) + Threshold adjust)**

Register Name:			REE taps 2 and 3 clip control register (!)
Register Mnemonic:			RX_REE_TAP2TON_CLIP
Register Address:			10_m_nnnn : 1011 : 10010
Bits	**Type**	**Reset**	**Description**
15:11	R/O	5'h0	Reserved
10:8	R/W	3'h5	**VGA target gain adjust multiplier**: Controls how much to multiply the VGA target gain adjust by, when calculating the tap threshold. The following are valid values, and the corresponding multiplier values. • 3'b000 : x 0.00 • 3'b001 : x 0.25 • 3'b010 : x 0.50 • 3'b011 : x 0.75 • 3'b100 : x 1.00 • 3'b101 : x 1.25 • 3'b110 : x 1.50 • 3'b111 : x 1.75 Note : By setting this field to 3'b000, the threshold value of the tap can be directly controlled by the threshold adjust field in this register.
7:5	R/O	3'h0	Reserved
4:0	R/W	5'h19	**Threshold adjust**: Controls how much the threshold can be adjusted by, after multiplying the VGA target gain adjust multiplier with the VGA target gain adjust. Note: This field is a positive number (not twos complement). **Note : This field must not be set to a value of 0.** **Note : The combination of the multiplier and threshold fields in this register controls the positive and negative thresholds for the taps, according to the following equation.** **Note: The logic associated with this limits the range of the magnitude to 0 to 31, before creating the positive and negative thresholds.** **+/- ((VGA target gain adjust * VGA target gain adjust multiplier) + Threshold adjust)**

Register Name:			REE control data mask register (!)
Register Mnemonic:			RX_REE_CTRL_DATA_MASK
Register Address:			10_m_nnnn : 1011 : 10011
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14	R/O	1'h1	**Ignore 1010 controller** Note: This is read only. This should never be disabled.
13	R/W	1'h0	**TX equalization evaluator**: When set to 1'b0, this REE component will be turned off when control data is being received, when either the rx_eq_training_data_valid signal going inactive, or a 1010 pattern is detected on the data signals by the REE.
12	R/W	1'h0	**TX post cursor control:** When set to 1'b0, this REE component will be turned off when control data is being received, when either the rx_eq_training_data_valid signal going inactive, or a 1010 pattern is detected on the data signals by the REE.
11	R/W	1'h0	**TX pre cursor control:** When set to 1'b0, this REE component will be turned off when control data is being received, when either the rx_eq_training_data_valid signal going inactive, or a 1010 pattern is detected on the data signals by the REE.
10	R/W	1'h0	**Short channel correction:** When set to 1'b0, this REE component will be turned off when control data is being received, when either the rx_eq_training_data_valid signal going inactive, or a 1010 pattern is detected on the data signals by the REE.
9	R/W	1'h0	**RX attenuation:** When set to 1'b0, this REE component will be turned off when control data is being received, when either the rx_eq_training_data_valid signal going inactive, or a 1010 pattern is detected on the data signals by the REE.
8	R/W	1'h0	**RX VGA gain:** When set to 1'b0, this REE component will be turned off when control data is being received, when either the rx_eq_training_data_valid signal going inactive, or a 1010 pattern is detected on the data signals by the REE.
7	R/W	1'h0	**RX offset correction coefficient:** When set to 1'b0, this REE component will be turned off when control data is being received, when either the rx_eq_training_data_valid signal going inactive, or a 1010 pattern is detected on the data signals by the REE.
6	R/W	1'h0	**RX peaking amp gain:** When set to 1'b0, this REE component will be turned off when control data is being received, when either the rx_eq_training_data_valid signal going inactive, or a 1010 pattern is detected on the data signals by the REE.
5	R/W	1'h0	**RX low frequency equalizer adaptive control:** When set to 1'b0, this REE component will be turned off when control data is being received, when either the rx_eq_training_data_valid signal going inactive, or a 1010 pattern is detected on the data signals by the REE.
4	R/O	1'h0	Reserved
3	R/O	1'h0	Reserved
2	R/W	1'h0	**RX tap 3:** When set to 1'b0, this REE component will be turned off when control data is being received, when either the rx_eq_training_data_valid signal going inactive, or a 1010 pattern is detected on the data signals by the REE.
1	R/W	1'h0	**RX tap 2:** When set to 1'b0, this REE component will be turned off when control data is being received, when either the rx_eq_training_data_valid signal going inactive, or a 1010 pattern is detected on the data signals by the REE.
0	R/W	1'h0	**RX tap 1:** When set to 1'b0, this REE component will be turned off when control data is being received, when either the rx_eq_training_data_valid signal going inactive, or a 1010 pattern is detected on the data signals by the REE.

Register Name:			REE coefficient FIFO diagnostic register (!)
Register Mnemonic:			RX_REE_FIFO_DIAG
Register Address:			10_m_nnnn : 1011 : 10100
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	**FIFO underflow** : Indicates when a FIFO underflow condition has occurred.
14	R/O	1'h1	**FIFO empty** :Indicates that the FIFO is empty.
13	R/W	1'h0	**FIFO output dequeue** : Writing a 1'b1 to this bit will dequeue the current values from the output of the FIFO. Once this register bit is written to a 1'b1, it must be written to a 1'b0 before it is used again to dequeue data.
12	R/W	1'h0	**FIFO output override enable** : Enables the FIFO output override functions. This bit must be set to 1'b1 before enabling the transmitter REE functions. When set, FIFO output dequeues can be performed from this register. When set, the TX equalization evaluator will not be able to dequeue data from the FIFO.
11	R/O	1'h0	**FIFO output data pre cursor increment** : Current value on the pre cursor increment bit on the FIFO output.
10	R/O	1'h0	**FIFO output data pre cursor decrement** : Current value on the pre cursor decrement bit on the FIFO output.
9	R/O	1'h0	**FIFO output data post cursor increment** : Current value on the post cursor increment bit on the FIFO output.
8	R/O	1'h0	**FIFO output data post cursor decrement** : Current value on the post cursor decrement bit on the FIFO output.
7	R/O	1'h0	**FIFO overflow** :Indicates when a FIFO overflow condition has occurred.
6	R/O	1'h0	**FIFO full** :Indicates that the FIFO is full.
5	R/W	1'h0	**FIFO input enqueue** : Writing a 1'b1 to this bit will enqueue the values of the FIFO input data bits in this register to their respective FIFO bits. Once this register bit is written to a 1'b1, it must be written to a 1'b0 before it is used again to dequeue data.
4	R/W	1'h0	**FIFO input override enable** : Enables the FIFO input override functions. This bit must be set to 1'b1 before enabling the transmitter REE functions. When set, FIFO input enqueues can be performed from this register. When set, the TX pre cursor control and TX post cursor control will not be able to enqueue data to the FIFO.
3	R/W	1'h0	**FIFO input data pre cursor increment** : Pre cursor increment value that will be enqueued to the FIFO by the FIFO input enqueue bit in this register.
2	R/W	1'h0	**FIFO input data pre cursor decrement** : Pre cursor decrement value that will be enqueued to the FIFO by the FIFO input enqueue bit in this register.
1	R/W	1'h0	**FIFO input data post cursor increment** : Post cursor increment value that will be enqueued to the FIFO by the FIFO input enqueue bit in this register.
0	R/W	1'h0	**FIFO input data post cursor decrement** : Post cursor decrement value that will be enqueued to the FIFO by the FIFO input enqueue bit in this register.

Register Name:			REE diagnostic control register (!)
Register Mnemonic:			RX_REE_DIAG_CTRL
Register Address:			10_m_nnnn : 1011 : 10101
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7	R/W	1'h0	**Analog tap disable**: When this bit is set to 1'b1, the rxda_dfe_tap_1_coef, rxda_dfe_tap_2_coef, and rxda_dfe_tap_3_coef signals being driven to the analog will be forced to all 0s.
6	R/W	1'h1	**Hold periodic equalization while RX idle**: When this bit is set to 1'b1, a detection of electrical idle on the receiver (rx_signal_detect is set to 1'b0), will disable the periodic REE general control state machine, and as a result freeze the current state of the parts of the REE that are controlled by it. When a high speed signal is later detected, the REE general control state machine will be enabled again.
5	R/W	1'h1	**Hold general control state machine 2 equalization while RX idle**: When this bit is set to 1'b1, a detection of electrical idle on the receiver (rx_signal_detect is set to 1'b0), will disable the REE general control state machine 2, and as a result freeze the current state of the parts of the REE that are controlled by it. When a high speed signal is later detected, the REE general control state machine will be enabled again.
4	R/W	1'h1	**Hold general control state machine 1 equalization while RX idle**: When this bit is set to 1'b1, a detection of electrical idle on the receiver (rx_signal_detect is set to 1'b0), will disable the REE general control state machine 1, and as a result freeze the current state of the parts of the REE that are controlled by it. When a high speed signal is later detected, the REE general control state machine will be enabled again.
3:2	R/O	2'h0	Reserved
1	R/W	1'h1	**Enable REE control clock on** : Enables the REE control clock. • 1'b0: Disabled • 1'b1: Enabled
0	R/W	1'h0	**Force REE function clock on** : When active, the REE function clock gate will allow the clock to run. Note: Because of how the REE reset functions are implemented, the REE controller clock must be enabled in order for REE function clock to be enabled. Therefore, if all of the REE control state machines are disabled, the Force REE controller lock on bit in this register must also be set, for this bit to take effect.

Register Name:			REE TX equalization evaluator control register (!)
Register Mnemonic:			RX_REE_TXEQEVAL_CTRL
Register Address:			10_m_nnnn : 1011 : 10110
Bits	Type	Reset	Description
15:2	R/O	14'h0	Reserved
1	R/W	1'h0	**TX equalization evaluation counter reset on gen mode change**: Controls if the incremental evaluation counter will be reset to its starting value when changing gen modes. When enabled, the reset will take place on the first equalization evaluation after a gen mode change has taken place. • 1'b0 : Gen mode change reset disabled. 1'b1 : Gen mode change reset enabled.
0	R/W	1'h0	**TX main coefficient direction change control**: This bit controls the function of the main coefficient direction change. • 1'b0 : The main coefficient will always return a value of no change (2'b00), no matter what the pre and post cursor coefficient direction changes are. 1'b1 : The main coefficient will generate a return value as a function of the pre and post cursor coefficient direction change. If an increment request exists on the pre or post cursor coefficient direction change, a decrement request will be returned for the main coefficient. Similarly, if a decrement request exists on the pre or post cursor coefficient direction change, an increment request will be returned for the main coefficient.

Register Name:			REE control state machine gen mode control register 1 (!)
Register Mnemonic:			RX_REE_SMGM_CTRL1
Register Address:			10_m_nnnn : 1011 : 10111
Bits	Type	Reset	Description
15:12	R/O	4'h0	Reserved
11	R/W	1'h0	**REE Periodic general control state machine enable standard mode 3**: This bit will control if the REE periodic general control state machine will run, when xcvr_standard_mode is set to 2'b11. 1'b0 : Disabled 1'b1 : Enabled
10	R/W	1'h1	**REE Periodic general control state machine enable standard mode 2**: This bit will control if the REE periodic general control state machine will run, when xcvr_standard_mode is set to 2'b10. 1'b0 : Disabled 1'b1 : Enabled
9	R/W	1'h1	**REE Periodic general control state machine enable standard mode 1**: This bit will control if the REE periodic general control state machine will run, when xcvr_standard_mode is set to 2'b01. 1'b0 : Disabled 1'b1 : Enabled
8	R/W	1'h0	**REE Periodic general control state machine enable standard mode 0**: This bit will control if the REE periodic general control state machine will run, when xcvr_standard_mode is set to 2'b00. 1'b0 : Disabled 1'b1 : Enabled
7	R/W	1'h1	**REE general control state machine 2 enable standard mode 3**: This bit will control if the REE general control state machine 2 will run, when xcvr_standard_mode is set to 2'b11. 1'b0 : Disabled 1'b1 : Enabled
6	R/W	1'h1	**REE general control state machine 2 enable standard mode 2**: This bit will control if the REE general control state machine 2 will run, when xcvr_standard_mode is set to 2'b10. 1'b0 : Disabled 1'b1 : Enabled
5	R/W	1'h1	**REE general control state machine 2 enable standard mode 1**: This bit will control if the REE general control state machine 2 will run, when xcvr_standard_mode is set to 2'b01. 1'b0 : Disabled 1'b1 : Enabled
4	R/W	1'h1	**REE general control state machine 2 enable standard mode 0**: This bit will control if the REE general control state machine 2 will run, when xcvr_standard_mode is set to 2'b00. 1'b0 : Disabled 1'b1 : Enabled
3	R/W	1'h0	**REE general control state machine 1 enable standard mode 3**: This bit will control if the REE general control state machine 1 will run, when xcvr_standard_mode is set to 2'b11. 1'b0 : Disabled 1'b1 : Enabled
2	R/W	1'h1	**REE general control state machine 1 enable standard mode 2**: This bit will control if the REE general control state machine 1 will run, when xcvr_standard_mode is set to 2'b10. 1'b0 : Disabled 1'b1 : Enabled
1	R/W	1'h1	**REE general control state machine 1 enable standard mode 1**: This bit will control if the REE general control state machine 1 will run, when xcvr_standard_mode is set to 2'b01. 1'b0 : Disabled 1'b1 : Enabled
0	R/W	1'h0	**REE general control state machine 1 enable standard mode 0**: This bit will control if the REE general control state machine 1 will run, when xcvr_standard_mode is set to 2'b00. 1'b0 : Disabled 1'b1 : Enabled

Register Name:			REE control state machine gen mode control register 2 (!)
Register Mnemonic:			RX_REE_SMGM_CTRL2
Register Address:			10_m_nnnn : 1011 : 11000
Bits	Type	Reset	Description
15	R/W	1'h0	**REE PCIe Gen 3 TX equalization state machine E path en standard mode 3**: This bit controls if the REE PCIE Gen 3 TX equalization state machine will enable the analog E path when xcvr_standard_mode is set to 2'b11. 1'b0 : Disabled 1'b1 : Enabled
14	R/W	1'h1	**REE PCIe Gen 3 TX equalization state machine E path en standard mode 2**: This bit controls if the REE PCIE Gen 3 TX equalization state machine will enable the analog E path when xcvr_standard_mode is set to 2'b10. 1'b0 : Disabled 1'b1 : Enabled
13	R/W	1'h0	**REE PCIe Gen 3 TX equalization state machine E path en standard mode 1**: This bit controls if the REE PCIE Gen 3 TX equalization state machine will enable the analog E path when xcvr_standard_mode is set to 2'b01. 1'b0 : Disabled 1'b1 : Enabled
12	R/W	1'h0	**REE PCIe Gen 3 TX equalization state machine E path en standard mode 0**: This bit controls if the REE PCIE Gen 3 TX equalization state machine will enable the analog E path when xcvr_standard_mode is set to 2'b00. 1'b0 : Disabled 1'b1 : Enabled
11	R/W	1'h0	**REE Periodic general control state machine E path en standard mode 3**: This bit controls if the REE periodic general control state machine will enable the analog E path when xcvr_standard_mode is set to 2'b11. 1'b0 : Disabled 1'b1 : Enabled
10	R/W	1'h1	**REE Periodic general control state machine E path en standard mode 2**: This bit controls if the REE periodic general control state machine will enable the analog E path when xcvr_standard_mode is set to 2'b10. 1'b0 : Disabled 1'b1 : Enabled
9	R/W	1'h1	**REE Periodic general control state machine E path en standard mode 1**: This bit controls if the REE periodic general control state machine will enable the analog E path when xcvr_standard_mode is set to 2'b01. 1'b0 : Disabled 1'b1 : Enabled
8	R/W	1'h0	**REE Periodic general control state machine E path en standard mode 0**: This bit controls if the REE periodic general control state machine will enable the analog E path when xcvr_standard_mode is set to 2'b00. 1'b0 : Disabled 1'b1 : Enabled

Register Name:			REE control state machine gen mode control register 2 (!)
Register Mnemonic:			RX_REE_SMGM_CTRL2
Register Address:			10_m_nnnn : 1011 : 11000
Bits	Type	Reset	Description
7	R/W	1'h0	**REE general control state machine 2 E path en standard mode 3**: This bit controls if the REE general control state machine 2 will enable the analog E path when xcvr_standard_mode is set to 2'b11. 1'b0 : Disabled 1'b1 : Enabled
6	R/W	1'h0	**REE general control state machine 2 E path en standard mode 2**: This bit controls if the REE general control state machine 2 will enable the analog E path when xcvr_standard_mode is set to 2'b10. 1'b0 : Disabled 1'b1 : Enabled
5	R/W	1'h0	**REE general control state machine 2 E path en standard mode 1**: This bit controls if the REE general control state machine 2 will enable the analog E path when xcvr_standard_mode is set to 2'b01. 1'b0 : Disabled 1'b1 : Enabled
4	R/W	1'h0	**REE general control state machine 2 E path en standard mode 0**: This bit controls if the REE general control state machine 2 will enable the analog E path when xcvr_standard_mode is set to 2'b00. 1'b0 : Disabled 1'b1 : Enabled
3	R/W	1'h0	**REE general control state machine 1 E path en standard mode 3**: This bit controls if the REE general control state machine 1 will enable the analog E path when xcvr_standard_mode is set to 2'b11. 1'b0 : Disabled 1'b1 : Enabled
2	R/W	1'h1	**REE general control state machine 1 E path en standard mode 2**: This bit controls if the REE general control state machine 1 will enable the analog E path when xcvr_standard_mode is set to 2'b10. 1'b0 : Disabled 1'b1 : Enabled
1	R/W	1'h1	**REE general control state machine 1 E path en standard mode 1**: This bit controls if the REE general control state machine 1 will enable the analog E path when xcvr_standard_mode is set to 2'b01. 1'b0 : Disabled 1'b1 : Enabled
0	R/W	1'h0	**REE general control state machine 1 E path en standard mode 0**: This bit controls if the REE general control state machine 1 will enable the analog E path when xcvr_standard_mode is set to 2'b00. 1'b0 : Disabled 1'b1 : Enabled

Register Name:			REE TX equalization evaluator pre-emphasis register (!)
Register Mnemonic:			RX_REE_TXEQEVAL_PRE
Register Address:			10_m_nnnn : 1011 : 11001
Bits	Type	Reset	Description
15:14	R/O	2'h0	Reserved
13:8	R/O	6'h00	**TX equalization evaluator pre-emphasis increment count**: Contains a count of the total number of pre-emphasis increment responses that have taken place during the TX equalization evaluator process.
7:6	R/O	2'h0	Reserved
5:0	R/O	6'h00	**TX equalization evaluator pre-emphasis decrement count**: Contains a count of the total number of pre-emphasis decrement responses that have taken place during the TX equalization evaluator process.

Register Name:			REE TX equalization evaluator post-emphasis register (!)
Register Mnemonic:			RX_REE_TXEQEVAL_POST
Register Address:			10_m_nnnn : 1011 : 11010
Bits	Type	Reset	Description
15:14	R/O	2'h0	Reserved
13:8	R/O	6'h00	**TX equalization evaluator post-emphasis increment count**: Contains a count of the total number of post-emphasis increment responses that have taken place during the TX equalization evaluator process.
7:6	R/O	2'h0	Reserved
5:0	R/O	6'h00	**TX equalization evaluator post-emphasis decrement count**: Contains a count of the total number of post-emphasis decrement responses that have taken place during the TX equalization evaluator process.

10.2.32 CDRLF Configuration Registers

Base address bits 15:5 :	10_m_nnnn : 0100
Address bits 4:0	**Register name and location of detailed description**
00000	CDRLF configuration register (!) on page 321
00001	CDRLF configuration register 2 (!) on page 322
00010	CDRLF configuration register 3 (!) on page 322
00011	CDRLF margin diagnostic register (!) on page 323
00100	CDRLF fast phase lock timer value register 0 (!) on page 323
00101	CDRLF fast phase lock timer value register 1 (!) on page 323
00110 - 00111	Reserved

Table 125: CDRLF configuration registers address space

Register Name:			CDRLF configuration register (!)
Register Mnemonic:			RX_CDRLF_CNFG
Register Address:			10_m_nnnn : 0100 : 00000
Bits	Type	Reset	Description
15	R/W	1’h0	**CDLRF reset hold**: When active (1’b1), the CDRLF will be held in reset beyond the time that it would normally be released by its asynchronous release signals. The CDRLF will be held in reset until this bit is deactivated (1’b0). **Note: This signal is intended be for verification purposes only. It should be set to 1’b0 for normal operation.**
14:12	R/W	3’h0	**CDRLF diagnostic mode control**: This field controls the information driven on the rx_pi_val_ln_{15:0}[7:0] signal, when in diagnostics mode.
11	R/W	1’h0	**CDRLF second order loop disable**: Activating (1’b1) this bit will disable the CDRLF second order loop. **Note: This signal is intended be for verification purposes only. It should be set to 1’b0 for normal operation.**
10:6	R/W	5’h04	**CDRLF second order loop sigma delta update rate**: This is the value that is added to or subtracted from the second order loop accumulator register when the serial data sample clock is detected as being out of phase with the serial data on a given parallel data word cycle. This is part of the CDRLF averaging function.
5	R/W	1’h0	**CDRLF first order loop disable**: Activating (1’b1) this bit will disable the CDRLF first order loop. **Note: This signal is intended to be for verification purposes only. It should be set to 1’b0 for normal operation.**
4:0	R/W	5’h0A	**CDRLF first order loop sigma delta update rate**: This is the value that is added to or subtracted from the first order loop accumulator register when the serial data sample clock is detected as being out of phase with the serial data on a given parallel data word cycle. This is part of the CDRLF averaging function.

Register Name:			CDRLF configuration register 2 (!)
Register Mnemonic:			RX_CDRLF_CNFG2
Register Address:			10_m_nnnn : 0100 : 00001
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14	R/W	1'h0	**CDRLF second order loop integrator max clear enable**: This signal enables the function in the CDRLF where the second order loop integrator is cleared when it reaches the maximum value.
13	R/O	1'h0	**CDRLF fast phase lock locked detected:** This register bit is the current status of the fphl_locked pin on the CDRLF, and indicates the fast phase lock process is complete.
12	R/W	1'h0	**CDRLF fast phase lock diagnostic enable:** This register bit can control the fphl_start pin on the CDRLF. for diagnostic purposes. If the CDRLF fast phase lock enabled by signal detect bit in this register is enabled, then either signal detect or this bit will activate the fphl_start pin.
11	R/W	1'h1	**CDRLF fast phase lock enabled by signal detect**: When active, signal detect will control the fphl_start pin on the CDRLF. This will cause fast phase lock to be enabled whenever the macro detects a transition from electrical idle to high speed data.
10	R/W	1'h0	**CDRLF reset on CDRLF PM Accumulator Max**: Activating (1'b1) this bit will force the CDRLF to be reset when the PM accumulator in the CDRLF reaches is maximum absolute value (the largest positive or negative value).
9	R/W	1'h1	**CDRLF freeze on electrical idle detect**: Activating (1'b1) this bit will force the CDRLF to be freeze in its current state when the receiver signal detect detects an electrical idle. When high speed data is detected, the freeze will be released.
8	R/W	1'h0	**CDRLF reset on electrical idle detect**: Activating (1'b1) this bit will force the CDRLF to be reset when the receiver signal detect detects an electrical idle. When high speed data is detected, the reset will be released.
7:6	R/O	2'h0	Reserved
5:0	R/W	6'h33	**CDRLF second order loop integrator threshold** : This value is the maximum magnitude the CDRLF second order loop integrator will be allowed to go to. **Note: This field must never be set to a value less than 2 (6'b000010).**

Register Name:			CDRLF configuration register 3 (!)
Register Mnemonic:			RX_CDRLF_CNFG3
Register Address:			10_m_nnnn : 0100 : 00010
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12'h0	Reserved
3	R/W	1'h0	**CDRLF data filter enable standard mode 3** : Enables the filter function for the data that feeds into the CDRLF from the deserializer, using the rxda_cdrlf_data_filter_en_n signal, when xcvr_standard_mode is set to 2'b11. • 1'b0 : Enable filter of 1010 patterns from the CDRLF. • 1'b1 : No CDRLF data filter function enabled.
2	R/W	1'h0	**CDRLF data filter enable standard mode 2** : Enables the filter function for the data that feeds into the CDRLF from the deserializer, using the rxda_cdrlf_data_filter_en_n signal, when xcvr_standard_mode is set to 2'b10. • 1'b0 : Enable filter of 1010 patterns from the CDRLF. • 1'b1 : No CDRLF data filter function enabled.
1	R/W	1'h1	**CDRLF data filter enable standard mode 1** : Enables the filter function for the data that feeds into the CDRLF from the deserializer, using the rxda_cdrlf_data_filter_en_n signal, when xcvr_standard_mode is set to 2'b01. • 1'b0 : Enable filter of 1010 patterns from the CDRLF. • 1'b1 : No CDRLF data filter function enabled.
0	R/W	1'h1	**CDRLF data filter enable standard mode 0** : Enables the filter function for the data that feeds into the CDRLF from the deserializer, using the rxda_cdrlf_data_filter_en_n signal, when xcvr_standard_mode is set to 2'b00. • 1'b0 : Enable filter of 1010 patterns from the CDRLF. • 1'b1 : No CDRLF data filter function enabled.

Register Name:			CDRLF margin diagnostic register (!)
Register Mnemonic:			RX_CDRLF_MGN_DIAG
Register Address:			10_m_nnnn : 0100 : 00011
Bits	**Type**	**Reset**	**Description**
15:3	R/O	13'h0	Reserved
2	W/O	1'h0	**CDRLF PI override down** : When the CDRLF PI override enable function is enabled, writing a 1'b1 to this bit will force a down to be generated in the CDRLF PI interface logic. **Note : This bit will be automatically cleared after the down request has been sent to the CDRLF PI interface logic. The down request is generated by changing the state of the cdrlf_pi_down_in signal in the CDRLF.**
1	W/O	1'h0	**CDRLF PI override up** : When the CDRLF PI override enable function is enabled, writing a 1'b1 to this bit will force an up to be generated in the CDRLF PI interface logic. **Note : This bit will be automatically cleared after the up request has been sent to the CDRLF PI interface logic. The up request is generated by changing the state of the cdrlf_pi_up_in signal in the CDRLF.**
0	R/W	1'h0	**CDRLF PI override enable** : Setting this bit to 1'b1 will enable the CDRLF PI override function, which will allow ups and downs to be forced to the CDRLF PI interface logic from the up and down override bits in this register. **Note : This bit must be set to 1'b1 before the CDRLF PI override up and down bits in this register are changed.**

Register Name:			CDRLF fast phase lock timer value register 0 (!)
Register Mnemonic:			RX_CDRLF_FPL_TMR0
Register Address:			10_m_nnnn : 0100 : 00100
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:4	R/W	4'h7	**Fast phase lock timer accumulate state time value** : Specifies the number of clock cycles minus 1 that the fast phase lock state machine will remain in the accumulate state.
3:0	R/W	4'h1	**Fast phase lock timer delay state time value** : Specifies the number of clock cycles minus 1 that the fast phase lock state machine will remain in the delay state.

Register Name:			CDRLF fast phase lock timer value register 1 (!)
Register Mnemonic:			RX_CDRLF_FPL_TMR1
Register Address:			10_m_nnnn : 0100 : 00101
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:8	R/W	4'h7	**Fast phase lock timer trigger 1 state time value** : Specifies the number of clock cycles minus 1 that the fast phase lock state machine will remain in the trigger state the first time it is in that state. The recommended value for this is the number of PI steps required to move the PI 1/4 UI.
7:4	R/W	4'h3	**Fast phase lock timer trigger 2 state time value** : Specifies the number of clock cycles minus 1 that the fast phase lock state machine will remain in the trigger state the second time it is in that state. The recommended value for this is the number of PI steps required to move the PI 1/8 UI.
3:0	R/W	4'h1	**Fast phase lock timer trigger 3 state time value** : Specifies the number of clock cycles minus 1 that the fast phase lock state machine will remain in the trigger state the third time it is in that state. The recommended value for this is the number of PI steps required to move the PI 1/16 UI.

10.2.33 Eye Surf Registers

Base address bits 15:5 :	10_m_nnnn : 0101
Address bits 4:0	**Register name and location of detailed description**
00000	Eye surf control register (!) on page 324
00001	Reserved
00010	Reserved
00011	Reserved
00100	Eye surf timer delay low register (!) on page 324
00101	Eye surf timer delay high register (!) on page 324
00110	Eye surf timer test low register (!) on page 324
00111	Eye surf timer test high register (!) on page 325
01000	Eye surf north south test point coordinate register (!) on page 325
01001	Eye surf east west test point coordinate register (!) on page 325
01010	Eye surf bit error count register (!) on page 325
01011 - 01111	Reserved

Table 126: Eye surf registers address space

Register Name:			Eye surf control register (!)
Register Mnemonic:			RX_EYESURF_CTRL
Register Address:			10_m_nnnn : 0101 : 00000
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Eye surf run**: Setting this bit to 1'b1 will initiate the eye surf process. This bit must remain set to 1'b1 until the eye surf done bit is set.
14	R/O	1'h0	**Eye surf done**: When this bit is set to 1'b1, the eye surf process has completed. This bit will be cleared after the eye surf run bit is cleared.
13:0	R/O	14'h0	Reserved

Register Name:			Eye surf timer delay low register (!)
Register Mnemonic:			RX_EYESURF_TMR_DELLOW
Register Address:			10_m_nnnn : 0101 : 00100
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'x0000	**Least significant 16 bits of the delay time**: The delay time specifies the number of clock cycles to wait between when a coordinate test point is set, and when to start testing the i and e data. **Note : It is not valid to set the total eye surf timer delay value to 0.**

Register Name:			Eye surf timer delay high register (!)
Register Mnemonic:			RX_EYESURF_TMR_DELHIGH
Register Address:			10_m_nnnn : 0101 : 00101
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'x0000	**Most significant 16 bits of the delay time**: The delay time specifies the number of clock cycles to wait between when a coordinate test point is set, and when to start testing the i and e data. **Note : It is not valid to set the total eye surf timer delay value to 0.**

Register Name:			Eye surf timer test low register (!)
Register Mnemonic:			RX_EYESURF_TMR_TESTLOW
Register Address:			10_m_nnnn : 0101 : 00110
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'x0000	**Least significant 16 bits of the test time**: The test time specifies the number of clock cycles to test the i and e data at a given coordinate test point. **Note : It is not valid to set the total eye surf timer test value to 0.**

Register Name:			Eye surf timer test high register (!)
Register Mnemonic:			RX_EYESURF_TMR_TESTHIGH
Register Address:			10_m_nnnn : 0101 : 00111
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'x0000	**Most significant 16 bits of the test time**: The test time specifies the number of clock cycles to test the i and e data at a given coordinate test point. **Note : It is not valid to set the total eye surf timer test value to 0.**

Register Name:			Eye surf north south test point coordinate register (!)
Register Mnemonic:			RX_EYESURF_NS_COORD
Register Address:			10_m_nnnn : 0101 : 01000
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8	R/W	1'h0	**Test point coordinate north south direction** : Indicates whether the desired test point is in the north or the south direction relative to the origin. • 1b1 : North • 1b0 : South
7	R/O	1'h0	Reserved
6:0	R/W	7'h00	**Test point coordinate north south offset** : Indicates how many steps in the north or south direction the desired test point is relative to the origin. **Note: In this implementation, only 5 least significant bits are used to control the sampler error DAC. Therefore, the most significant bits of this field are not used.**

Register Name:			Eye surf east west test point coordinate register (!)
Register Mnemonic:			RX_EYESURF_EW_COORD
Register Address:			10_m_nnnn : 0101 : 01001
Bits	**Type**	**Reset**	**Description**
15:9	R/O	7'h0	Reserved
8	R/W	1'h0	**Test point coordinate east west direction** : Indicates whether the desired test point is in the east or the west direction relative to the origin. • 1b1 : East • 1b0 : West
7:5	R/O	3'h0	Reserved
4:0	R/W	5'h00	**Test point coordinate east west offset** : Indicates how many steps in the east or west direction the desired test point is relative to the origin.

Register Name:			Eye surf bit error count register (!)
Register Mnemonic:			RX_EYESURF_ERRCNT
Register Address:			10_m_nnnn : 0101 : 01010
Bits	**Type**	**Reset**	**Description**
15:0	R/O	16'h0000	**Test point bit error count** : The total number of bit errors that were detected for a given run of the eye surf function. This register is only valid after the eye surf done bit in the Eye surf control register (!) on page 324 is active, and will be cleared when the done bit is cleared in response to the run bit being cleared.

10.2.34 Receiver BIST Controller Registers

One should note that the BIST registers should not be accessed when the diagnostic power island is switched off. Additionally, these registers do not retain their state when the diagnostic power island is switched off and are reset when the power island is switched back on.

Base address bits 15:5 :	10_m_nnnn : 0101
Address bits 4:0	**Register name and location of detailed description**
10000	Receiver BIST control register on page 326
10001	Receiver BIST sync count register on page 326
10010	Receiver BIST user defined data write register on page 327
10011	Receiver BIST error count register on page 327
10100 - 11111	Reserved

Table 127: Receiver BIST controller registers address space

Register Name:			Receiver BIST control register **Note: This register is on the diagnostic power island, and cannot be written to when this power island is switched off.**
Register Mnemonic:			RX_BIST_CTRL
Register Address:			10_m_nnnn : 0101 : 10000
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:8	R/W	4'h0	**Receiver BIST mode**: Controls which mode the BIST will operate in. The value of this field must match the corresponding field for the receive BIST controller. The following are the values used for this field, and what BIST mode they correspond to. • 4'b0000 : User defined data FIFO • 4'b0001 - 4'b0111 : Reserved • 4'b1000 : PRBS7 : 7 bit LFSR • 4'b1001 : PRBS15 : 15 bit LFSR • 4'b1010 : PRBS23 : 23 bit LFSR • 4'b1011 : PRBS31 : 31 bit LFSR • 4'b1100 - 4'b1111 : Reserved
7:5	R/O	3'h0	Reserved
4	R/W	1'h0	**Receiver BIST error reset**: Writing this bit is set to a 1'b1 will hold the error indicators in the receive BIST logic in reset. When this signal goes active, the rx_bist_status pin, rx_bist_err_toggle pin, and receive BIST error count field in this register will be cleared. When this bit is cleared, these error indicators will be released from reset and operate normally.
3:2	R/O	2'h0	Reserved
1	W/O	1'h0	**Receiver BIST user defined data FIFO clear**: Writing a 1'b1 to this bit will clear the receiver BIST user defined data FIFO. Note : This bit is automatically cleared after it is written to. Note : This clear function simply resets the FIFO pointers. It does not clear the contents of the FIFO.
0	R/W	1'h0	**Receiver BIST enable**: This bit enables the receiver BIST function. **Note : The remaining bits in this register must be stable when changing this bit. Therefore, it is best to enable and disable this function using a read / modify / write of this bit.**

Register Name:			Receiver BIST sync count register **Note: This register is on the diagnostic power island, and cannot be written to when this power island is switched off.**
Register Mnemonic:			RX_BIST_SYNCCNT
Register Address:			10_m_nnnn : 0101 : 10001
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h0000	**Receiver BIST sync count**: This field controls the value of the RX BIST sync count. The sync count indicates the number of consecutive received data words with no BIST bit errors that must be received in order for the RX BIST module to be considered synced to the input data stream. It is recommended that, when using a clean channel, this register be set to 16'h00FA. This will require the RX BIST to receive 250 consecutive words of data to confirm a BIST sync, which is enough time to allow the CDRLF to lock.

Register Name:			Receiver BIST user defined data write register **Note: This register is on the diagnostic power island, and cannot be written to when this power island is switched off.**
Register Mnemonic:			RX_BIST_UDDWR
Register Address:			10_m_nnnn : 0101 : 10010
Bits	**Type**	**Reset**	**Description**
15:10	R/O	6'h0	Reserved
9:0	W/O	10'h000	**Receiver BIST user defined data**: Writing a data word to this field will result in that data word being placed in the next available position in the receiver BIST user defined data FIFO. Note, when in 20 bit mode, all 10 of these bits are used. When in 16 bit mode, only the least significant 8 bits are used. Note: The FIFO in this implementation is 18 words deep. It is up to the user to not write more than 18 words of data into this FIFO. Exceeding this amount of data will generate unpredictable results.

Register Name:			Receiver BIST error count register **Note: This register is on the diagnostic power island, and cannot be written to when this power island is switched off.**
Register Mnemonic:			RX_BIST_ERRCNT
Register Address:			10_m_nnnn : 0101 : 10011
Bits	**Type**	**Reset**	**Description**
15:0	R/O	16'h0000	**Receiver BIST error count:** Indicates the number of BIST errors that have been observed by the receive BIST logic, since the last time the BIST error indicator logic was reset or restarted. This counter increments up to a maximum value of 16'hFFFF. Note : Due to the nature of the BIST PRBS error detection logic, when a bit error occurs in PRBS mode, two or three errors will be counted.

10.2.35 Receiver Signal Detect Filter Registers

Base address bits 15:5 :	**10_m_nnnn : 0100**
Address bits 4:0	**Register name and location of detailed description**
10000	Receiver signal detect filter high to low filter timer register (!) on page 327
10001	Receiver signal detect filter high to low delay timer register (!) on page 328
10010	Receiver signal detect filter high to low min timer register (!) on page 328
10011	Receiver signal detect filter high to low init timer register (!) on page 328
10100	Receiver signal detect filter low to high filter timer register (!) on page 328
10101	Receiver signal detect filter low to high delay timer register (!) on page 328
10110	Receiver signal detect filter low to high min timer register (!) on page 328
10111	Receiver signal detect filter low to high init timer register (!) on page 329

Table 128: Receiver signal detect filter registers address space

Register Name:			Receiver signal detect filter high to low filter timer register (!)
Register Mnemonic:			RX_SIGDET_HL_FILT_TMR
Register Address:			10_m_nnnn : 0100 : 10000
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h0C	**Signal detect filter high to low filter timer value**: This is the value loaded into the filter timer in the signal detect high to low filter circuit.

Register Name:			Receiver signal detect filter high to low delay timer register (!)
Register Mnemonic:			RX_SIGDET_HL_DLY_TMR
Register Address:			10_m_nnnn : 0100 : 10001
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h00	**Signal detect filter high to low delay timer value**: This is the value loaded into the delay timer in the signal detect high to low filter circuit.

Register Name:			Receiver signal detect filter high to low min timer register (!)
Register Mnemonic:			RX_SIGDET_HL_MIN_TMR
Register Address:			10_m_nnnn : 0100 : 10010
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h00	**Signal detect filter high to low min timer value**: This is the value loaded into the min timer in the signal detect high to low filter circuit.

Register Name:			Receiver signal detect filter high to low init timer register (!)
Register Mnemonic:			RX_SIGDET_HL_INIT_TMR
Register Address:			10_m_nnnn : 0100 : 10011
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h28	**Signal detect init timer value**: This is the value loaded into the initialization timer in the signal detect filter high to low filter circuit.

Register Name:			Receiver signal detect filter low to high filter timer register (!)
Register Mnemonic:			RX_SIGDET_LH_FILT_TMR
Register Address:			10_m_nnnn : 0100 : 10100
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h04	**Signal detect filter low to high filter timer value**: This is the value loaded into the filter timer in the signal detect low to high filter circuit. This should be set to 5 less than the number of clocks of desired filter time. This takes into account the synchronizing of the signal to be filtered (4 clocks) and the time to load the filter timer (1 clock). Therefore, the smallest granularity of filtering available here is 5 clocks. This is more of an issue of latency through the filter circuit as the signal initially rises.

Register Name:			Receiver signal detect filter low to high delay timer register (!)
Register Mnemonic:			RX_SIGDET_LH_DLY_TMR
Register Address:			10_m_nnnn : 0100 : 10101
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h00	**Signal detect filter low to high min timer value**: This is the value loaded into the min timer in the signal detect low to high filter circuit.

Register Name:			Receiver signal detect filter low to high min timer register (!)
Register Mnemonic:			RX_SIGDET_LH_MIN_TMR
Register Address:			10_m_nnnn : 0100 : 10110
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h00	**Signal detect filter high to low min timer value**: This is the value loaded into the min timer in the signal detect high to low filter circuit.

Register Name:			Receiver signal detect filter low to high init timer register (!)
Register Mnemonic:			RX_SIGDET_LH_INIT_TMR
Register Address:			10_m_nnnn : 0100 : 10111
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:0	R/W	6'h28	**Signal detect init timer value**: This is the value loaded into the initialization timer in the signal detect filter high to low filter circuit.

10.2.36 Receiver LFPS Detect Filter Registers

Base address bits 15:5 :	10_m_nnnn : 0100
Address bits 4:0	**Register name and location of detailed description**
11000	Receiver LFPS detect minimum pulse distance counter register (!) on page 329
11001	Receiver LFPS detect no signal counter register (!) on page 329
11010	Receiver LFPS detect ramp down counter register (!) on page 329
11011	Receiver LFPS detect minimum pulse duration register (!) on page 330
11100	Receiver LFPS detect diagnostic control register (!) on page 330
11101 - 11111	Reserved

Table 129: Receiver LFPS detect filter registers address space

Register Name:			Receiver LFPS detect minimum pulse distance counter register (!)
Register Mnemonic:			RX_LFPSDET_MD_CNT
Register Address:			10_m_nnnn : 0100 : 11000
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12'h0	Reserved
3:0	R/W	4'h5	**Minimum pulse distance counter value (MD)**: Specifies the minimum pulse distance for a valid LFPS sequence. Note: MD must be set to a value that is greater than MP.

Register Name:			Receiver LFPS detect no signal counter register (!)
Register Mnemonic:			RX_LFPSDET_NS_CNT
Register Address:			10_m_nnnn : 0100 : 11001
Bits	**Type**	**Reset**	**Description**
15:2	R/O	14'h0	Reserved
1:0	R/W	2'h0	**No signal counter value (NS)**: Specifies the number of clock cycles where pulse_high and pulse_low are inactive before declaring no signal. Note: a value of 2'b00 is not a valid value for this field.

Register Name:			Receiver LFPS detect ramp down counter register (!)
Register Mnemonic:			RX_LFPSDET_RD_CNT
Register Address:			10_m_nnnn : 0100 : 11010
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12'h0	Reserved
3:0	R/W	4'h7	**Ramp down counter value (RD)**: Species the number of clock cycles that are used in the LFPS detect ramp down process. This signal should correspond to the MD, to reduce LFPS distortion. Note: The actual ramp down time is a function of the detection of no signal when lfps_detected is driven active. If no signal is detected, the ramp down time is the value in this field minus the value of the no signal counter value field in the RX_LFPSDET_NS_CNT register. Otherwise, the value in this field is used as is.

Register Name:			Receiver LFPS detect minimum pulse duration register (!)
Register Mnemonic:			RX_LFPSDET_MP_CNT
Register Address:			10_m_nnnn : 0100 : 11011
Bits	**Type**	**Reset**	**Description**
15:3	R/O	13'h0	Reserved
2:0	R/W	3'h7	**Minimum pulse duration (MP)**: Specifies the minimum number of clock cycles required for a given LFPS pulse to be driven active to be considered part of a valid LFPS burst.

Register Name:			Receiver LFPS detect diagnostic control register (!)
Register Mnemonic:			RX_LFPSDET_DIAG_CTRL
Register Address:			10_m_nnnn : 0100 : 11100
Bits	**Type**	**Reset**	**Description**
15:3	R/O	13'h0	Reserved
2	R/W	1'h0	**Disable pulse none MD check**: When active (1'b1), the check that tests if pulse_none is driven inactive while the MD check is being performed will be disabled.
1	R/W	1'h0	**LFPS detect override enable**: When active (1'b1), the LFPS detect override bit in this register will drive the LFPS detect output directly.
0	R/W	1'h0	**LFPS detect override**: When enabled by the LFPS detect override enable bit in this register, this bit will drive the LFPS detect output directly.

10.2.37 Transceiver Clock Measurement Module Registers

Base address bits 15:5 :	10_m_nnnn : 1110
Address bits 4:0	**Register name and location of detailed description**
00000	Clock frequency measurement control register on page 330
00001	Test clock selection register on page 331
00010	Reference clock timer value register on page 331
00011	Test clock counter value register on page 331
00100 - 11111	Reserved

Table 130: Transceiver clock measurement module registers address space

Register Name:			Clock frequency measurement control register
Register Mnemonic:			XCVR_CMSMT_CLK_FREQ_MSMT_CTRL
Register Address:			10_m_nnnn : 1110 : 00000
Bits	**Type**	**Reset**	**Description**
15	R/W	1'h0	**Run test clock measurement**: Activating (1'b1) this bit will run the test clock measurement process. This bit must remain active until the test clock measurement process is complete, as indicated by the test clock measurement done bit in this register. To start another measurement process, this bit must first be driven inactive then driven active again. **Note: Both of the clocks used to generate the ref_clk and test_clk clocks must be active prior to setting or clearing this bit.** **Note: The values in the Test clock selection register and Reference clock timer value register must be set prior to activating this bit.**
14	R/O	1'h0	**Test clock measurement done**: This bit will be set to 1'b1 when the test clock measurement process is complete. It will be cleared by the deactivation of the start test clock measurement bit in this register.
13:0	R/O	14'h0	Reserved

Register Name:			Test clock selection register
Register Mnemonic:			XCVR_CMSMT_TEST_CLK_SEL
Register Address:			10_m_nnnn : 1110 : 00001
Bits	**Type**	**Reset**	**Description**
15:3	R/O	13'h0	Reserved
2:0	R/W	3'h0	**Test clock select**: This field drives the test_clk_select pin, in order to control an external MUX for selecting between multiple test clocks to measure. The following is the encoding for this field, and the clock each value selects. • 3'b000 : Transmitter data clock • 3'b001 : Serializer clock • 3'b010 : Receiver data clock • 3'b011 : Deserializer clock • 3'b100 : Receiver 2X data clock • 3'b101 : Deserializer 2X clock • 3'b110 : Signal detect clock • 3'b111 : Reserved

Register Name:			Reference clock timer value register
Register Mnemonic:			XCVR_CMSMT_REF_CLK_TMR_VALUE
Register Address:			10_m_nnnn : 1110 : 00010
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:0	R/W	12'h000	**Reference clock timer value** : This specifies the amount of time, in reference clock cycles, to count test clock cycles. This value minus 1 is loaded into the reference clock timer. A value of 0 for this field is not valid when running this function.

Register Name:			Test clock counter value register
Register Mnemonic:			XCVR_CMSMT_TEST_CLK_CNT_VALUE
Register Address:			10_m_nnnn : 1110 : 00011
Bits	**Type**	**Reset**	**Description**
15:12	R/O	4'h0	Reserved
11:0	R/O	12'h000	**Test clock counter value**: When the test clock measurement process is complete, the value in this field specifies the number of test clock cycles that were counted in the time specified by the reference clock timer value. This field is only valid while the test clock measurement done bit in the Clock frequency measurement control register is active.

10.2.38 Receiver Control And Diagnostics Registers

Base address bits 15:5 :	10_m_nnnn : 1111
Address bits 4:0	**Register name and location of detailed description**
00000	Receiver DFE control register (!) on page 332
00001	DFE amp fine tuning register (!) on page 333
00010	DFE amp fine tuning 2 register (!) on page 334
00011	DFE amp fine tuning 3 register (!) on page 335
00100	REE DAC control register (!) on page 335
00101	Nyquist control register (!) on page 336
00110	Low frequency equalizer tuning register (!) on page 336
00111	Reserved
01000	RX signal detect tuning and control register (!) on page 337
01001	Sample and hold signal detect calibration code register (!) on page 338
01010	Signal detect test register (!) on page 338
01011	Reserved
01100	RX sampler diagnostic control register (!) on page 339
01101	Sample and hold RX sampler latch calibration I predictive positive code register (!) on page 339
01110	Sample and hold RX sampler latch calibration I predictive negative code register (!) on page 339
01111	Sample and hold RX sampler latch calibration Q predictive positive code register (!) on page 339
10000	Sample and hold RX sampler latch calibration Q predictive negative code register (!) on page 340
10001	Sample and hold RX sampler latch calibration E predictive positive code register (!) on page 340
10010	Sample and hold RX sampler latch calibration E predictive negative code register (!) on page 340
10011	Reserved
10100	PI rate selection register (!) on page 341
10101	PI capacitor selection register (!) on page 342
10110	PI tuning register (!) on page 343
10111	Reserved
11000	RX loopback controller register (!) on page 343
11001	Receiver control reset diagnostic register (!) on page 344
11010 - 11101	Reserved
11110	Receiver digital cover your alternatives register (!) on page 344
11111	Receiver analog cover your alternatives register on page 344

Table 131: Receiver control and diagnostics registers address space

Register Name:			Receiver DFE control register (!)
Register Mnemonic:			RX_DIAG_DFE_CTRL
Register Address:			10_m_nnnn : 1111 : 00000
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12’h0	Reserved
3	R/W	1’h0	**Receiver DFE low frequency equalization enable value standard mode 3**: This bit controls the rxda_dfe_lfeq_en signal going to the analog, which is used to enable the receiver DFE low frequency equalization analog circuits, when xcvr_standard_mode is set to 2’b11. • 1’b0 : Disabled 1’b1 : Enabled
2	R/W	1’h1	**Receiver DFE low frequency equalization enable value standard mode 2**: This bit controls the rxda_dfe_lfeq_en signal going to the analog, which is used to enable the receiver DFE low frequency equalization analog circuits, when xcvr_standard_mode is set to 2’b10. • 1’b0 : Disabled 1’b1 : Enabled
1	R/W	1’h0	**Receiver DFE low frequency equalization enable value standard mode 1**: This bit controls the rxda_dfe_lfeq_en signal going to the analog, which is used to enable the receiver DFE low frequency equalization analog circuits, when xcvr_standard_mode is set to 2’b01. • 1’b0 : Disabled 1’b1 : Enabled
0	R/W	1’h0	**Receiver DFE low frequency equalization enable value standard mode 0**: This bit controls the rxda_dfe_lfeq_en signal going to the analog, which is used to enable the receiver DFE low frequency equalization analog circuits, when xcvr_standard_mode is set to 2’b00. • 1’b0 : Disabled 1’b1 : Enabled

Register Name:			DFE amp fine tuning register (!)
Register Mnemonic:			RX_DIAG_DFE_AMP_TUNE
Register Address:			10_m_nnnn : 1111 : 00001
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:12	R/W	3'h4	**DFE constant gm bias tune**: Adjusts the constant gm bias. • 3'b000: -34 uA • 3'b001 • 3'b010 • 3'b011 • 3'b100: - 50 uA • 3'b101 • 3'b110 • 3'b111: -62 uA
11	R/W	1'h1	**DFE VGA constant gm bias enable**: Enables the VGA constant gm bias. This bit drives the rxda_v-ga_cnst_gm_en signal going to the analog. • 1'b0 : Disabled. • 1'b1 : Enabled.
10:8	R/W	3'h5	**DFE VGA amp current adjust**: Adjusts the current for the DFE VGA amp, using the rxda_vga_current_adj signal, as specified below. Note: The percentages are approximate values. • 3'b000: +30% (maximum) • ... • 3'b101: 0% • ... • 3'b111: -30% (minimum)
7	R/W	1'h1	**DFE peaking constant gm bias enable**: Enables the peaking constant gm bias. This bit drives the rxda_peak_cnst_gm_en signal going to the analog. • 1'b0 : Disabled. • 1'b1 : Enabled.
6:4	R/W	3'h5	**DFE peaking amp current adjust**: Adjusts the current for the DFE peaking amp, using the rxda_peak_current_adj signal, as specified below. Note: The percentages are approximate values. • 3'b000: +30% (maximum) • ... • 3'b101: 0% • ... • 3'b111: -30% (minimum)
3	R/W	1'h1	**DFE summing constant gm bias enable**: Enables the summing constant gm bias. This bit drives the rxda_sum_cnst_gm_en signal going to the analog. • 1'b0 : Disabled. • 1'b1 : Enabled.
2:0	R/W	3'h5	**DFE summing amp current adjust**: Adjusts the current for the DFE summing amp, using the rxda_-sum_current_adj signal, as specified below. Note: The percentages are approximate values. • 3'b000: +30% (maximum) • ... • 3'b101: 0% • ... 3'b111: -30% (minimum)

Register Name:			DFE amp fine tuning 2 register (!)
Register Mnemonic:			RX_DIAG_DFE_AMP_TUNE_2
Register Address:			10_m_nnnn : 1111 : 00010
Bits	**Type**	**Reset**	**Description**
15:14	R/O	2'h0	Reserved
13:12	R/W	2'h0	**Receiver peaking amp common mode adjust**: Adjusts the common mode voltage for the receiver peaking amp, by driving the rxda_fe_pkamp_cm_adj signal to the analog. • 2'b00 : Nominal common mode voltage. • 2'b01 : Reduces the common mode voltage by one step. • 2'b10 : Increases the common mode voltage by one step. • 2'b11 : Reserved
11	R/W	1'h1	**DFE low frequency equalizer constant gm bias enable**: Enables the low frequency equalizer constant gm bias. This bit drives the rxda_lfeq_cnst_gm_en signal going to the analog. • 1'b0 : Disabled. 1'b1 : Enabled.
10:8	R/W	3'h4	**DFE low frequency equalizer current adjust**: Adjusts the current for the DFE low frequency equalizer, using the rxda_lfeq_current_adj signal, as specified below. Note: The percentages are approximate values. • 3'b000: +30% (maximum) • ... • 3'b101: 0% • ... 3'b111: -30% (minimum)
7	R/W	1'h0	**DFE peaking amp active inductor boost**: Enables the active inductors boost function in the peaking amp, for high data rates. This bit controls the rxda_peak_actind_en signal going to the analog. • 1'b0 : Disabled 1'b1 : Enabled
6	R/W	1'h0	Reserved - Spare
5	R/W	1'h0	**DFE VGA stage 2 active inductor boost**: Enables the active inductors boost function in stage 2 of the VGA, for high data rates. This bit controls the rxda_vga_st2_actind_en signal going to the analog. • 1'b0 : Disabled • 1'b1 : Enabled
4	R/W	1'h0	**DFE RX Tap 1 DAC Range Select**: Controls the tap 1 DAC range in the analog DFE. This bit controls the rxda_dfe_tap_1_range signal going to the analog. • 1'b0: Normal operation 1'b1: The tap 1 DAC range is increased.
3:2	R/O	2'h0	Reserved
1:0	R/W	2'h1	**DFE RX amp current adjust**: Adjusts the mix of constant-gm and External Current for RX front end amplifiers. This controls the rxda_cnstgm_ext_sel signal going into the analog. The encoding of the rxda_cnstgm_ext_sel signal, as a function of the values in this register is specified below. • 2'b00 => 3'b000 : gm_bias = 25u, ext = 25u, total = 50u • 2'b01 => 3'b001 : gm_bias = 30u, ext = 20u, total = 50u • 2'b10 => 3'b011 : gm_bias = 35u, ext = 15u, total = 50u • 2'b11 => 3'b111 : gm_bias = 40u, ext = 10u, total = 50u •

Register Name:			DFE amp fine tuning 3 register (!)
Register Mnemonic:			RX_DIAG_DFE_AMP_TUNE_3
Register Address:			10_m_nnnn : 1111 : 00011
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12'h0	Reserved
3	R/W	1'h0	**DFE VGA stage 1 active inductor boost enable standard mode 3**: This bit enables the active inductors boost function in stage 1 of the VGA, for high data rates, when xcvr_standard_mode is set to 2'b11. This bit controls the rxda_vga_st1_actind_en signal going to the analog. • 1'b0 : Disabled • 1'b1 : Enabled
2	R/W	1'h0	**DFE VGA stage 1 active inductor boost enable standard mode 2**: This bit enables the active inductors boost function in stage 1 of the VGA, for high data rates, when xcvr_standard_mode is set to 2'b10. This bit controls the rxda_vga_st1_actind_en signal going to the analog. • 1'b0 : Disabled • 1'b1 : Enabled
1	R/W		**DFE VGA stage 1 active inductor boost enable standard mode 1**: This bit enables the active inductors boost function in stage 1 of the VGA, for high data rates, when xcvr_standard_mode is set to 2'b01. This bit controls the rxda_vga_st1_actind_en signal going to the analog. • 1'b0 : Disabled • 1'b1 : Enabled
0	R/W	1'h0	**DFE VGA stage 1 active inductor boost enable standard mode 0**: This bit enables the active inductors boost function in stage 1 of the VGA, for high data rates, when xcvr_standard_mode is set to 2'b00. This bit controls the rxda_vga_st1_actind_en signal going to the analog. • 1'b0 : Disabled • 1'b1 : Enabled

Register Name:			REE DAC control register (!)
Register Mnemonic:			RX_DIAG_REE_DAC_CTRL
Register Address:			10_m_nnnn : 1111 : 00100
Bits	**Type**	**Reset**	**Description**
15:3	R/O	13'h0	Reserved
2	R/W	1'h1	**DFE Offset DAC enable**: Enables the DFE offset DAC associated with the VGA amp. • 1'b0: Disabled. • 1'b1: Enabled.
1	R/W	1'h0	**DFE Offset DAC attenuation**: Adds attenuation to the DFE offset DAC associated with the VGA amp. • 1'b0: No attenuation. • 1'b1: DAC gain attenuated by 40%.
0	R/W	1'h0	**DFE DAC attenuation**: Adds attenuation to the DFE DACs associated with the summing amp. • 1'b0: No attenuation. 1'b1: DAC gain attenuated by 30%.

Register Name:			Nyquist control register (!)
Register Mnemonic:			RX_DIAG_NQST_CTRL
Register Address:			10_m_nnnn : 1111 : 00101
Bits	**Type**	**Reset**	**Description**
15:12	R/W	4'h0	**RX nyquist select value standard mode 3**: This field specifies the receiver DFE nyquist frequency select value on the rxda_dfe_nqst_sel signal driven to the analog, when xcvr_standard_mode is set to 2'b11. • 4'b1011 : 4.00 GHz • 4'b1001 : 2.50 GHz • 4'b1000 : 1.25 GHz
11:8	R/W	4'hB	**RX nyquist select value standard mode 2**: This field specifies the receiver DFE nyquist frequency select value on the rxda_dfe_nqst_sel signal driven to the analog, when xcvr_standard_mode is set to 2'b10. • 4'b1011 : 4.00 GHz • 4'b1001 : 2.50 GHz • 4'b1000 : 1.25 GHz
7:4	R/W	4'h9	**RX nyquist select value standard mode 1**: This field specifies the receiver DFE nyquist frequency select value on the rxda_dfe_nqst_sel signal driven to the analog, when xcvr_standard_mode is set to 2'b01. • 4'b1011 : 4.00 GHz • 4'b1001 : 2.50 GHz • 4'b1000 : 1.25 GHz
3:0	R/W	4'h8	**RX nyquist select value standard mode 0**: This field specifies the receiver DFE nyquist frequency select value on the rxda_dfe_nqst_sel signal driven to the analog, when xcvr_standard_mode is set to 2'b00. • 4'b1011 : 4.00 GHz • 4'b1001 : 2.50 GHz • 4'b1000 : 1.25 GHz

Register Name:			Low frequency equalizer tuning register (!)
Register Mnemonic:			RX_DIAG_LFEQ_TUNE
Register Address:			10_m_nnnn : 1111 : 00110
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7:6	R/W	2'h3	**RX low frequency equalization zero frequency value standard mode 3**: This field specifies the receiver zero frequency setting for the low frequency equalization function, by thermometer encoding the value of this field and driving it on the rxda_dfe_lfeq_zero_freq signal going into the analog, when xcvr_standard_-mode is set to 2'b11. The following is the encoding of this field. • 2'b00 : 150 MHz • 2'b01 : 200 MHz • 2'b10 : 250 MHz • 2'b11 : 300 MHz
5:4	R/W	2'h2	**RX low frequency equalization zero frequency value standard mode 2**: This field specifies the receiver zero frequency setting for the low frequency equalization function, by thermometer encoding the value of this field and driving it on the rxda_dfe_lfeq_zero_freq signal going into the analog, when xcvr_standard_-mode is set to 2'b10. The following is the encoding of this field. • 2'b00 : 150 MHz • 2'b01 : 200 MHz • 2'b10 : 250 MHz • 2'b11 : 300 MHz
3:2	R/W	2'h1	**RX low frequency equalization zero frequency value standard mode 1**: This field specifies the receiver zero frequency setting for the low frequency equalization function, by thermometer encoding the value of this field and driving it on the rxda_dfe_lfeq_zero_freq signal going into the analog, when xcvr_standard_-mode is set to 2'b01. The following is the encoding of this field. • 2'b00 : 150 MHz • 2'b01 : 200 MHz • 2'b10 : 250 MHz • 2'b11 : 300 MHz
1:0	R/W	2'h0	**RX low frequency equalization zero frequency value standard mode 0**: This field specifies the receiver zero frequency setting for the low frequency equalization function, by thermometer encoding the value of this field and driving it on the rxda_dfe_lfeq_zero_freq signal going into the analog, when xcvr_standard_-mode is set to 2'b00. The following is the encoding of this field. • 2'b00 : 150 MHz • 2'b01 : 200 MHz • 2'b10 : 250 MHz • 2'b11 : 300 MHz

Register Name:			RX signal detect tuning and control register (!)
Register Mnemonic:			RX_DIAG_SIGDET_TUNE
Register Address:			10_m_nnnn : 1111 : 01000
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14	R/W	1'h0	**Signal detect calibration half gain select**: Controls the resolution of each step in the signal detect calibration code by adjusting the gain of signal detect offset correction, by driving the rxda_sd_cal_halfgain signal going to the analog, as specified below. • 1'b0: 8mV resolution per calibration code step. • 1'b1: 4mV resolution per calibration code step.
13:12	R/W	2'h1	**Signal detect filter function select**: Selects which of the two RX signal detect filter functions are enabled. As specified below. • 2'b00 : Reserved • 2'b01 : RX low to high filter disabled, RX high to low filter enabled. • 2'b10 : RX low to high filter enabled, RX high to low filter disabled. • 2'b11 : RX low to high filter enabled, RX high to low filter enabled.
11:8	R/O	4'h0	Reserved
7	R/W	1'h0	**Receiver signal detect invert samplers**: Inverts the behavior of the rxda_sd_pulse_high and rxda_sd_-pulse_low samplers, by driving the rxda_sd_invert_samplers signal to the analog. • 1'b0 : Normal mode : The rxda_sd_pulse_high sampler outputs a 1'b1 if the output of the comparator has been above the positive threshold at any time during the last clock period. Similar behavior exists for the rxda_sd_pulse_low sampler in this mode. • 1'b1 : Inverted mode : The rxda_sd_pulse_high sampler outputs a 1'b1 if the output of the comparator has been above the positive threshold for the entire duration of the last clock period. Similar behavior exists for the rxda_sd_pulse_low sampler in this mode.
6	R/W	1'h0	**Receiver signal detect squelch pulse none**: Enable the squelch function for the rxda_sd_pulse_none signal, by driving the rxda_sd_squelch_pulse_none signal to the analog. This debug feature could allow the detection of LFPS signals having slow transition times (e.g. 4 nSec) and low frequency differential noise (e.g. 500 MHz). The later can create noise glitches taking place over man than 1/2 cycle when DCD is taken into consideration. • 1'b0 : Disabled • 1'b1 : Enabled
5	R/W	1'h0	**Receiver signal detect one comparator mode**: Enables one comparator mode, as a power reduction option, by driving the rxda_sd_onecomp_mode_en signal to the analog. • 1'b0 : Both signal detect comparators functioning. • 1'b1 : One signal detect comparator functioning.
4	R/W	1'h0	**Receiver signal detect DC coupled path enable**: Enables a DC coupled path to the signal detect for verification and testability purposes, by driving the rxda_sd_dcpath_en signal to the analog. • 1'b0 : Disabled • 1'b1 : Enabled
3	R/O	1'h0	Reserved
2:0	R/W	3'h2	**Signal detect level**: Sets the reference voltage level at which the comparators will detect a signal by driving the rxda_sd_siglvl_n signal going to the analog. The following specifies the encoding of this signal. • 3'b000 : 8'b11111110 : Min • 3'b001 : 8'b11111101 • ... • 3'b110 : 8'b10111111 • 3'b111 : 8'b01111111 : Max

Register Name:			Sample and hold signal detect calibration code register (!)
Register Mnemonic:			RX_DIAG_SH_SIGDET
Register Address:			10_m_nnnn : 1111 : 01001
Bits	**Type**	**Reset**	**Description**
15:13	R/O	3'h0	Reserved
12	R/O	1'h0	**Signal detect 1 up**: Signal detect 1 calibration up signal value, as it is currently captured in the sample and hold latches.
11:8	R/O	4'h0	**Signal detect 1 code**: Signal detect 1 calibration code signal value, as it is currently captured in the sample and hold latches.
7:5	R/O	3'h0	Reserved
4	R/O	1'h0	**Signal detect 0 up**: Signal detect 0 calibration up signal value, as it is currently captured in the sample and hold latches.
3:0	R/O	4'h0	**Signal detect 0 code**: Signal detect 0 calibration code signal value, as it is currently captured in the sample and hold latches.

Register Name:			Signal detect test register (!)
Register Mnemonic:			RX_DIAG_SD_TEST
Register Address:			10_m_nnnn : 1111 : 01010
Bits	**Type**	**Reset**	**Description**
15:4	R/O	12'h0	Reserved
3	R/W	1'h0	**LFPS detected low test bit**: This bit can be used to detect if the rx_lfps_detect is driven to a low state. The following procedure can be used to detect this condition. • Write this bit to a 1'b1. • Read this bit. • If the value of the read bit is 1'b0, rx_lfps_detect is driven to a low state between the write and the read. Note: If rx_lfps_detect is already driven to a low state at the time of the write, the bit will not be set to 1'b1.
2	R/W	1'h0	**LFPS detected high test bit**: This bit can be used to detect if the rx_lfps_detect is driven to a high state. The following procedure can be used to detect this condition. • Write this bit to a 1'b1. • Read this bit. • If the value of the read bit is 1'b0, rx_lfps_detect is driven to a high state between the write and the read. Note: If rx_lfps_detect is already driven to a high state at the time of the write, the bit will not be set to 1'b1.
1	R/W	1'h0	**Signal detected low test bit**: This bit can be used to detect if the rx_signal_detect is driven to a low state. The following procedure can be used to detect this condition. • Write this bit to a 1'b1. • Read this bit. • If the value of the read bit is 1'b0, rx_signal_detect is driven to a low state between the write and the read. Note: If rx_signal_detect is already driven to a low state at the time of the write, the bit will not be set to 1'b1.
0	R/W	1'h0	**Signal detected high test bit**: This bit can be used to detect if the rx_signal_detect is driven to a high state. The following procedure can be used to detect this condition. • Write this bit to a 1'b1. • Read this bit. • If the value of the read bit is 1'b0, rx_signal_detect is driven to a high state between the write and the read. Note: If rx_signal_detect is already driven to a high state at the time of the write, the bit will not be set to 1'b1.

Register Name:			RX sampler diagnostic control register (!)
Register Mnemonic:			RX_DIAG_SAMP_CTRL
Register Address:			10_m_nnnn : 1111 : 01100
Bits	**Type**	**Reset**	**Description**
15:2	R/O	14'h0	Reserved
1	R/W	1'h0	**RX sampler latch range extend**: Controls the range of the RX sampler calibration, by driving the rxda_sampler_latch_cal_range_ext signal to the analog. • 1'b0 : Normal range • 1'b1 : Extended range (adds approximately 1.4 mV per step or 21 mV total.
0	R/W	1'h1	**Analog sampler rxda_dfe_0p5ui_mode_en signal control**: Selects which delayed I data is to be used to unroll the Q data in the sampler. • 1'b0: The I data that occurs 1.5 UI before the Q data is used to select the proper Q sample. • 1'b1: The I data that occurs 0.5 UI before the corresponding Q data is used to select the proper Q sample. **Note: The signal controlled by this register is forced to 1'b0 when RX sampler latch calibration is running.**

Register Name:			Sample and hold RX sampler latch calibration I predictive positive code register (!)
Register Mnemonic:			RX_DIAG_SH_SLC_IPP
Register Address:			10_m_nnnn : 1111 : 01101
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:8	R/O	7'h0	**RX sampler latch calibration I even positive code**: RX sampler latch calibration I even positive code signal value, as it is currently captured in the sample and hold latches.
7	R/O	1'h0	Reserved
6:0	R/O	7'h0	**RX sampler latch calibration I odd positive code**: RX sampler latch calibration I odd positive code signal value, as it is currently captured in the sample and hold latches.

Register Name:			Sample and hold RX sampler latch calibration I predictive negative code register (!)
Register Mnemonic:			RX_DIAG_SH_SLC_IPM
Register Address:			10_m_nnnn : 1111 : 01110
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:8	R/O	7'h0	**RX sampler latch calibration I even negative code**: RX sampler latch calibration I even negative code signal value, as it is currently captured in the sample and hold latches.
7	R/O	1'h0	Reserved
6:0	R/O	7'h0	**RX sampler latch calibration I odd negative code**: RX sampler latch calibration I odd negative code signal value, as it is currently captured in the sample and hold latches.

Register Name:			Sample and hold RX sampler latch calibration Q predictive positive code register (!)
Register Mnemonic:			RX_DIAG_SH_SLC_QPP
Register Address:			10_m_nnnn : 1111 : 01111
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:8	R/O	7'h0	**RX sampler latch calibration Q even positive code**: RX sampler latch calibration Q even positive code signal value, as it is currently captured in the sample and hold latches.
7	R/O	1'h0	Reserved
6:0	R/O	7'h0	**RX sampler latch calibration Q odd positive code**: RX sampler latch calibration Q odd positive code signal value, as it is currently captured in the sample and hold latches.

Register Name:			Sample and hold RX sampler latch calibration Q predictive negative code register (!)
Register Mnemonic:			RX_DIAG_SH_SLC_QPM
Register Address:			10_m_nnnn : 1111 : 10000
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:8	R/O	7'h0	**RX sampler latch calibration Q even negative code**: RX sampler latch calibration Q even negative code signal value, as it is currently captured in the sample and hold latches.
7	R/O	1'h0	Reserved
6:0	R/O	7'h0	**RX sampler latch calibration Q odd negative code**: RX sampler latch calibration Q odd negative code signal value, as it is currently captured in the sample and hold latches.

Register Name:			Sample and hold RX sampler latch calibration E predictive positive code register (!)
Register Mnemonic:			RX_DIAG_SH_SLC_EPP
Register Address:			10_m_nnnn : 1111 : 10001
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:8	R/O	7'h0	**RX sampler latch calibration E even positive code**: RX sampler latch calibration E even positive code signal value, as it is currently captured in the sample and hold latches.
7	R/O	1'h0	Reserved
6:0	R/O	7'h0	**RX sampler latch calibration E odd positive code**: RX sampler latch calibration E odd positive code signal value, as it is currently captured in the sample and hold latches.

Register Name:			Sample and hold RX sampler latch calibration E predictive negative code register (!)
Register Mnemonic:			RX_DIAG_SH_SLC_EPM
Register Address:			10_m_nnnn : 1111 : 10010
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:8	R/O	7'h0	**RX sampler latch calibration E even negative code**: RX sampler latch calibration E even negative code signal value, as it is currently captured in the sample and hold latches.
7	R/O	1'h0	Reserved
6:0	R/O	7'h0	**RX sampler latch calibration E odd negative code**: RX sampler latch calibration E odd negative code signal value, as it is currently captured in the sample and hold latches.

Register Name:			PI rate selection register (!)
Register Mnemonic:			RX_DIAG_PI_RATE
Register Address:			10_m_nnnn : 1111 : 10100
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:12	R/W	3'h0	**PI rate selection standard mode 3**: This field is used to scale the power of the CML cells inside the PI to tune it for a given data rate, by driving the rxda_pi_rate_sel signal going into the analog, when xcvr_standard_mode is set to 2'b11. The following is the encoding of this field. • 3'b000: 1/2X power scaling for the CML cells. This should be the default setting for 2.5G data rates. • 3'b001:1X power scaling for the CML cells. This should be the default setting for 5G data rates. • 3'b010: Reserved • 3'b011: 2X power scaling for the CML cells. This should be the default setting for 10G data rates. • 3'b100 - 3'b111: Reserved
11	R/O	1'h0	Reserved
10:8	R/W	3'h3	**PI rate selection standard mode 2**: This field is used to scale the power of the CML cells inside the PI to tune it for a given data rate, by driving the rxda_pi_rate_sel signal going into the analog, when xcvr_standard_mode is set to 2'b10. The following is the encoding of this field. • 3'b000: 1/2X power scaling for the CML cells. This should be the default setting for 2.5G data rates. • 3'b001:1X power scaling for the CML cells. This should be the default setting for 5G data rates. • 3'b010: Reserved • 3'b011: 2X power scaling for the CML cells. This should be the default setting for 10G data rates. • 3'b100 - 3'b111: Reserved
7	R/O	1'h0	Reserved
6:4	R/W	3'h1	**PI rate selection standard mode 1**: This field is used to scale the power of the CML cells inside the PI to tune it for a given data rate, by driving the rxda_pi_rate_sel signal going into the analog, when xcvr_standard_mode is set to 2'b01. The following is the encoding of this field. • 3'b000: 1/2X power scaling for the CML cells. This should be the default setting for 2.5G data rates. • 3'b001:1X power scaling for the CML cells. This should be the default setting for 5G data rates. • 3'b010: Reserved • 3'b011: 2X power scaling for the CML cells. This should be the default setting for 10G data rates. • 3'b100 - 3'b111: Reserved
3	R/O	1'h0	Reserved
2:0	R/W	3'h1	**PI rate selection standard mode 0**: This field is used to scale the power of the CML cells inside the PI to tune it for a given data rate, by driving the rxda_pi_rate_sel signal going into the analog, when xcvr_standard_mode is set to 2'b00. The following is the encoding of this field. • 3'b000: 1/2X power scaling for the CML cells. This should be the default setting for 2.5G data rates. • 3'b001:1X power scaling for the CML cells. This should be the default setting for 5G data rates. • 3'b010: Reserved • 3'b011: 2X power scaling for the CML cells. This should be the default setting for 10G data rates. • 3'b100 - 3'b111: Reserved

Register Name:			PI capacitor selection register (!)
Register Mnemonic:			RX_DIAG_PI_CAP
Register Address:			10_m_nnnn : 1111 : 10101
Bits	**Type**	**Reset**	**Description**
15	R/O	1'h0	Reserved
14:12	R/W	3'h0	**PI capacitor selection standard mode 3**: This field is used to implement waveform shaping inside the PI CML cells, by changing the capacitive loading. As the data rate decreases, this is used to increase cap loading, to maintain semi-sinusoidal wave shapes inside the PI CML input buffers and mixers, by driving the rxda_pi_cap_sel signal going into the analog, when xcvr_standard_mode is set to 2'b11. The following is the encoding of this field. • 3'b000: Data rates > 1.25 Gbps • 3'b001: Data rates <= 1.25 Gbps • 3'b010 - 3'b111: Reserved •
11	R/O	1'h0	Reserved
10:8	R/W	3'h0	**PI capacitor selection standard mode 2**: This field is used to implement waveform shaping inside the PI CML cells, by changing the capacitive loading. As the data rate decreases, this is used to increase cap loading, to maintain semi-sinusoidal wave shapes inside the PI CML input buffers and mixers, by driving the rxda_pi_cap_sel signal going into the analog, when xcvr_standard_mode is set to 2'b10. The following is the encoding of this field. • 3'b000: Data rates > 1.25 Gbps • 3'b001: Data rates <= 1.25 Gbps • 3'b010 - 3'b111: Reserved •
7	R/O	1'h0	Reserved
6:4	R/W	3'h0	**PI capacitor selection standard mode 1**: This field is used to implement waveform shaping inside the PI CML cells, by changing the capacitive loading. As the data rate decreases, this is used to increase cap loading, to maintain semi-sinusoidal wave shapes inside the PI CML input buffers and mixers, by driving the rxda_pi_cap_sel signal going into the analog, when xcvr_standard_mode is set to 2'b01. The following is the encoding of this field. • 3'b000: Data rates > 1.25 Gbps • 3'b001: Data rates <= 1.25 Gbps • 3'b010 - 3'b111: Reserved •
3	R/O	1'h0	Reserved
2:0	R/W	3'h0	**PI capacitor selection standard mode 0**: This field is used to implement waveform shaping inside the PI CML cells, by changing the capacitive loading. As the data rate decreases, this is used to increase cap loading, to maintain semi-sinusoidal wave shapes inside the PI CML input buffers and mixers, by driving the rxda_pi_cap_sel signal going into the analog, when xcvr_standard_mode is set to 2'b00. The following is the encoding of this field. • 3'b000: Data rates > 1.25 Gbps • 3'b001: Data rates <= 1.25 Gbps • 3'b010 - 3'b111: Reserved •

Register Name:			PI tuning register (!)
Register Mnemonic:			RX_DIAG_PI_TUNE
Register Address:			10_m_nnnn : 1111 : 10110
Bits	**Type**	**Reset**	**Description**
15:8	R/O	8'h0	Reserved
7	R/W	1'h0	**Receiver CML to CMOS rate select value standard mode 3**: This bit will drive the rxda_c2c_rate_sel signal going into the analog, when xcvr_standard_mode is set to 2'b11. • 1'b0 : High resistance • 1'b1 : Low resistance
6	R/W	1'h1	**Receiver CML to CMOS rate select value standard mode 3**: This bit will drive the rxda_c2c_rate_sel signal going into the analog, when xcvr_standard_mode is set to 2'b11. • 1'b0 : High resistance • 1'b1 : Low resistance
5	R/W	1'h1	**Receiver CML to CMOS rate select value standard mode 3**: This bit will drive the rxda_c2c_rate_sel signal going into the analog, when xcvr_standard_mode is set to 2'b11. • 1'b0 : High resistance • 1'b1 : Low resistance
4	R/W	1'h0	**Receiver CML to CMOS rate select value standard mode 3**: This bit will drive the rxda_c2c_rate_sel signal going into the analog, when xcvr_standard_mode is set to 2'b11. • 1'b0 : High resistance • 1'b1 : Low resistance
3:1	R/O	3'h0	Reserved
0	R/W	1'h1	**PI current select**: Selects either the external based bias or the poly based bias for the PI, by driving the rxda_pi_cur_sel signal going into the analog. • 1'b0: Poly based bias • 1'b1: External based bias

Register Name:			RX loopback controller register (!)
Register Mnemonic:			RX_DIAG_LPBK_CTRL
Register Address:			10_m_nnnn : 1111 : 11000
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5:4	R/W	2'h0	**Recovered clock loopback select**: Selects which recovered clock to use when recovered clock loopback is enabled. • 2'b00: I clock. • 2'b01: Q clock • 2'b10: E clock.
3:0	R/W	4'h0	**Attenuation settings**: Sets the attenuation for the ISI generation loopback filter, as specified below. • 4'b0000 : 6 dB • 4'b0001 • ... • 4'b0101 : 13 dB • ... • 4'b1000 : 16 dB • ... • 4'b1100 : 23 dB • ... • 4'b1101 • 4'b1110 : Highest attenuation • 4'b1111 : High impedance

Register Name:			Receiver control reset diagnostic register (!)
Register Mnemonic:			RX_DIAG_RST_DIAG
Register Address:			10_m_nnnn : 1111 : 11001
Bits	**Type**	**Reset**	**Description**
15:6	R/O	10'h0	Reserved
5	R/O	1'h0	Current state of the rxda_clk_reset_n reset.
4	R/O	1'h0	Current state of the rx_dig_reset_n reset.
3	R/O	1'h0	Current state of the rxda_cdrlf_reset_n reset.
2	R/O	1'h0	Current state of the rx_ree_fcn_reset_n reset.
1	R/O	1'h0	Current state of the rx_ree_ctrl_reset_n reset.
0	R/O	1'h0	Current state of the rx_lfps_det_filter_reset_n reset.

Register Name:			Receiver digital cover your alternatives register (!)
Register Mnemonic:			RX_DIAG_DCYA
Register Address:			10_m_nnnn : 1111 : 11110
Bits	**Type**	**Reset**	**Description**
15:0	R/W	16'h0	Reserved - spare

Register Name:			Receiver analog cover your alternatives register
Register Mnemonic:			RX_DIAG_ACYA
Register Address:			10_m_nnnn : 1111 : 11111
Bits	**Type**	**Reset**	**Description**
15:8	R/W	8'h0	Reserved - spare
7	R/W	1'h0	**Serial loopback sampler setup time reduction select (TX only lane implementation)**: For TX only lane implementations, this bit can be used to reduce the setup time (providing more hold margin) for the serial loopback data. This function is in place, in case there are unexpected timing issues, due to the lack of a CDRLF in the transmit only lane implementations. • 1'b0: Default setup time • 1'b1: Reduced setup time
6:2	R/W	5'h0	Reserved - spare
1	R/W	1'h0	**Force receiver signal detect clock enable**: This bit is used to force the receiver signal detect clock (rxda_sd_clk) driven from the analog to be enabled, independent of whether the analog signal detect function is enabled or not. • 1'b0: Clock not forced to be enabled • 1'b1: Clock forced to be enabled
0	R/W	1'h0	**Full rate clock duty cycle correction enable**: Enables the analog duty cycle correction function for the full rate clock that drives the divide by 2 circuit to generate the half rate I and Q clocks. • 1'b0: Enabled • 1'b1: Disabled

Change Log

Version	Description of changes
0.1	Release Date: November 20, 2017 • Preliminary version of user guide
0.2	Release date: January 22, 2018 • Updated chapters to reflect design spec changes and updates
0.3	Release Date: March 20, 2018 • Updated chapters to reflect design spec and programmer's guide changes (SER-14878)
0.4	Release Date: June 08, 2018 • Updated chapters to reflect design spec and programmer's guide changes (SER-15866)
0.5	Release Date: December 05, 2018 • Updated chapters include PHY and PMA overview section, CDB register descriptions, and recent design spec updates
0.6	Release Date: February 19, 2019 • Updated revision ID register.
0.7	Release Date: March 25, 2019 • Updated CDB register definitions, PHY configuration section, operations, procedures, options.
0.8	Release Date: May 06, 2019 • Updated versions of chapters reflect changes and updates made to design specs and programmer's guide
0.9	Release Date: July 10, 2019 • Updated versions of chapters reflect changes and updates made to design specs and programmer's guide
0.10	Release Date: December 10, 2019 • Updated overview section and chapters to reflect changes and updates made to design spec and programmer's guide
0.11	Release Date: September 02, 2020 • R130 version of user guide • Reflects updates implemented in PHY and PMA specs
0.12	Release Date: May 25, 2021 • Reflects updates implemented in PHY and PMA specs
0.13	Release Date: August 24, 2021 • Reflects updates implemented in PHY and PMA specs
0.14	Release Date: y 30, 2023 • Reflects updates implemented in PHY and PMA specs

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